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PSS®Express 33.9 GUI Users Guide

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PART 1: THE WORKING CASE

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SIEMENS

Chapter 1

Introduction

1.1 Overview

The Siemens PTI Power System Simulator (PSS®Express) is a package of programs for studies of power system transmission network and generation performance in both steady-state and dynamic conditions. PSS®Express handles power flow, fault analysis (balanced and unbalanced), network equivalent construction, and dynamic simulation. Using PSS®Express, the engineer can handle a wide range of investigations for the planning and operation of modern electric power systems.

The *PSS®E Program Operation Manual* is a comprehensive working guide to PSS®Express's capabilities. It documents:

- The operation and behavior of the various functional modules available in PSS®Express.
- The formats of the various input data files read by PSS®Express.
- The output produced by PSS®Express's functional modules

This manual, the *PSS®Express GUI Users Guide*, describes the operation of PSS®Express through its **Graphical User Interface** (GUI). The GUI is the primary interface to PSS®Express; it is also the interface through which new users are first exposed to PSS®Express. This manual is structured to help the user become familiar with the available tools and the manner in which they can be used for power system analytical investigations. It liberally references the *PSS®E Program Operation Manual* and other manuals supplied with the PSS®Express package (see PSS®E Documentation).

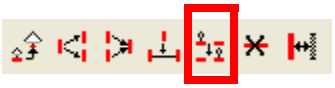
The functions and analyses available through the interface include:

- Power flow and related network functions
- One-line diagrams
- Optimal power flow
- Fault analysis
- Dynamic simulation
- Linear network analysis
- Transmission pricing and open access
- Network equivalencing
- Program automation

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1.2 Document Conventions

The following conventions are used in PSS®Express manuals:

Examples	Description
Power Flow > Solution > Parameters…	Navigation path in a PSS®Express GUI menu
[Solution Parameters] dialog	Interactive dialog in Graphical User Interface (GUI)
[Spreadsheet] **[Diagram]**	Interface windows and views
`SOLV,OPT` `LIST`	User entry the line mode in the Command Line Interface (CLI) dialog
[F10], **[Enter]**	Keys found on a standard computer keyboard
[OK], **[…]**, **[Close]**	Action buttons available on a dialog
	GUI toolbar button that starts an activity sequence
Set tap ratios to Unity	Activity option
Print outaged branches	Program option
`GENERATOR CONVERSION COMPLETED`	Message sent to **Progress** tab
`<quantity>, <bus number>`	Variable in message sent to **Progress** tab
`ENTER OUTPUT DEVICE CODE:`	Line mode dialog request/response
Progress tab, **Report** tab	Activity results displayed in **[Output]** view
OPEN	PSS®Express activity name with hyperlink to source information
Section 5.15.2, Figure 12-4	Cross-reference with hyperlink to source information
Creating a Power Flow Raw Data File	Short reference with hyperlink to source information
PSS®E Program Operation Manual, Section 18.14, Listing Saved Case or Snapshot Files	Full reference with hyperlink to source information
	Indicates report output example clipped to fit the page
PSS®E Program Application Guide	PSS®Express manual reference
	Indicates additional information of interest.
	Indicates important information.
n.a.	Abbreviation meaning "not applicable".

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1.3 Startup Options

Use one of the following methods to start the program:

- From the Windows® Start menu, select ***Programs > PSSE 32 > PSS®Express 32***.

  This is the default location as established during program installation. Other applications within suite of tools may also be initiated from the Windows® Start menu.

- From the Windows® Explorer application, double-click the psse32.exe file.

  C:\Program Files\PTI\PSSE32\PSSBIN is the default directory location of psse32.exe. An alternate directory location may be specified during program installation.

- From the Windows® Start menu, select ***Programs > PSSE 32 > PSSE-32 Command Prompt*** and enter *psse32.exe* (or simply *psse32*) at the DOS prompt, followed by **[Enter]**. See *PSS®E Program Operation Manual* Section 3.2.1, Startup Command Options for arguments that may be used in the startup sequence.

  These arguments may be appended to the psse32.exe path name specified in the Target field of the **[Properties]** window, opened by right-clicking on the PSS®Express desktop shortcut icon (see Figure 1-1).

- Start a PSSE-32 Command Prompt and type prog params where:

  prog = name of program (e.g., IPLAN32) that is to be executed.

  params = whatever start-up parameters are appropriate for the program.

- Double-click a previously defined PSS®Express shortcut icon located on the desktop.

⚠ Do not try to close the copyright window that opens during PSS®Express-32 startup.

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1.4 Establishing the Working Directory

PSS®Express always operates out of a working directory. By default, this is the EXAMPLE subdirectory under its main directory. You are not restricted to this directory or, for that matter, to a single working directory. Refer to *PSS®E Program Operation Manual*, Section 2.1, Directories and Files Overview.

A simple way to administer a variety of working directories is to create a shortcut icon for each one. Alter the properties of the shortcut to indicate the directory in which the program is to start. In Figure 1-1 the directory path specified in the *Start in:* field may be set to an alternate location.

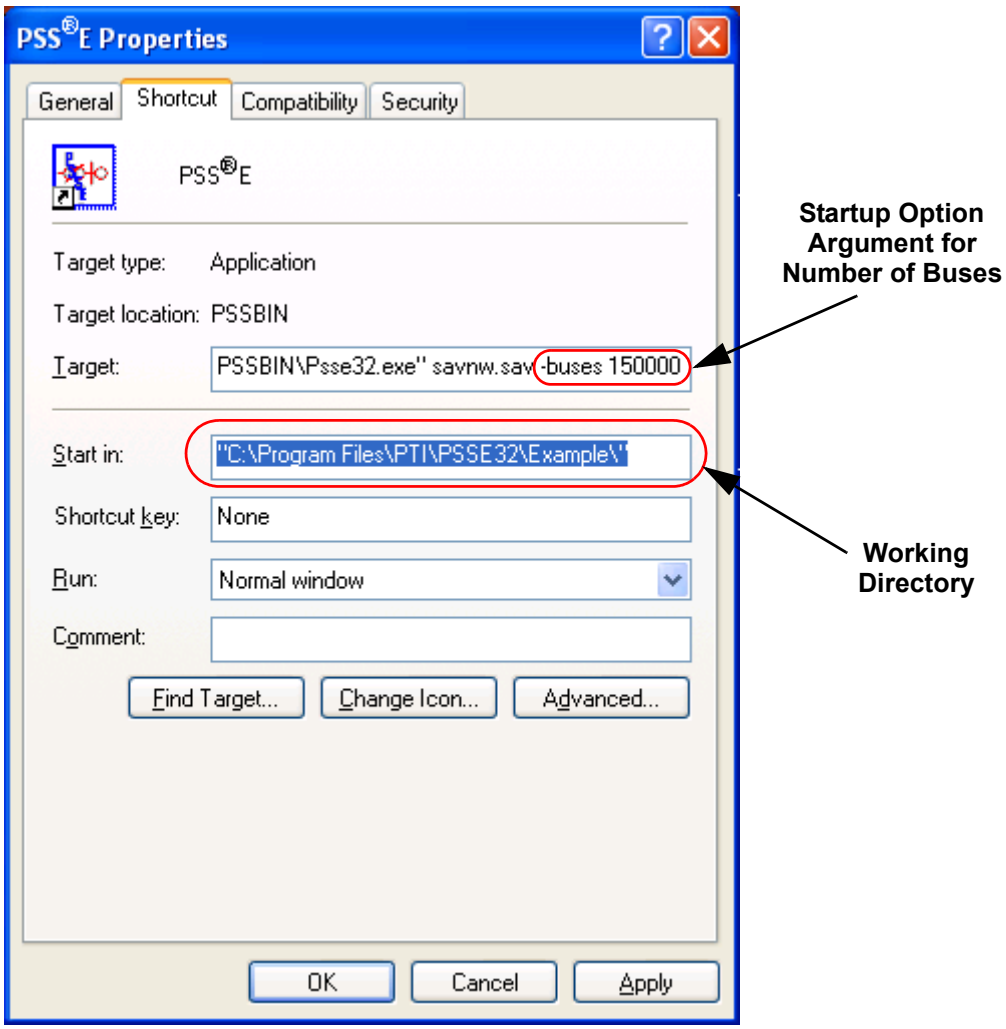


Figure 1-1. Properties Dialog

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1.5 Changing Program Preferences

Edit > Preferences…

The **[Program Preferences]** dialog allows the specification of options when either **[Spreadsheet]** or **[Diagram]** is active.

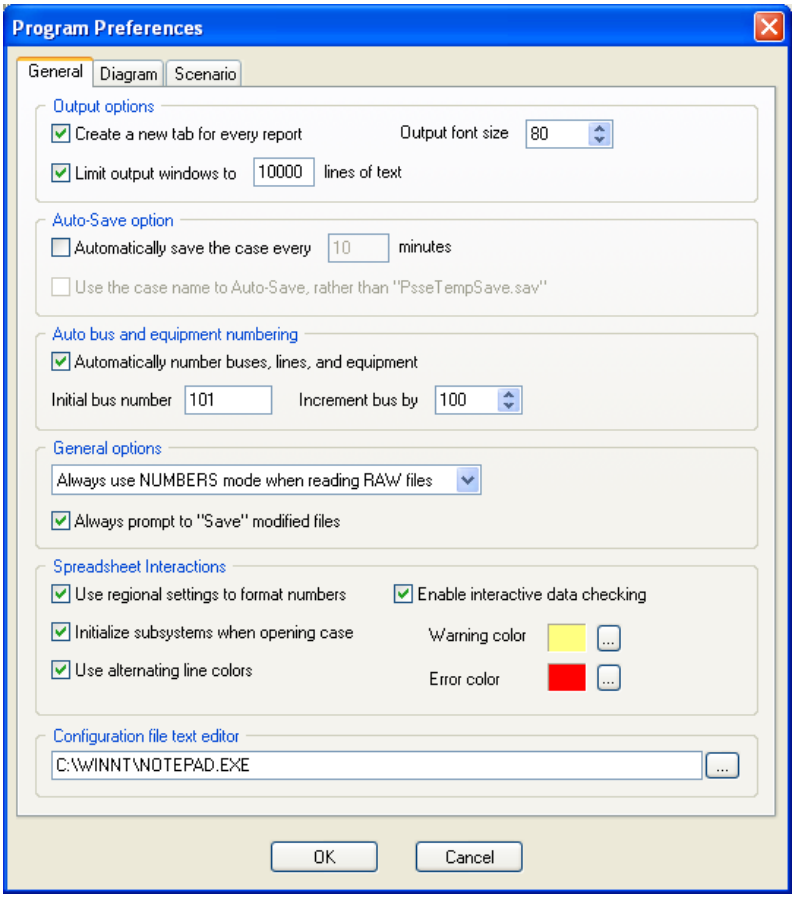


Figure 1-2. Program Preferences Dialog

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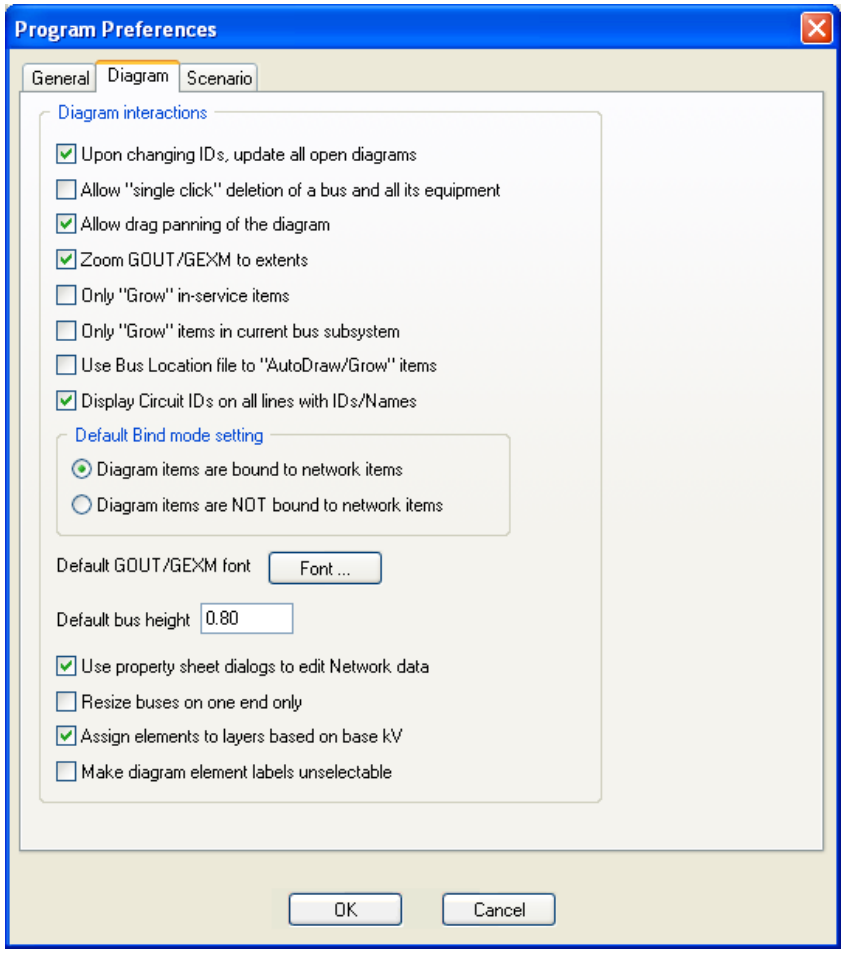


Figure 1-3. Program Preferences Dialog - Diagram Tab

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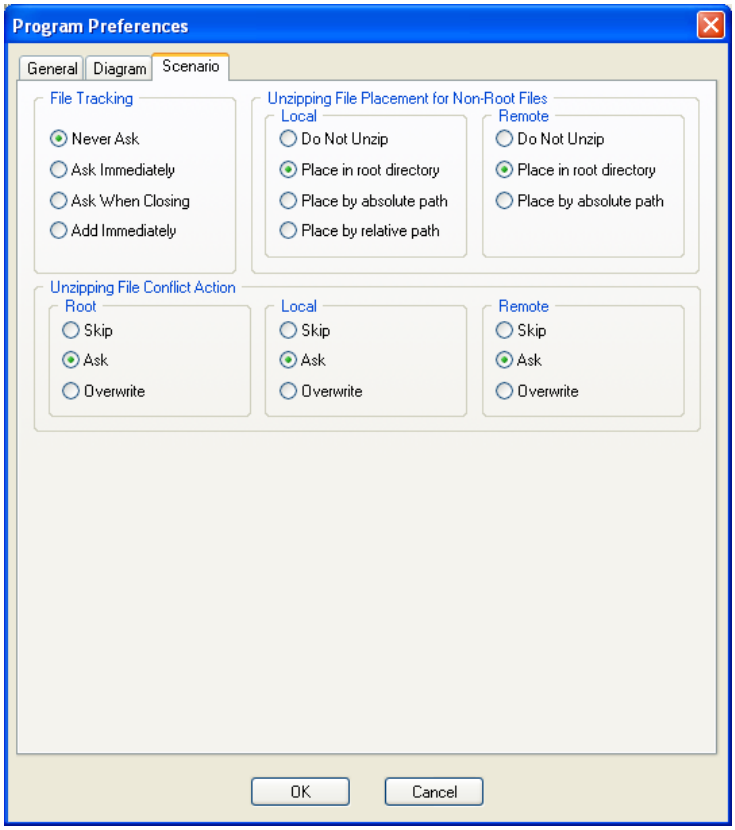


Figure 1-4. Program Preferences Dialog - Scenario Tab

The dialog (Figure 1-2, Figure 1-3, Figure 1-4) contains several groups of options, as defined in the following table.

Preference Type	Enabled Options
Output Options	Create a new tab for every report
	Limit output windows to <number> lines of text
Auto-Save option	Automatically save the case every <number> minutes
	Use the case name to Auto-Save, rather than PsseTempSave.sav
Auto bus and equipment numbering	Automatically number buses, lines, and equipment If you create a new case, this option automatically numbers the buses as they are added, starting with the *Initial bus number* and incrementing subsequent buses as specified.

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Preference Type	Enabled Options
General options	Bus data format options: • Always use NUMBERS mode when reading RAW files **OR** • Always use NAMES mode when reading RAW files **OR** • Always prompt me when reading RAW files The option for manual specification of bus names or bus numbers whenever a Power Flow Raw Data File is read into the working case opens the **[Read Power Flow Raw Data]** dialog, where the *For input, use bus names* option may be selected when specifying the PSS®Express version number that created the RAW file.
	Always prompt to save modified files
Spreadsheet interactions	Use regional settings to format numbers Initialize subsystems when opening case Use alternating line colors Enable interactive data checking Warning color Error color
Configuration file text editor	Open the directory of executable files (*.exe) to select a text editor that can be launched wherever an input configuration text file is specified. If a text file editor is specified, all the **[Edit...]** buttons used to edit configuration files in the various dialogs will be enabled.
Diagram interactions	Assign elements to layers based on base kV Make diagram element labels unselectable This option serves two purposes: 1. The initial selectability of newly created labels that are related to a diagram element (i.e. busbar, branch) will be based off of this option 2. Toggling this option will apply the chosen selectability to all applicable labels. Leaving the option in the same state it was in when opening the dialog will have no effect. N.B. This option will not apply to labels brought in from a saved diagram, only toggling the option will have an effect on current labels.
	Upon changing IDs, update all open diagrams Changing an ID in **[Spreadsheet]** automatically changes an ID in **[Diagram]**.
	Allow single-click deletion of a bus and all its equipment A specified bus on a diagram is deleted, along with all its equipment, either by using the **[Delete]** key or selecting *Delete* from the right-click menu.
	Allow drag panning of the diagram The mouse pointer turns to Pan mode when the left mouse button is clicked on the diagram, held down, and dragged. If disabled, a selection box is formed around that portion of the diagram.

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Preference Type	Enabled Options
Diagram interactions (*continued*)	Zoom GOUT/GEXM to extents A bus that is displayed in the diagram using the GOUT/GEXM activity will fill the entire **[Diagram]** area.
	Only *Grow* in-service items Only those items currently in-service will be drawn during a Grow operation.
	Only *Grow* items in current bus subsystem Only those items in the current bus subsystem will be drawn during a Grow operation.
	Use Bus Location file to *Grow* items If a Bus Location file has been opened, any Grow or Auto-Draw operations will use the Bus Location file when placing buses on the diagram.
	Display Circuit IDs on all lines with IDs/Names Circuit IDs are displayed for every line in **[Diagram]** that is bound to a line in **[Spreadsheet]**.
	Allow *Grow* from GOUT/GEXM A diagram generated from GOUT/GEXM can run the *Grow* operation. This method is not the same as the auto draw grow method. When the user specifies the *Grow* operation for a bus, all lines connected to it and all buses connected to those lines are drawn. In other words, *Grow* in GOUT redraws the specified bus.
	Default Bind mode setting • Diagram items are bound to network items **OR** • Diagram items are NOT bound to network items When creating a one-line diagram, the addition of network items into **[Diagram]** also populate **[Spreadsheet]** with data as you build a new network. Otherwise, the new network items display only in **[Diagram]**.
	Default GOUT/GEXM font Select a font for labels created in GOUT/GEXM diagrams.
	Default Bus Height Specify length of the bus, in inches, when either *Auto-Draw* or *Grow* operations are in effect in a one-line diagram.
	Default *Grow* Spacing Specify distance between buses, in inches, when either *Auto-Draw* or *Grow* operations are in effect in a one-line diagram.
	Use Dialogs to edit Diagram Items When this option is selected, double-clicking a network item in either **[Diagram]** or **[Network Tree]** will pop up a dialog to use to edit the data, rather than jumping to the appropriate spreadsheet entry.
	Resize buses on one end only When selected in the Diagram view, buses will resize from one end only, not both ends.
File Tracking	Specify when the Scenario prompts for adding new files

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Preference Type	Enabled Options
Unzipping File Placement for Non-Root Files	Select placement for files outside of the root directory
Unzipping File Conflict Action	Select action to take when a file conflict is detected when unzipping files

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1.6 Defining the Program Size

The *PSS®E Program Operation Manual*, Section 3.3.1, Specifying the Maximum Number of Buses describes the dimensional capacities of the various elements at standard size levels. The default bus size level, established during program installation, is defined at startup.

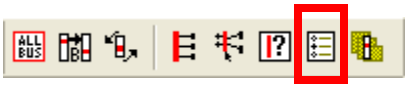
An alternate size level may be specified by selecting ***Misc > Change program settings (OPTN)…***, and entering the desired size level in the *Startup bus dimension* field. The new bus size level will take effect **[OK]** is clicked. An alternative is to specify the *-buses <number of buses >* argument (see Figure 1-1).

When opening a Saved Case File, activity CASE automatically redimensions PSS®Express if the Saved Case exceeds PSS®Express's current capacity limits.

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1.7 Changing Program Settings

Activity OPTN



Misc > Change program settings (OPTN)…

Most PSS®Express calculation and reporting functions recognize one or more program settings. When PSS®Express is installed on the system, default program settings are established; see **[Program Settings]** dialog in Figure 1-5. Refer to *PSS®E Program Operation Manual*, Section 3.3.3, Program Run-Time Option Settings.

1.7.1 Extended Bus Names

When the *Bus input* option *Extended Name* is enabled in **[Program Settings]**, extended bus names **must** be specified in data fields designating buses on PSS®Express activity dialogs. The extended bus name is formed by concatenating the 12-character bus alphanumeric name and the bus base voltage. The various selectors available by clicking **[Select...]** next to these input fields honor the *Bus input* option setting.

The requirements to specify an extended bus name are:

1. It must be enclosed in single quotes.
2. The twelve-character bus name, including any trailing blanks, must be the first twelve characters.
3. The bus base voltage in kV must immediately follow the bus name. Up to six characters may be used.

Thus, valid forms of an extended bus name include '*aaaaaaaaaaaavvvvvv*' and '*aaaaaaaaaaaavv*'.

See *PSS®E Program Operation Manual*, Extended Bus Names for details on the use of extended bus names in Power Flow Raw Data Files. Note that the use of extended bus names in these files is governed by an option of the input activity; it is independent of the *Bus input* option setting.

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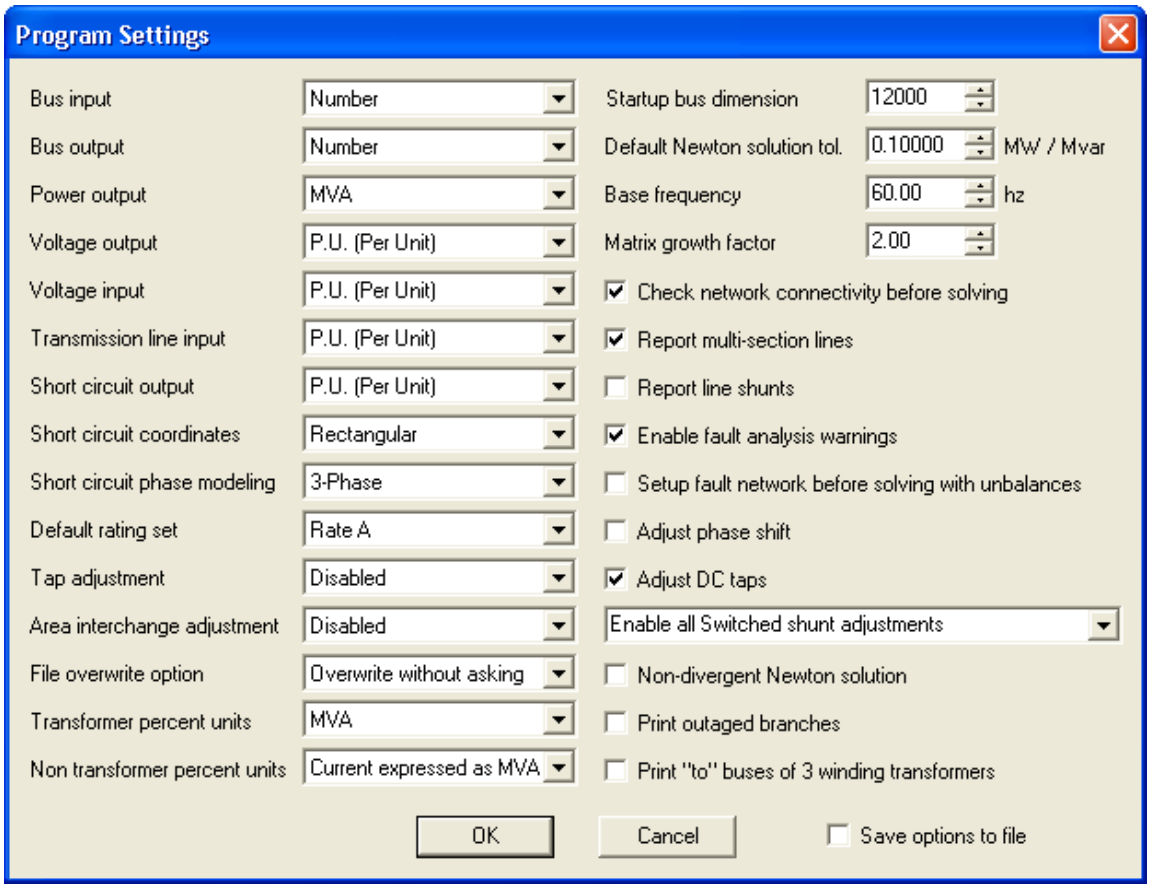


Figure 1-5. Program Settings Dialog

See *PSS®E Program Operation Manual* Section 18.10, Changing PSS®E Program Settings for field definitions.

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1.8 Displaying and Resetting Timing Statistics

Activity TIME

Misc > Display timing statistics (TIME)

This activity obtains and displays execution time statistics during a work session. It initializes a timer and routes a message to the **Progress** tab. For example:

```
FRI, MAY 21 2004 08:36 - TIMER INITIALIZED
```

On subsequent runs of this activity, a summary of elapsed time, in seconds, since the previous run and cumulative times from the point at which the timers were last initialized are printed. In addition, two additional system statistics are printed, one of which indicates CPU utilization. The output is displayed in the following format:

```
FRI, MAY 21 2004  15:21   ELAPSE    USER      KERNEL
  SINCE LAST "TIME"       XX.XXX    XX.XXX    XX.XXX
  CUMULATIVE              XX.XXX    XX.XXX    XX.XXX
```

The timing process is not sensitive to any interrupt control code options.

At any time during a work session in PSS®Express, if ***Misc > Reset timing statistics to zero (TIME,INIT)*** is selected, the timers are initialized and the same initialization message is sent to the **Progress** tab.

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1.9 The User Interface

The interface provides the following components (see Figure 1-6):

- **Tree View:** All network items are represented as selectable elements in a hierarchical list. Items in the list are organized by data type and are displayed in expandable/collapsible folders.
- **Spreadsheet View:** Importing a power flow case file or raw data file populates the spreadsheet with network data. Tabs along the bottom of the **[Spreadsheet]** allow specification of the various data categories.
- **Output View:** Progress, alerts/warnings, and report tabs are displayed in this expandable window. Alerts and warnings also appear in red text in the progress stream.
- **Diagram View:** Creates one-line diagrams in Slider format. In addition to showing power flow results, **[Diagram]** facilitates the building of new diagrams and networks bus by bus. For existing power flow cases, this view enables the *growing* of one-line diagrams by automatically drawing specified buses and all their equipment and connected buses. **[Diagram]** view opens only when a diagram is opened or created.
- **Toolbars:** Allows convenient selection of analytical tools, creation of one-line diagrams, generation of reports, specification of subsystems, and view management.
- **Main Menu:** Provides access to file handling, interface views, analytical functions, automation tools, I/O formatting, toolbar organization and online help.
- **Status Bar:** Provides information related to the diagram status and operating mode.
- **Command Line Interface (CLI) Window**: Provides a field for command line input and a pull-down list for reviewing command history. Another pull-down list provides selection of the desired command language.

Each analytical activity is available directly from pull-down menus and user-customizable toolbars. With some exceptions, the functional activities of the old interface exist in the new interface. For convenience, the traditional activity names are shown on many of the menu items.



All view windows can be individually re-sized and located anywhere on the interface between the **Status** bar and the toolbars. The user can open and close (hide) **[Tree]**, **[Output]**, and **[Command Line]** and the **Status** bar by toggling the *View* menu items. An alternative way to close the view is to right-click within the view and select the *Hide* option. Right-clicking in **[Tree]** or **[Output]** brings up a menu with *Allow Docking* option. Toggling this option ON and OFF allows the corresponding view to float as a window or be docked to the main window. When undocked, the two views can be moved at will and resized. **[Diagram]** and **[Spreadsheet]** cannot be undocked but can be controlled using the standard Windows® controls (i.e., Minimize, Maximize, Move, etc.).

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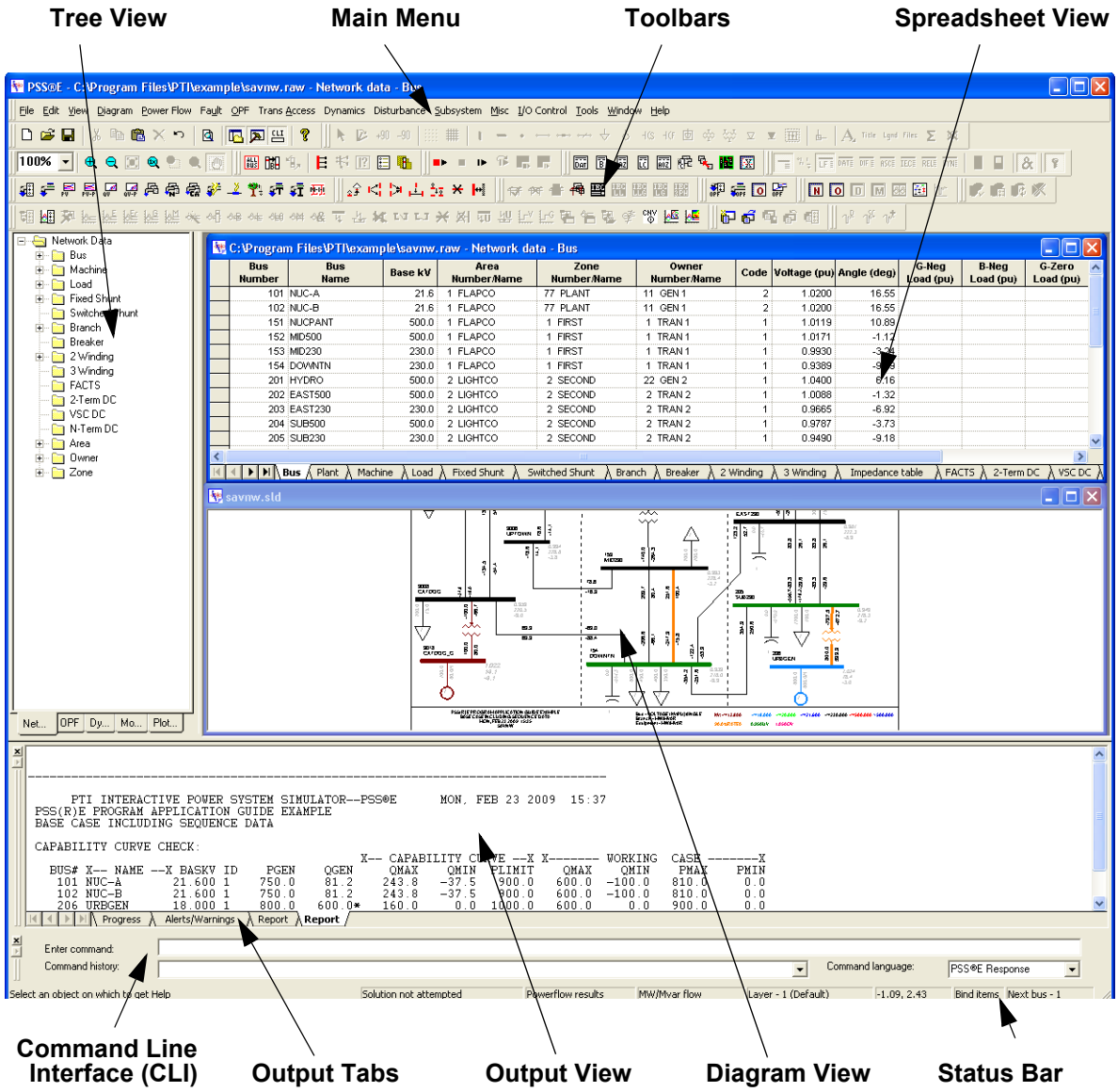


Figure 1-6. Key Elements of the Interface

The default view arranges **[Diagram]** and **[Spreadsheet]** views in a stack. Data changes can be made only on the active window, that is, the top view. Click the appropriate button to make the desired view active (see Figure 1-7).

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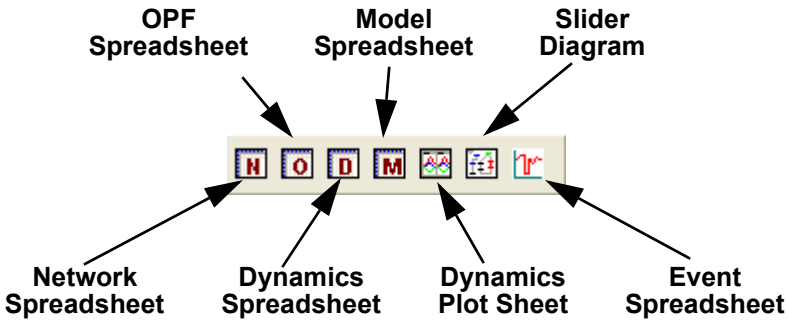


Figure 1-7. View Activation Buttons

For convenience, Appendix A of this manual provide summaries of the toolbar options. A complete description of **[Spreadsheet]** may be found in Chapter 2, **[Diagram]** in Chapter 3, and **[Tree]** in Chapter 4.

The **Status** bar displays explanatory text while you are using PSS®Express. For example, when moving the mouse arrow over a toolbar button the function of that button is displayed in the left portion of the **Status** bar.

[Output] is used to display program information, dialog, errors, and warning messages. It is also used to display analysis output formatted as text reports.

Standard Microsoft Windows® capabilities for selecting and copying text to the clip board or saving it to an external file are supported in both views. This allows for easy transition between PSS®Express and external applications such as Microsoft Excel® or Microsoft Word®.

The Copy/Paste functions relate to the type of window involved. Specifically, when you copy or cut from **[Diagram]**, you can only paste back into **[Diagram]**. When working from **[Spreadsheet]**, when you copy a cell or group of cells, you can paste the results into **[Spreadsheet]** or an outside application like Microsoft Excel®. Further, you can copy a piece of your network from a diagram into PSS®Express's clipboard, you are allowed to copy a collection of cells from a Spreadsheet view into PSS®Express clipboard at the same time; PSS®Express will keep track of both copies. Afterwards, when you paste back onto the spreadsheet or an external application like Notepad, the software will know to paste the copied data from the spreadsheet. If you paste on the diagram, the software will paste the copied data from the diagram.

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1.9.1 Using the Command Line



View > CLI Window

Many users remain faithful to the origins of PSS®Express when the Command Line Interface (CLI) was used for all interactive dialogs with the program. Command line input continues to be available by selecting this menu option. The resulting command line input window supports the use of legacy PSS®Express activity names, batch (BAT_) commands, and Python commands (see Figure 1-8).

> Line Mode commands entered in the Command Line Interface (CLI) are handled by a Line Mode Interpreter (LMI); Batch Commands are handled by the Batch Command interpreter, and Python Commands are passed to the embedded Python interpreter. When using the Line Mode, prompts are provided that resemble the traditional Line Mode prompts; when an activity is completed the "Activity?" prompt is presented. When an incomplete Batch Command is entered, a prompt of the form *API-name:* is presented.

To enter batch commands or activity names, select *PSS®Express Response* from the pull-down list.

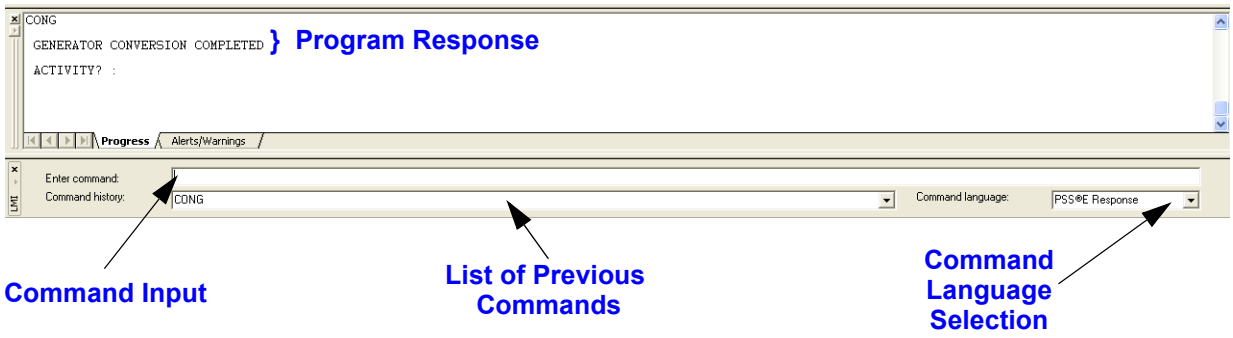


Figure 1-8. Command Line Interface Operation

An interactive session is run by typing in activity names and responding to program responses in the traditional manner. The user input must be in the *Enter command* field. Program responses are seen in the **Progress** tab.

Using the command line input field will lock out access to some aspects of the program until the current command being executed is completed. The lock out aspect depends on the API being invoked.

Re-selecting ***View > CLI Window*** closes **[Command Line]**. If the *Auto-Save option* has been specified in **[Program Preferences]**, the **Progress** tab displays the time saved and location of the saved case.

The manual *PSS®E Command Line Interface (CLI) Users Guide* describes the operation of PSS®Express using its command line interface.

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1.10 Creating a Working Case

Selecting ***File > New*** facilitates the creation of a new diagram, a new power flow case, a new diagram and power flow case, or a plot sheet. Figure 1-9a shows the **[New]** dialog.

If a *Network case* option is specified, then the **[Build New Case]** dialog opens for initiation of data input (Figure 1-9b). After the system *Base MVA* is entered and two (optional) *Heading* lines are input, a blank **[Spreadsheet]** view is displayed. A new case can be built entirely in **[Spreadsheet]** or jointly with **[Diagram]** (see Chapter 2 and Chapter 3).

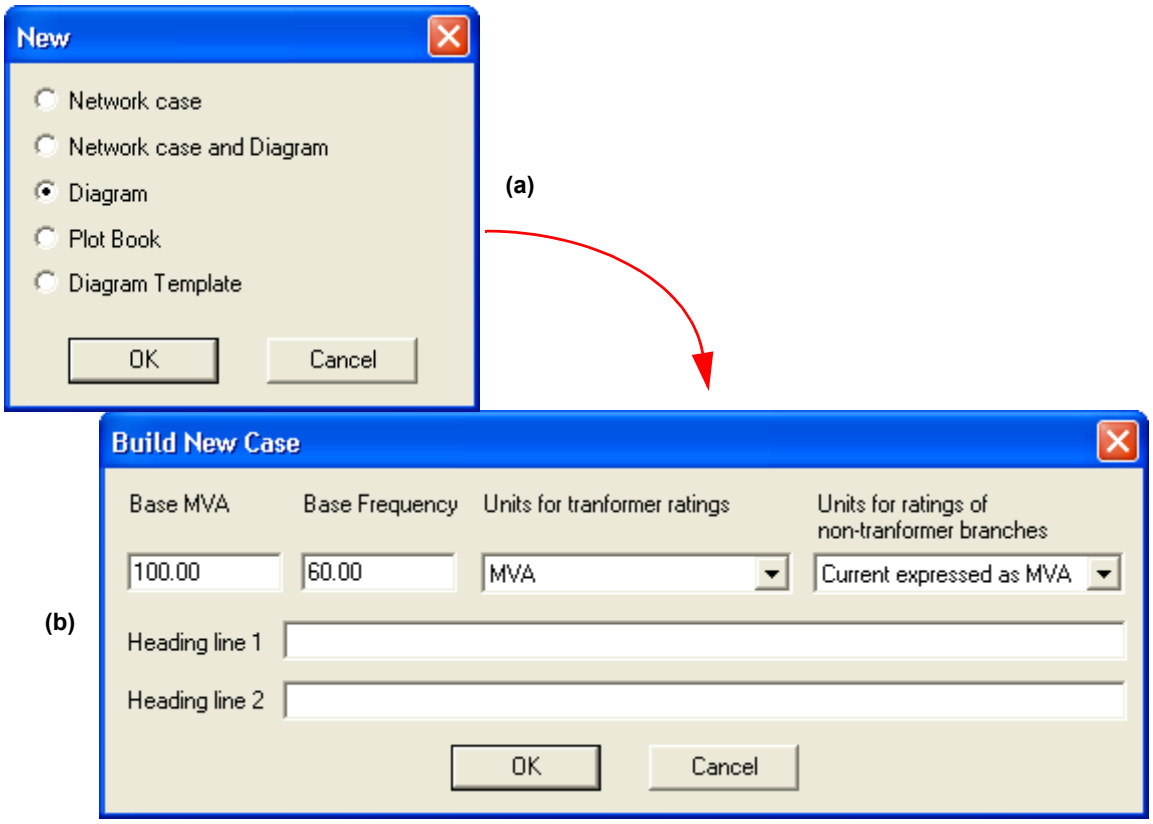


Figure 1-9. New and Build New Case Dialogs

The system stores the heading lines as comments if the case is exported. They are used to document the case.

A new working case can be created merely by typing data elements directly into a blank spreadsheet. This method would typically be limited to only small cases or to establish a small network model into which significantly more data could be pasted into or imported.

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1.11 Data Input

The representation of power networks in PSS®Express comprises several data categories of network and equipment elements, each of which requires a particular type of data. The data categories and the order in which they are input are described in the *PSS®E Program Operation Manual,* Section 5.2.1, Power Flow Raw Data File Contents. In addition to the basic data set describing the network elements, PSS®Express requires other data and control element files for specific applications. Not all the data described in Part 2: PSS®Express Activities are needed for all applications, and some data can be defaulted.

The bulk power flow data input facility ***File > Open…*** imports hand-typed power flow source data from a correctly formatted Power Flow Raw Data File (.raw) and enters it into the power flow working case, rearranging it from its original format into a computationally oriented data structure in the process (see Section 6.3) .

All data is read in free format with data items separated by a comma or one or more blanks. Tabbed delimited data items are not recommended. The ***File > Open…*** selection may also import binary saved case files (*.sav) containing power flow data as well as solution values and related options (see Section 6.1).

A list of all data formats available for input into a new case follows. **[Spreadsheet]** cells are populated by each data set automatically, according to PSS®Express activity requirements.

Saved Case file (*.sav)	Section 10.1
Slider Binary file (*.sld)	Section 3.15
Power Flow Raw Data file (*.raw)	Section 10.4
Power Flow Data file, Options (*.raw)	Section 6.4, Section 6.3.2
Dynamics Snapshot Data file (*.snp)	Section 20.6
Dynamics Model Raw Data file (*.dyr)	Section 20.1
Add Dynamics Model Data (*.dyr)	Section 20.1.1
Dynamics Snapshot Raw Data file (*.srs)	Section 20.3
Bus Location Data file (*.loc)	Section 3.6.8
Sequence Data file (*.seq)	Section
Optimal Power Flow Data file (*.rop)	Section 19.7
Transactions Raw Data file (*.mwm)	SECTION DELETED
Machine Impedance Data file (*.rwm)	Section 10.3
Slider XML file (*.sldxml)	Section 3.15
Ucte Data file (*.uct)	*UCTE Data Exchange Format Application Guide and Manual*
Event Study file (*.evs)	Section 25.1
Channel Output file (*.out)	Section 20.4

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1.11.1 Resetting Dialog Options

Edit > Reset all dialog options to defaults

During operation of PSS®Express, the last used options specified in each dialog are saved. However, the user may want to go back to the default dialog options but does not remember what they are. This function restores all options specified in the dialog boxes to their respective defaults.

1.11.2 Closing the Application

File > Exit

This function closes the application. If the default option *Always prompt to save modified files* is enabled in **[Program Preferences]**, you will see a prompt requiring confirmation to save changes to the working case. If this option is disabled, all files are closed without saving, regardless of their status, modified or un-modified. This setting is preserved between from the current work session to the next.

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Chapter 2

Spreadsheet View

2.1 Introduction

All network data components (e.g., buses, lines, loads) are represented on worksheet-style tabs in **[Spreadsheet]** view. At program startup **[Spreadsheet]** is displayed only after a raw data or saved case file is opened.

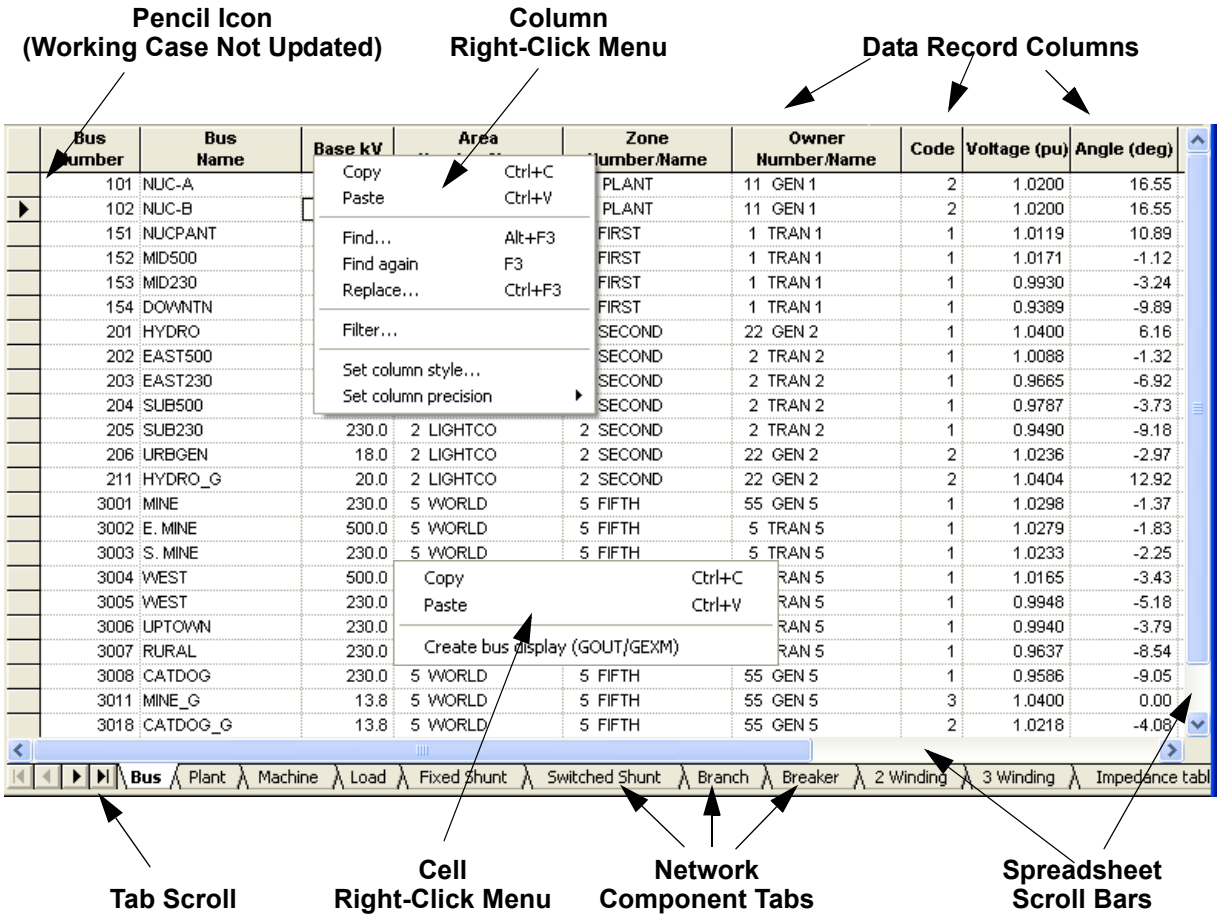


Figure 2-1. Spreadsheet View

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Spreadsheets are constructed to display data in arrangements that are convenient to review and modify for specific PSS®Express activities. Although many of the column headings are abbreviated for easier scanning, the following lists are unabbreviated.

[Network Spreadsheet] is the default view for the interface and remains open once it is populated. Tabs are provided to review and modify data for the following network component types:

Bus	Plant	Machine	Load
Fixed Shunt	Switched Shunt	Branch	Breaker/Switch
2-Winding Transformer	3-Winding Transformer	Impedance Table	FACTS Device
2-Terminal dc Line	VSC dc Line	Multi-terminal dc Line	Area
Inter-area Transfer	Owner	Zone	Multi-section Line
Mutual Coupling			

For more information, refer to Section 2.3, Network Spreadsheet.

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2.2 Editing and Formatting the Spreadsheet

ITo record data changes made in spreadsheet cells, either point to another row or press **[Enter]**. The pencil symbol disappears.

IIf a column is grayed out on a spreadsheet tab, the cells cannot be modified from this location. For instance, a bus number can be changed using the *Bus* tab on the Network Spreadsheet, but it cannot be changed from the *Plant*, *Machine*, *Load*, etc., tabs. Read-only columns can be changed to a different color, if desired (see *Set column style* option below).

2.2.1 Menu Options

The right-click menu supports standard Windows® commands, such as copy and paste actions. Multiple rows can be selected by holding the **[Shift]** key to include all rows dragged or holding the **[Ctrl]** key to select row-by-row. It is possible to copy and paste data to and from another application, such as Microsoft Excel®.

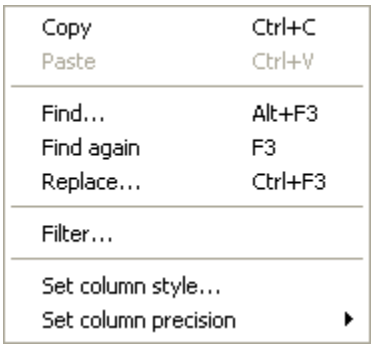


Figure 2-2. Spreadsheet Column Right-Click Menu

Bus numbers and other identifiers are not copied as this would result in duplicate data items in the network. Instead, the next available unused identifier is placed in the field. If the copied area exceeds the number of rows available, PSS®Express will automatically create the extra network elements required (and generate appropriate bus numbers).

The column heading right-click menu (Figure 2-2) provides access to the **[Find]** and **[Replace]** dialogs (also available through the *Edit Menu*).

Sorting and filtering capabilities are provided to increase usability, especially with large systems.

Any data edits performed within a filtered spreadsheet will automatically be reflected in the original unfiltered sheet. This allows the use of a reduced data set on which to perform edits.

Spreadsheet appearance and data entry precision can also be customized using this menu. Highlight the entire column to use these features.

Set column style opens the **[Cells]** dialog (Figure 2-3), with tabs to change the font, color, and border of the cells in the highlighted column. *Set column precision* opens an option list that permits data precision up to 6 decimal places.

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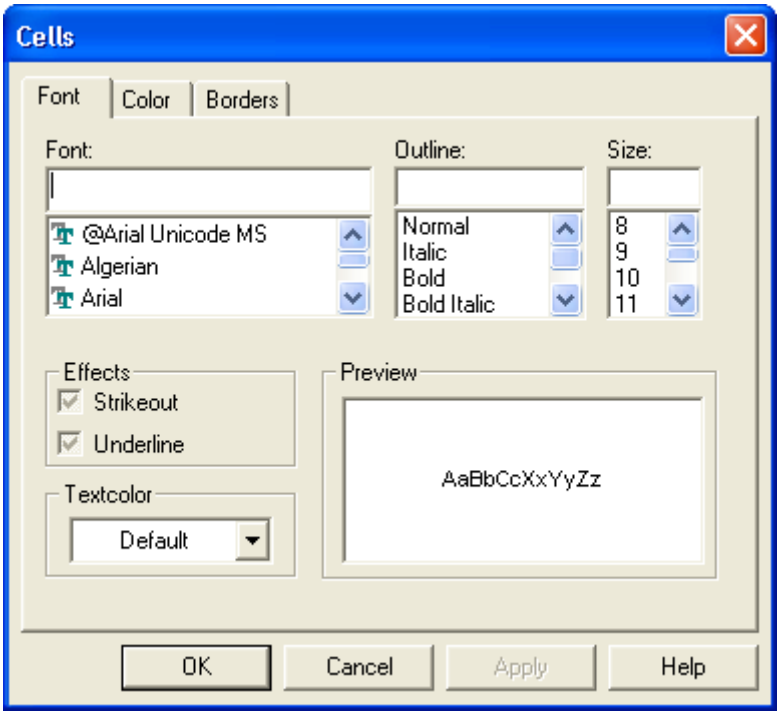


Figure 2-3. Cells Dialog

Color coding as a data visibility option is also available for those columns that are grayed out, that is, not available for edit on a particular worksheet. However, only the appearance of the column changes; the data record can be edited only from the appropriate worksheet (for example, *Bus Number* from the Bus tab; *Pgen* from the Machine tab).

The toolbar options to zoom in/out and return to 100% are active in **[Spreadsheet]**, although magnification about 150% becomes illegible unless the row height is also increased.

The right-click menu for bus data records also contains the option ***Create bus display (GOUT/GEXM)***. Refer to Section 3.11, Creating a Bus Display for details.

File Menu options include options to set up printing formats for the spreadsheet data. See Section 2.2.5, Print Setup for details.

In order to access *File Menu* options, **[Spreadsheet]** must be the active view. Click in any cell to make it active if the desired option does not appear in the menu.

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2.2.2 Managing Columns

For data spreadsheets that contain a large number of columns it may be useful to lock the identification fields in place when scrolling to view the other columns. Highlight the columns and select ***Edit > Freeze Columns***. Use ***Edit > Unfreeze Columns*** to undo the action.

Some data columns may hold data that is of little or no importance to the analysis the user is doing. In order to help the user organize the column so they can be viewed easily,

Columns can be hidden from view if they are not important to the current analysis. Highlight the columns and select ***Edit > Hide Columns***. Use ***Edit > Unhide Columns*** to unhide all columns that were previously hidden.

To rearrange the order of the columns, highlight the column and drag it. While the column is being dragged, a red line indicates where the column will be dropped when the mouse button is released.

To reset **[Spreadsheet]** to the default layout, select ***Edit > Reset Active Spreadsheet*** or ***Edit > Reset All Spreadsheets***, as appropriate.

2.2.3 Header/Footer Setup

File > Spreadsheet Header/Footer Setup…

The **[Header / Footer]** dialog (Figure 2-4) provides settings for header and footer spacing, the alignment, page numbering and the style, size and other font characteristics. If the *Save settings to profile* option is enabled, then the saved profile will be activated at the next PSS®Express work session.

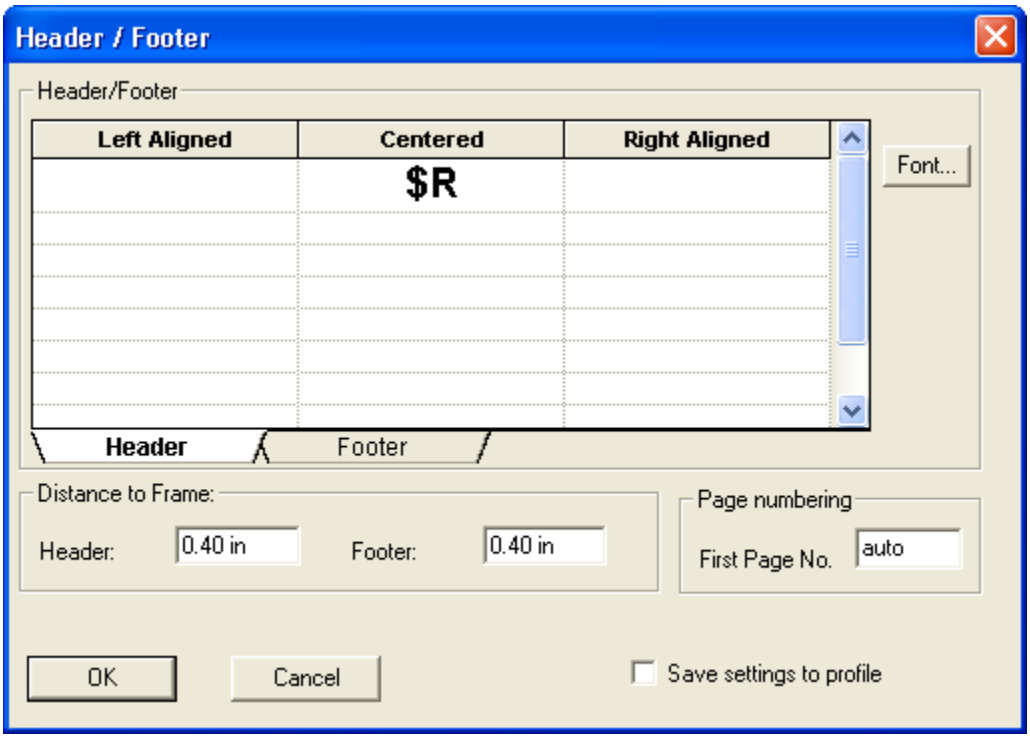


Figure 2-4. Spreadsheet Header / Footer Dialog

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2.2.4 Page Setup

File > Spreadsheet Page Setup…

The **[Page Setup]** dialog (Figure 2-5) provides settings for margins and grid lines. If the *Save settings to profile* option is enabled, then the saved profile will be activated at the next PSS®Express work session.

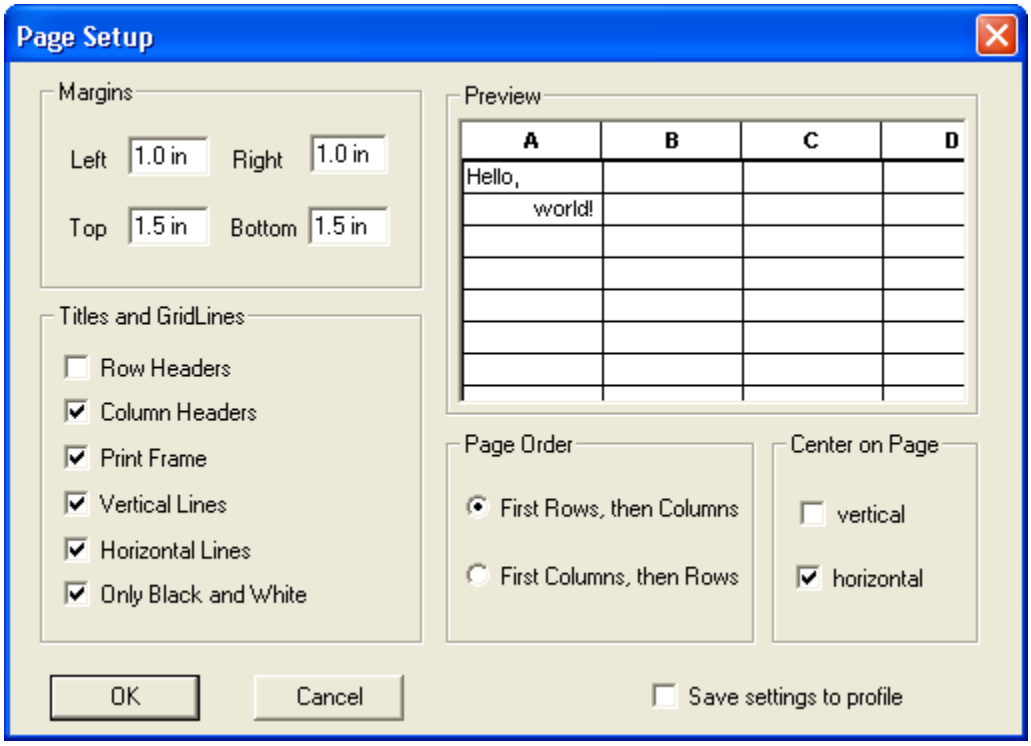


Figure 2-5. Spreadsheet Page Setup Dialog

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2.2.5 Print Setup

File > Print Setup…

The **[Print Setup]** dialog (Figure 2-6) is a standard Windows® dialog.

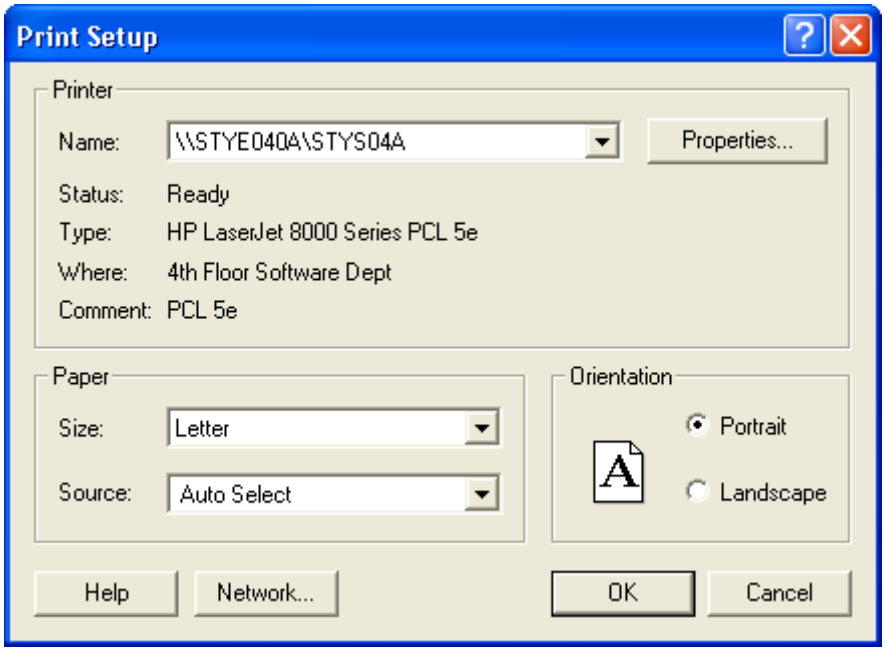


Figure 2-6. Print Setup Dialog

The ***File > Print Preview*** action provides view options of each page, using the [Next Page] button, and **[Zoom]** options from full page view to 100%. To return to **[Spreadsheet]** without printing, press the keyboard **[Esc]** key.

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2.3 Network Spreadsheet

[Network Spreadsheet] is synchronized with the bus subsystem selector (see Figure 2-9) so that a subset of the data can be viewed during a work session. It can be minimized, but you cannot close it unless you close the working case.

As shown in Figure 2-7, the *Bus* tab is the default worksheet; it displays the bus data records in the working case. Data records corresponding to each instance of the following network elements are available on separate tabs for review and modification:

- Bus
- Plant
- Machine
- Load
- Fixed Shunt
- Switched Shunt
- Branch
- Breaker/Switch
- 2-Winding Transformer
- 3-Winding Transformer
- Impedance Table
- FACTS Device
- 2-Terminal dc Line
- VSC dc Line
- N-Terminal dc Line
- Area
- Inter-area Transfer
- Owner
- Zone
- Multi-section line
- Mutual Coupling

Bus Number	Bus Name	Base kV	Area Number/Name	Zone Number/Name	Owner Number/Name	Code	Voltage (pu)	Angle (deg)	G-Neg Load (pu)	B-Neg Load (pu)	G-Zero Load (pu)	B-Zero Load (pu)
101	NUC-A	21.6	1 FLAPCO	77 PLANT	11 GEN 1	2	1.0200	16.55				
102	NUC-B	21.6	1 FLAPCO	77 PLANT	11 GEN 1	2	1.0200	16.55				
151	NUCPANT	500.0	1 FLAPCO	1 FIRST	1 TRAN 1	1	1.0119	10.89				
152	MID500	500.0	1 FLAPCO	1 FIRST	1 TRAN 1	1	1.0171	-1.12				
153	MID230	230.0	1 FLAPCO	1 FIRST	1 TRAN 1	1	0.9930	-3.24				
154	DOWNTN	230.0	1 FLAPCO	1 FIRST	1 TRAN 1	1	0.9389	-9.89				
201	HYDRO	500.0	2 LIGHTCO	2 SECOND	22 GEN 2	1	1.0400	6.16				
202	EAST500	500.0	2 LIGHTCO	2 SECOND	2 TRAN 2	1	1.0088	-1.32				
203	EAST230	230.0	2 LIGHTCO	2 SECOND	2 TRAN 2	1	0.9665	-6.92				
204	SUB500	500.0	2 LIGHTCO	2 SECOND	2 TRAN 2	1	0.9787	-3.73				
205	SUB230	230.0	2 LIGHTCO	2 SECOND	2 TRAN 2	1	0.9490	-9.18				
206	URBGEN	18.0	2 LIGHTCO	2 SECOND	22 GEN 2	2	1.0236	-2.97				
211	HYDRO_G	20.0	2 LIGHTCO	2 SECOND	22 GEN 2	2	1.0404	12.92				
3001	MINE	230.0	5 WORLD	5 FIFTH	55 GEN 5	1	1.0298	-1.37				
3002	E. MINE	500.0	5 WORLD	5 FIFTH	5 TRAN 5	1	1.0279	-1.83				
3003	S. MINE	230.0	5 WORLD	5 FIFTH	5 TRAN 5	1	1.0233	-2.25				
3004	WEST	500.0	5 WORLD	5 FIFTH	5 TRAN 5	1	1.0165	-3.43				
3005	WEST	230.0	5 WORLD	5 FIFTH	5 TRAN 5	1	0.9948	-5.18				
3006	UPTOWN	230.0	5 WORLD	5 FIFTH	5 TRAN 5	1	0.9940	-3.79				
3007	RURAL	230.0	5 WORLD	5 FIFTH	5 TRAN 5	1	0.9637	-8.54				
3008	CATDOG	230.0	5 WORLD	5 FIFTH	55 GEN 5	1	0.9586	-9.05				
3011	MINE_G	13.8	5 WORLD	5 FIFTH	55 GEN 5	3	1.0400	0.00				
3018	CATDOG_G	13.8	5 WORLD	5 FIFTH	55 GEN 5	2	1.0218	-4.08				

Bus | Plant | Machine | Load | Fixed Shunt | Switched Shunt | Branch | Breaker | 2 Winding | 3 Winding | Impedance table | FACTS | 2-Term DC | VSC DC | N-Term

Figure 2-7. Network Spreadsheet View, Bus Tab

Network elements may be added, modified, or deleted in the appropriate worksheet. If a subsystem selector has been activated to reduce the amount of network data presented at any one time, read-only fields are colored light gray to distinguish them from editable fields. See Section 2.3.2, Creating a Bus Subsystem and Section 2.3.3, Using the Subsystem Menu.

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2.3.1 Revising the Working Case

In order to revise the working case using [Network Spreadsheet], it must be the active view. Choose the most convenient way to make it active from among the following options:

- Toolbar selection 

- Double-clicking the item to be revised in [Network Tree] or [Diagram]

Most data fields may be edited directly by clicking in the cell and entering in the desired value. Editable area, owner, and zone cells have **[Area Selection]**, **[Owner Selection]**, and **[Zone Selection]** dialogs available by double-clicking in the cell.

Data fields that contain check boxes (for example, in-service, metered, and auto-adjust options) are considered to be enabled when checked and disabled when unchecked.

Some data cells require specification from a list of valid entries (for example, I/O code for a winding, impedance, or admittance). Clicking in the cell displays a pull-down list from which one value may be specified.

2.3.2 Creating a Bus Subsystem

A bus subsystem can be created directly from the bus data spreadsheet by highlighting the bus data records to be included and selecting the ***Create bus subsystem*** option from the right-click menu (Figure 2-8a). [Spreadsheet] will then show only that bus subsystem (Figure 2-8b).

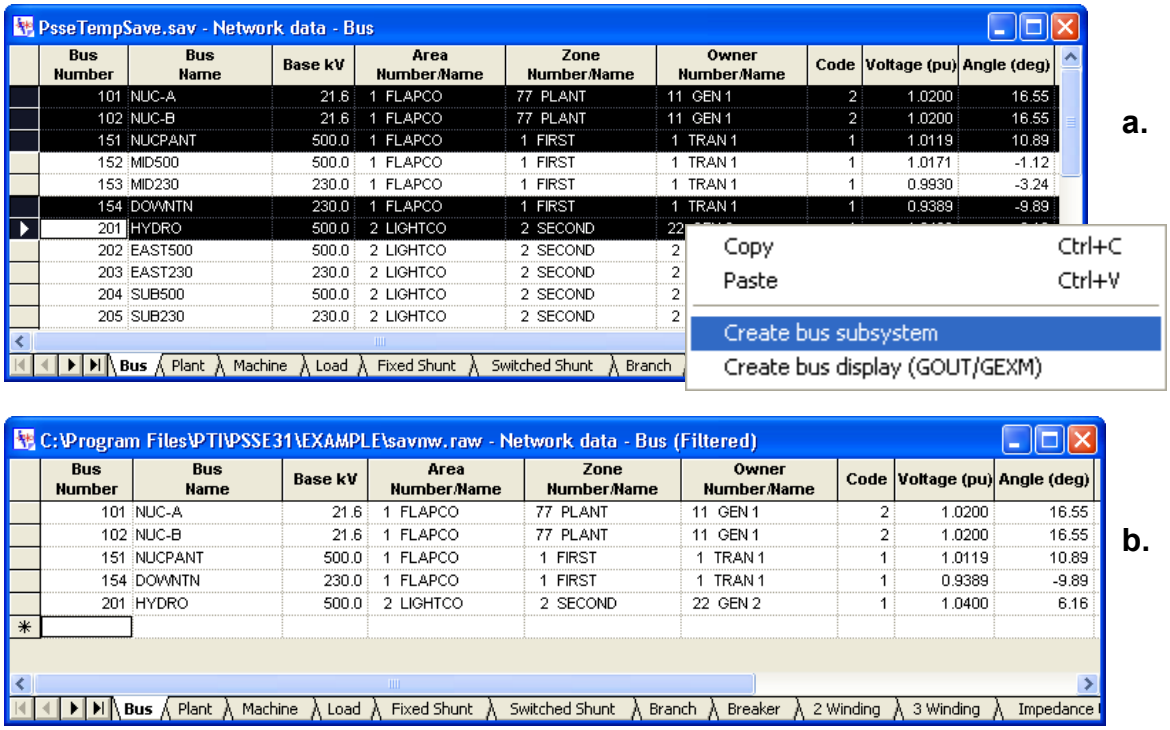
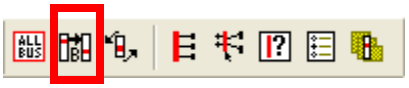


Figure 2-8. Creating a Bus Subsystem

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2.3.3 Using the Subsystem Menu



Subsystem > Bus...

A bus subsystem can also be created using the **[Bus Subsystem Selector]** dialog (Figure 2-9). Using this method, a subsystem can be built using filters by area, owner, zone, base kV, or bus numbers. Each category is available on a separate tab. For Areas, Owners, Zones, and Buses use the appropriate lists to specify the subsystem. Click **[Reset]** to return all entries in the *Selected* list to the *Unselected* list.

The *Buses* tab provides a **[Filter]** button that opens the **[Filter Bus Subsystem Selection]** dialog (Figure 2-10), an additional mechanism to exclude buses from the selection process in very large systems.

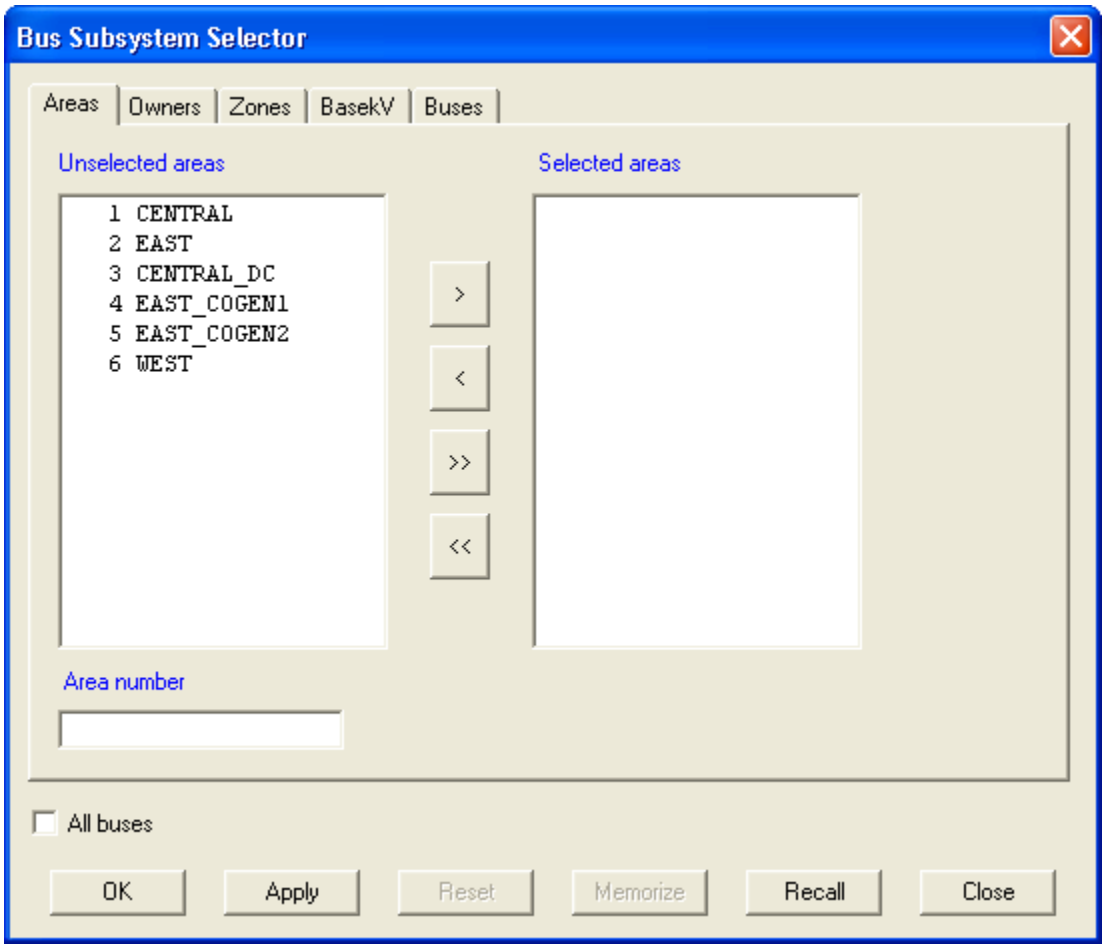


Figure 2-9. Bus Subsystem Selector Dialog

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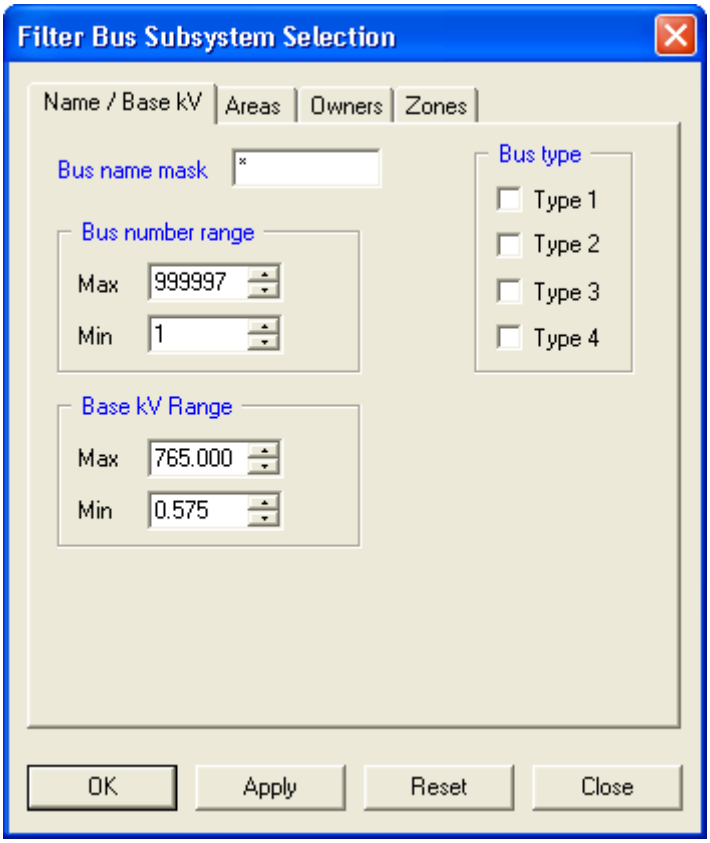


Figure 2-10. Filter Bus Subsystem Selection Dialog

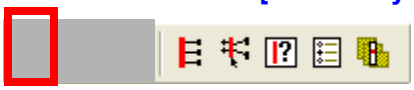
After specifying the subsystem, click **[Apply]**. The tab with a specified subsystem will be marked with an asterisk (for example *Buses* *) until the selected component is returned to the unselected list.

Click **[Memorize]** to preserve a specific subsystem for later recollection. The **[Save As]** dialog requires entering a filename for the Bus Subsystem file (*.sbs).

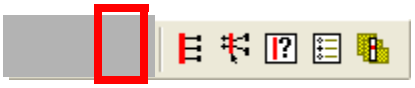
Click **[Recall]** to open a saved Bus Subsystem file (*.sbs).

Click **[OK]** to close the dialog and refresh **[Spreadsheet]**. Only data corresponding to the subsystem is displayed, until a new subsystem selection is made.

To return the entire working case to **[Spreadsheet]**, select ***Subsystem > Bus...*** and select the *All buses* check box from **[Bus Subsystem Selector]** or click the ALL BUS icon.



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Subsystem > Primary bus subsystem

This function toggles between two subsystems. Under normal operation, only one subsystem is created at a time. However, a second subsystem will be created when, in **[Diagram]**, information is requested for an element that is not in the current subsystem. The creation of the second subsystem allows the first subsystem to be preserved.

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2.4 Exporting Spreadsheet Data



File > Export > Spreadsheet Tab to text file…

This activity exports all data from an open spreadsheet to a tab-delimited file. The file selection window in the **[Save As]** dialog lists only Text Files (*.txt). Specify the filename; this can be a new file or a previously-built file to be over-written.

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2.5 Undoing an Action

Edit > Undo Record

This action returns a PSS®Express operation to its state previous to the run of an automation file (see Section 14.3, Executing Automation Files). It is necessary when adding a new record to any of the spreadsheets and the application generates an error message about missing information. At certain points, this can lead to application lockup. The Add operation cannot continue until the missing data is provided and the missing data cannot be provided because the application keeps notifying the user of the problem. At this point, ***Edit > Undo Record*** will reset the application back to the state before the Add operation was begun. The missing data can then be added in other spreadsheets and the item added again. Once the data has been entered, however, the *Undo Record* operation will have no effect.

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Chapter 3

Diagram View

3.1 Introduction

[Diagram] view (Figure 3-1) is used to create and modify one-line diagrams and to display a variety of results, such as:

- Power flow results (Section 3.8)
- Network impedances (Section 3.9)
- Graphical differences between power flow cases (Section 3.10)
- Fault analysis results (Section 3.11)

To open **[Diagram]**, open an existing diagram (***File > Open >*** *Slider Binary file (*.sld)*), or begin a new diagram (***File > New >*** *Diagram*).

As you add new elements to a diagram, **[Network Spreadsheet]** (Chapter 2) and **[Network Tree]** (Chapter 4) are automatically updated to reflect the addition. To move from **[Diagram]** to **[Network Spreadsheet]**, double-click the desired diagram component. **[Network Spreadsheet]** opens to the specified data tab and highlights the row of data for that component.

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Figure 3-1. Diagram View with Slider Diagram

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3.2 Specifying Diagram Properties

Diagram > Properties…

The **[Diagram Properties]** dialog (Figure 3-2), available only when **[Diagram]** is the active view, supports user-configurable settings, such as units, background colors, grid spacing, and zoom / pan thresholds.

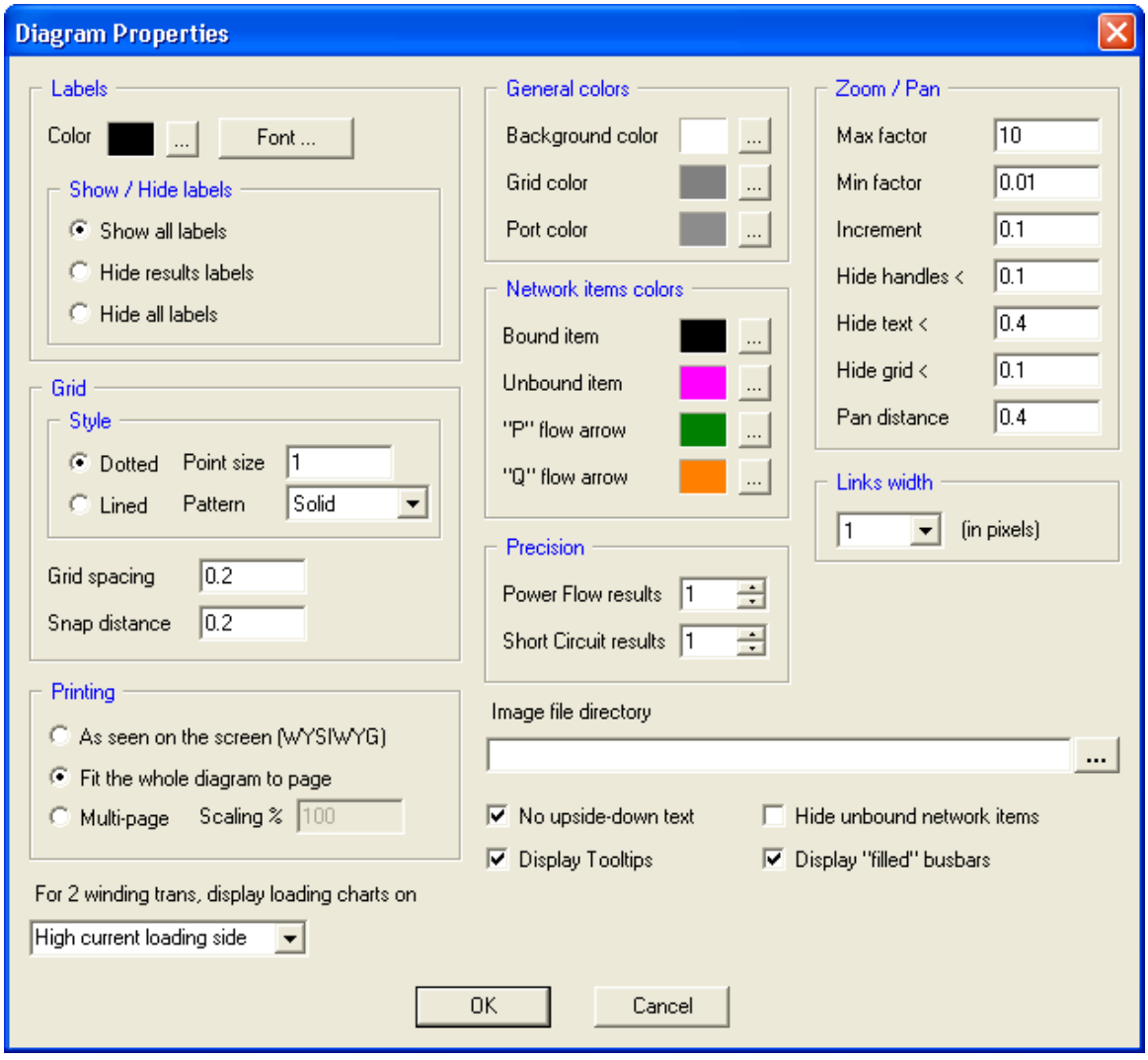


Figure 3-2. Diagram Properties Dialog

Labels: Select from a variety of font styles, sizes and colors for labels. Options for hiding or showing all labels and hiding results labels are also available. Click **[…]** to open the **[Color]** dialog (Figure 3-3) or click **[Font]** to open the **[Font]** dialog (Figure 3-4) to change label color or font.

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Figure 3-3. Color Dialog

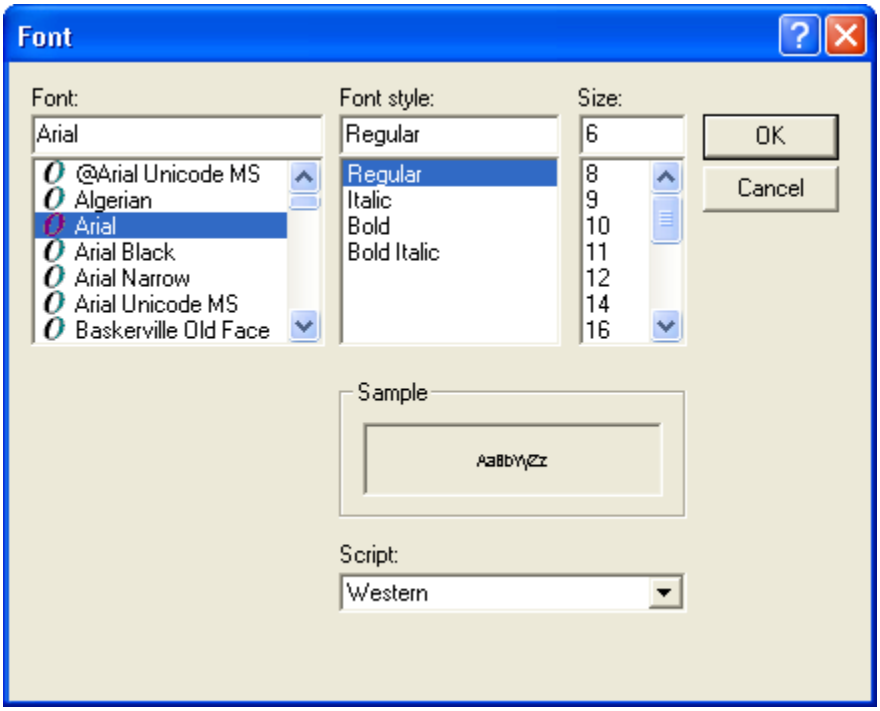


Figure 3-4. Font Dialog

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Grid: An underlying grid pattern may be made visible. Line style and width, grid spacing, and snap distance, may be specified. For very dense diagrams the snap distance must be small enough to ensure that items snap to the desired grid point instead of one further away.

General Colors: Defines background, grid, and port colors. Click **[…]** to open the **[Color]** dialog (Figure 3-3).


The port symbol is a small circle that appears at a location where a line or equipment connects to a bus. Ports appear as connection points when selecting lines or equipment and move to the new bus location when changes are made from **[Diagram]**.

Network Items Colors: Defines colors for bound and unbound diagram items and power flow direction. Click **[…]** to open the **[Color]** dialog (Figure 3-3).

All network items in **[Diagram]** that represent real network data are drawn in the bound color (Section 3.6.4), unless the color has been over-ridden by range checking coloration (Figure 3-46). All network items added to a diagram that do not represent data in the network are drawn in the unbound color.

The P flow arrow and the Q flow arrow display animated flow directions for real and reactive power for power flow results (Figure 3-46).

Precision: Defines the number of decimal places to be used in representing power flow and short circuit results on a diagram.

Zoom/Pan: Enter values for zoom factors directly in the fields provided. In addition, it is possible to hide some diagram items when the zoom factor is small and the diagram becomes crowded, including the handles that appear at each end of a bus when it is selected.


These values set limits. The diagram can be expanded, reduced and panned using the Zoom Toolbar buttons. Computers equipped with a mouse-wheel can pan vertically using just the mouse-wheel. Holding down the **[Shift]** key while moving the mouse-wheel will pan horizontally. Holding down the **[Ctrl]** key while moving the mouse-wheel will zoom. Keyboard arrow keys may also be used to pan the image.

Image File Directory: Click **[…]** to open the dialog specifying a default path for commonly used images. These images may be imported to the background layer of **[Diagram]**.

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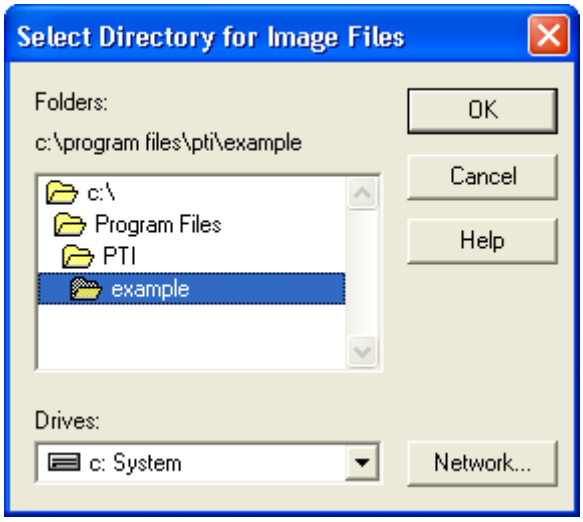


Figure 3-5. Select Directory for Image Files Dialog

Printing: Three options are available for printing a diagram:

- *As seen on the screen*: *WYSIWYG* (What you see is what you get.) prints only the visible portion of the one-line diagram. If you have zoomed in to see greater detail, this might be only a portion of a bus. Verify the desired print area by selecting ***File > Print Preview***.
- *Fit the whole diagram to page*: The entire one-line diagram is printed on one page, independent of whether or not the entire diagram is shown in **[Diagram]**.
- *Multi-page*: The entire diagram is printed on multiple pages, independent of whether or not the entire diagram is visible in **[Diagram]**. In addition, a muiltiple-page diagram may be scaled. This is useful when the one-line diagram is large.

The user must specify loading chart placement for 2-winding transformers from the following options:

- High current loading side
- Low current loading side
- Both sides

Other diagram options may be enabled using checkboxes. If *Display Tooltips* is enabled, moving the mouse cursor over a diagram item opens a pop-up note that provides basic information, such as bus number, name, and base kV.

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3.3 Creating a New Power Flow Case using Diagram View

After starting a new case and creating a blank [Spreadsheet] (see Creating a Working Case), select ***File > New*** and the *Diagram* option. Click **[OK]** to create a blank [Diagram]. Then network elements can be added graphically instead of typing data into the spreadsheet.

The first task is to specify whether to number the buses automatically; see Changing Program Preferences. For example, select ***Edit > Preferences*** to open **[Program Preferences]**, partially shown in Figure 3-6. Here the buses are numbered starting at 1 in increments of 100.

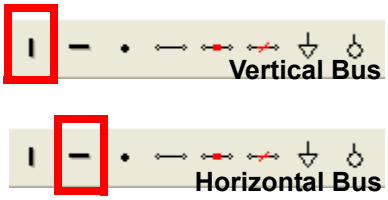
The example built in this section will have:

- A generator at bus 1 and bus 201.
- A non transformer branch between bus 1 and 101.
- A non transformer branch between bus 101 and 102.
- A load at bus 101.



Figure 3-6. Example: Automatic Bus Numbering

3.3.1 Adding Buses



To add buses to the diagram, and consequently the database, select a *Bus* toolbar button (vertical or horizontal) in the Diagram toolbar. Click [Diagram] in those locations where the buses are to be placed. They can be moved and otherwise adjusted later. If *Bind Items* is enabled for the Diagram (Section 3.6.4, Binding Data) then, as buses are added to the diagram, they will also be added to the spreadsheet with default values; in other words, they are bound to the spreadsheet. If *Use property sheet dialogs to edit Network data* is selected under *Program Preferences* (Section 1.5, Changing Program Preferences) then the *Datasheet Dialog* for the placed bus will come up so that the default values may be modified, however this behavior can be suppressed by holding down the **[Shift]** key while placing buses. Default values can be subsequently modified by editing in [Spreadsheet] as well.

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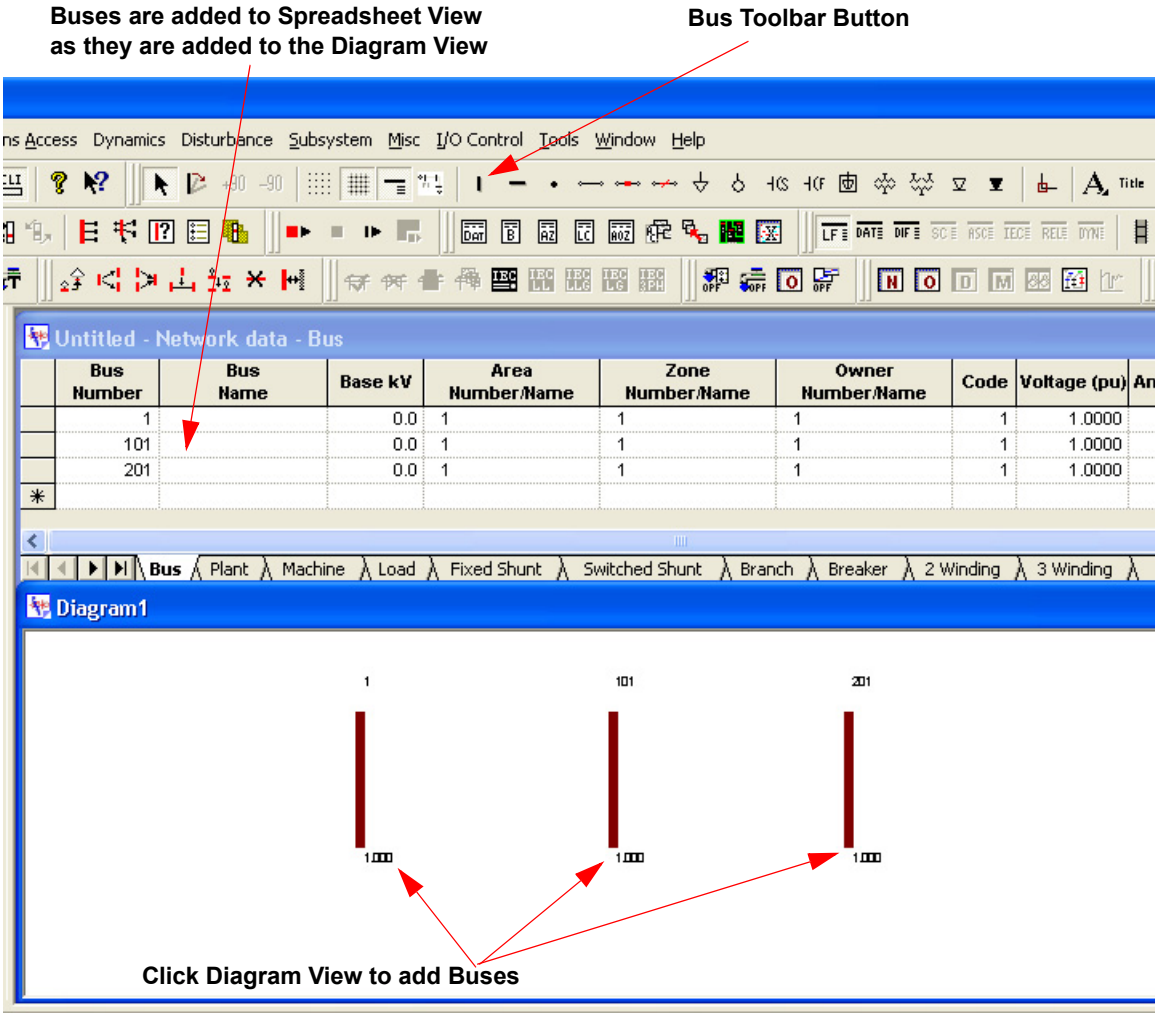
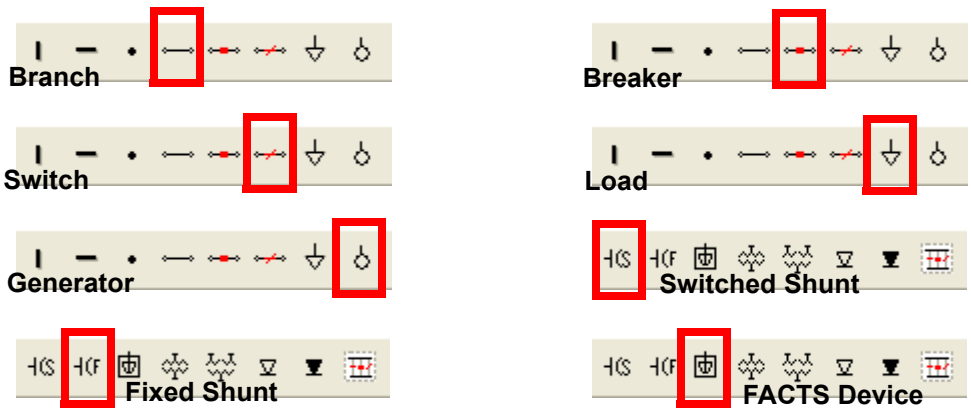
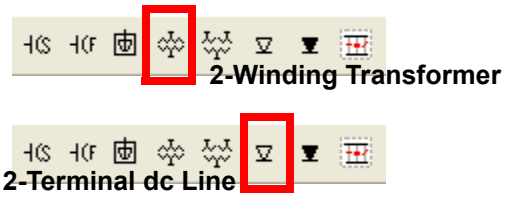


Figure 3-7. Adding Buses to the Database Using the Diagram View

3.3.2 Adding Network Elements



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With the buses roughly positioned, select the *Branch* button and drag the mouse between two buses to represent branch connections. Select the *Breaker* button or the *Switch* button and drag the mouse to a branch to represent these connections. Select the *Load* button and drag the mouse from a bus to an adjacent empty area. Select the *Generator* button and drag the mouse between two buses. If *Bind Items* is enabled for the Diagram (Section 3.6.4, Binding Data) then, as items are added to the diagram, they will also be added to the spreadsheet with default values. If *Use property sheet dialogs to edit Network data* is selected under Program Preferences (Section 1.5, Changing Program Preferences) then the *Datasheet Dialog* for the placed bus will come up so that the default values may be modified, however this behavior can be suppressed by holding down the **[Shift]** key while placing items. The data can also be edited in **[Spreadsheet]** to change default values. Figure 3-8 summarizes an example.

As each element is added, a summary of the case building activity is displayed in the **Progress** tab.

See Section 3.15, Closing the Diagram View for options to save the diagram.

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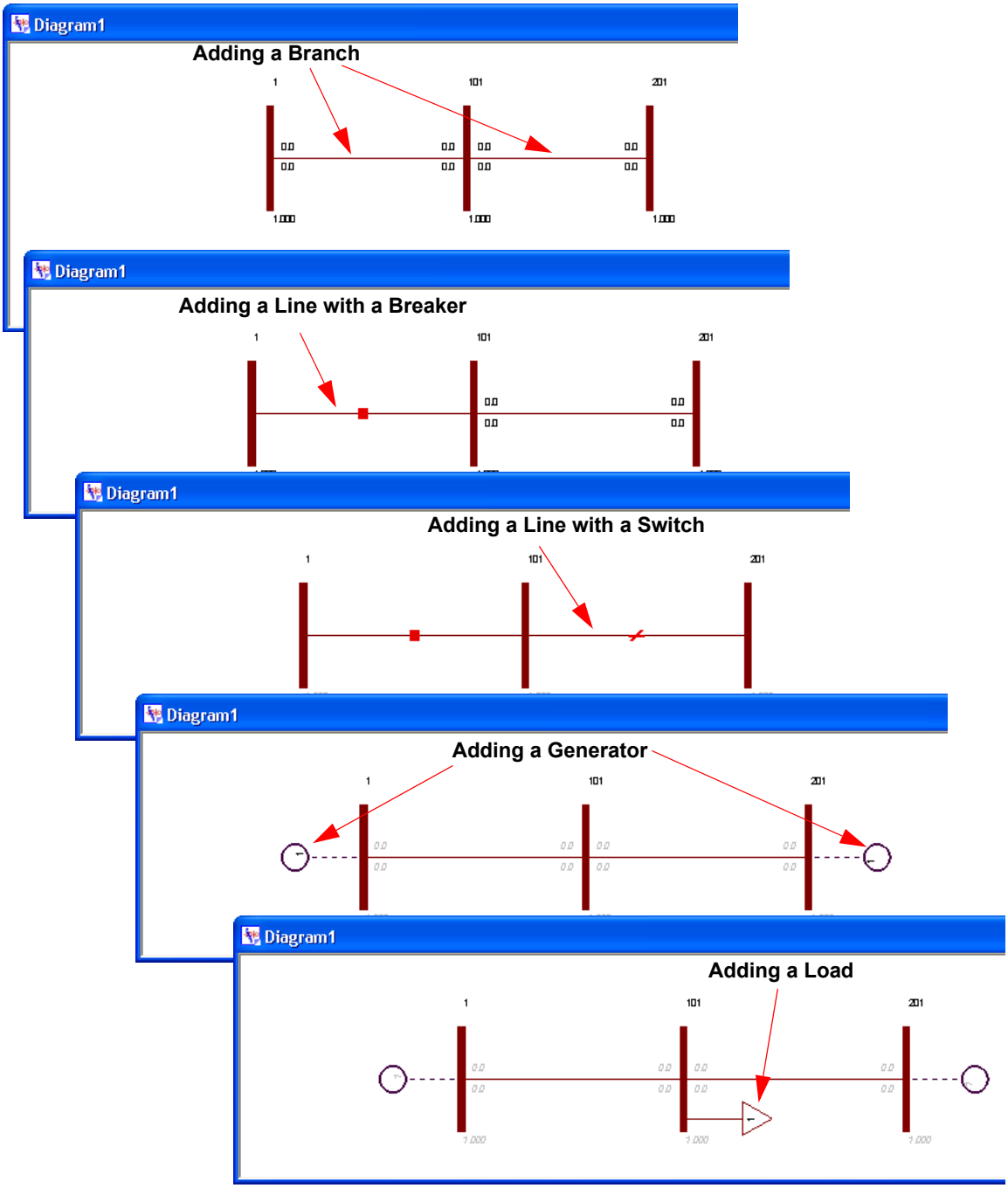


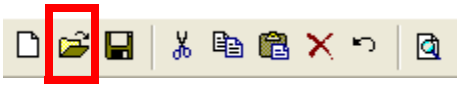
Figure 3-8. Adding Network Elements to the Diagram

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3.4 Opening or Importing Existing Diagram Files

Diagram data may be supplied in one of two formats: the old PSS®Express DRAW format or the new Slider file format created by the current version of PSS®Express, the file type that will be used for future versions.

3.4.1 Slider Diagram File



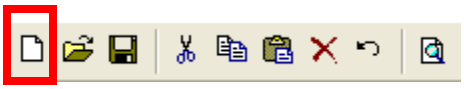
File > Open…

From the **[Open]** file selector window, select *Slider Binary File (*.sld)*. Highlight the desired file and click **[Open]**. The resulting diagram will be displayed in a new diagram window.

3.4.2 Importing a DRAW File

Activity DRAW

Requirements / Prerequisites
If bus voltages and/or line flows are to be printed, validly specified power flow case solved to an acceptable mismatch tolerance. Draw Data file (*.drw)



File > New…

From the **[New]** dialog, select the *Diagram* option to create a blank **[Diagram]**. Then select ***File > Import > DRAW file…***. A list of available *.drw files will be provided. Highlight the desired file and click **[Open]**.

DRAW coordinate data files **cannot** be opened using ***File > Open*** from the Main Menu.

The open DRAW file can be saved as a Slider Binary file (*.sld). Select ***File > Save As*** and rename the file as a slider file.

Although existing PSS®Express DRAW files can be imported, diagrams can be saved only in the slider format (*.sld). The new diagrams are *not* backward compatible with the PSS®Express DRAW/DRED activities used to display and edit one-line diagrams in earlier versions of PSS®Express.

3.4.3 Importing an Image File

File > Import > Image file…

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Image files may be imported into a diagram if they are formatted as *.bmp, *.jpg, *.gif, or *.png files. From the **[Import Image File]** selector window, select the desired file type. Highlight the desired file and click **[Open]**. The image will be displayed in the active diagram layer.

To assign image attributes, right-click the image and select ***Image Properties…***. The **[Image Properties]** dialog permits layer reassignment (see Section 3.13.3, Diagram Layers), scaling, and options to select and move the image in the diagram as if it were a network element. However, there is no network data associated with the image.

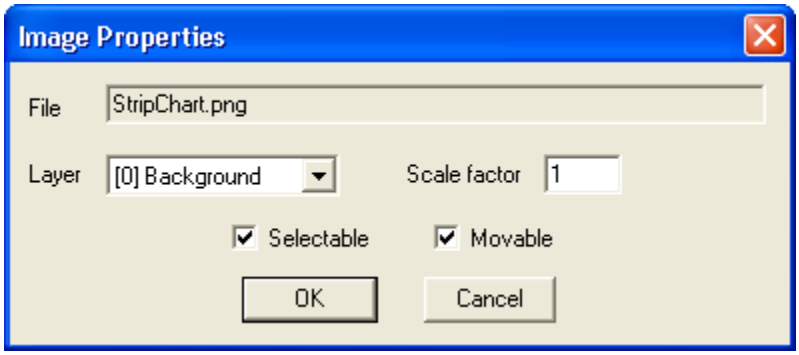


Figure 3-9. Image Properties Dialog

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3.5 Diagram Templates



Diagram templates can be used to build a library of common network configurations to be used over and over when creating new network elements. The diagram template function was initially created for use with the Substation Reliability Assessment (SRA) option.

As engineers add detailed bus/breaker/switch substation configurations to their networks for evaluation, the same configurations repeat over and over. The diagram template allows you to create the basic substation configuration once and then use it to reproduce the substation repeatedly in the network.

Diagram templates can be created using buses, branches, switches, breakers, loads etc. The diagram template is created and saved without network data associated, only the topology of the network elements is maintained (see Figure 3-10 for an example). When a diagram template is inserted in a slider diagram, the network topology is created in the network and default data is assigned to each new element.

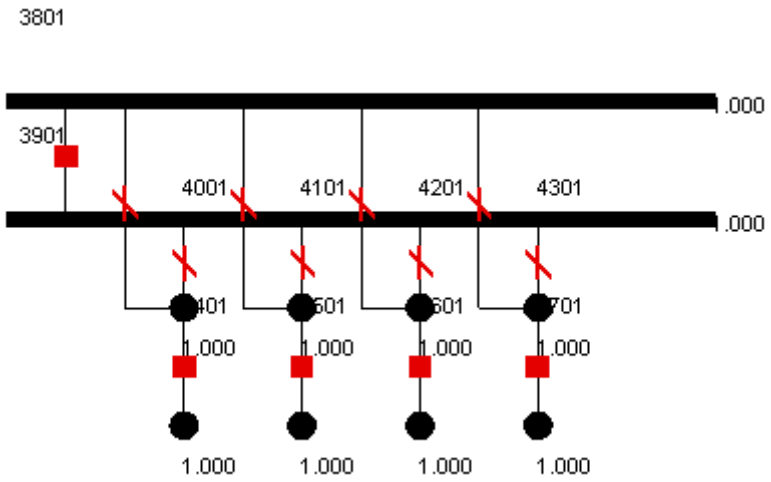


Figure 3-10. Diagram Template, Double Bus and Single Breaker, 4 Circuits

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3.6 Revising a Diagram

3.6.1 Using Data Records

Data record dialogs for individual network components may be used to revise the working case. To activate the data record option, select ***Edit > Preferences...***, select *Use Dialogs to edit Diagram Items* on the **[Program Preferences]** dialog, and click **[OK]** to update preferences. Thereafter, data records are activated whenever a diagram component is double-clicked.

When all changes have been entered in the data record, click **[OK]** to update the data record in the diagram and the network spreadsheet. When the working case is saved, the component data record is permanently updated.

Data records are available for bus, machine, load, and branch data records starting with PSS®Express-32. An inventory of currently available data records is provided in Appendix B.

3.6.2 Using Diagram Menus

Various menus are available by right-clicking in **[Diagram]**, with or without a network item selected. Selecting a network item and right-clicking opens a menu that is specific to the type of network element selected. For example, selecting a bus and right-clicking opens the following menu (Figure 3-11).

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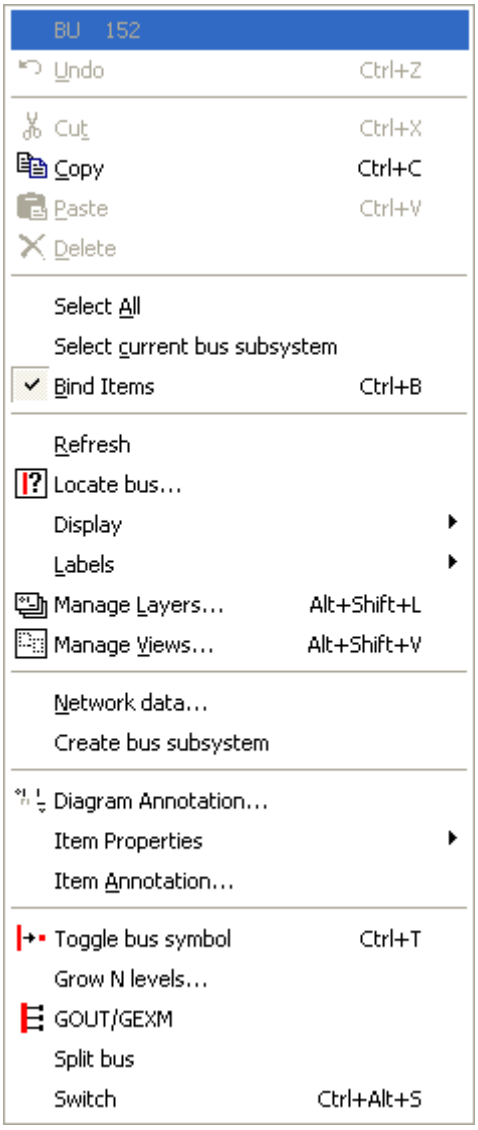


Figure 3-11. Example of Diagram View Right-click Menu

3.6.3 Editing Functions

All cutting, copying and pasting actions in **[Diagram]** use the application diagram clipboard. Diagram items placed on the application diagram clipboard may only be used in another **[Diagram]** within the same instance of PSS®Express.

If the bind option is enabled (Section 3.6.4), all changes made in in **[Diagram]** are reflected in **[Network Spreadsheet]**.

Two options are available from the right-click menu:

- *Select All*: Selects all items in the diagram.

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- *Select current bus subsystem*: Select a network item to specify. This option selects all items that are in that bus subsystem. If no bus subsystem has been specified, *Select All* occurs.

The remaining right-click menu functions are sensitive to the bind option.

If the bind option has been enabled for *Cut* and *Copy* functions, both the graphic representation and the corresponding model information are available from the diagram clipboard. If it has not been enabled, only the graphic representation is available from the diagram clipboard.

If the bind option has been enabled for the *Paste* function, and the copied area exceeds the number of existing elements, PSS®Express will automatically create the extra network elements required (and generate appropriate bus numbers).

If the bind option has been enabled for the *Delete* function, all network elements removed from the diagram are also removed from the working case.

Unless the edited diagram has been saved, ***Edit > Undo*** will reverse the last 30 actions, undoing both **[Diagram]** and network changes.


To copy a diagram from PSS®Express to another application, such as Microsoft® Word, the system clipboard is used. Select ***Edit > Copy to clipboard***. This action copies everything visible in **[Diagram]** to the clipboard; the user cannot select specific elements. To paste the clipboard contents to another application, use the paste function in that application. Note that the system copy is a screen capture of the **[Diagram]** view; and if components have been highlighted, the handles will be visible in the pasted copy.

3.6.4 Binding Data

Selecting *Bind Items* toggles the binding mode, where all selected diagram items are synchronized with their associated network data.

When *Bind Items* is enabled, a check mark is displayed in the menu. If a network item is deleted in the diagram, it will be deleted from the working case. Consequently, there will be no data displayed in either **[Network Tree]** or **[Network Spreadsheet]**; and the item symbol will disappear from the diagram. Similarly, if network items are pasted into **[Diagram]**, then a corresponding network item will be added to the working case and will appear in both **[Network Tree]** and **[Network Spreadsheet]**.

The *Bind Items* option also affects diagram elements deleted from **[Network Tree]**. The delete action removes the network item from the working case and, consequently, from **[Network Spreadsheet]**. However, in **[Diagram]** the symbol representing the network item remains in view, but its color changes to the previously specified unbound color, indicating that the item is no longer part of the working case.

Conversely, if the Bind Items option is not selected when a diagram item is deleted, it is deleted only from the diagram but remains part of the power flow case. It will thus still be available in both **[Network Tree]** and **[Diagram]**.

Network components drawn in **[Diagram]** that *do not* correspond to existing items in the network will be displayed in an unbound item color. Network components drawn in **[Diagram]** that *do* correspond to existing items in the network will be displayed in bound item colors, unless over-ridden by range checking coloration. Both the bound and unbound colors can be set by the user in the **[Diagram Properties]** dialog (Section 3.2).

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3.6.5 Displaying and Modifying Network Elements

Refresh: This right-click menu function redraws all items in the diagram.

Locate bus…: The **[Select Bus]** dialog initiates a search for the location of a specified bus. If the bus exists in the diagram, it is centered in **[Diagram]** and highlighted. If a requested bus is not in the diagram, a pop-up message is displayed.

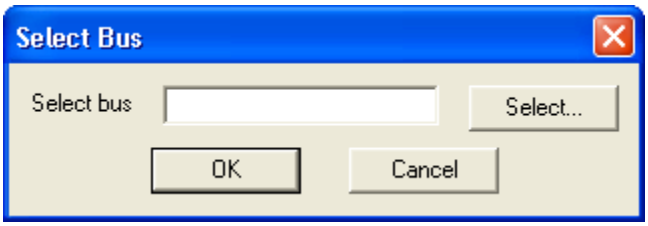


Figure 3-12. Select Bus Dialog

Display: Diagrams may be comprised of multiple layers, where each layer has an associated display list, a list of graphical objects. Individual network items are drawn in display-list order. If it is desirable to have a particular item drawn after or before another, for example, where two diagram items occupy the same general area in **[Diagram]** and one needs to be drawn over the other, layers become necessary. For each of the following functions, a network element must be highlighted.

Selecting ***Display > Bring to Front*** moves the specified item to the end of the display list. This object is drawn last, on top of the other graphical objects in the same layer.

Selecting ***Display > Send to Back*** moves the selected item to the beginning of the display list. This object is drawn first, and other items in the same layer should be drawn on top of this network item.

Selecting ***Display > Center*** moves the specified item to the center of **[Diagram]**.

Selecting ***Display > Assign to Layer*** opens the **[Select Layer]** dialog to move the selected item to a predefined layer (see Figure 3-69).

Grouping: Diagram items may be grouped into logical units. These units can then be modified as a single entity. Grouping items together is particularly helpful when moving a collection of diagram elements. A group cannot consist of only one item.

Diagram items can belong to only one group at a time. If you attempt to reassign an item to another group, it is removed from the existing group before being added to the new group. Define a block to be grouped by dragging the mouse on a portion of the diagram. Or define the group one item at a time by using **[Ctrl]**+ left-mouse-click to selectively add items to the group. Select ***Grouping > Group*** to build a group out of all the specified diagram items.

To return a group to its individual items, highlight the group by clicking any item. Then select ***Grouping > Ungroup***.

The system manages annotations (labels) associated with a given device as if it were grouped with the device. If the device is deleted, the related annotation is also deleted. If the selected device is moved, the related annotation is also moved. Therefore, adding a device to a larger group, in effect, adds all the related annotation to the larger group as well. This grouping behavior automatically moves annotations when the device is moved (called autopositioning), but can be disabled

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by selecting ***Item Properties > Auto Position***. When autopositioning is disabled, the annotation must be explicitly selected to be added to the group during the grouping operation.

Labels: The user can show or hide diagram labels (also called annotations). In order to be activated, the ***Labels > Toggle label selectability*** option is used to open the following dialog.



Figure 3-13. Make Labels Selectable Dialog



After selectability is activated, use the toggle button in the toolbar to show/hide the labels.

Manage Layers…: The **[Layers]** dialog (Figure 3-67) provides the means to create layers for a specified diagram. Each layer has its own *Visibility* options, as well as an option to turn selectability on and off. See Section 3.13.3, Diagram Layers.

Manage Views…: The **[Saved Views]** dialog (Figure 3-14) provides provides the means to create and retrieve a variety of one-line diagram presentations. These may be desirable for re-displaying unique views of the data, for example, results at different zoom levels from different parts of the network. If specific views are needed on a regular basis, then those views can be stored for rapid retrieval.

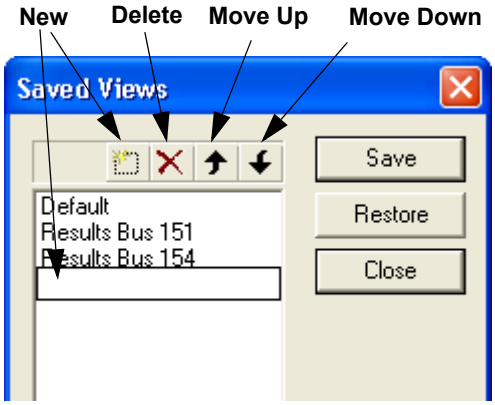


Figure 3-14. Saved Views Dialog

Click the new view icon, enter a name for the current view, and click **[Save]**. The title of the view appears in the dialog list. Double-click the title to edit the view. Highlight the desired view and click

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[Restore] to reposition the diagram to the saved view. Highlight the desired view and use the up and down arrows to re-order the list. Any view except *Default* may be deleted.

Network Data: Highlight the desired network item and select this function to activate **[Network Spreadsheet]**, which displays the appropriate data tab and highlights the appropriate row. The data can then be examined and manipulated as desired. Only one network item at a time may be selected. See also Section 3.6.1, Using Data Records.

Create bus subsystem: Define a block to be defined by dragging the mouse on a portion of the diagram and select this function to define a bus subsystem. Both **[Spreadsheet]** and **[Network Tree]** will be updated to reflect the new subsystem sselection.

Diagram Annotation: This function opens the **[Powerflow Data Annotation]** dialog (Figure 3-44) where diagram annotations (that is, branch, bus, equipment labels and diagram title) and diagram range checking options can be set. These are described in Section 3.8.1 and Section 3.8.2.

Item Properties > *Auto Position*: If no network item is specified, this function opens the **[Enable Auto Position]** dialog, where the auto-position mode for the entire diagram may be set. If only one equipment item or annotation is specified (i.e. machine, load, capacitor), then the *Auto Position* option can be toggled ON or OFF for the selected element.

If auto-positioning is enabled, the equipment or annotation connected to a bus will orient itself perpendicular to the bus. Toggling this property OFF allows the diagram item to be positioned in any orientation (see Figure 3-15).

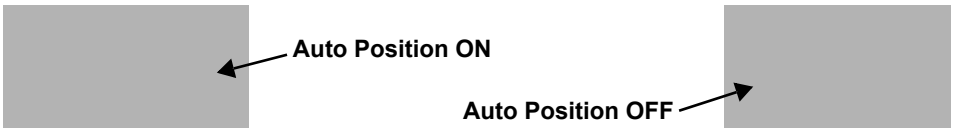


Figure 3-15. Impact of Auto-Positioning

Item Properties > *Auto Relink*: If no bus is specified, this function opens the **[Enable Auto Relink]** dialog, where the auto-relink mode for all buses in the diagram may be set. If only one busbar/bus node is specified, then the *Auto Relink* option can be toggled ON or OFF for the selected bus.

If auto-relinking is enabled, moving or re-sizing a busbar causes the attached branches and transformers to re-attach themselves for a perceived optimal representation. Toggling this property OFF allows the busbar to be moved or re-sized without affecting the attachment of branches and transformers.

Item Properties > *Auto Rotate*: When a non-bus item is specified, this function opens the **[Enable Auto Rotate]** dialog, which allows equipment connected to the bus to rotate when the bus is rotated. Toggling this property OFF keeps the original position of the equipment when the bus is rotated.

Item Properties > *Font*: When a label (annotation) is specified, this function opens the **[Font]** dialog (Figure 3-4) from which a new text font, style, and point size for the selected item and its result labels can be applied.

Item Properties > *Line Style / Color*: When a line is specified, this function opens the **[Line Style]** dialog (Figure 3-16) from which the line type (solid or dotted), width, and color may be changed. This function may also be used to change the color of annotations or primitive shapes. Primitive

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shapes include straight lines, arcs, circles, polygons, etc., available on the Diagram Toolbar with the other network diagram elements.

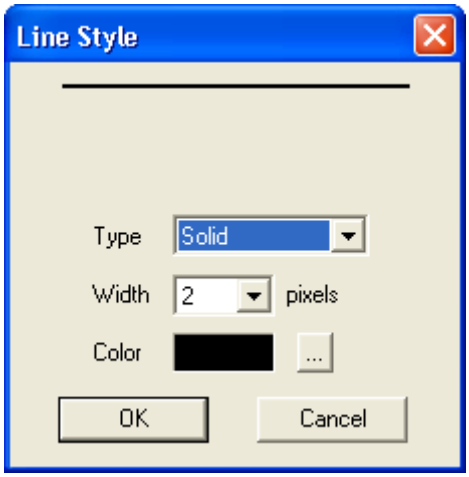


Figure 3-16. Line Style Dialog

Item Properties > *Visible*: When a network item is specified, it may be made invisible using this function. The item is still available, as its handles appear when the location of the item is clicked. Thus, an invisible network item may be made visible using this function.

Item Properties > *Unbind*: When one or more diagram elements are specified, they may be disassociated from the corresponding network item, overriding the previously enabled binding. In order to re-establish binding, the network model item in **[Network Tree]** and the diagram element must be reselected. Then select the *Bind* option from the **[Network Tree]** right-click menu.

Item Properties > *Location*: Certain diagram elements allow the specification of GIS coordinates. This is useful for providing locations for network elements for use with Google Maps.

Item Properties > *Map String*: This is used to edit the mapping string used to associate a diagram item with a specific network element. Most users will have no need to alter this field, but certain Python APIs can be used to set and query these map strings and perform operations based on results.

Item Annotation: When a network item is specified, an item-specific dialog provides the means to change the way its annotation is displayed. For example, the **[Bus Annotation Properties]** dialog (Figure 3-29) is displayed when the specified item is a bus. You may choose to apply the changes to the specified bus or to all buses. See also Section 3.8.1, Power Flow Annotation, Section 3.9.1, Impedance Annotation, Section 3.10.1, Comparison Annotation. The items at the bottom of the **[Diagram]** right-click menu are customized to the element(s) selected. Figure 3-17 shows the difference between specifying one or two buses and lines. Other functions are available when another type of equipment is selected (i.e., load, machine, etc). This sections provides a description of all possible functions.

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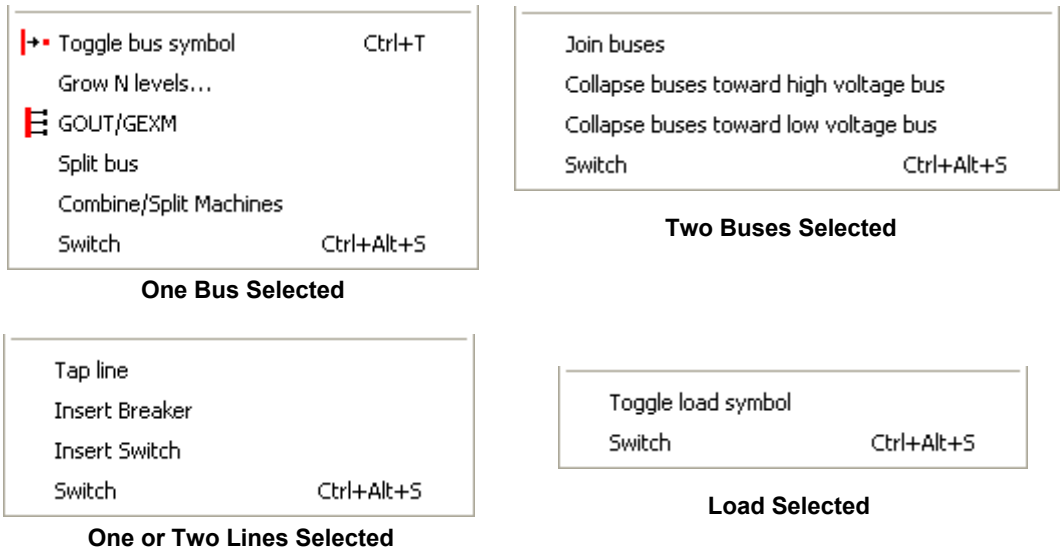


Figure 3-17. Customized Right-Click Menu

Toggle bus symbol: This function toggles the specified bus between the busbar and busnode representations (see Figure 3-18).



Figure 3-18. Alternative Bus Symbols

Grow N Levels…: The **[Specify Grow Levels]** dialog specifies a method for drawing elements connected to the bus (see Section 3.7).

Split bus: The **[Split Bus]** dialog requires a new bus number, name and base kV for the additional bus. Click **[OK]** to split the bus, preserving the existing one and adding the new one to the diagram. Equipment and lines can then be dragged to the new bus. See also Section 7.4, Splitting Buses.

Join buses: This function is available when two buses have been specified. The **[Join Buses]** dialog requires specification of an option to handle *Line shunts of deleted in-service branches*. Click **[OK]** to join the two buses into a single retained bus. See also Section 7.3, Joining Buses.

Switch: This function changes the status of a network component. Visual representation of out-of-service equipment is specified using **[Powerflow Data Annotation]** (see Section 3.8.2, Range Checking). A summary of the change is routed to the **Progress** tab (Figure 3-19). When using this function to disconnnect a bus, its type code is set to 4 and all ac branches, dc lines, and series FACTS devices connected to the bus are represented as out-of-service.

Toggle load symbol: This function toggles the specified load symbol size.

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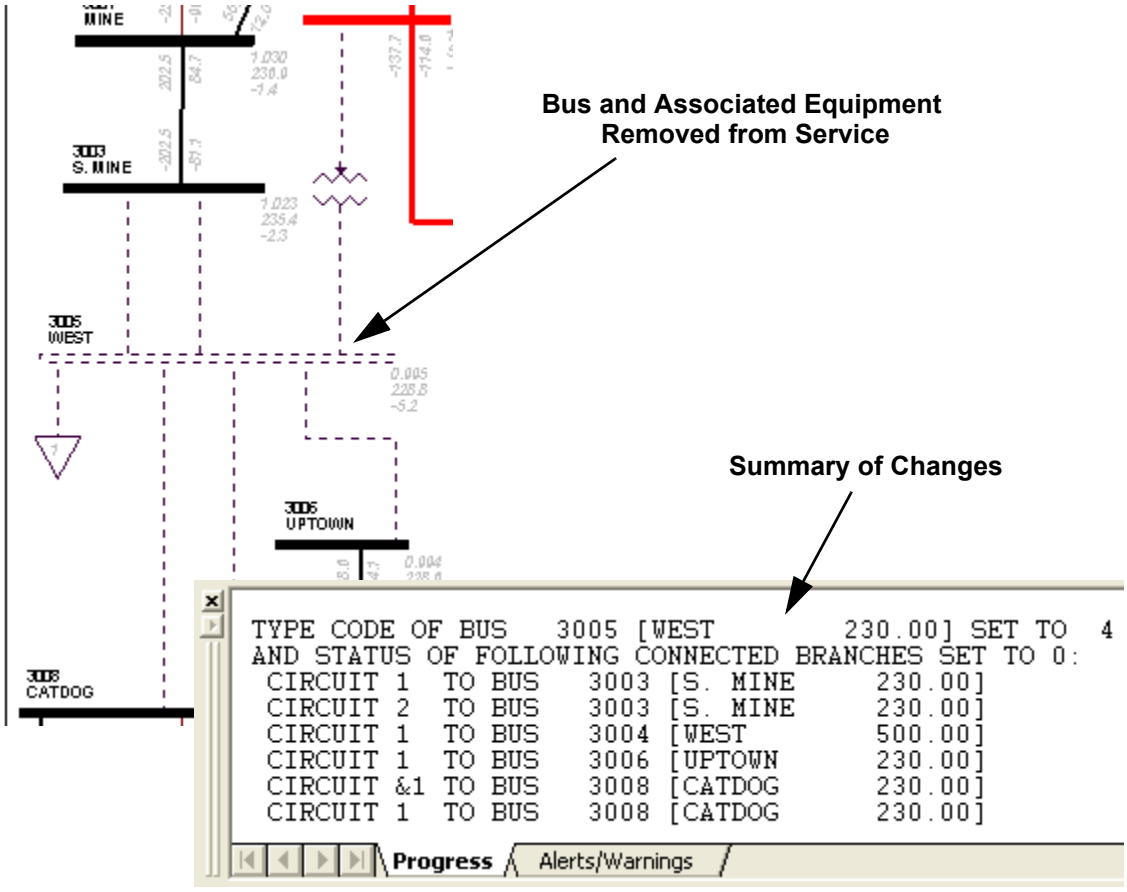


Figure 3-19. Example of Switch Function to Out-of-Service

For bus connected capacitors and reactors the *Switch* option is not available. To switch the status of capacitors and reactors, double-click the network item (or right-click and select ***Network data…***) to edit the status field of the corresponding data record in **[Spreadsheet]**.

Combine/Split Machines: This function is available when a single bus with at least one connected machine is selected. When performed, it combines multiple machine diagram items at the bus into a single summing machine symbol, where results are summed for total generation at the bus. A summing machine symbol has no machine ID in the center of the symbol, whereas a machine diagram item representing a single machine has a machine ID in the center of the symbol.

If only a summing machine symbol exists on the bus, selecting the bus and the *Combine/Split Machines* function will remove the summing machine symbol and replace it with one or more machine diagram items representing the individual machines at the bus.

Combine/Split Loads: This function is available when a single bus with at least one load is selected. Its action is comparable to *Combine/Split Machines*, affecting the bus load.

Combine/Split Fixed Shunts: This function is available when a single bus with at least one fixed shunt is selected. Its action is comparable to *Combine/Split Machines*, affecting the fixed shunt.

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Collapse buses towards high voltage bus: This function is available when two buses are selected. It is used to visually change the layout in **[Diagram]**, and has no effect on the actual network. When performed, all components on the lower voltage bus appear to be connected to the high voltage bus, and the lower voltage bus is removed, as well as any connections between the two buses. Only **[Diagram]** is affected; these components are not removed from the actual network. If an analysis is performed, all components of the two buses are included in the calculations.

Collapse buses towards low voltage bus: This function is available when two buses are selected. It is used to visually change the layout in **[Diagram]**, and has no effect on the actual network. When performed, all components on the lower voltage bus appear to be connected to the low voltage bus, and the higher voltage bus is removed, as well as any connections between the two buses. Only **[Diagram]** is affected; these components are not removed from the actual network. If an analysis is performed, all components of the two buses are included in the calculations.

Tap Line: This function is available when a line is selected. The **[Tap Line]** dialog requires specification of the tap location and identification of the new bus. Any nontransformer branch may be tapped. Click **[OK]** to tap the line. See also Section 7.5, Tapping a Line.

Insert Breaker: This function is available when a line is selected. This is used to insert a dummy bus node and a specially identified zero impedance line that is used to model a breaker in substation reliability analysis. The **[Insert a Breaker]** dialog (Figure 3-20) requires specification of the dummy bus created to build the breaker. Click **[OK]** to add the breaker.

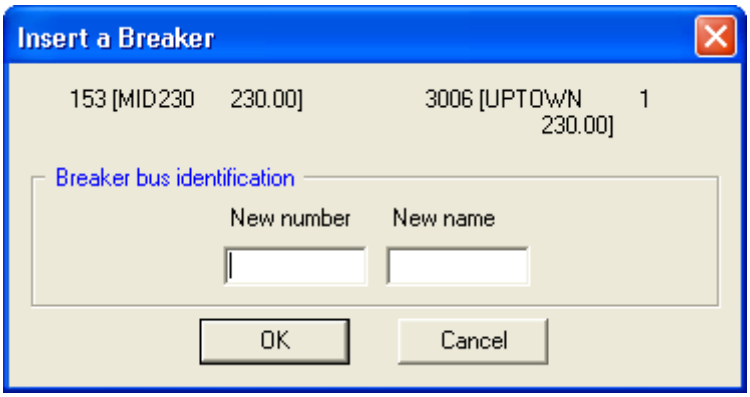


Figure 3-20. Insert a Breaker Dialog

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Insert Switch: This function is available when a line is selected. This is used to insert a dummy bus node and a specially identified zero impedance line that is used to model a switch in substation reliability analysis. The **[Insert a Switch]** dialog (Figure 3-21) requires specification of the dummy bus created to build the switch.

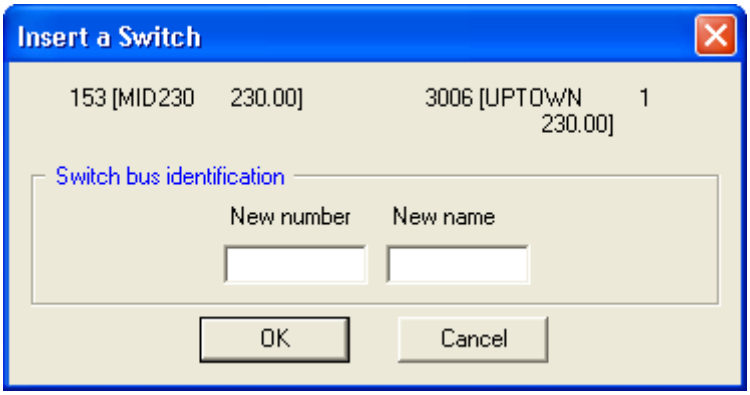


Figure 3-21. Insert a Switch Dialog

3.6.6 Drawing a Missing Bus

If a bus is present in the working case but is not drawn in the active diagram, it is indicated in the expanded **[Network Tree]** without the busbar symbol (see Figure 4-2). Right-clicking the missing bus activates a *Draw* option. Selecting Draw from the pop-up menu draws the bus and all its equipment in the active **[Diagram]**.

3.6.7 Renumbering Buses in the Diagram

File > Renumber Buses in Diagram (RNFI)…

The *Renumber Buses in Diagram* option is available only when working in **[Diagram]**. It opens the **[Bus number translation file]** selection window for the required Bus Number Translation file (*.trn). The creation of this file is described in Section 7.15, Renumbering Buses. The file contains records of old bus numbers and the corresponding new numbers.

3.6.8 Specifying Bus Locations from a File

File > Open…

Loading a Bus Location Data File (*.loc) from the **[Open]** dialog allows the bus locations in the diagram to be specified from the same Cartesian or GIS coordinates across repeated analyses or different working case subsystem configurations.

File data is used to position any buses drawn or *Grown* for which an entry exists in the file. Those buses in the working case that are not defined in the file are assigned default locations in the diagram.

The same file is used to reposition buses in an existing diagram; select ***Diagram > Update bus locations*** after the file is loaded. If no Bus Location file has been loaded, this function is disabled. Buses may be specified for repositioning. If buses are selected, only those buses will be evaluated for update. If no buses are selected, all buses in the diagram will be evaluated for update.

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Bus Location Data File Contents

Bus positions on a diagram may be specified in a Bus Location file that can be accessed whenever a bus is created. These positions can be specified as x/y coordinates or as GIS coordinates in an hour, minutes, seconds format. The format of the file is as follows:

CARTESIAN | GEOPHYSICAL

> The first record in the file describes the format of all bus position records. CARTESIAN specifies that bus positions are provided as x/y locations on the diagram. GEOPHYSICAL specifies that bus positions are provided in GIS form as two pairs of hour, minute, seconds inputs.

The next N lines specify the bus positions. The Bus Location file **does not** require an entry for every bus in the case. If a bus position record is not found in the file when positioning a bus on the diagram, default bus positioning is used.

The format of a CARTESIAN bus position record is as follows:

```
BusId, X location, Y location, Rotation, Length
```

where:

BusId	Either a bus number or a quoted extended bus name.
X location	The bus X location, in inches, on the diagram.
Y location	The bus Y location, in inches, on the diagram.
Rotation	The optional bus rotation, in degrees.
Length	The optional length of the busbar, in inches.

The format of a GEOPHYSICAL bus position record is as follows:

```
BusId, X location, Y location, Rotation, Length
```

where:

BusId	Either a bus number or a quoted extended bus name.
X location	The bus X location, in degrees latitude (see format, below).
Y location	The bus Y location, in degrees longitude (see format, below).
Rotation	The optional bus rotation, in degrees.
Length	The optional length of the busbar, in inches.

Degrees are specified as a set of 1 to 3 real numbers (degrees, minutes, seconds) followed by a direction (E, W, N, S) (the direction can be omitted if all three numbers are specified). Positive values are interpreted as E or N, negative values as W or S. The degree parts can be separated by spaces, commas, apostrophes, or by the specific unit markers for degree, minutes, or seconds (in which case they must be used in order).

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For example, the following all represent the same value:

79°30’36W

-79 -30 -36

79.51W

-79.51,,,

The Bus Location file can also contain location entries for branches and two winding transformers found in the network.

Branch and two winding transformers records are indicated by the BRANCH record appearing in the file.

BRANCH

The format of a CARTESIAN branch position record is as follows:

FromBusId	Either a bus number or a quoted extended bus name of the from bus.
ToBusId	Either a bus number or a quoted extended bus name of the to bus.
CircuitId	Circuit Id of branch.
X From loc	The from bus X location, in inches, on the diagram.
Y From loc	The from bus Y location, in inches, on the diagram.
X KP1 loc	The first knee point X location, in inches, on the diagram.
Y KP1 loc	The first knee point Y location, in inches, on the diagram.
X KPn loc	The nth knee point X location, in inches, on the diagram.
Y KPn loc	The nth knee point Y location, in inches, on the diagram.
X To loc	The to bus X location, in inches, on the diagram.
Y To loc	The to bus Y location, in inches, on the diagram.

The format of a GEOPHYSICAL branch position record is as follows:

FromBusId	Either a bus number or a quoted extended bus name of the from bus.
ToBusId	Either a bus number or a quoted extended bus name of the to bus.
CircuitId	Circuit Id of branch.
X From loc	The from bus X location, in degrees latitude (see format, above).
Y From loc	The from bus Y location, in degrees latitude (see format, above).
X KP1 loc	The first knee point X location, in degrees latitude (see format, above).
Y KP1 loc	The first knee point Y location, in degrees latitude (see format, above).
X KPn loc	The nth knee point X location, in degrees latitude (see format, above).
Y KPn loc	The nth knee point Y location, in degrees latitude (see format, above).
X To loc	The to bus X location, in degrees latitude (see format, above).
Y To loc	The to bus Y location, in degrees latitude (see format, above).

If there are no kneepoints associated with the branch, there will be no X & Y KPn entries in the record.

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The format of a CARTESIAN two winding transformer position record is as follows:

FromBusId	Either a bus number or a quoted extended bus name of the from bus.
ToBusId	Either a bus number or a quoted extended bus name of the to bus.
CircuitId	Circuit Id of two winding transformer.
X From loc	The from bus X location, in inches, on the diagram.
Y From loc	The from bus Y location, in inches, on the diagram.
X Sym loc	The two winding transformer symbol X location, in inches, on the diagram.
Y Sym loc	The two winding transformer symbol Y location, in inches, on the diagram.
X KP1 loc	The first knee point X location, in inches, on the diagram.
Y KP1 loc	The first knee point Y location, in inches, on the diagram.
X KPn loc	The nth knee point X location, in inches, on the diagram.
Y KPn loc	The nth knee point Y location, in inches, on the diagram.
X To loc	The to bus X location, in inches, on the diagram.
Y To loc	The to bus Y location, in inches, on the diagram.

The format of a GEOPHYSICAL two winding transformer position record is as follows:

FromBusId	Either a bus number or a quoted extended bus name of the from bus.
ToBusId	Either a bus number or a quoted extended bus name of the to bus.
CircuitId	Circuit Id of two winding transformer.
X From loc	The from bus X location, in degrees latitude (see format, above).
Y From loc	The from bus Y location, in degrees latitude (see format, above).
X Sym loc	The two winding transformer symbol X location, in degrees latitude (see format, above).
Y Sym loc	The two winding transformer symbol Y location, in degrees latitude (see format, above).
X KP1 loc	The first knee point X location, in degrees latitude (see format, above).
Y KP1 loc	The first knee point Y location, in degrees latitude (see format, above).
X KPn loc	The nth knee point X location, in degrees latitude (see format, above).
Y KPn loc	The nth knee point Y location, in degrees latitude (see format, above).
X To loc	The to bus X location, in degrees latitude (see format, above).
Y To loc	The to bus Y location, in degrees latitude (see format, above).

If there are no kneepoints associated with the two winding transformer, there will be no X & Y KPn entries in the record.

3.6.9 Moving Diagram Elements

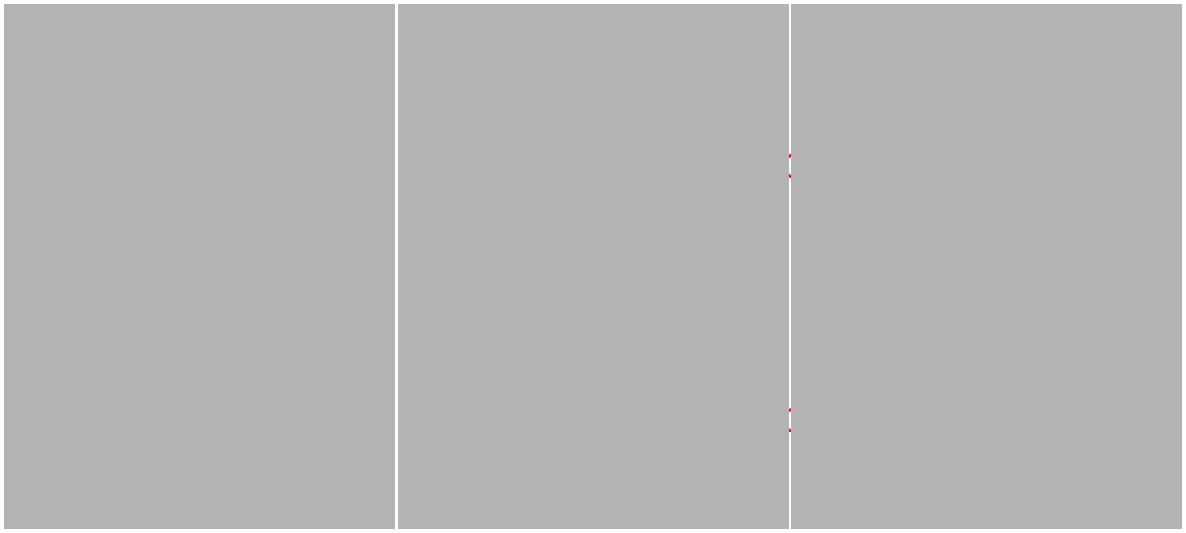
Existing diagram items can be moved with mouse actions. Buses can be moved and their length changed. Equipment and lines can be repositioned on a bus or moved to another bus. If the *Bind Items* option is selected, then moving equipment and lines to another bus in **[Diagram]** will also move the equipment and lines within the network data.

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To move or resize a bus, select the bus in the diagram with a mouse click, (see Figure 3-22a). If the **[Program Preferences]** dialog has been specified to *Resize buses on one end only*, the double-arrow mouse pointer will indicate this.

To move the bus to a different location, select the desired bus and position the mouse over the busbar. The mouse pointer will change to a four-arrow cursor. Drag the bus to the desired position (Figure 3-22b).

To make the bus longer or shorter, select the desired bus and position the mouse over one end of the bus. The pointer will change to a double sided arrow. Drag the bus end handle to the desired length (Figure 3-22c).



(a) Select Bus to Move **(b) New Position** **(c) Grab Bus End to Re-size**

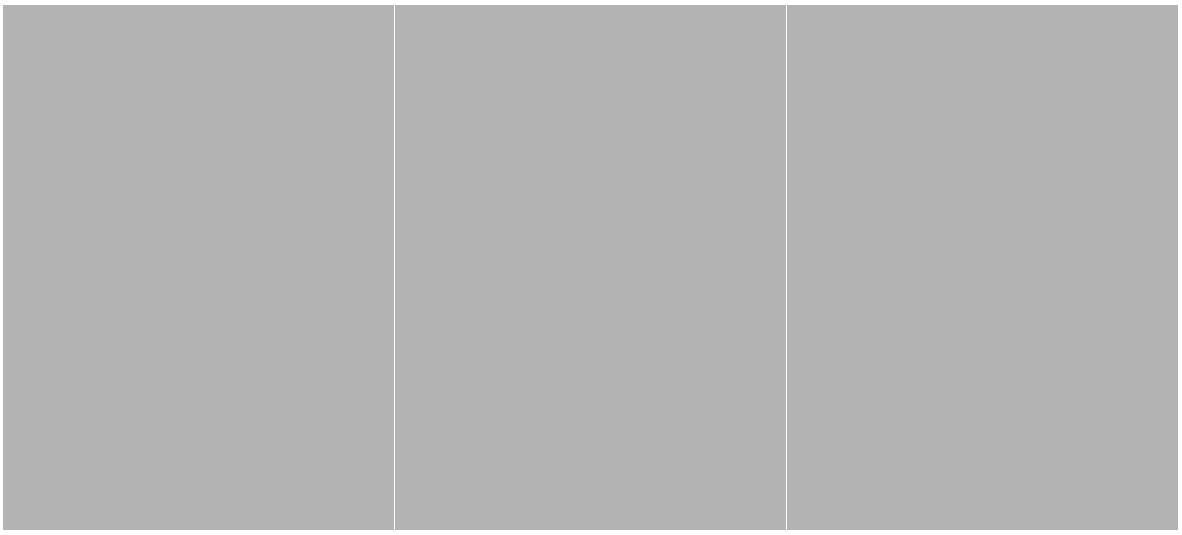
Figure 3-22. Moving and Resizing Buses

To move existing equipment and lines:

1. Click the diagram component you wish to move; a bus end connection is illuminated with a circle (Figure 3-23a).
2. Drag the end to the new attachment point. Valid attachment points will be illuminated with circles as the mouse passes over them (Figure 3-23b).
3. When the end illuminates the desired attachment point, release the mouse to connect the diagram component (Figure 3-23c).

The capacitor symbol direction from the bus can be controlled. Drag the connection circle to the desired edge of the busbar; the crosshairs in the circle will guide you.

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(a) Selecting **(b) Dragging to New Position** **(c) New position**

Figure 3-23. Moving Equipment

Not only can equipment and lines be moved to another location on a bus but they can also be moved to another bus, using the same technique. If *Bind Items* is selected, (see Section 3.6.4) moving equipment and lines to another bus in **[Diagram]** will also move the equipment and lines in **[Network Tree]**. The change will also be reflected in **[Spreadsheet]**. Consequently, this is another form of modifying not only the power flow topology but also the location of equipment.

3.6.10 Changing Status

In addition to using the right-click *Switch* function (Section 3.6.5), to toggle equipment to and from in-service status, the connection between **[Diagram]** and **[Spreadsheet]** may be useful. As shown in the following example, double-clicking a component in the diagram opens the spreadsheet tab and highlights the row containing that component's data. In-service status may be enabled or disabled, as desired.

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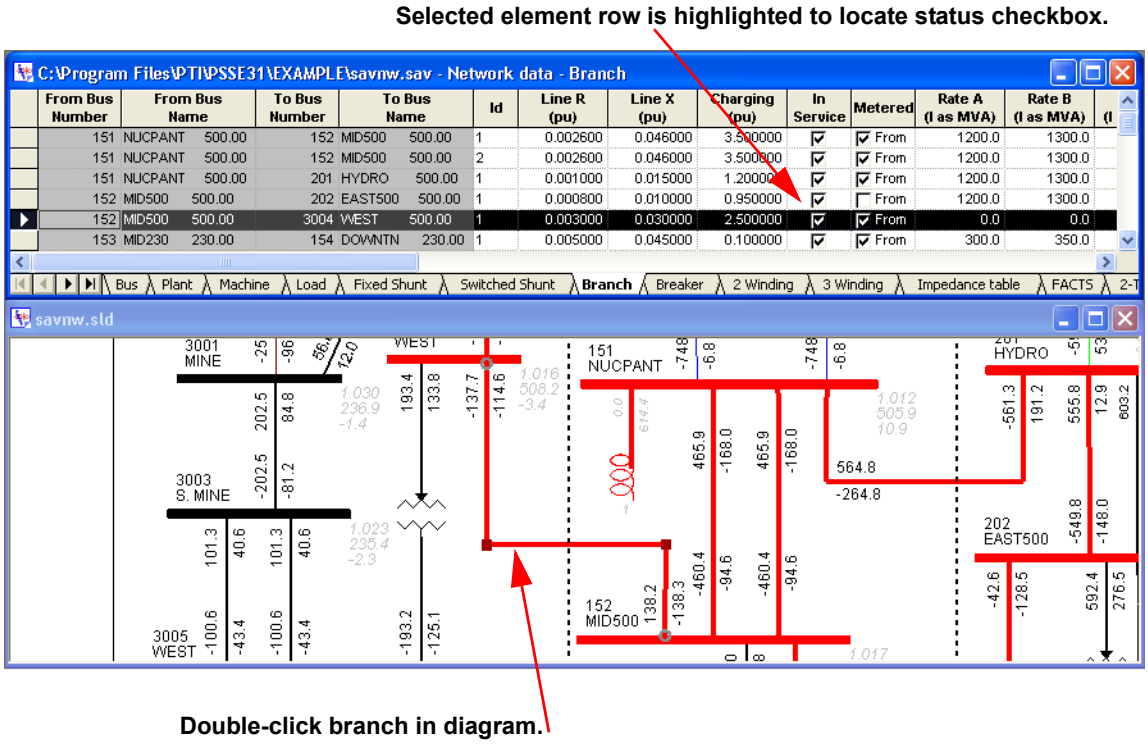


Figure 3-24. Changing Line Status from Diagram View

3.6.11 Displaying ISO Symbols

Diagram > Display ISO Symbols

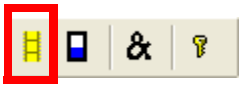
This option modifies the images of network components, replacing PSS®Express symbols with standard ISO symbols. Figure 3-25 shows the standard PSS®Express transformer symbol, compared with the ISO symbol.



PSS®Express Symbols **ISO Symbols**

Figure 3-25. Examples of Transformer Symbols

3.6.12 Animating Flows



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This function animates branch flows on the active **[Diagram]** using the flows from the last power flow solution (Figure 3-26).

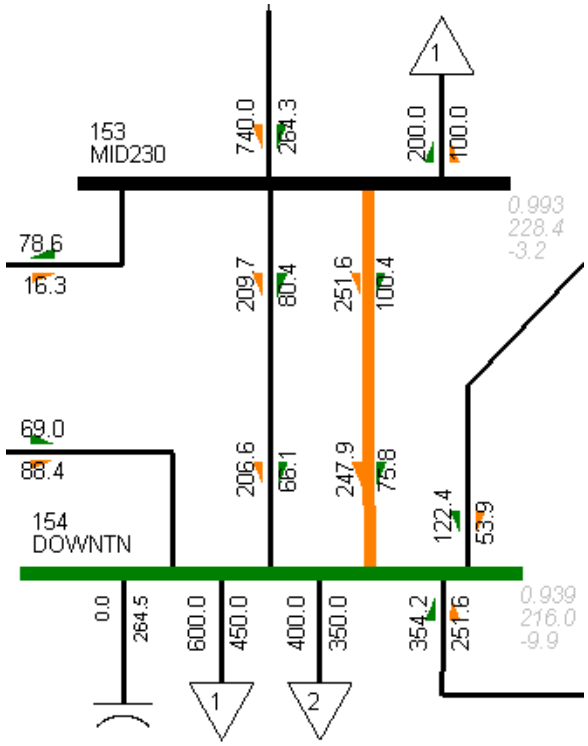


Figure 3-26. Animated Flow

3.6.13 Viewing Current Loadings



This function displays line loading graphs using values from the last solution (Figure 3-27).

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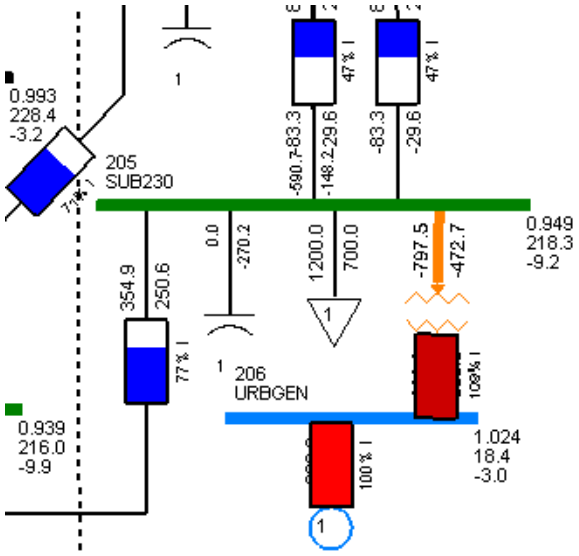


Figure 3-27. Current Loading Blocks Indicating Flow Levels

3.6.14 Multi-Section Line Reporting



This function displays or hides the dummy buses that constitute the terminals of multiple sections that make up a single line between buses.



(a) Reporting OFF



(b) Reporting On

Figure 3-28. Multi-Line Section Reporting ON and OFF

3.6.15 Item Annotation

Right-clicking on a bus in **[Diagram]** opens the **[Bus Annotation Properties]** dialog (Figure 3-29). The label displayed, as well as its positioning, can be specified. Furthermore, you can choose to

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suppress all annotation at the selected bus or to apply the annotation properties specified for the single bus to all buses by selecting the *Apply the selection option to all buses* checkbox.

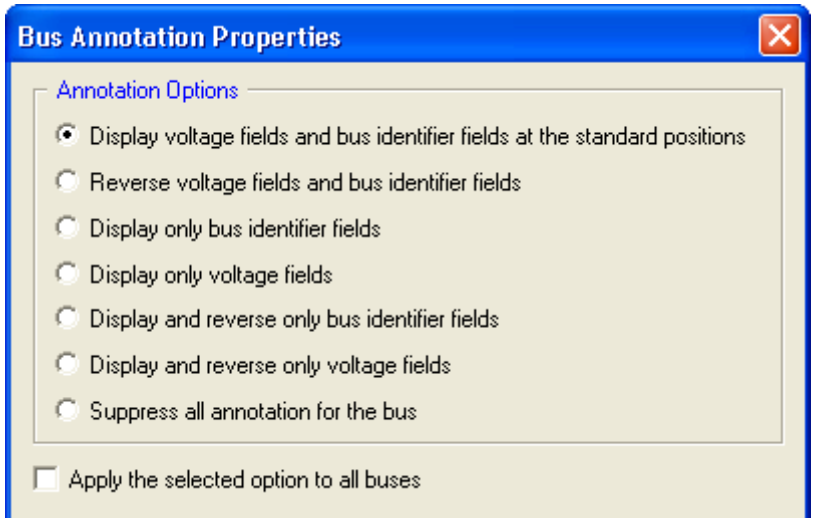


Figure 3-29. Bus Annotation Properties Dialog

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Right-clicking on a load or a shunt capacitor/reactor opens the appropriate annotation properties dialog (see Figure 3-30). You can choose the type of image or annotation, suppression of annotation, and application of the properties to all shunt or load items.

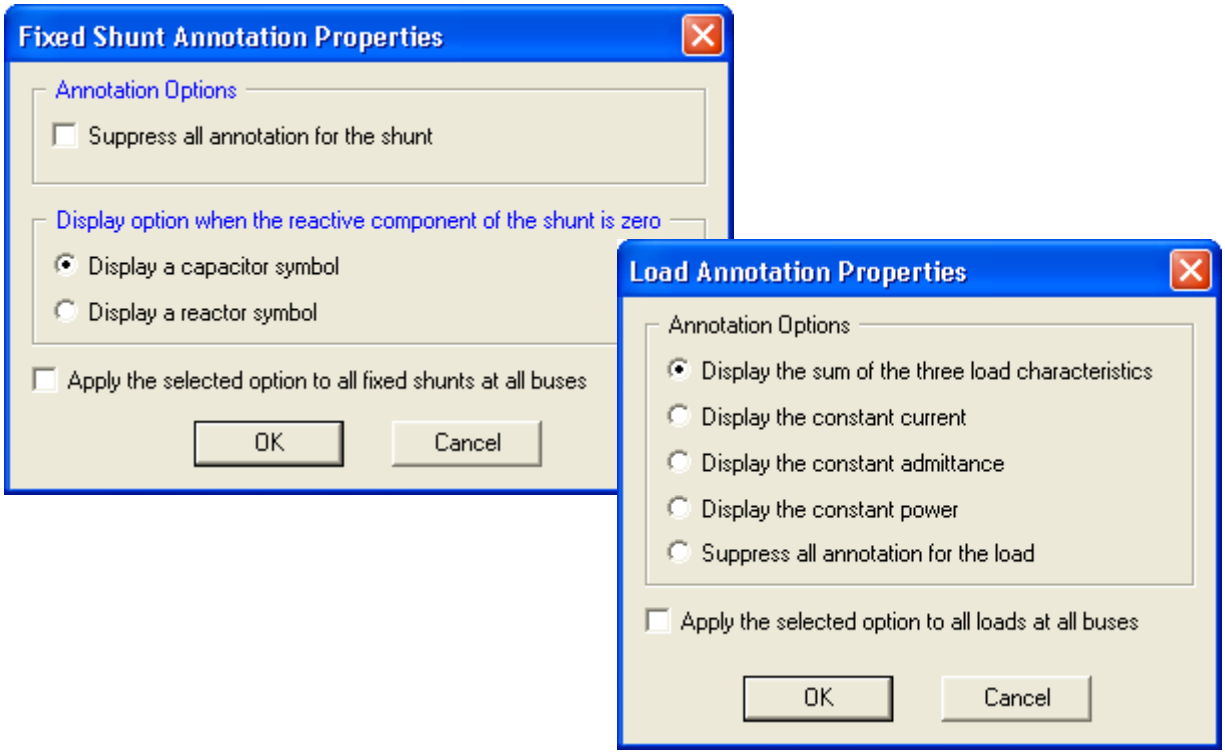


Figure 3-30. Annotation Properties Options for Shunts and Loads

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Right-clicking on a branch or 2-winding transformer transformer (the transformer symbol itself, not the transformer line) and selecting ***Item Annotation...*** opens the **[Branch Annotation Properties]** or **[Two-Winding Transformer Annotation Properties]** (Figure 3-31) dialogs, respectively. These dialogs are similar, except that when selecting a nontransformer branch, only the *Flow Annotation Options* are available, with the option to draw a series capacitor symbol or circuit ID.

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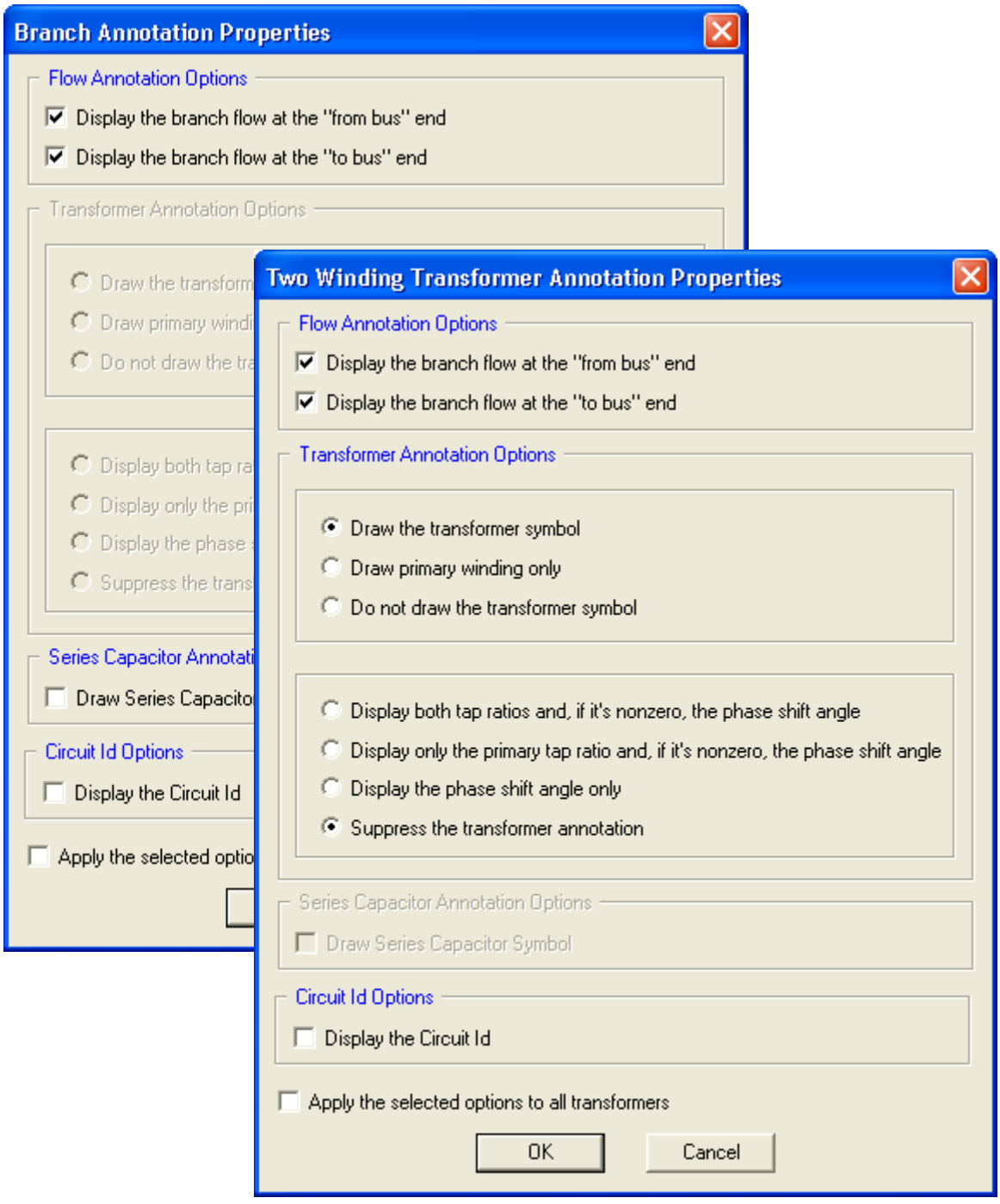


Figure 3-31. Branch / Two Winding Transformer Annotation Properties Dialogs

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A separate dialog is provided for three-winding transformers, **[Three-Winding Transformer Annotation Properties]** (Figure 3-32).

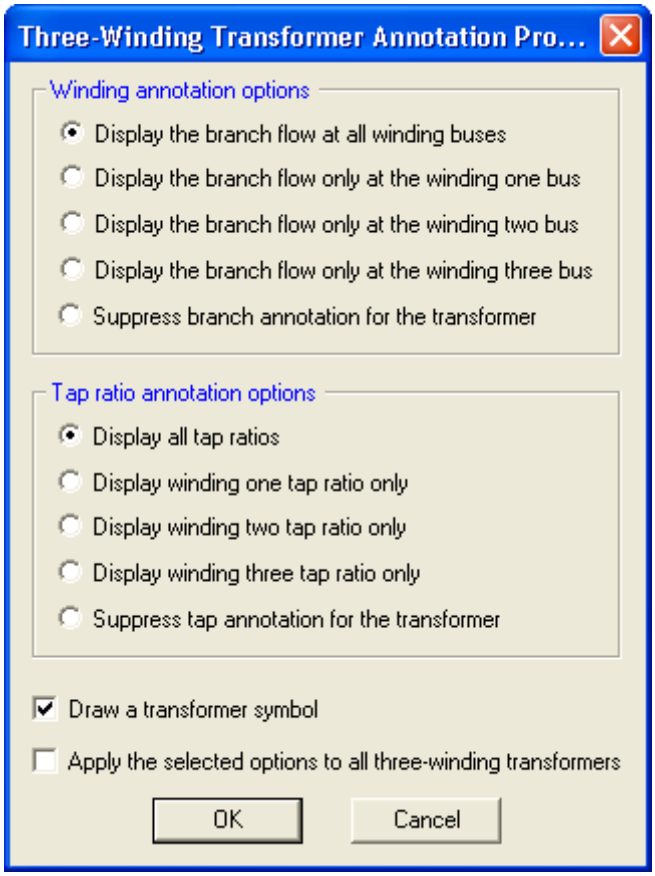


Figure 3-32. Three-Winding Transformer Annotation Properties Dialog

Dialogs are also available for annotating generators and multi-terminal dc lines (Figure 3-33) and FACTS devices (Figure 3-34).

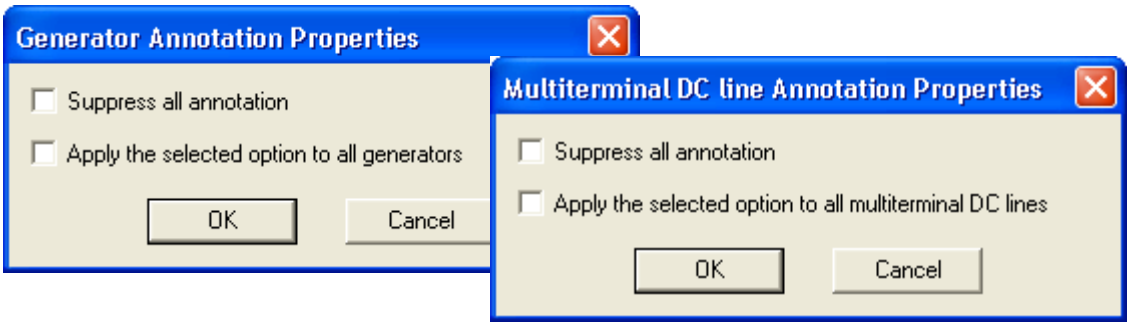


Figure 3-33. Other Annotation Properties Dialogs

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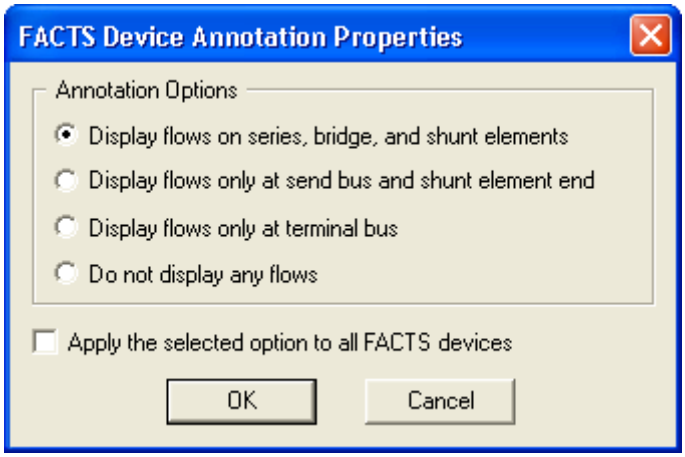


Figure 3-34. FACTS Device Annotation Properties Dialog

3.6.16 Contouring



Diagram > Contouring > Enable Contouring

The **[Contour Settings]** dialog provides the means to control contouring used to visualize voltage violations and magnitude for bus, line or machine elements.

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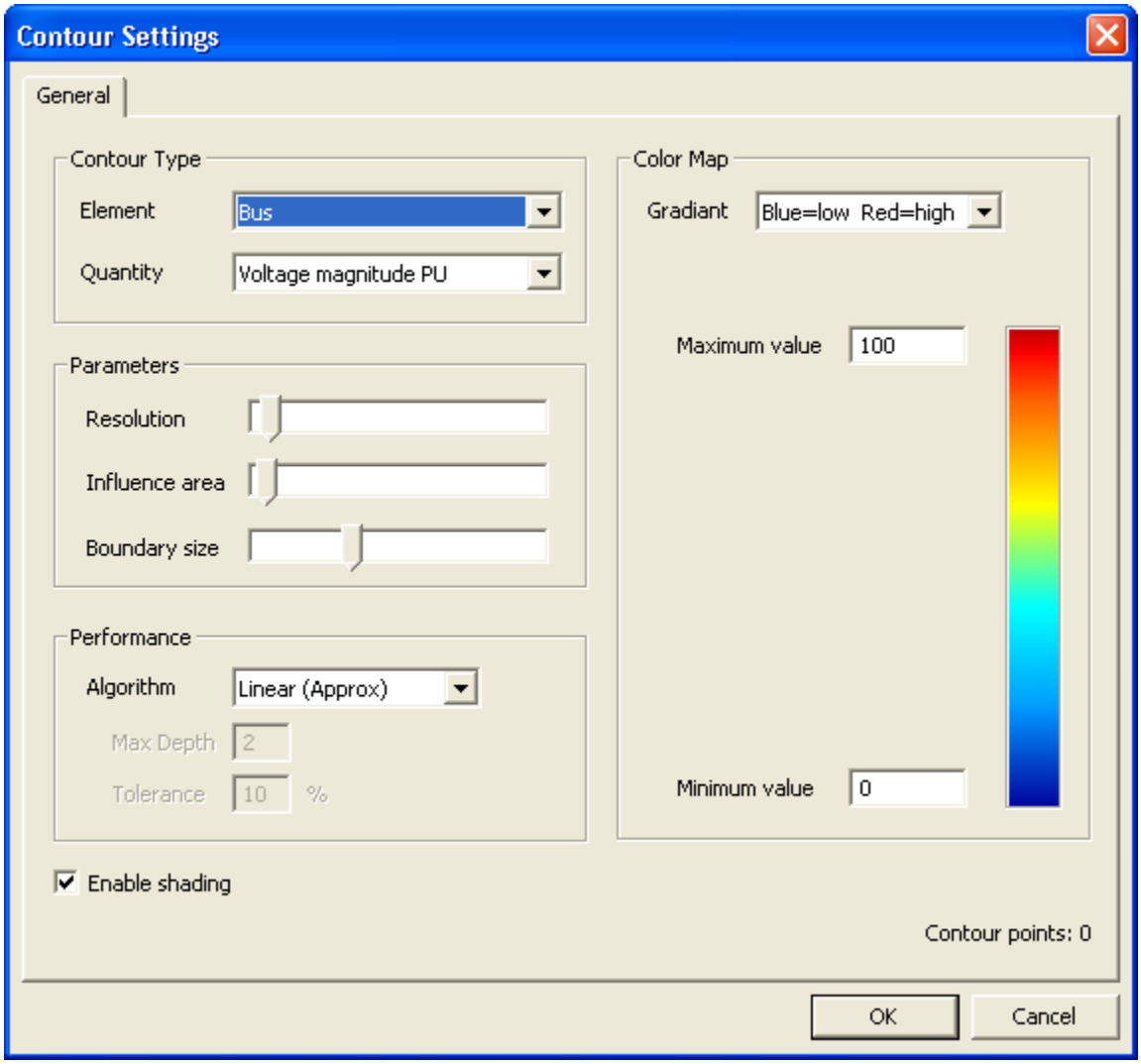


Figure 3-35. Contour Settings Dialog

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One of the following quantities must be specified along with the contour type:

Bus Element

- Voltage Magnitude, pu
- Voltage Magnitude, kV
- Angle deg
- Angle rad
- Load P
- Load Q
- Load MVA
- Area
- Owner
- Zone
- Scheduled Voltage

Line Element

- MW Flow
- MVar Flow
- MVA Flow
- % Rating
- Ampe
- pu Current

Machine Element

- QGen
- QMax
- QMin

The three contour parameters are selected from a sliding scale.

Resolution

Range 10 - 2000. To compute values across the entire contouring region, a virtual grid is super-imposed on the contour points. The value at each grid point is interpolated using actual values at the contour points. The weights are inversely proportional to the square of the distance from each contour point. Values at the grid points are then mapped to a color using the selected colormap before rendering is performed. A larger resolution results in a finer grid and more accurate results but also increases processing time.

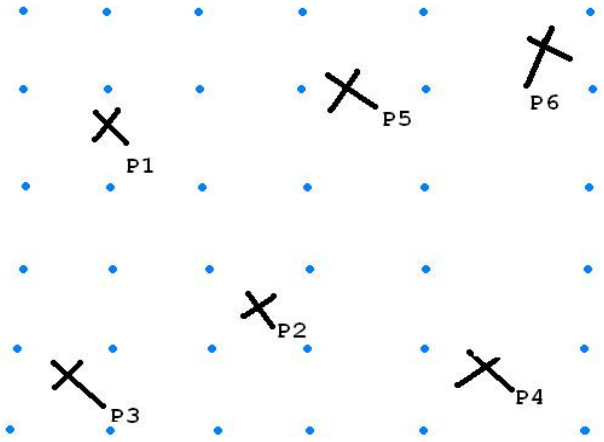


Figure 3-36. 6 X 6 Virtual Grid Superimposed on Contour Points P1 – P6

Influence Area

Range 1 - 100. Each contour point impacts a circular region, centered at the point with radius proportional to the influence setting. In the current implementation, the radius is equal to the square of the influence. A larger influence results in more blending and can be used for general overviews. Smaller settings are applied to highlight variations as each grid point is affected by fewer contour points. By decreasing influence, each grid point will be affected less by contour points farther away.

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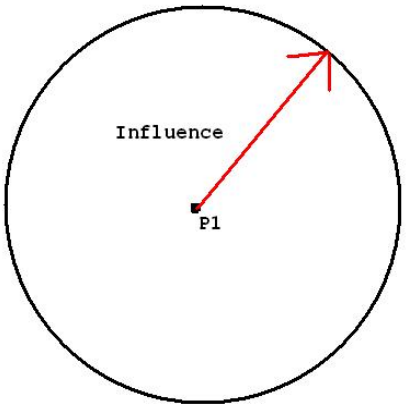


Figure 3-37. Circle of Influence around Contour Point P1

Boundary Size

Range 0 - 3. The typical setting is 0 or 1.

The user must specify one of the following interpolation methods.

Linear Approximation

The Linear Approximation method iterates over all contour points. A given contour point contributes to each grid point that lies within its circle of influence. Larger circles take longer to process therefore processing time is sensitive to changes in influence rather than resolution. This algorithm is faster than the Linear method while providing accuracy and is the current default. The order of the algorithm is O(m) where m is the number of contour points.

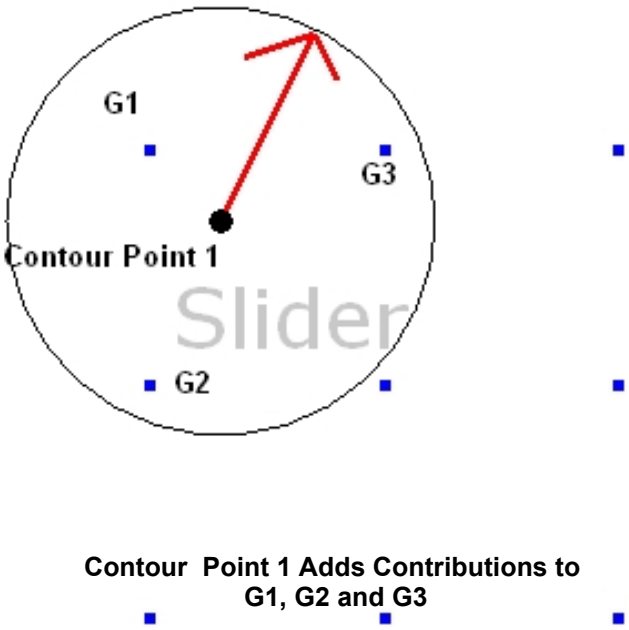
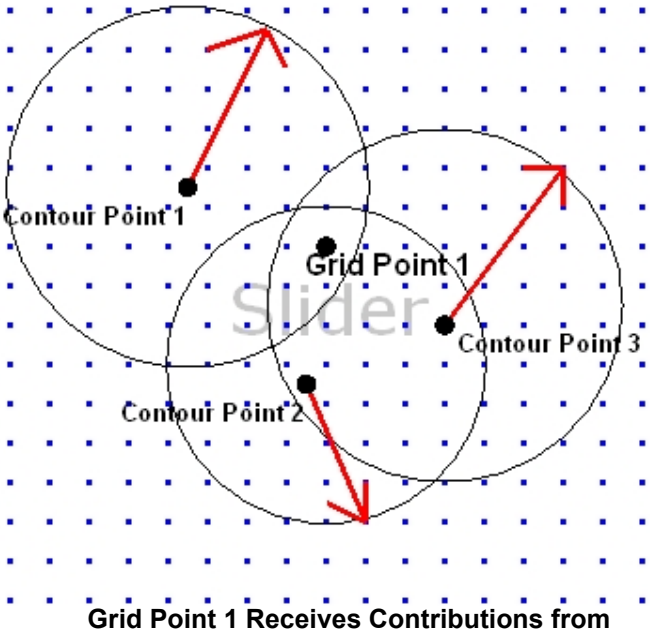


Figure 3-38. Linear Approximation Method

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Linear

The Linear Method is the most accurate. Processing speed depends on the resolution of the grid. Values at each grid point are interpolated using all contour points within a certain distance. The contribution from each contour point is the value of the physical quantity being plotted weighted (inversely) by the square of its distance from the grid point. Thus the speed of the algorithm is O(n^2 * m) where n is the desired resolution and m is the number of contour points. The time taken by the Linear Method is therefore very sensitive to the resolution of the grid.



Grid Point 1 Receives Contributions from Contour Points 1, 2, 3

Figure 3-39. Linear Method

Quad Tree

This method subdivides the diagram into uneven sized grid cells with finer subdivision occurring in areas that exhibit greater variation. Processing efforts are focused on regions that show signs of larger differences in the physical quantity being plotted. This occurs at the expense of other areas that have more uniform values. The QuadTree method generally results in a more blended diagram. For small values of depth, contours are generated very quickly.

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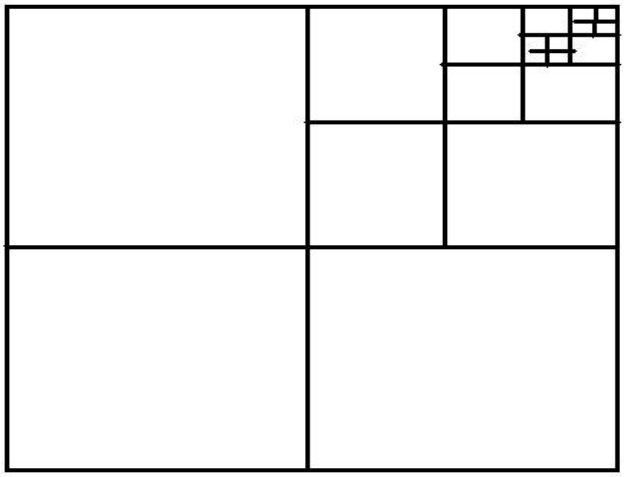


Figure 3-40. Non-Uniform Partitioning using the QuadTree Method

The Depth and Tolerance settings are required for the quad tree interpolation method.

Depth

The *Depth* setting determines the maximum level of subdivision that will occur. Further partitioning of grid cells will not occur even if the current difference is greater than the user selected tolerance value.

Tolerance Value

The *Tolerance Value* is needed to end subdivision. For a given grid cell, the difference between the maximum and minimum at the corners is calculated. If it is less than the specified tolerance, no further splitting occurs. Otherwise, the grid cell is recursively subdivided further.

The *Enable Shading* option can be disabled to create a non-shaded image. This is useful when viewing regions of finer subdivision resulting from the QuadTree partitioning.

The Color Map options determine which values are to be displayed and the colors assigned to them. the *Maximum* and *Minimum* values are used as a filter. Any points that lie outside this range will be removed. The discarded quantities don't contribute to interpolated values at the grid points. This feature can be used to see variations within a desired range. The *Gradient* options translate interpolated values at grid points into colors. The maximum value is mapped to the topmost color and the minimum is gets assigned to the bottom color. The user selects from the following color options:

- Red (dark to light)
- Green (dark to light)
- Blue (dark to light)
- GrayScale (dark to light)
- Blue=low Red=high (default)
- Red=low Blue=high

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Helpful Hints

- The Linear Approximation method is recommended unless extreme accuracy is needed or only a generalized picture is desired.
- When using the Linear Approximation, start with the default influence setting. Smaller values highlight detailed variations better. Increase influence slowly and only as necessary.
- Larger influence values result in more blending and tell a more generalized story.
- When using the Linear method start with the default value for resolution, increase it slowly if any square-like artifacts appear. Once the artifacts are gone, a further increase in resolution can be tried if desired but after a certain point the law of diminishing returns goes into effect.
- Both Linear and Linear Approximation are sensitive to the number of contour points.
- For diagrams with large extents, larger influence values are needed. If at first no contours appear, try increasing the influence while keeping the resolution constant. If the contours appear but exhibit square-like grid cells, increase the resolution till you see a better picture.
- Use a Bounding Box of 0 or 1, especially for diagrams with large extents.
- To get a generalized view of a large area, use larger values for influence.
- As a general recommendation, start with defaults for influence and resolution. Increase each parameter only as needed while keeping the other constant to see the difference in the picture.
- Use the QuadTree method to get quick generalized pictures.
- Processing time is a combination of several factors in all cases. However the speed of the Linear Approximation method is affected mostly by the influence setting while that of the Linear method is determined by the resolution.
- Busbars and Branches are represented by several contour points, placed equally along the length. When plotting quantities for Busbars or Branches and using a minimum influence (value of 1) an oval effect is seen around the element being plotted. This is the union of all the circles centered at the contour points.

3.6.17 Customizing Tooltips

Diagram > Customize Tooltips > Show item ID only

Diagram > Customize Tooltips > Show item current results

Diagram > Customize Tooltips > Show item expanded parameters

Tooltips are the pop-up informational boxes that appear when the mouse cursor is held over a component in the diagram. The *Display Tooltips* option in the **[Diagram Properties]** dialog must be enabled for tooltips to function (see Section 3.2). If the option to display tooltips is enabled, one of the three types of information can be displayed in the tooltip (see Figure 3-41). The example shows possible tooltips for a bus.

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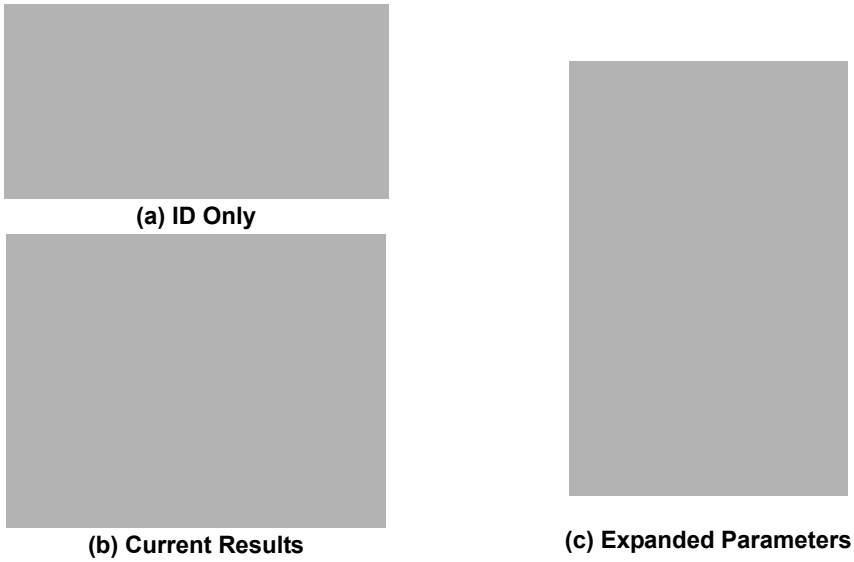
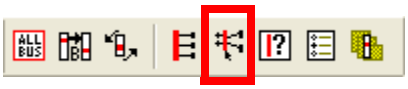


Figure 3-41. Alternative Tooltip Presentations for a Bus

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3.7 Using Auto-Draw



One of the features of **[Diagram]** is the ability to automatically build or expand one-line diagrams of power flow cases bus-by-bus or by groups of buses. This facilitates the rapid population of a blank **[Diagram]** with a one-line diagram of the open power flow case.

Using the *Auto Draw* function will place a selected bus on the diagram graphically, and automatically draw adjacent equipment, including adjacent buses. The Auto Draw feature also allows the specification of the number of bus levels to extend out of the first bus. Auto Draw will draw and connect these buses and neighboring facilities of all of these buses.

The user has two options to draw a diagram quickly. The first approach involves the following steps:

1. Open a blank diagram (see Section 3.3).
2. Select the *Auto Draw* button and click in **[Diagram]** to set the bus position.

   The **[Select Bus for Auto-Draw]** dialog (Figure 3-42) opens.
3. Select or specify the bus to grow and the number of levels to grow out from this bus.

   For example, using the *savnw.sav* power flow case, bus 101 has been selected and *grown* two levels (see Figure 3-42).

The bus, all its equipment, all lines, transformers, and attached buses are then laid out in **[Diagram]**. In the example, Bus 101 bus and its one neighboring bus are drawn, along with their attached equipment and connecting transformer branch.

If a Bus Location file has been opened and the *Use Bus Location file to AutoDraw/Grow items* option on the **[Program Preferences]** dialog has been selected (see Section 3.6.8), the buses will be placed at the locations specified in the Bus Location file. If the bus location data for the bus does not exist or the option is disabled, then the bus will be placed at the default location.

The process could be continued by selecting locations on the diagram and bus numbers from the **[Select Bus for Auto-Draw]** dialog. If, in this example, bus 152 were the next one selected and one level was grown out of this bus, the diagram would grow to include the other attached buses to bus 152, which are buses 153, 202, and 3004, their connected equipment, and the branches from these buses back to bus 152.

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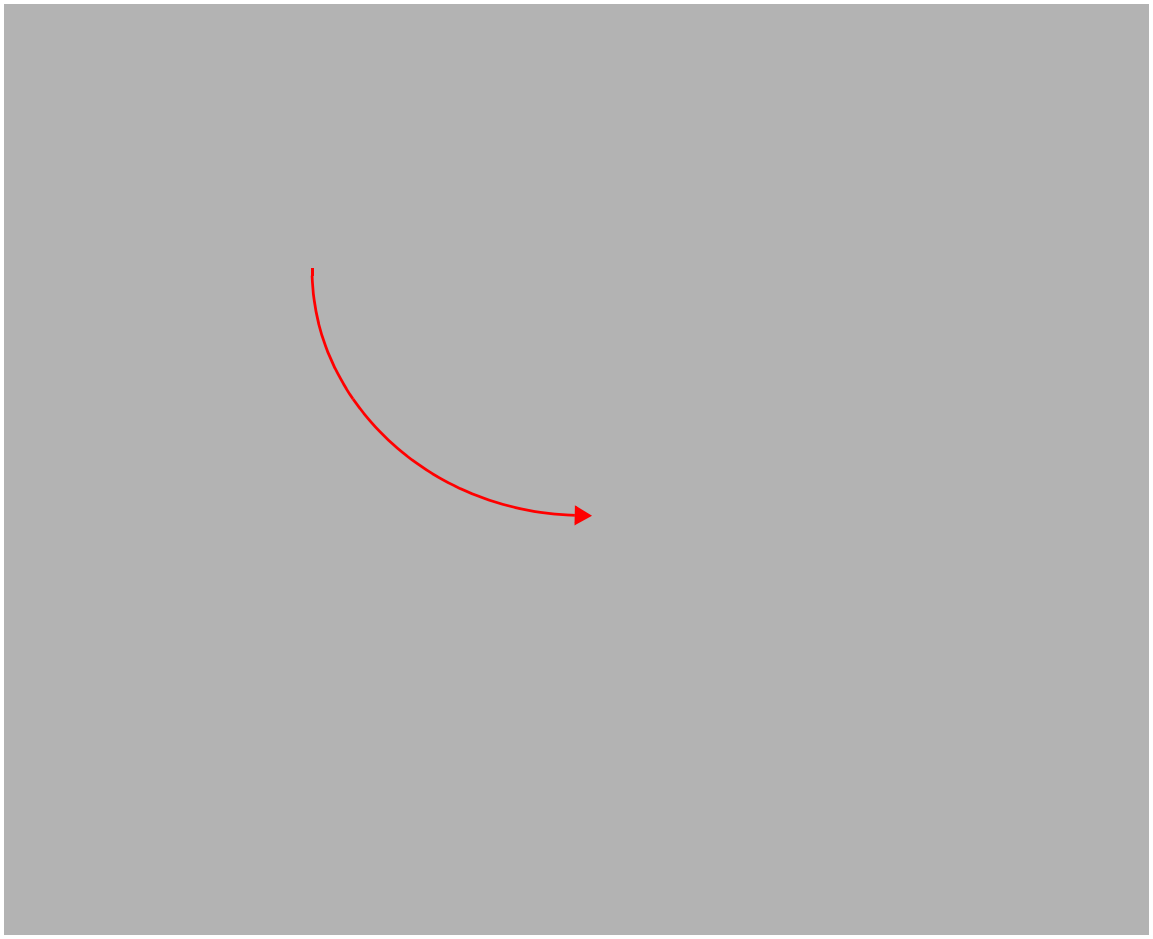


Figure 3-42. Drawing Bus 101 and Buses Two Levels Out, using the Auto Draw Toolbar Button

The second approach to start drawing a diagram is to reverse the order of actions, as follows:

1. Select (highlight) the desired starting bus in **[Network Tree]**.
2. Click the *Auto Draw* button from the Diagram Toolbar.
3. Click in **[Diagram]** to place the bus.

If the first bus selected in **[Network Tree]** was bus 101, the result would be bus 101 and 151 plus all equipment attached to either bus. The process could be continued by selecting other buses from **[Network Tree]**.

If there is a bus or buses already drawn on the one-line diagram it is possible to use the *Grow* option from **[Network Tree]** to add more one more level of buses. Also, the ***Grow N Levels...*** option can be used to add multiple levels of buses to the diagram. Right-click the bus in **[Diagram]** (see Figure 3-11).

Selecting ***Grow N Levels...*** on either of these menus will initiate the drawing of all buses attached to the bus selected, along with the connected equipment and the branches back to the selected bus

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N levels out from the bus. This method assists in rapidly creating a one-line diagram of the open power flow case.

If a connecting bus or branch already exists on the diagram when a Grow is performed, any new connections will be made to the existing equipment.

3.7.1 Bus Orientation

Diagram > Auto Draw style > Vertical buses

Diagram > Auto Draw style > Horizontal buses

When buses are added to **[Diagram]** using *Auto Draw* or by selecting the *Grow* option described in Section 3.6, the buses will be drawn either horizontally or vertically. This function permits shifting between vertical and horizontal bus representations as the one-line diagram is being developed

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3.8 Displaying Power Flow Results



Diagram > Results > Powerflow results

Superimposing power flow results on a one-line diagram is an effective means of presentation. This section will highlight some of those features on the assumption that a slider diagram (**.sld)* has been created. The following examples use the *savnw.sav* and *savnw.sld* files provided in the Example folder of the PSS®Express installation.

A portion of the network power flow results can be seen in Figure 3-43. Here the annotation is selected to show:

- Bus name and number with voltages shown in pu and kV
- Branch flows at to and from ends
- Transformer taps on both sides
- MW and Mvar flows on branches and in equipment
- Signs as opposed to arrows to show flow directions. Plus signs indicate flows out of a bus and minus signs indicate flows into a bus.



Figure 3-43. Power Flow Results in Diagram View

3.8.1 Power Flow Annotation

Diagram > Annotation…

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Results of a power flow analysis may be modified as a whole for presentation using a set of annotation dialogs. For example, the user can choose to show all the information in Figure 3-43 for all buses, branches and equipment, including generation, or to globally prevent showing of particular types of information (such as no branch flows on all branches) or to select particular buses, lines and equipment to have their annotation suppressed.

The right-click menu applied to a selected diagram item opens a specific annotation dialog for that item (see also Sections 3.6.5 and 3.6.15).

When power flow results are shown, the **[Powerflow Data Annotation]** dialog provides a *Diagram Annotations* tab (Figure 3-44) to modify the presentation for printing or to save to a file.

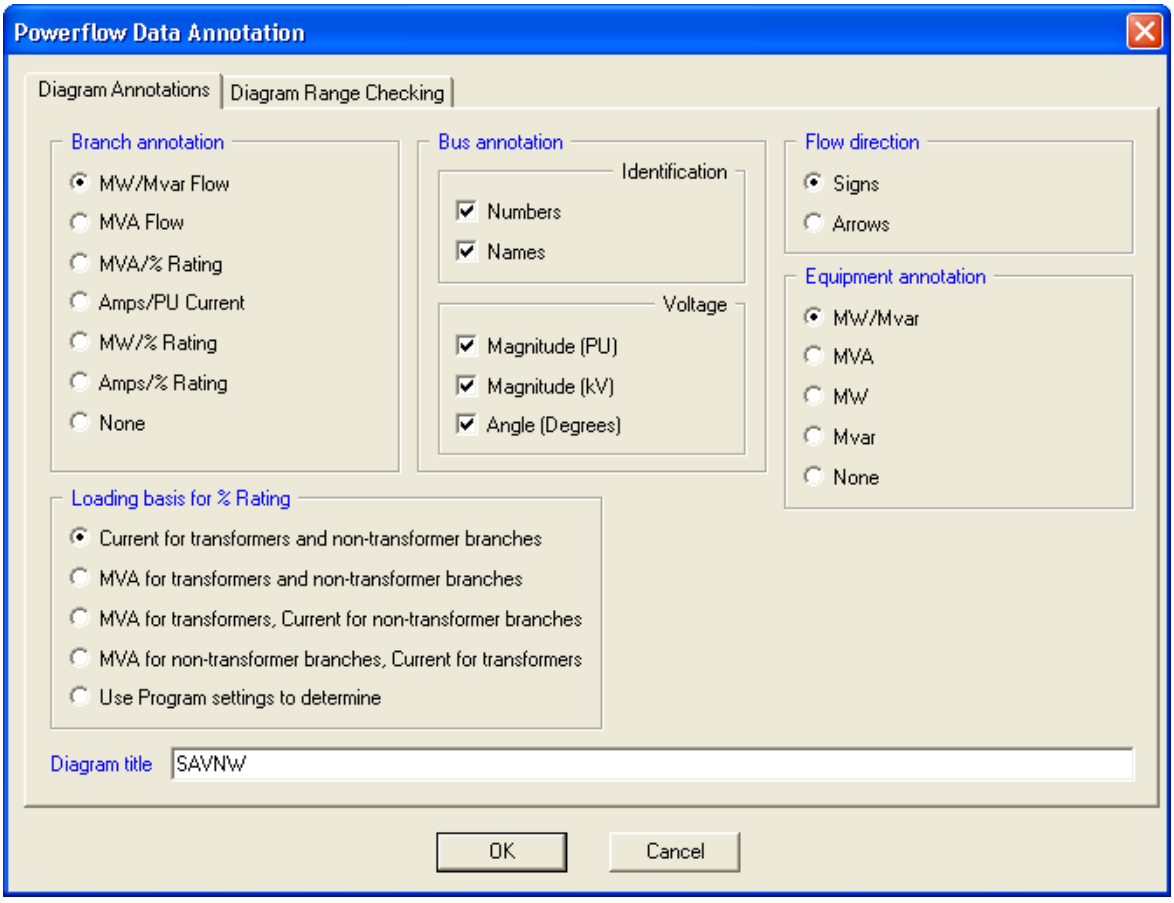


Figure 3-44. Power Flow Data Annotation Dialog

Options include the type of results annotation information to be displayed for branches, buses and equipment. The user can represent flows on the branches with either signs or arrows (see Figure 3-45). The real power, in each case is shown above the branch.

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(a) Signs **(b) Arrows**

Figure 3-45. Signs and Arrows to Display Flow

3.8.2 Range Checking

The **[Powerflow Data Annotation]** dialog also provides a *Diagram Range Checking* tab (Figure 3-46), which is a very useful tool. Not only can **[Diagram]** show numerical results for bus voltage, line flow and equipment loading, but color-coding can be set up to provide rapid identification of problems. It is possible to highlight branches that are overloaded and bus voltages that are either higher than a selected high limit or lower than a selected low limit.

In addition, the user can choose to over-ride the color selected for bound items and select colors for each voltage level in the network. This makes it easy to identify networks at different voltage levels.

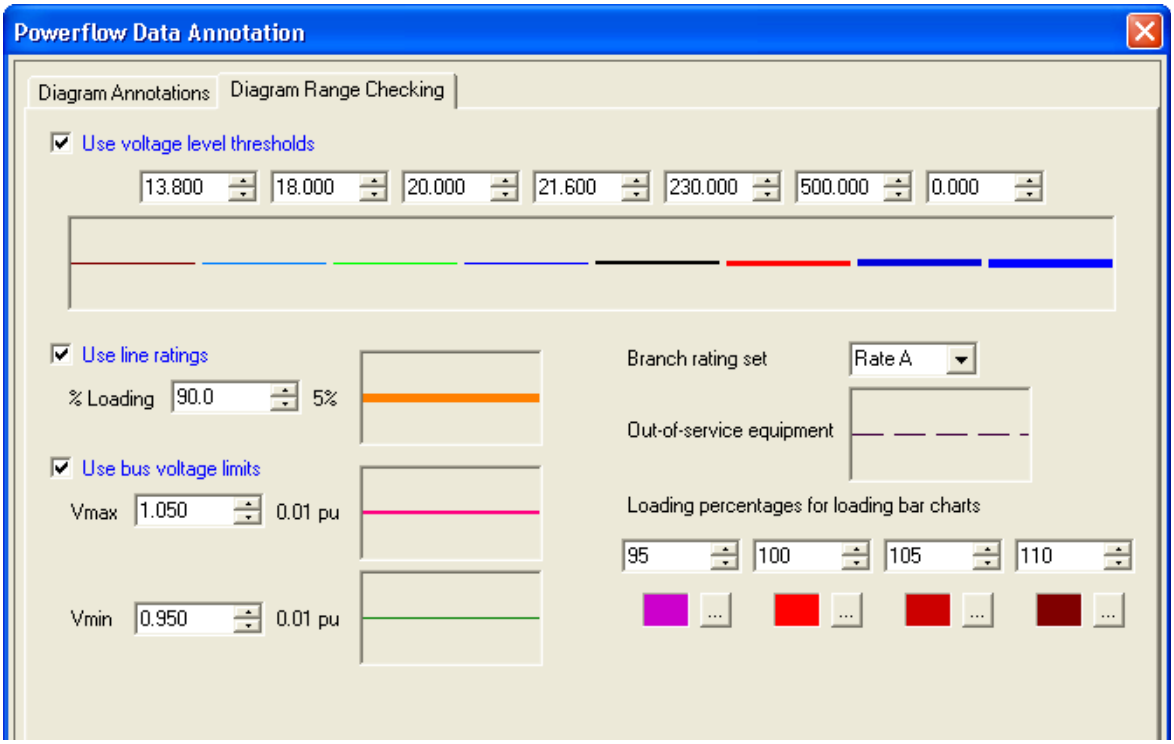


Figure 3-46. Diagram Range Checking Tab

Use voltage level thresholds: When enabled, this option indicates equipment and lines based on voltage level. Threshold levels may be specified by entering a value directly in the input field or using the scroll buttons. Click the sample line corresponding to each threshold to open a **[Voltage level]** dialog for each threshold (Figure 3-47) to change line styles and colors. Click the palette

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button to open the **[Color]** dialog (Figure 3-3).

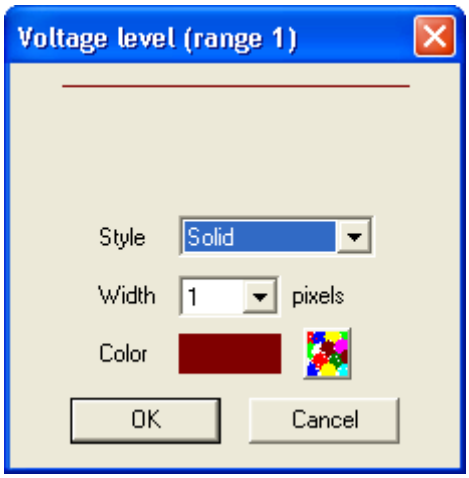


Figure 3-47. Voltage Level Dialog

Use line ratings: When enabled, this option indicates lines that are loaded above the specified percent loading value. Click the sample line to open a **[Line ratings appearance]** dialog (Figure 3-48) to change line style and color. Click the palette button to open the **[Color]** dialog (Figure 3-3).

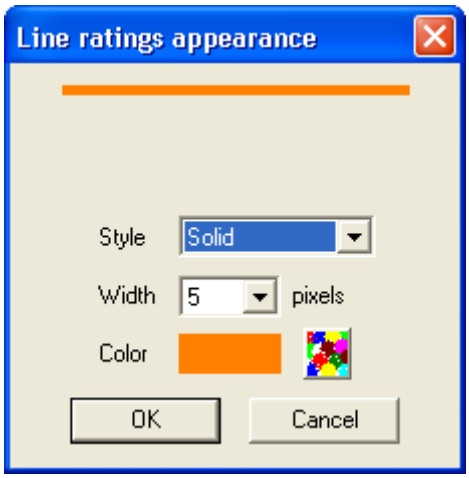


Figure 3-48. Line Ratings Appearance Dialog

Branch Rating Set: This option provides a pull-down list of branch ratings to which the % loading value is to be applied. Rate A is typically used for normal power flow conditions while Rate B or C would be used to examine contingency conditions where loadings are often allowed to be higher.

Use bus voltage limits: When enabled, this option indicates buses with voltages above or below specified maximum and minimum voltage levels. Click the sample line to open the **[Bus voltage limit appearance]** dialog (Figure 3-49) to change line style and color. Click the palette button to open the **[Color]** dialog (Figure 3-3).

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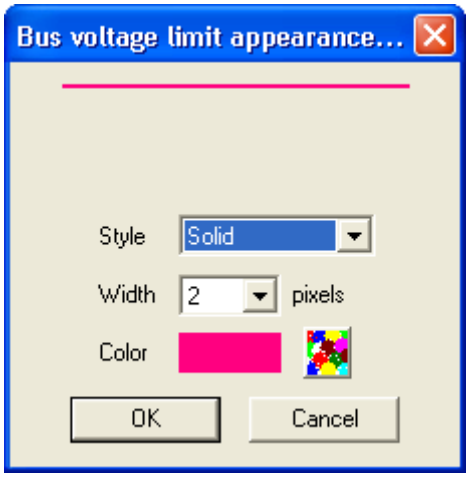


Figure 3-49. Bus Voltage Limit Appearance Dialog

Out-of-service equipment: This setting indicates status of out-of-service equipment. Click the sample line to open the **[Out-of-service equipment appearance]** dialog (Figure 3-50) to change line style and color. Click the palette button to open the **[Color]** dialog (Figure 3-3).

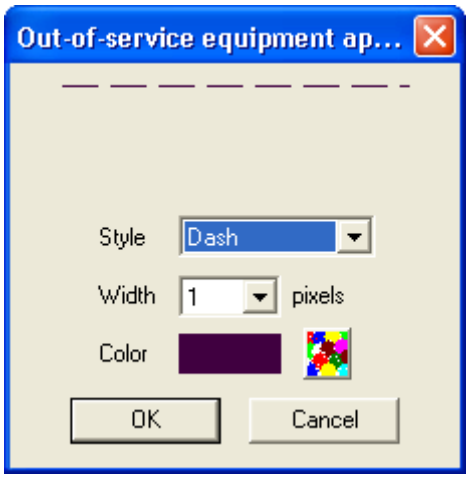


Figure 3-50. Out-of-Service Equipment Appearance Dialog

Loading percentages for loading bar charts: These fields are used to control the percentage at which overload colors are applied to the loading bar charts. Overload colors by default progress from purple to red to deeper shades of red. Click **[…]** to open the **[Color]** dialog to modify these colors.

The following example shows that that buses 206 and 3018 (and others) are highlighted for having voltages in excess of the high limit, specified at 1.02 pu. Bus 203 and 205 are highlighted for having voltages below the low limit, 0.97 pu.

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The circuit one between buses 153 and 154 (and others) are highlighted to show their loading is in excess of 90% of their Rate A.



Figure 3-51. Power Flow Diagram showing Effect of Range Checking

3.8.3 Adding a Summation Record



The user can clarify the power flow results visualization by including a summation block for flow information. This could be useful to report current results when testing a variety of network conditions. The **[Edit Summation]** dialog provides fields for a title line for the summation and a variety of system results to sum. For example, these can be generator output and line flows. The example in Figure 3-52 sums the difference between generator output at bus 101 and the flow on circuit 1 of the line from bus 151 to 201.

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The summation block is placed in **[Diagram]** at the point selected by the user. Subsequently, it can be edited and/or moved to a different location.

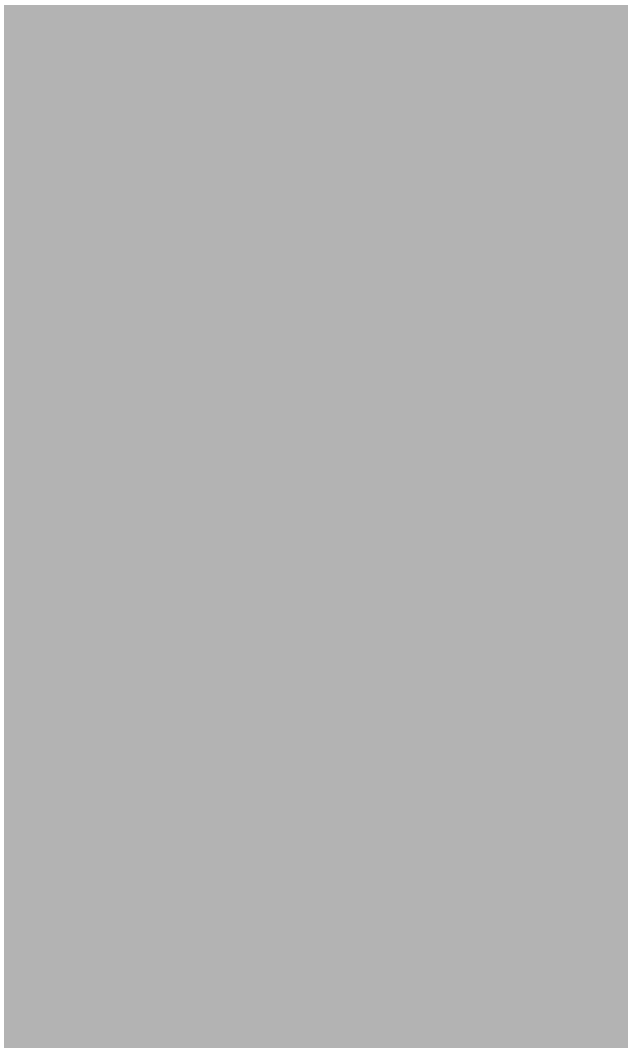


Figure 3-52. Edit Summation Dialog

To cancel the summation activity, press the **[Esc]** key. Refer to *PSS®E Program Operation Manual*, Summation Block Structure for additional information.

3.8.4 Adding a Report Node



The user can clarify the power flow results visualization by including a report node to display any type of information that may be of importance. This could be useful to report current results when testing a variety of network conditions.

The report node is Python based. Python code can be written to calculate and return any value, using existing Python APIs, that may be of interest to the user.

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The **[Edit Report Node]** dialog provides fields for a title line for the report node, passing arguments to Python, defining the Python function, and defining the Python module that contains the Python function. The example in Figure 3-53 shows the definition of a report node called *Area Summary:*, using the Python function *area_summary*, in the Python module *pssgrpg*, passing *1* as an argument.

The report node is placed in **[Diagram]** at the point selected by the user. Subsequently, it can be edited and/or moved to a different location.

To cancel the report node activity, press the **[Esc]** key.

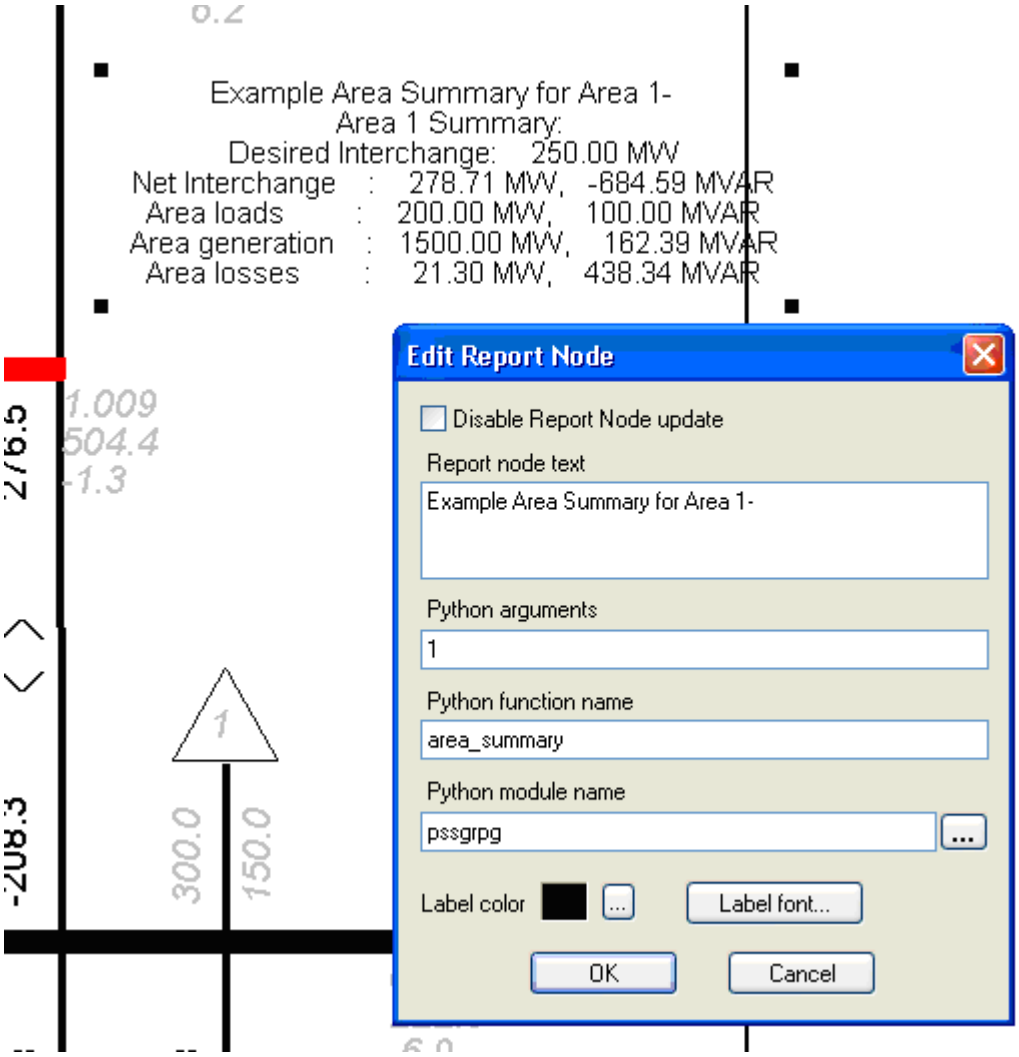


Figure 3-53. Example of Report Node

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3.9 Displaying Impedance Data

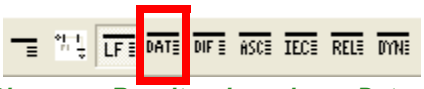


Diagram > Results > Impedance Data

This function displays impedance and equipment rating data. By default the diagram displays the generator scheduled power and the reactive power limits. Bus information includes the bus number, name and base kV. Line and transformer information shows the R, X and B values, as appropriate, and the transformer tap information is included.



Figure 3-54. Example of Impedance Data in a One-Line Diagram

3.9.1 Impedance Annotation

Diagram > Annotation... opens the **[Impedance Data Annotation]** dialog (Figure 3-55).

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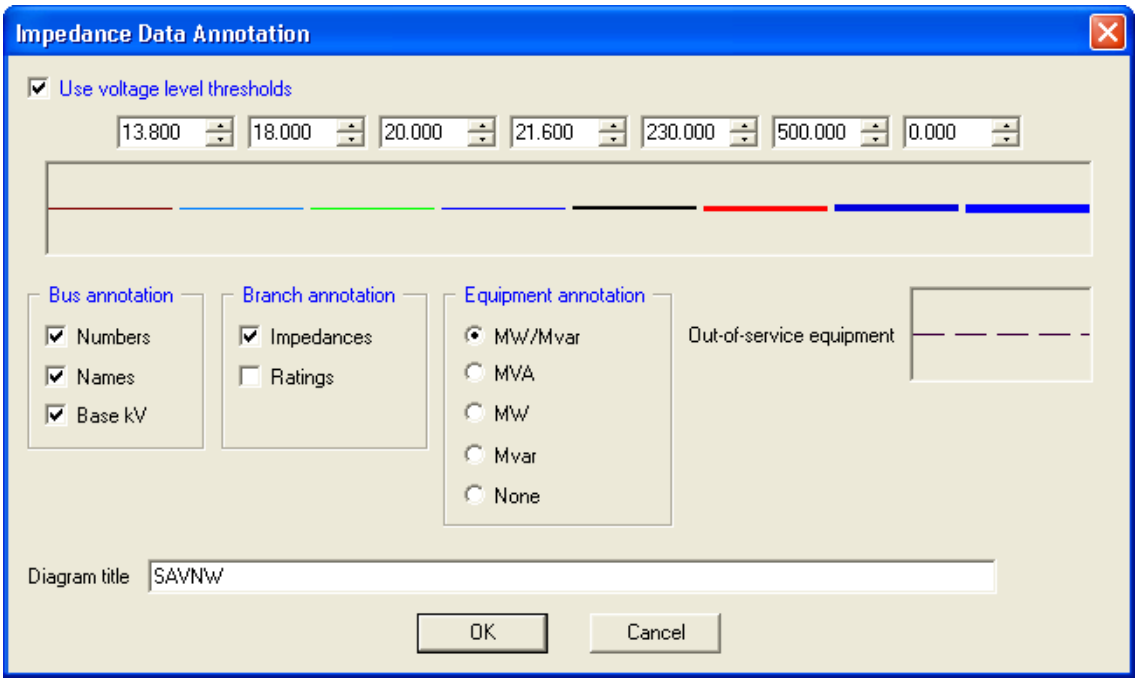


Figure 3-55. Impedance Data Annotation Dialog

Bus, branch, and equipment annotations, as well as voltage level thresholds are available in the impedance data annotation.

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3.10 Displaying Graphical Difference Data

Activity GDIF

Requirements / Prerequisites
Validly specified power flow case. Saved Case file (*.sav) Both cases solved to an acceptable mismatch level.



Diagram > Results > Graphical Difference Data…

The **[Compare Cases on a Diagram]** dialog (Figure 3-56) requires the specification of a saved case against which to compare the working case. Click **[…]** to open the selection window for the required *Saved Case file* (*.sav). Click **[OK]** to perform the activity. A summary of the comparison is routed to the **Report** tab.

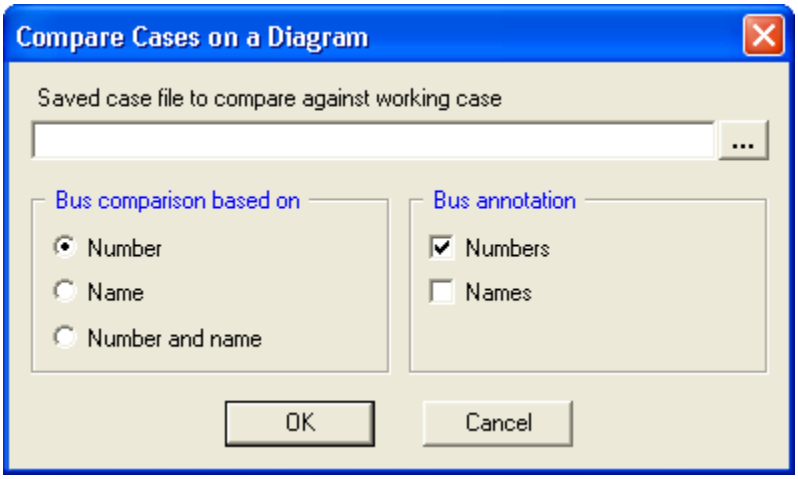


Figure 3-56. Compare Cases on a Diagram Dialog

Differences in solution results and bus boundary conditions between the current power flow case and the selected saved case are displayed. Differences are always calculated as comparison case values minus power flow case values. At each bus in the bus comparison list, voltage differences in per unit and phase angle differences in degrees are shown. All other difference values are shown in MW and Mvar. An example diagram is shown in (Figure 3-57).

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Figure 3-57. Example of Case Comparison Differences

The difference fields are left blank for equipment items present in the power flow case, but not in the comparison case. Those items present in the comparison case that are not included in the power flow case are omitted from the diagram.

Load and shunt differences include voltage sensitivity effects.

Differences in flow into a converter bus of DC line n are shown if all of the following conditions apply:

- DC line n is present in both cases.
- The converter bus is in the bus compare list.
- The same converter bus is specified in both cases.

Two-terminal dc lines are annotated according to the annotation properties currently set in the diagram.

Differences in the sending end bus shunt element of FACTS device n are shown if all of the following conditions apply:

- FACTS device n is present in both cases.
- The sending end bus is in the bus compare list.
- The same sending end bus is specified in both cases.

If the above conditions are satisfied and FACTS device n has a series element in the power flow case, differences in series flow at the sending end bus are also shown.

Differences in series flow at the terminal end bus are shown if all of the following conditions apply:

- FACTS device n is present and has a series element in both cases.
- The terminal end bus is in the bus compare list.
- The same terminal end bus is specified in both cases.

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3.10.1 Comparison Annotation

Diagram > Annotation... opens the **[Graphical Case Comparison Data Annotation]** dialog (Figure 3-58).

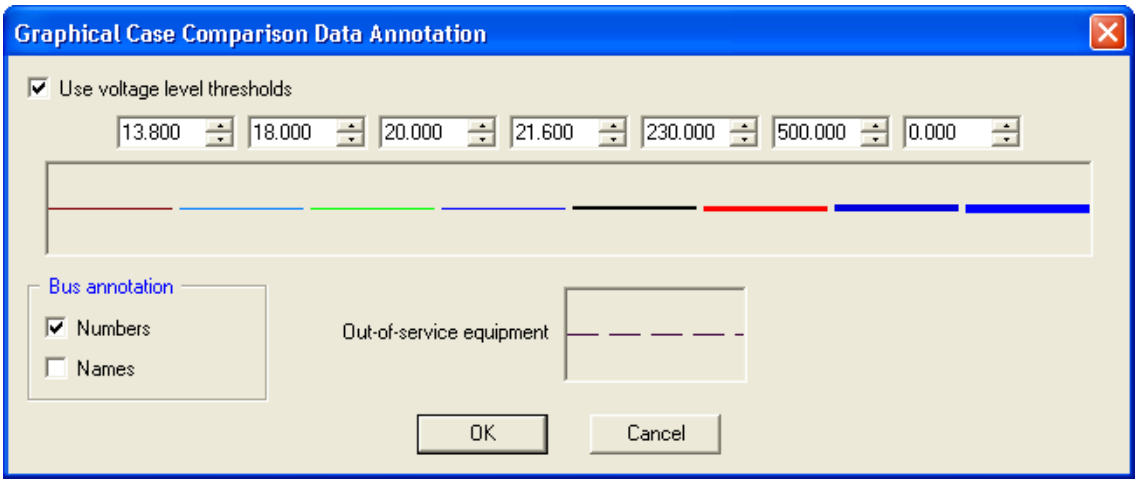


Figure 3-58. Graphical Case Comparison Data Annotation Dialog

Only bus annotation and voltage thresholds are available in the case comparison annotation.

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3.11 Creating a Bus Display

Activity GOUT/GEXM

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch level.



Diagram > Generate a graphical power flow bus display

This function provides the means to create a one-line drawing starting with a single bus. Enter the desired bus number in the **[Select Bus]** dialog (Figure 3-59) and click **[OK]**. A diagram of power flow results is generated for the selected bus, showing associated lines and equipment and distant to buses. All distant buses are represented as nodes (Figure 3-60).

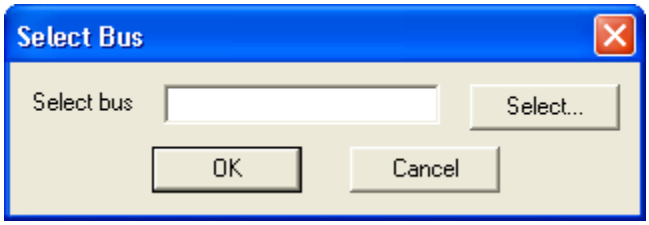


Figure 3-59. Select Bus Dialog

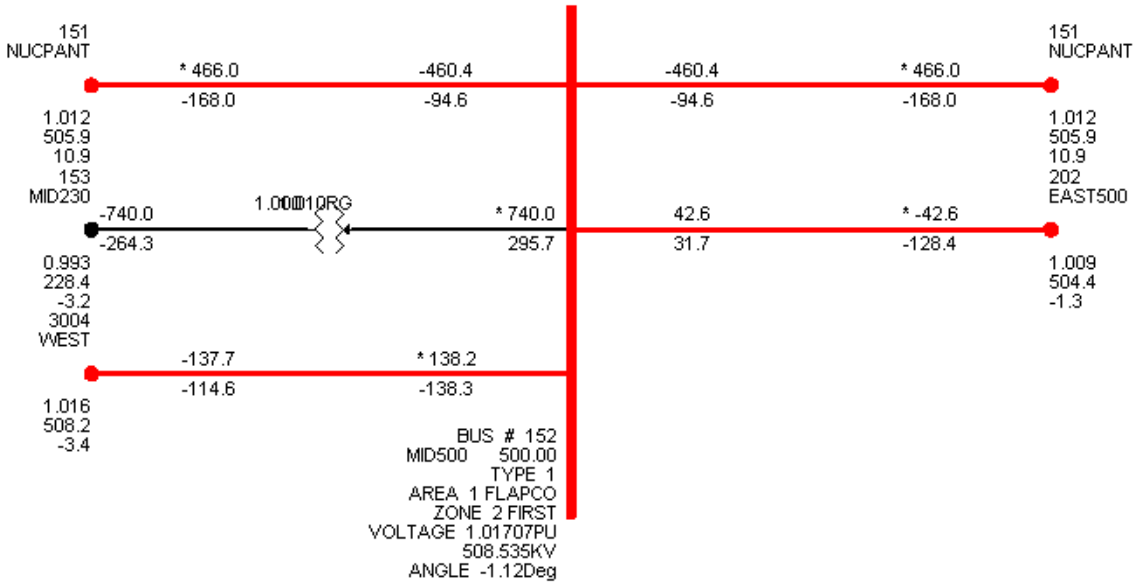


Figure 3-60. Example of Single Bus Display, Power Flow Results

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All **[Diagram]** functions are available using the single bus display. Thus, selecting ***Diagram > Results > Impedance data*** and the *Current loadings* icon from the active diagram in Figure 3-60 produces the example shown in Figure 3-61. Switching and data changing functions are available for any data display.

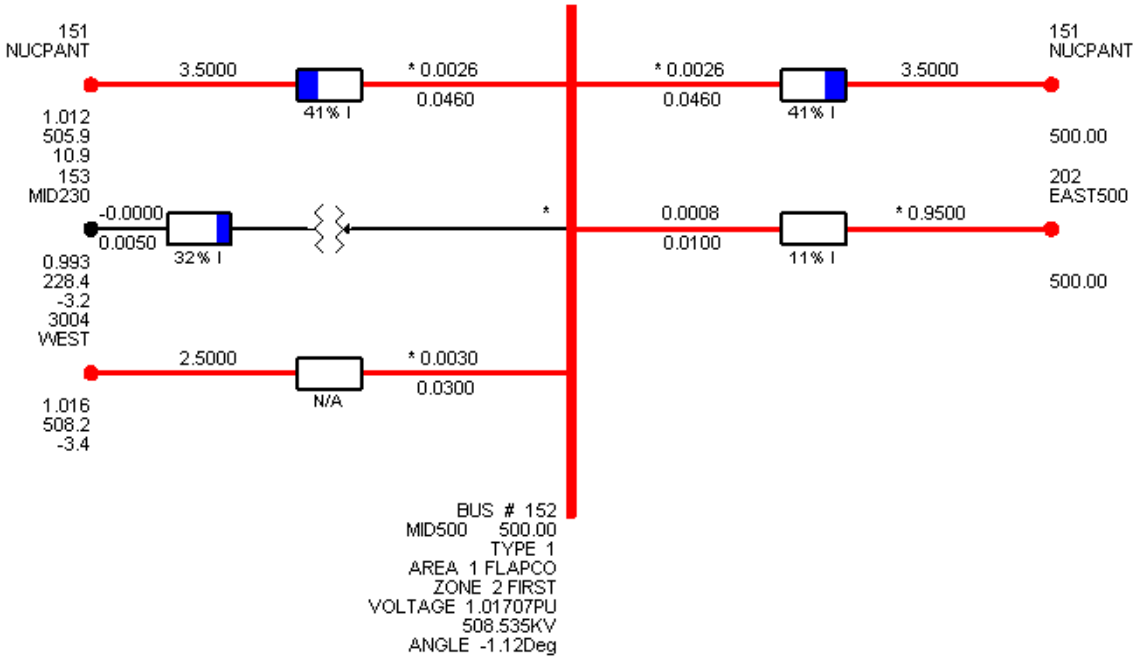


Figure 3-61. Example of Single Bus Display, Impedance Data

The single bus display is also available from **[Network Spreadsheet]**. Select the desired *Bus Number* on the *Buses* tab, right-click, and select ***Create bus display (GOUT/GEXM)***. The default view (*Power Flow Results*) is displayed.

3.11.1 Single Bus Display Labels and Symbols

Default labels for the single bus display depend upon the type of bus specified.

If the selected bus is a generator bus controlling the voltage at a remote bus, the remote bus number, name, and base voltage will be tabulated with the other selected bus information. Further, if the active and reactive power mismatch at the bus exceed 0.5 MVA or kVA, according to the power output option in effect, the mismatch information will be listed.

Each machine at a selected bus is separately displayed with the machine identifier printed inside of the generator symbol. In the *Power Flow Results* view, the machine loading is shown along with one of the H, L and R flags indicating the current reactive loading condition. In the *Impedance Data* view, the machine's defined power setting and reactive limits are shown.

Load and shunt elements are represented with actual loadings when shown in the *Power Flow Results* view view and ratings when shown in the *Impedance Data* view. Load identifiers are shown within the symbol.

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Branches are drawn from the top of the page in ascending to bus order (numeric or alphabetic according to the bus output option in effect). The to bus number and name are drawn on the extreme right of the branch line adjacent to the to bus busbar. Placing the cursor on the branch will pop up the line from and to bus numbers/names and the circuit identifier. In the *Power Flow Results* view flows are shown as active and reactive power leaving the from bus (the selected bus) and as active and reactive power arriving at the to bus end. In the *Impedance Data* view, the R, X and B data quantities are shown.

For two-winding transformer branches, two transformer symbols are drawn between the flow and the to bus. Off-nominal turns ratio and any nonzero phase shift angle are displayed between the flow and the first transformer symbol and between transformer symbols.

Any three-winding transformer connected to the bus is drawn similar to the two-winding transformer except one more branch and bus are displayed, and off-nominal turns ratio and any nonzero phase shift angle are shown only for the winding connected to the bus box.

When the multisection line reporting option is enabled (see Changing Program Settings), the far end to bus (rather than the closest dummy bus) of each multisection line connected to the bus being displayed is shown as its to bus.

An asterisk (*) is drawn at the metered end of each branch. If a branch is a member of a multisection line grouping and the multisection line reporting option is enabled, the asterisk indicates the metered end of the line section adjacent to the bus being displayed and a plus sign (+) designates the metered end of the multisection line grouping.

Any dc lines connected to the bus are drawn before any AC branches in ascending DC line number order, with two-terminal lines listed first, followed by any multi-terminal lines. Power flow conventions for dc lines are as in the Bus based report output format. Alpha, gamma, and the converter transformer off-nominal turns ratios are displayed and tagged as in the Bus based reports. For multi-terminal lines, no to bus end conditions are listed.

Any FACTS device with no series element which is connected to the selected bus is drawn to the left of the bus. It is illustrated with a straight line drawn from the displayed bus and connects to a rectangular box, which is attached to a shunt symbol. The rectangular box contains a power flow direction arrow pointing in the direction of the shunt. The FACTS device number is displayed above the device.

Any series FACTS device connected to the selected bus is drawn to the right of the bus with a line connecting that bus and its associated terminal/send bus. The series element is illustrated by a circle containing an embedded arrow and is located midway on the connecting line. The arrow indicates the power flow direction (sending end bus to terminal end bus direction). The FACTS device number is displayed to the right of the connected bus.

For series FACTS devices with non-zero shunt current and/or bridge active power transfer limits, when the selected bus is the sending end bus, two additional lines are drawn to represent the shunt and bridge element connections. The first line is drawn to the right of the sending end bus underneath the series element connecting line and stops at a point midway between the sending end bus and the terminal end bus. Another line is drawn from that point upward to the series element circle.

3.11.2 Navigating the Network Using the Single Bus Display

It is possible to proceed through the network graphically to examine power flow conditions and data or to check topology.

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To go from the initially selected bus to one of its to buses, double click the to bus node. The diagram will then feature that bus and its connections. Figure 3-62 shows the result of double-clicking from bus to bus.

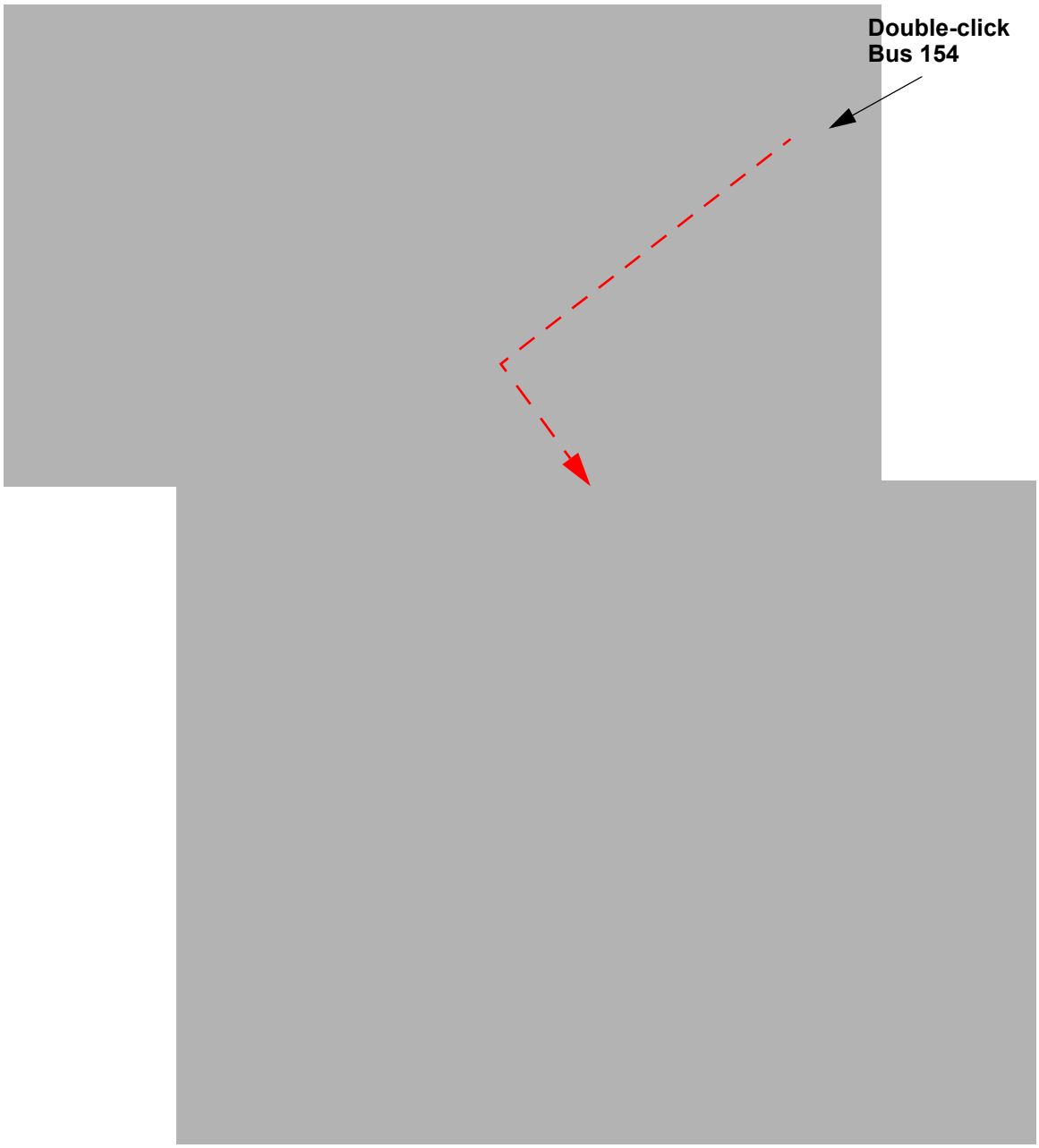


Figure 3-62. Scrolling through the Network

3.11.3 Converting Grow Mode

Convert Grow mode from GOUT to Auto-Draw

Single bus display diagrams are used for quickly traversing the network. A diagram originally created with GOUT/GEXM will respond to *growing* elements differently from a diagram created using Auto-Draw or Grow (see Section 3.7, Using Auto-Draw). During a Grow operation they simply

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display everything connected to the bus. The Grow operations pays no attention to whether or not the element exists in **[Diagram]**; the item is simply created again to lend clarity to the layout.

This function can be used to convert the *grow* behavior of a diagram from GOUT/GEXM to an Auto-Draw style *grow* where each specific element only is created once.

This option is available only when a **[Diagram]** is the active view.

Additional Information
PSS®E Program Operation Manual, Section 11.33, Displaying Power Flow Solution on the Diagram Section 11.34, Displaying Power Flow Data on the Diagram

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3.12 Generating a Graphical Report

Activity GRPG

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch level if network data is to be included on the drawing.



Diagram > Generate a graphical report (GRPG)…

This activity generates the old GRPG-style report. The **[Graphical Report Data file]** selection window requires entering the filename of an existing Graphical Report Data file (*.grp). The **[Terminal Read]** dialog (Figure 3-63) then requires a code for the output device. The most common response is 26, directing the output to the screen. Future releases will provide support for GPRG style reporting directly from **[Diagram]**.

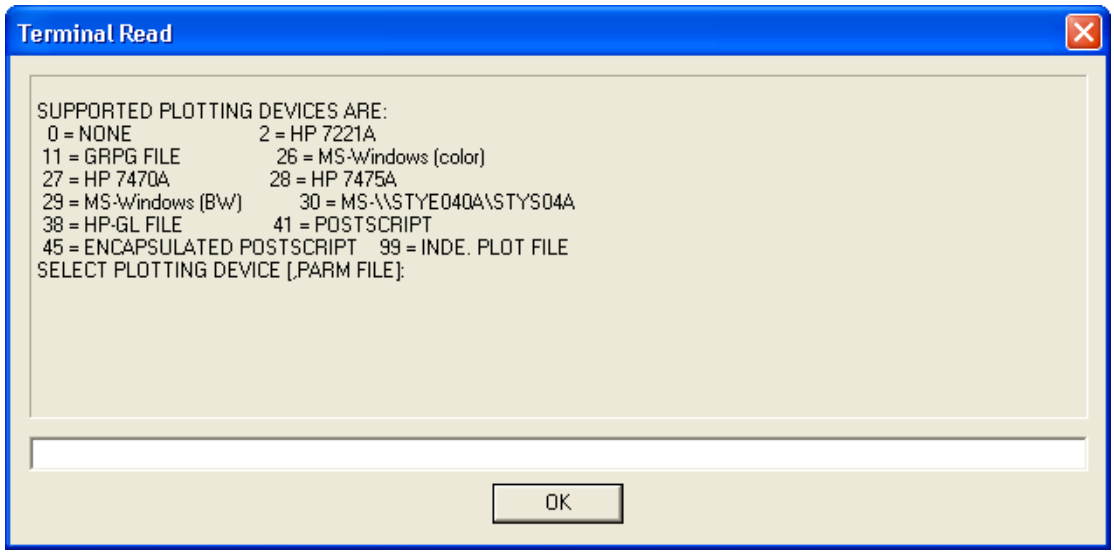


Figure 3-63. Terminal Read Dialog

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.

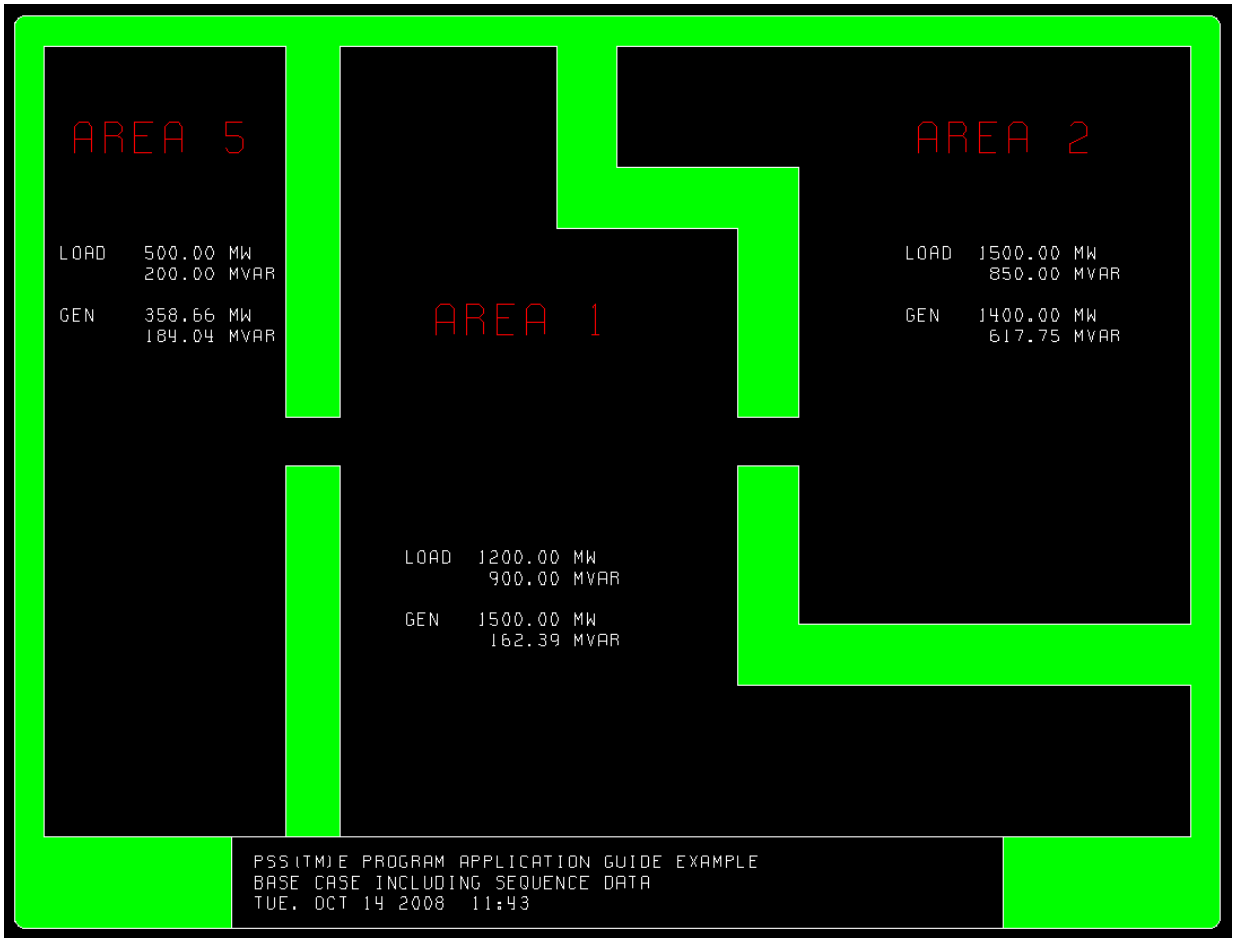


Figure 3-64. GRPG-Style Graphical Report

Additional Information
PSS®E Program Operation Manual, Section 11.30, Generating a Graphical Report

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3.13 Diagram Management

3.13.1 Diagram Contents

Diagram > Contents

The **[Diagram Contents]** screen presents an itemized list of the number of components in the entire diagram, including diagram items, annotations, images, and text annotations added by the user (Figure 3-65).

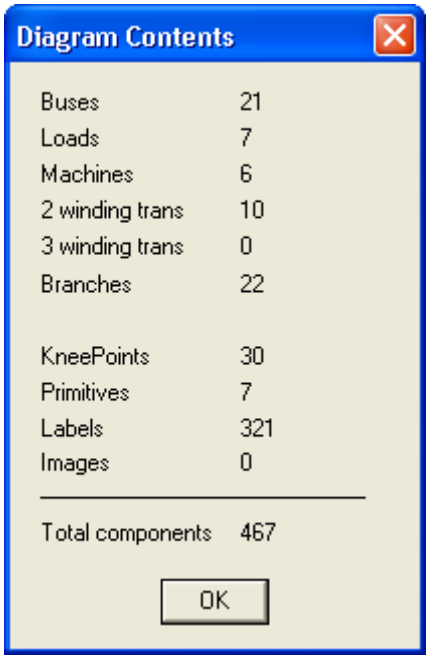


Figure 3-65. Diagram Contents

3.13.2 Errors/Missing Items

Diagram > Check

This function examines the entire network and diagram for apparent errors and missing items. The results are displayed in the **Progress** tab (Figure 3-66).

To examine the network by subsystem, select ***Diagram > Check by Subsystem***.



Figure 3-66. Results Generated by the Check Menu Option

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3.13.3 Diagram Layers

Diagram > Manage Layers…

A simple way to visualize layers is to imagine **[Diagram]** as an infinite number of sheets of clear mylar, stacked on top of another. Diagram items are drawn on a single sheet of mylar, that is, they belong to a particular layer. When all sheets are laid down, **[Diagram]** is seen in its entirety.

This function opens the **[Layers]** dialog (Figure 3-67).

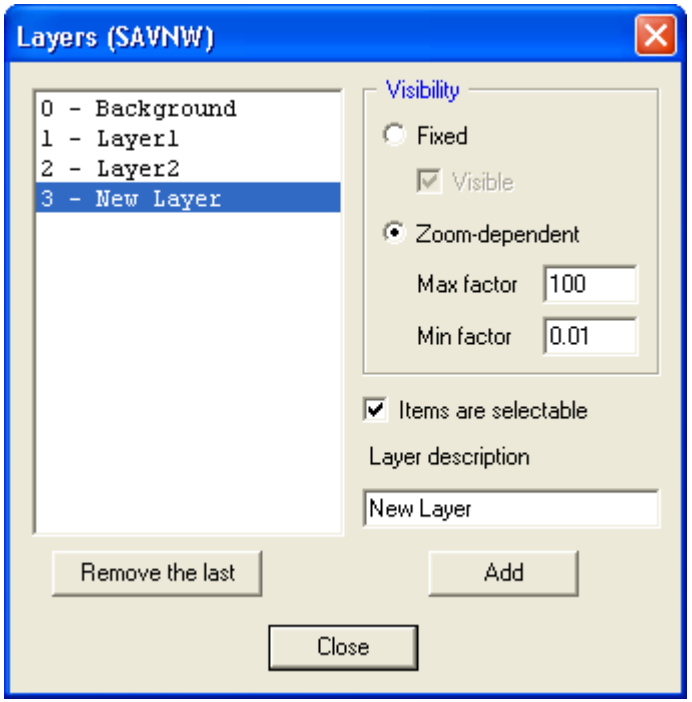


Figure 3-67. Layers Dialog

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PSS®Express provides the capability to view selected layers. For example, layers could be defined for base voltages of 110 kV, 200 kV, and 300 kV. The diagram items that correspond to these base voltages could then be created on the appropriate layer. To view the entire network, all layers would be made visible. To view only the 200 kV elements, the 100 kV and 300 kV layers would be made invisible, leaving only the 200 kV layer visible.

At a minimum, **[Diagram]** contains two layers:

- **Layer 0:** The background layer with any imported images.
- **Layer 1:** The default layer.

You can change attributes in these layers, but they cannot be removed from **[Diagram]**. All new diagram items added to **[Diagram]** are created on the active layer, which is the single layer specified.

Even if all layers are visible, only one is considered the active layer. It is displayed in the **Status** bar at the bottom of the PSS®Express interface when **[Layers]** is closed (Figure 3-68).

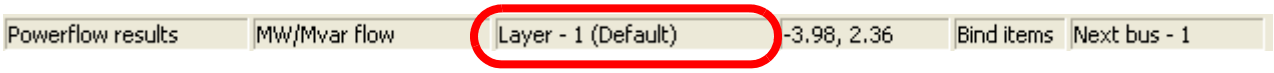


Figure 3-68. Status Bar Showing Active Layer

Click **[Add]** to specify a new layer. You may create a unique title by modifying the *Layer description* field. Clicking another layer or **[Close]** saves the layer attributes, which may be modified. Highlight a layer and click **[Remove the last]** to delete it.

The visibility of the layers depends on the options selected in **[Layers]**.

Fixed: The layer is in the diagram file. It is visible if the *Visible* checkbox is selected.

Zoom-dependent: The current zooming factor determines layer visibility. If the current zooming factor is within the minimum and maximum zooming factors specified, then the layer is visible. If the zooming factor is outside the range, then the layer is invisible.

The *Items are selectable* checkbox enables the ability to highlight specific components of a layer. To avoid the inadvertent manipulation of certain diagram components, place them on a separate layer and deselect the checkbox.

To set the active layer, go to ***Diagram > Set Active Layer...*** and select the desired layer from the pull-down list in the **[Select Layer]** dialog (Figure 3-69).

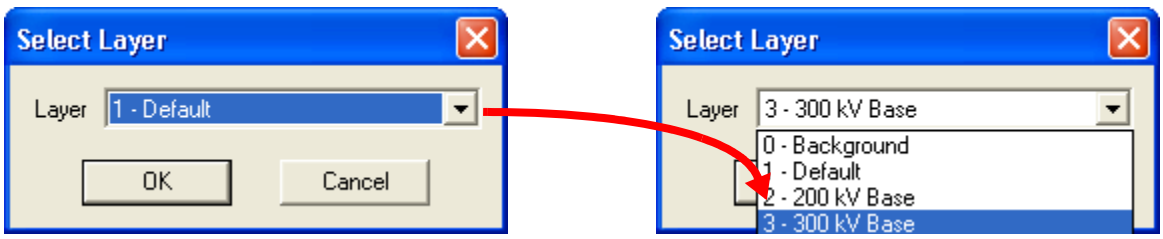


Figure 3-69. Setting the Active Layer

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To assign network components to a layer, select the component and right-click to display the menu. Select ***Display > Assign to Layer...*** and select the desired layer from the pull-down list in **[Select Layer]**.

This method can be used to assign diagram items to a layer one at a time, or a group of items can be selected and all be assigned to a layer. In the following example, all diagram items from Area 1 of the *savnw.sav* power flow case have been selected and assigned to a layer with the name Area 1

To see the entire one-line diagram, select all layers to be visible (see Figure 3-70). If the Area 1 layer is selected to be *not* visible, the one-line diagram will show only the diagram items belonging to the *Default* layer (see Figure 3-71).



Figure 3-70. All Layers Visible from Layers Dialog

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Figure 3-71. Only Default Visible

Layer assignment may also be made from **[Network Tree]**. Right-clicking the network folder opened to an expanded list will display a pop-up menu with an ***Assign all items to Layer*** option. This function opens **[Select Layer]**. When the layer is selected, all drawn items in the expanded list are re-assigned and **[Diagram]** is refreshed. This tool can be useful for assigning network items to layers for an existing or imported diagram.

3.13.4 Property Overrides

Diagram > Select all items with property overrides

This function selects all items in the diagram that have had color, line style, or line width established in Specifying Diagram Properties or Range Checking replaced by **[Color]** or **[Line Style]** dialogs (Figure 3-3 and Figure 3-16).

Diagram > Remove property overrides from selected items

This function removes any property overrides from selected items in the diagram. The line style, width and color will return to those specified by Specifying Diagram Properties or Range Checking.

3.13.5 Cleaning the Diagram

Diagram > Clean

This function performs a general diagram cleanup. As bugs are discovered in existing diagrams, new cleaning modes are added to this function and released in program patches so that users will be able to repair their diagrams. Repeatedly executing this activity will have no effect on a diagram, it addresses only specific diagram issues. The current issues addressed are:

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1. Makes sure any name extension is separated by a ' ' from the end of the internally generated name.
2. Removes any unattached Branch or Radial symbols.
3. Checks for Summations that have lost their Identity strings.
4. Fixes any bad annotation tags found on branches and two winding transformers.

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3.14 Exporting Data

3.14.1 Image File

File > Export > Diagram Image…

You may export the visible area of an active diagram to either a Bitmap or a JPEG file. The **[Export Image As]** dialog requires the name of the file in which to save the image and navigation to the desired directory for file storage.

When the JPEG file type is selected, the export process requires the specification of the *JPEG quality* level on a scale of 1 to 100 (see Figure 3-72). Some experimentation may be necessary to find the desired level of quality, balanced with file size.

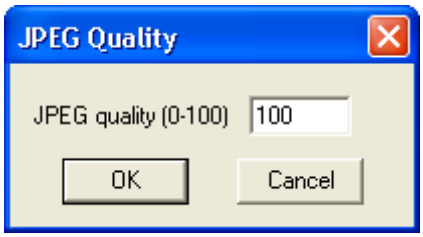


Figure 3-72. JPEG Quality Dialog

3.14.2 Bus Location File

File > Export > Bus Locations…

You may export the x/y coordinates from all buses, branches, and two winding transformers in the active diagram to a text file. Bus, branch and two winding transformer locations are recorded as Cartesian coordinates. The **[Select bus location output file]** dialog requires the name of the file (*.loc) in which to save the data and navigation to the desired directory for file storage.

3.14.3 Google Earth Locations File

File > Export > Google Earth Locations…

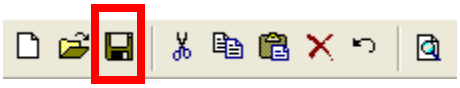
You may export slider diagrams to KML (Keyhole Markup Language) format. Geographic Coordinate information in the WGS-84 (World Geodetic System) is required for graphical elements that will be exported. The KML file can then be opened and manipulated in Google Earth. This feature allows a power network to be viewed against the backdrop of location specific, geographic data. The **[Select Google Earth data output file]** dialog requires the name of the file (*.kml) in which to save the data and navigation to the desired directory for file storage.

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3.15 Closing the Diagram View

File > Close

The *Close* option provides the opportunity to save any changes made to the diagram since it was last opened.



File > Save

The *Save* option over-writes a previous version of the file. If a new diagram is to be saved, the **[Save As]** dialog requires a filename and the specification of file type: either Slider Binary file (*.sld), the standard format, or Slider XML file (*.sldxml), an XML text file.

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Chapter 4

Data Selection Tree View

4.1 Introduction

The [Tree] view provides a hierarchical, expandable and collapsible list view of the network, OPF, dynamics, model, and plot data in the working case (see Figure 4-1). Folders can be expanded or collapsed to control the amount of data visible. Only one folder may be expanded at a time.

A right-click in a blank area of [Tree] (without selecting a data category) will display a menu from which hiding/exposing and docking/undocking the view can be selected.

Hide: Closes the view. The user must select ***View > Tree View*** to reopen it.

Allow Docking: Toggles between docking the view and making it a free-floating window.

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4.2 Network Tree

[Network Tree] provides a quick glance at all network data in the system. It is synchronized with the bus subsystem selector to enable the user to reduce the amount of data presented at any one time. **[Network Tree]** is also synchronized with **[Network Spreadsheet]** and **[Diagram]**, reflecting their current contents.

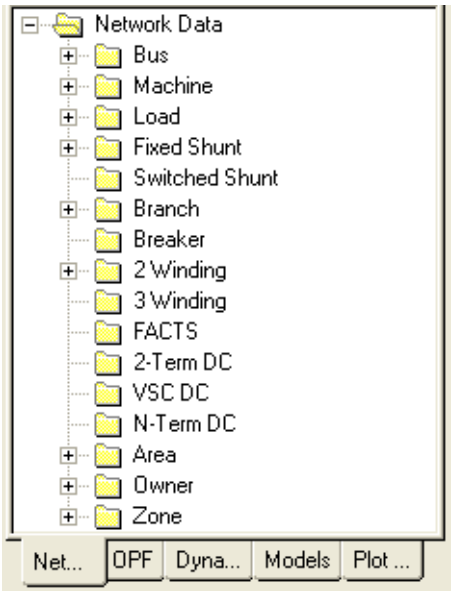


Figure 4-1. Network Tree View

When **[Diagram]** is active, the data items in **[Network Tree]** are updated to reflect whether a particular item is drawn in the diagram. If the item is not drawn, then the symbol to the left of the item is blank. If the item is drawn, the symbol to the left of the item is filled with the symbol belonging to the data category (i.e., busbar, load symbol, etc.).

When **[Diagram]** is active, data record property sheets can be opened by double-clicking a data items in **[Network Tree]**. See Section 3.6.1, Using Data Records.

The symbols in **[Network Tree]** are refreshed whenever **[Diagram]** is made active. A network item can be drawn in one diagram and not another, so the symbols provide a visual cue indicating whether the network item is drawn in the active diagram (see Figure 4-2). When **[Spreadsheet]** is active, all symbols are replaced with blanks.

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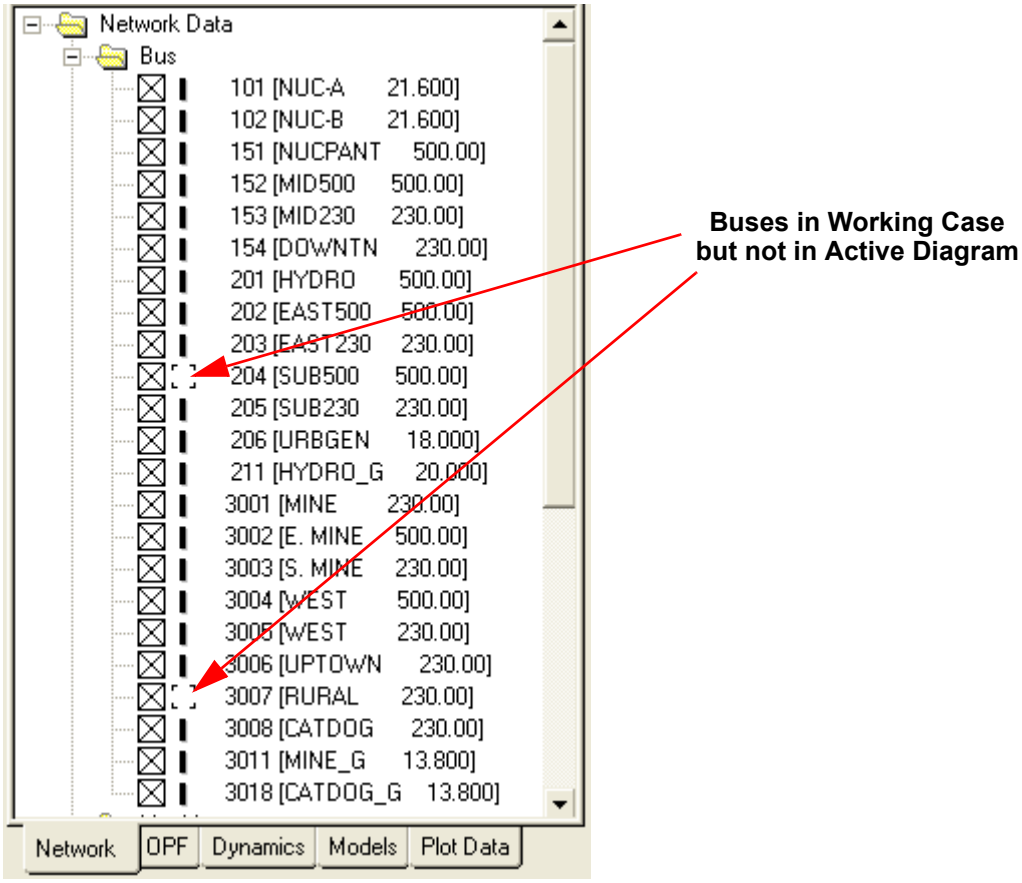
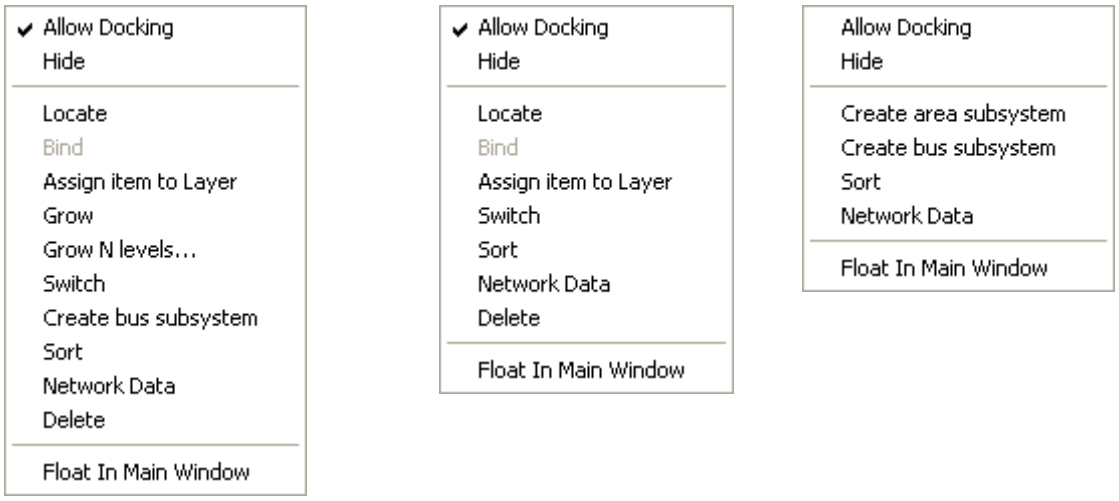


Figure 4-2. Tree View, Buses in Active Diagram

Various menus are available by selecting and right-clicking a network component. Menus are specific to the type of network element selected (see Figure 4-3).

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Bus Right-Click Menu **Machine Right-Click Menu** **Area Right-Click Menu**

Figure 4-3. Some Tree View Menu Options

Switch: Toggles in-service status; that is, disconnects the device.

Sort: All items in the selected data category are sorted in ascending alphanumeric order.

Network Data: Points to the selected network component in **[Network Spreadsheet]** on the relevant data tab, and highlights its data row.

Dynamics Data: Points to the selected network component in **[Dynamics Spreadsheet]** on the relevant data tab, and highlights its data row.

Delete: Deletes the network component from the model.

Create bus subsystem: Filter the model by a set of selected buses.

Create area subsystem: Filter the model by a set of selected areas.

Create zone subsystem: Filter the model by a set of selected zones.

Create owner subsystem: Filter the model by a set of selected owners.

If **[Diagram]** is the active view, then the right-click menu for the selected element provides additional options. These are described in detail in Section 3.6.4, Binding Data, Section 3.6.5, Displaying and Modifying Network Elements, Section 3.6.6, Drawing a Missing Bus, Section 3.7, Using Auto-Draw, and Section 3.13.3, Diagram Layers.

Double-clicking a component activates **[Network Spreadsheet]**, opens the correct data tab, and highlights its data row.

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4.3 OPF Tree

The OPF tab in the Tree View (Figure 4-4) provides a quick glance at all OPF data in the working case.

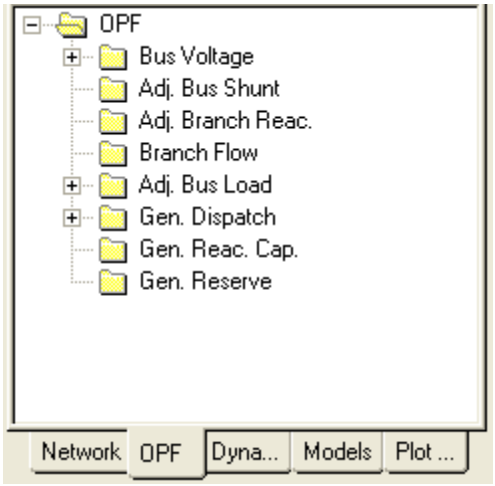


Figure 4-4. OPF Tree View

All network elements that have OPF data associated with them will have an entry in the appropriate folder in the OPF tab.

Double-clicking a data element (i.e., a particular bus) activates **[OPF Spreadsheet]**, opens the correct data tab, and highlights the spreadsheet data row for the selected item.

[OPF Tree] displays only network elements for which OPF data has been defined, either through opening an Optimal Power Flow data file or by initializing OPF data in the current working case (see Section 19.2, Data Initialization for OPF).

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4.4 Dynamics Tree

The Dynamics tab in the Tree View (Figure 4-5) provides a quick glance at all Dynamics data in the working case. The Dynamics Tab is synchronized with the bus subsystem selector to enable the user to reduce the amount of Dynamics data presented at any one time.

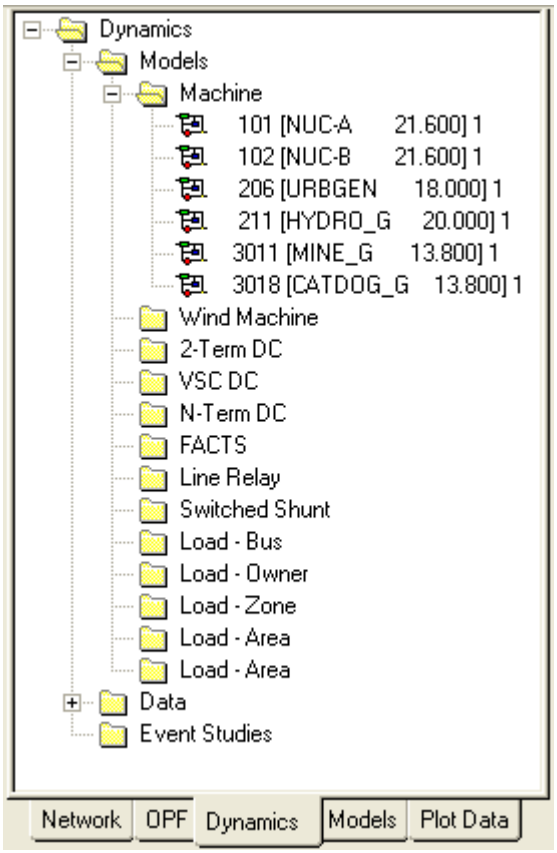


Figure 4-5. Dynamics Tree View

All network elements that have Dynamics models associated with them will have an entry in the appropriate folder in the Dynamics tab.

Double-clicking a data element (i.e., a particular machine) activates **[Dynamics Spreadsheet]**, opens the correct data tab, and highlights the spreadsheet data row for the selected item.

[Dynamics Tree] only displays network elements for which Dynamics models have been defined, either through opening a Snapshot or DYRE file or through **[Dynamics Spreadsheet]**.

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4.5 Models Tree

The Models tab in the Tree View provides a quick glance at all Dynamics models present in the current network (see Figure 4-6).

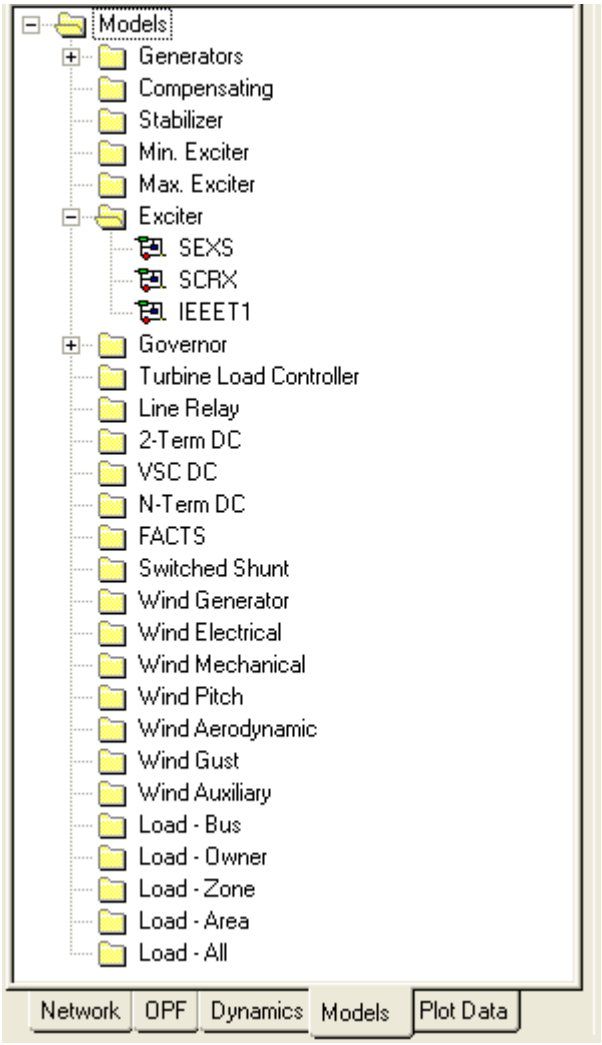


Figure 4-6. Models Tree View

The Models tab is organized by Dynamics model types e.g. Generators, Stabilizers etc. Each model type contains a list of all models of that type present in the current network. The example in Figure 4-6 shows that there are three types of exciter models present in the current network, SEXS, SCRX and IEEET1.

Double-clicking a model activates **[Models Spreadsheet]** opened to the CONS tab of the selected model.

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4.6 Plot Tree

The Plot tab in the Tree View provides a quick glance at all channels defined in a channel output file (see Figure 4-7). Folders can be expanded or collapsed to control the amount of data visible.

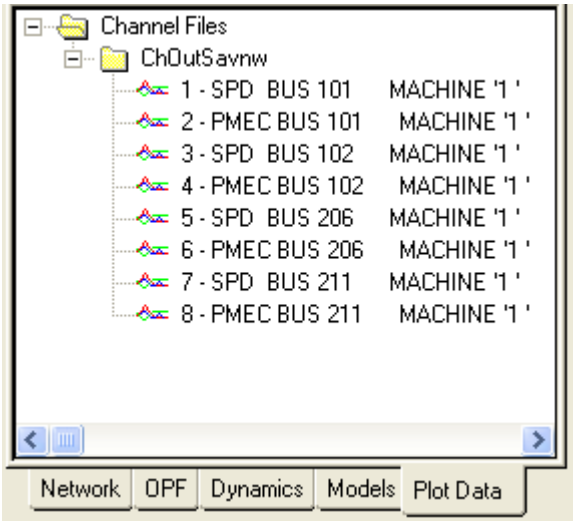


Figure 4-7. Plot Tree

Further details on using the integrated plot package can be found in Chapter 22, Dynamic Simulation PlotPackage.

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Chapter 5

Tools

The *Tools Menu* allows the user to designate active toolbars, to select the icons (activity commands) that are to be shown on each toolbar, to define Custom Toolbar buttons, to assign model search paths, and to create the User Dynamics DLL. Toolbar icons are indexed in Appendix A.

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5.1 Customizing Toolbars

Tools > Customize Toolbars…

The **[Customize]** dialog Toolbars tab (Figure 5-1) allows arrangement of the toolbars to suit your modeling requirements. Toolbars can be turned on or off, and icons can be added or removed from them. Custom toolbars can also be created.

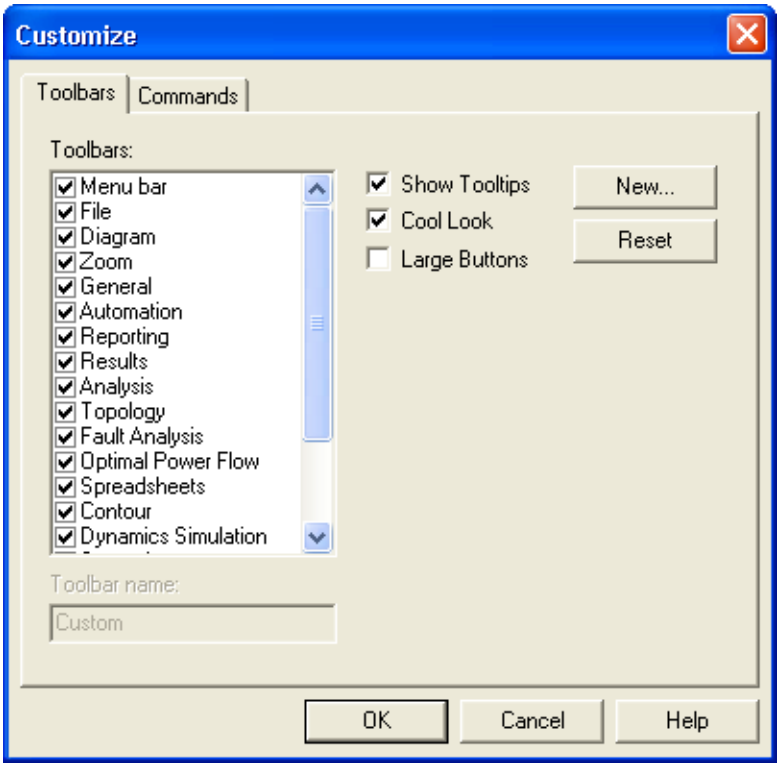


Figure 5-1. Customize Dialog, Toolbars Tab

Use the checkboxes to enable/disable displays of the default toolbar arrangements. You may choose whether or not to show tooltip messages identifying each button's function when the mouse pointer is held over it.

Toolbars can be rearranged in any position on the interface. They can also be dragged off the toolbar location and converted into floating windows.

Many of the toolbar buttons (or toolbar commands) are duplicates of menu options and will open the same dialog.

The Commands tab (Figure 5-2) displays the individual toolbar categories and shows the buttons currently assigned to each toolbar. Click an icon to provide a description of its command function. The example displays *Show or hide the output bar*.

To remove a button from an active toolbar, drag the icon from the toolbar to the *Buttons* area on the Commands tab. To add a button to an active toolbar, by drag the icon from the the *Buttons* area on the Commands tab to the desired location on the toolbar.

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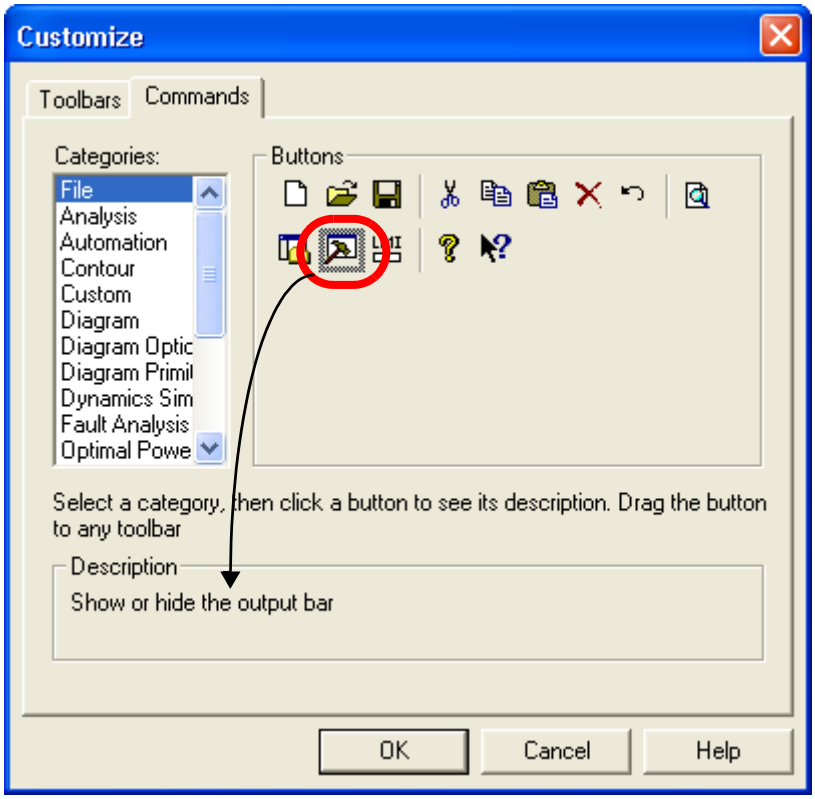


Figure 5-2. Customize Dialog: Commands Tab

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5.2 Configuring Custom Toolbar Buttons

Tools > Configure Custom Toolbar Buttons…

The **[Customize Toolbar Buttons]** dialog lists available custom toolbar buttons and active toolbar buttons (see Figure 5-3). These can be defined to execute user-specified commands.

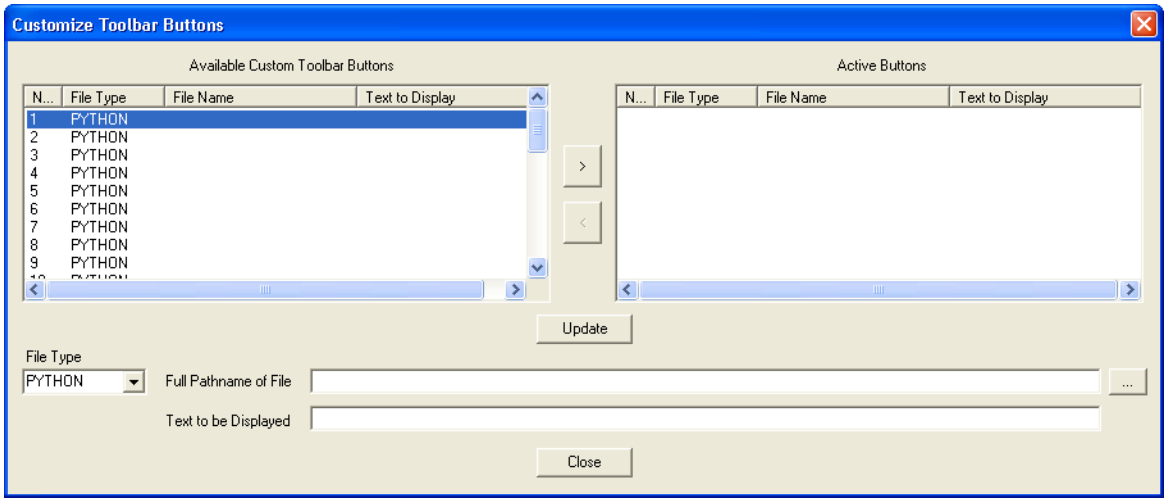


Figure 5-3. Customize Toolbar Buttons Dialog

Thirty custom toolbar buttons are available. Each button requires a file type and a file name.

To define a toolbar button, highlight a button and assign a *File Type*, one of the following Automation file types to be run when the button is clicked:

- PYTHON - A Python script file (*.py) (default)
- RESPONSE - A Response file (*.idv)
- IPLAN - An IPLAN program (*.irf)

Click **[…]** to open the **[Open]** selection window to specify the filename of the Automation file to be executed when clicking the corresponding custom toolbar button. Make sure the *Full Pathname of File* appears in the field.

You may specify a description (tooltip) of the operation to be performed when the corresponding custom toolbar button is clicked in the optional *Text To Display* field. This description pops up when the mouse cursor is held over the custom toolbar button. If a tooltip is not specified, the filename associated with the button is displayed as a tooltip.

Click **[Update]** to apply the changes that you've made.

The arrows in the center of the dialog move buttons from being available to being active. Figure Figure 5-4 shows the results of several defined buttons, two of which are in the active list.

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Figure 5-4. Defined Toolbar Buttons

The button must be in the *Active Buttons* list to be available on the toolbar. The others will be grayed out (Figure 5-5).



Figure 5-5. Custom Toolbar Showing Defined Buttons

These custom toolbar button definitions are preserved in a file named “Toolbar.prm” found in the Windows® Document and Settings directory for the active user.

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5.3 Creating Custom Toolbars

A custom designed toolbar can be created by selecting the **[New]** button in the Toolbars tab of the **[Customize]** dialog (see Figure 5-1). A new New Toolbar dialog window (Figure 5-6) will be displayed into which a name for the new toolbar may be entered. Click **[OK]** to display the new toolbar in the list of toolbars and in the PSS®Express interface as a small, free-floating toolbar.

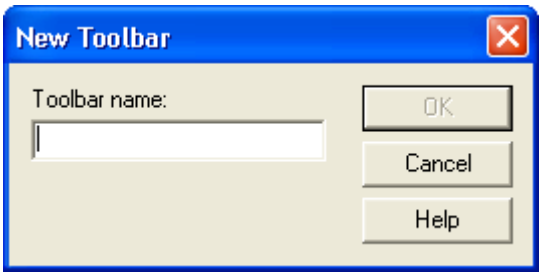


Figure 5-6. New Toolbar Dialog

To add command buttons to the new toolbar, click the Commands tab and drag the desired toolbar buttons to the new toolbar window. The new toolbar window can be docked to the toolbar area by dragging the window to the desired location on the toolbar and releasing the mouse.

To remove a user-defined toolbar from the list of toolbars on the Toolbars tab, highlight the desired toolbar to be deleted and click **[Delete]**. The standard toolbars can be reset only to their default settings.

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PART 2: PSS®EXPRESS ACTIVITIES

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Chapter 6

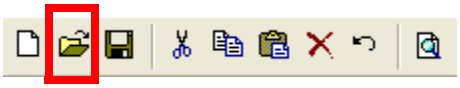
Power Flow Data Entry

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6.1 Retrieving a Power Flow Saved Case File

Activity CASE

Requirements / Prerequisites
Saved Case file (*.sav) Section 10.1, Creating a Saved Case File



File > Open…

The case retrieval activity CASE restores the contents of a previously saved power flow Saved Case File into the working case. The contents of the working case are overwritten while the contents of the specified Saved Case File are unchanged.

The file selection window in the **[Open]** dialog lists only those files available when the *Files of type:* field has been scrolled to the desired file type.

For this activity, scroll to *Save Case file (*.sav)*. Highlight the desired file and click **[Open]**. After the data is read into the working case, **[Spreadsheet]** is updated to include the new working case data.

Additional Information
PSS®E Program Operation Manual, Section 5.1, Retrieving a Power Flow Saved Case File *PSS®E Application Program Interface (API)*, Section 1.50, CASE Section 6.2, Listing Saved Case Filenames

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6.2 Listing Saved Case Filenames

Activity SHOW

Requirements / Prerequisites
none

File > File information (SIZE/SHOW/BUSN)...

Selecting the *Powerflow Save cases* option from **[File Information]** produces a report of power flow case filenames in the current working directory.

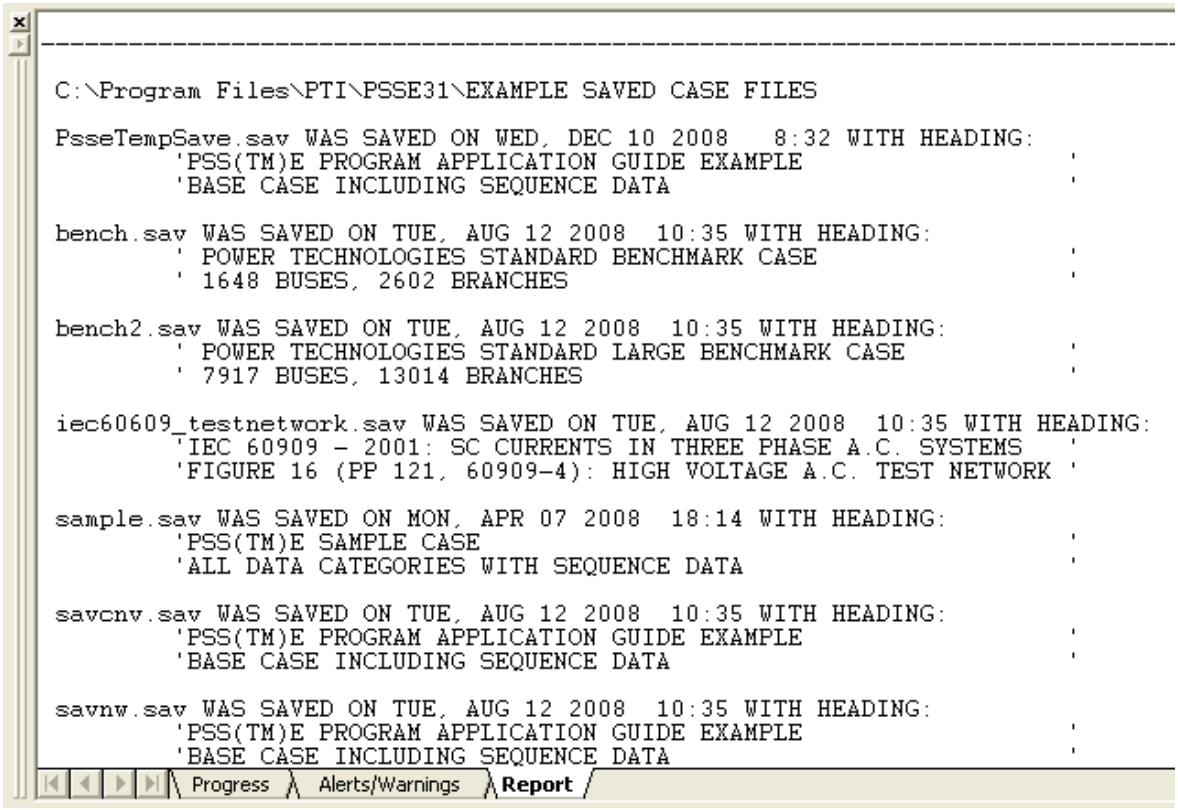


Figure 6-1. Example of Power Flow Saved Case File Report

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6.3 Reading Power Flow Raw Data into the Working Case

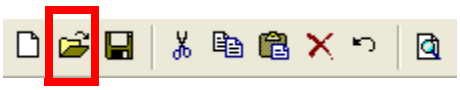
The bulk power flow data input activity READ picks up hand-typed power flow source data and enters it into the power flow working case, rearranging it from its original format into a computationally oriented data structure in the process. The source data records are input from a Power Flow Raw Data file.

After the data is read into the working case, **[Spreadsheet]** is updated to include the new working case data.

6.3.1 Standard Data Input

Activity READ

Requirements / Prerequisites
If reading change data, the working case must contain a validly specified power flow case. A Power Flow Raw Data File is available.



File > Open…

The file selection window in the **[Open]** dialog lists only those files available when the *Files of type:* field has been scrolled to the desired file type.

For a standard READ of a Power Flow Raw Data File in the format of the PSS®Express release indicated in the file, scroll to *Power Flow Raw Data file (*.raw)*, highlight the desired file, and then click **[Open]**. In the specified Power Flow Raw Data file, fields designating ac buses on load, generator, fixed shunt, branch, transformer, area, two-terminal dc line, VSC dc line, multi-terminal dc line, multi-section line, FACTS device, and switched shunt data records **must** be specified as bus numbers.

A standard READ may also be initiated by using the *Power Flow Raw Data file, Options (*.raw)* entry in the *Files of type:* list. Highlight the desired file and then click **[Open]**. In the resulting **[Read Power Flow Raw Data]** dialog, select *Standard (READ)* as the *Power flow data input application.* In the *RAW file version* fields, the current or a prior PSS®Express release corresponding to the format of the Power Flow Raw Data File must be indicated. When the revision number is specified in the file, specify the current version; if some other version is specified, it will take precedence over that specified as REV on the first record of the Power Flow Raw Data File. The *For input, use bus names* check box **must** be checked if the selected Power Flow Raw Data File contains any bus data fields specified as extended bus names. Then click **[OK]**.

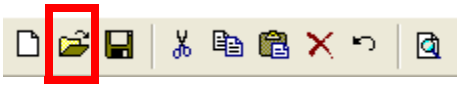
Additional Information
PSS®E Program Operation Manual, Section 5.2, Reading Power Flow Raw Data into the Working Case *PSS®E Application Program Interface (API)* Section 1.258, READ Section 1.259, READRAWVERSION See also: Section 10.4, Creating a Power Flow Raw Data File

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6.3.2 Subsystem Data Input

Activity READ

Requirements / Prerequisites
If adding data, the working case must contain a validly specified power flow case.. A Power Flow Raw Data File is available.



File > Open…

The file selection window in the **[Open]** dialog lists only those files available when the *Files of type:* field has been scrolled to the desired file type.

For a subsystem READ of a Power Flow Raw Data File, scroll to *Power Flow Data file, Options (*.raw)*, highlight the desired file, and click **[Open]**.

In the resulting **[Read Power Flow Raw Data]** dialog (Figure 6-2), select *Stubsystem (READ,OPT)* as the *Power flow data input application*. In the *RAW file version* fields, the current or a prior PSS®Express release corresponding to the format of the Power Flow Raw Data File must be indicated. When the revision number is specified in the file, specify the current version; if some other version is specified, it will take precedence over that specified as REV on the first record of the Power Flow Raw Data File. The *For input, use bus names* check box **must** be checked if the selected Power Flow Raw Data File contains any bus data fields specified as extended bus names.

The *Input by Subsystem* fields allow the selection of the options described in Section 5.2.4, Subsystem READ of the *PSS®E Program Operation Manual*. Data input may be limited to any of the following options:

- Add only data within subsystem
- Add only tie lines from subsystem
- Add subsystem data + tie lines

The various subsystem selection checkboxes enable the specification of the subsystem by base voltage, area, owner, and/or zone. Clicking **[Select]** for an active option opens the **[Area Subsystem Selector]**, **[Owner Subsystem Selector]**, or **[Zone Subsystem Selector]** dialog, as appropriate.

With all options selected, click **[OK]** to begin reading the data into the working case.

Additional Information
PSS®E Program Operation Manual, Section 5.2, Reading Power Flow Raw Data into the Working Case *PSS®E Application Program Interface (API)* Section 1.260, READSUB Section 1.261, READSUBRAWVERSION See also: Section 10.4, Creating a Power Flow Raw Data File

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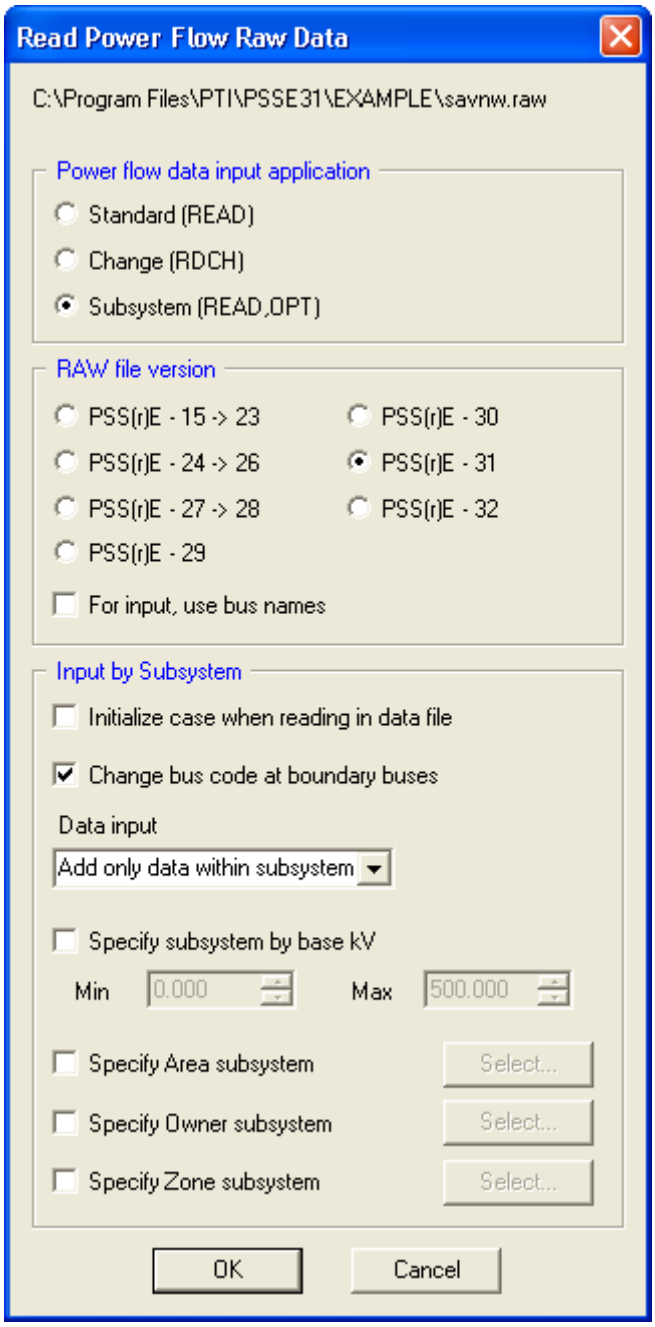


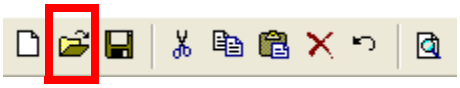
Figure 6-2. Read Power Flow Raw Data Dialog, Subsystem

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6.4 Reading / Changing Power Flow Data

Activity RDCH

Requirements / Prerequisites
The working case must contain a validly specified power flow case. A Power Flow Raw Data File is available.



File > Open…

The bulk power flow data input and modification activity RDCH picks up manually-entered power flow source data and enters it into the power flow working case. The source data records are in the form of a Power Flow Raw Data File *except* that the Case Identification Data records are omitted; that is, the first data record is expected to be a bus data record.

The file selection window in the **[Open]** dialog lists only those files available when the *Files of type:* field has been scrolled to the desired file type.

For activity RDCH, scroll to *Power Flow Data file, Options (*.raw)*, highlight the desired file, and click **[Open]**.

In the resulting **[Read Power Flow Raw Data]** dialog (Figure 6-3), select *Change (RDCH)* as the *Power flow data input application*. In the *RAW file version* fields, the current or a prior PSS®Express release corresponding to the format of the Power Flow Raw Data File must be indicated. The *For input, use bus names* check box **must** be checked if the selected Power Flow Raw Data File contains any bus data fields specified as extended bus names. Then click **[OK]**.

After the data is read into the working case, **[Spreadsheet]** is updated to include the new working case data.

Additional Information
PSS®E Program Operation Manual, Section 5.7, Reading / Changing Power Flow Data *PSS®E Application Program Interface (API)* Section 1.254, RAWD_2 Section 1.256, RDCHRAWVERSION See also: Section 10.4, Creating a Power Flow Raw Data File

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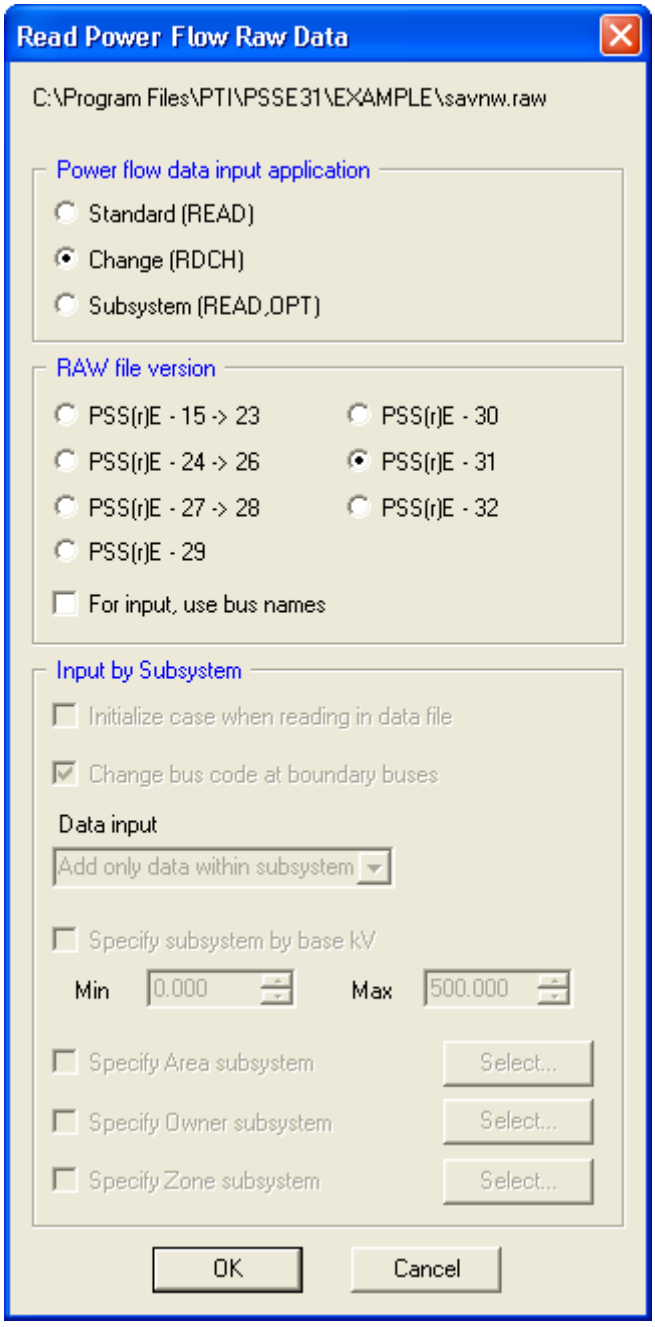


Figure 6-3. Read Power Flow Raw Data Dialog, RDCH

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6.5 Selecting Extended Bus Name Input Format

Misc > Select extended bus name input format

The extended bus name format was changed at PSS®Express Version 30 to accommodate longer bus names. This activity allows the user to select the input format for older versions of PSS®Express. All application functions that process buses based on the extended bus name will use the selected format for processing. This setting is not preserved between runs of the application.

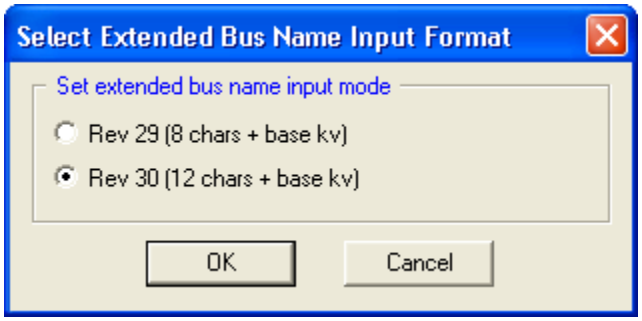


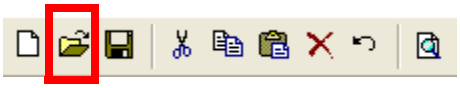
Figure 6-4. Select Extended Bus Name Input Format Dialog

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6.6 Adding Machine Impedance Data

Activity MCRE

Requirements / Prerequisites
The working case must contain a validly specified power flow case. Machine Impedance Data file (*.rwm)



File > Open…

The machine impedance data input activity MCRE enters source data records from a Machine Impedance Data File into the power flow working case.

The file selection window in the **[Open]** dialog lists only those files available when the *Files of type:* field has been scrolled to the desired file type.

For activity MCRE, scroll to *Machine Impedance Data file (*.rwm)*. Highlight the desired file and click **[Open]**.

In the resulting **[Machine Impedance Data]** dialog, select the options for the setting of the status of each new machine and for the treatment of existing machines with no input record. Then click **[OK]**.

After the data is read into the working case, **[Spreadsheet]** is updated to include the new working case data.

Additional Information
PSS®E Program Operation Manual, Section 5.4, Adding Machine Impedance Data *PSS®E Application Program Interface (API)*, Section 1.153, MCRE See also: Section 10.3, Saving Machine Impedance Data

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6.7 Managing Case Titles

6.7.1 Importing a Long Title

Activity RETI

Requirements / Prerequisites
The working case must contain a non-null case.

File > Import > Long Title (RETI)…

The long title data input activity RETI reads a data file containing up to 16 lines of alphanumeric data and places them into the long title. The previous content of the long title is overwritten.

Activity RETI opens a file selection dialog with the title **[Select file containing long case title]**. The file selector contains all entries in your working directory (see Establishing the Working Directory). Select the desired file and click **[Open]** to import the long title into the working case.

Additional Information
PSS®E Program Operation Manual, Section Activity, TREA
PSS®E Application Program Interface (API), Section 1.271, RETI

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6.7.2 Changing the Case Title and the Long Title

Activity CHTI

Requirements / Prerequisites
The working case must contain a non-null case.

File > Case titles, short & long (CHTI)…

The **[Case Titles]** dialog provides fields where the two-line case title and the 16-line long title may be edited using standard windows techniques.

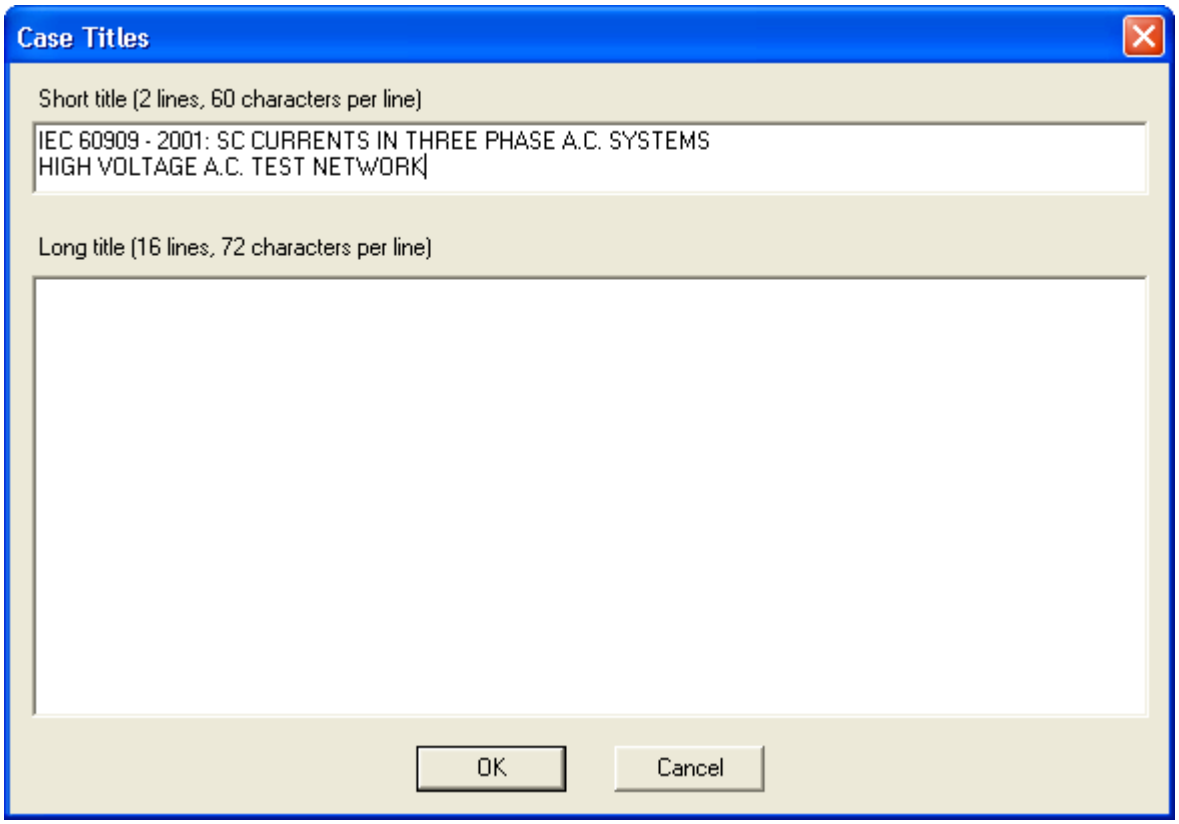


Figure 6-5. Case Titles Dialog

Additional Information
PSS®E Program Operation Manual, Section 5.34.2, Changing the Long Title *PSS®E Application Program Interface (API)* Section 2.11, CASE_TITLE_DATA Section 2.27, LONG_TITLE_DATA

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Chapter 7

Power Flow Data Modification

PSS®Express allows the user to modify the original network model in a variety of ways. Network elements can be added and removed either on an individual basis or in bulk. Network buses can be renumbered. Transmission resistance elements can be updated. Separate power flow files can be merged. Generation dispatch can be based on economic data. Existing network topology can be modified.

7.1 Changing Service Status and Power Flow Parametric Data

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

PSS®Express permits the user to change all service status, control mode, and other parametric data associated with equipment represented in the working case. In the GUI, the primary means of changing individual data items is the **[Spreadsheet]** (refer to Spreadsheet View).

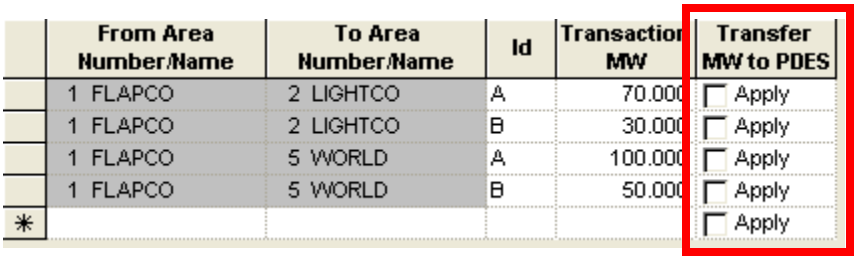
7.1.1 Inter-area Transfer Data Changes

On the *Inter-area Transfer* tab, the *Apply* check box provides the ability to apply changes in the transfer MW to the desired interchanges of the from and to areas (refer to *PSS®E Program Operation Manual*, Section 5.9.3, Interarea Transfer Data Changes).

If this box is checked for the row of an inter-area transfer, a change to the *Transaction MW* of that inter-area transfer results in a corresponding change to the desired area net interchange values of the from and to areas.

If this box is unchecked, desired area net interchange values are not updated to reflect any inter-area transfer change.

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	From Area Number/Name	To Area Number/Name	Id	Transaction MW	Transfer MW to PDES
	1 FLAPCO	2 LIGHTCO	A	70.000	Apply
	1 FLAPCO	2 LIGHTCO	B	30.000	Apply
	1 FLAPCO	5 WORLD	A	100.000	Apply
	1 FLAPCO	5 WORLD	B	50.000	Apply
*					Apply

Figure 7-1. Inter-area Transfer Spreadsheet, Transfer MW

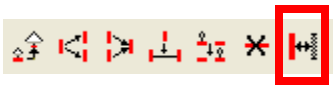
Additional Information
PSS®E Program Operation Manual, Section 5.9, Changing Service Status and Power Flow Parametric Data

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7.2 Electrically Disconnecting/Reconnecting a Bus

Activity DSCN/RECN

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Changing > Disconnect / Reconnect bus (DSCN/RECN)…

The bus disconnection activity DSCN automates the data changes required to electrically isolate a bus. The bus reconnection activity RECN automates the data changes required to electrically reconnect a bus.

The **[Disconnect / Reconnect Bus]** dialog (Figure 7-2) requires specification of the desired action and the bus to be connected. The bus number or extended bus name, as established by the bus input program option setting (refer to Saved Case Specific Option Settings of the *PSS®E Program Operation Manual*), may be entered in the input field. Alternatively, **[Select...]** may be used to open the **[Bus Selection]** dialog (Figure 7-3) where the desired bus may be specified by double-clicking on an entry in the selection list; click **[Filter]** to open a **[Bus Filter]** dialog (Figure 7-4) to apply filtering criteria to limit the number of buses in the selection list.

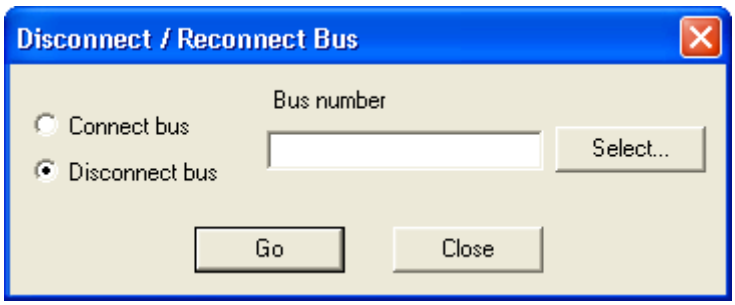


Figure 7-2. Disconnect/Reconnect Bus Dialog

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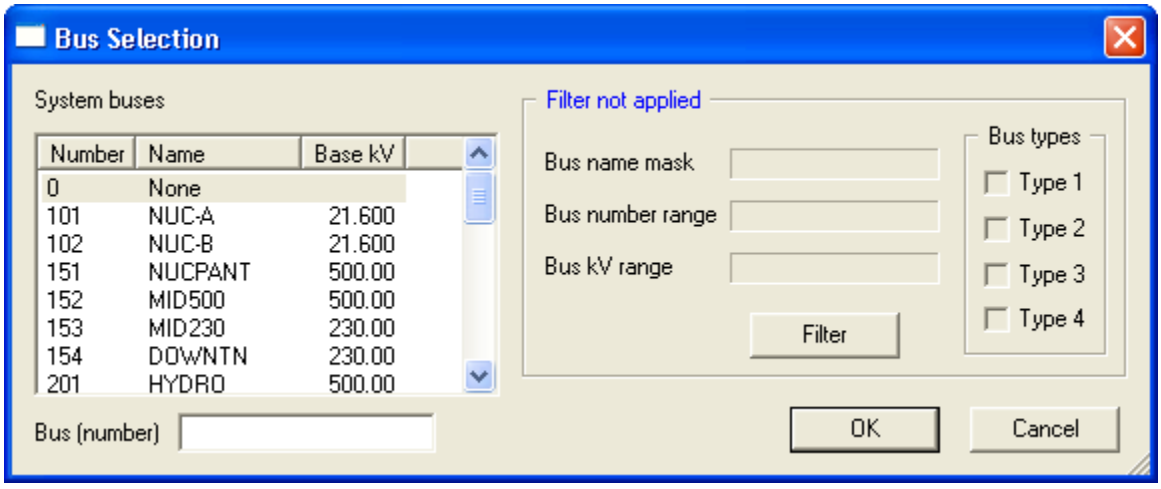


Figure 7-3. Bus Selection Dialog

Click **[Filter]** to open the **[Bus Filter]** dialog (Figure 7-4).

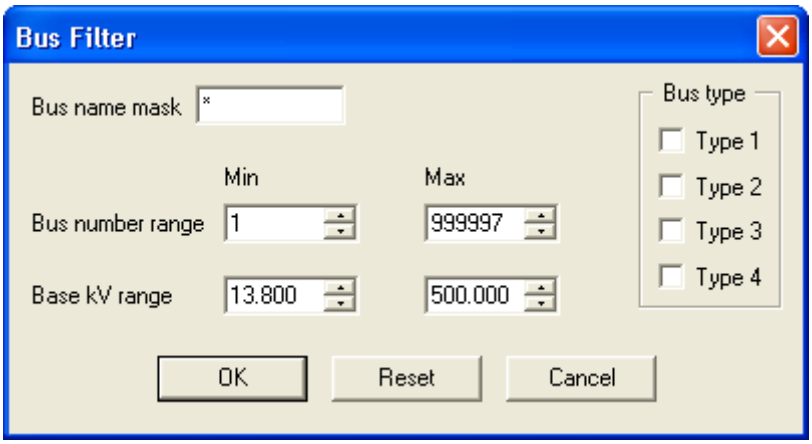


Figure 7-4. Bus Filter Dialog

Select one or more of the filtering criteria from among:

- bus name mask
- bus number range
- base voltage range
- bus type code

Click **[OK]** to go back to **[Bus Selection]** with the selected filtering options enforced.

Click **[Go]** on **[Disconnect / Reconnect Bus]** to implement the selected action.

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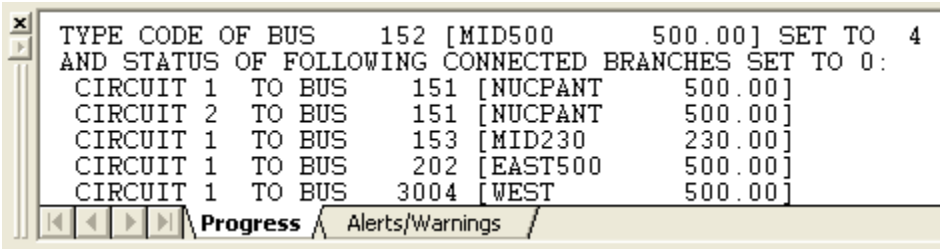


Figure 7-5. Example of Disconnect Bus Output

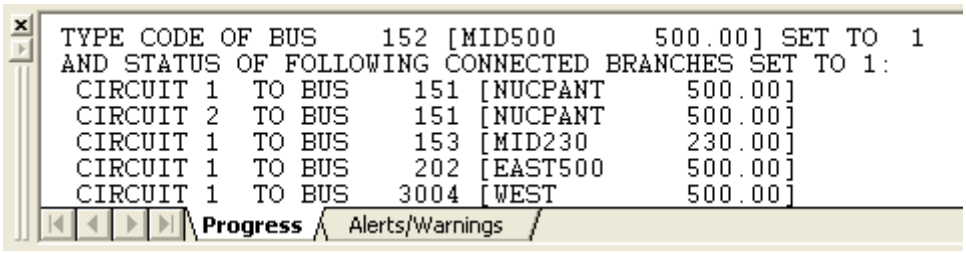


Figure 7-6. Example of Reconnect Bus Output

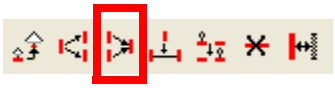
Additional Information
PSS®E Program Operation Manual, Section 5.11, Electrically Disconnecting a Bus Section 5.12, Electrically Reconnecting a Bus *PSS®E Application Program Interface (API)* Section 1.73, DSCN Section 1.262, RECN

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7.3 Joining Buses

Activity JOIN

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Changing > Join buses (JOIN)…

The bus joining activity JOIN enables the user to combine pairs of buses, retaining the identity of one of the two buses.

The **[Join Buses]** dialog (Figure 7-7) requires identification of the two buses to be joined and the specification of the line shunt treatment option. Either bus numbers or extended bus names, as established by the bus input program option setting (refer to Saved Case Specific Option Settings of the *PSS®E Program Operation Manual*), are required and may be entered directly in the input fields. Alternatively, **[Select...]** adjacent to each input field may be used to open the **[Bus Selection]** dialog (Figure 7-3) where the desired bus may be specified by double-clicking on an entry in the selection list; click **[Filter]** to open a **[Bus Filter]** dialog (Figure 7-4) to apply filtering criteria to limit the number of buses in the selection list.

Click **[Go]** to complete the joining process.

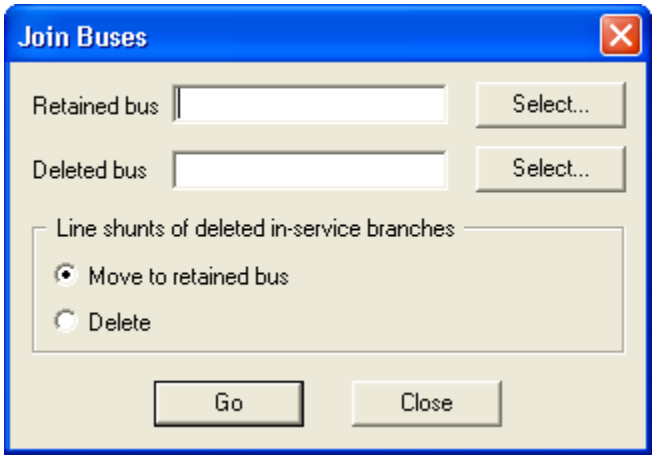


Figure 7-7. Join Buses Dialog

7.3.1 JOIN Example

As an example, using the *savnw.sav* power flow case, the buses ’154 DOWNTN’ and ’3008 CATDOG’ will be joined. In Figure 7-9 it can be seen that bus 154 has two loads (600+j450 MVA and 400+j350 MVA) and bus 3008 has one load (200 + j75 MVA) at nominal voltage. The buses are joined by one 230 kV line without line shunts.

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If bus 154 is the retained bus, the following topological changes are implemented as a result of joining buses 154 and 3008:

- The load at bus 3008 is moved to bus 154 and its identifier is changed from ’1’ to ’3’.
- The branch connecting buses 154 and 3008 is removed.
- The transformer from bus 3008 to bus 3018 is rerouted so that it now connects buses 154 and 3018.
- The non-transformer branch from bus 3008 to bus 3005 is rerouted so that it now connects buses 154 and 3005.
- The mullti-section line from bus 3008 to bus 3005 is rerouted so that it now connects buses 154 and 3005, and its multi-section line member from bus 3008 to 3007 is rerouted so that it now connects buses 154 and 3007.

The summary output shown in Figure 7-8 reports the load identifier change. Figure 7-10 shows the **[Diagram]** view of the retained bus 154 after bus 3008 is joined with it. The topological changes are shown on the diagram.

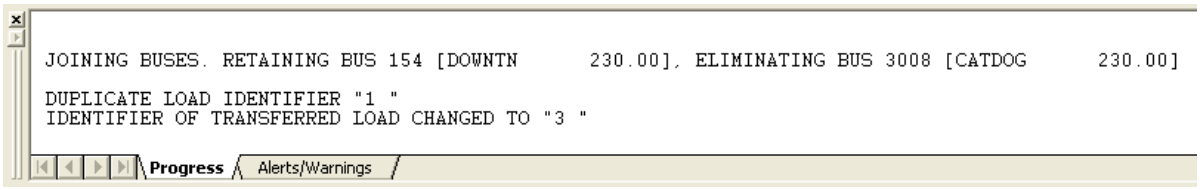


Figure 7-8. Example of Join Buses Output

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Figure 7-9. Original Topology

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Figure 7-10. Revised Topology After Joining Buses 154 and 3008

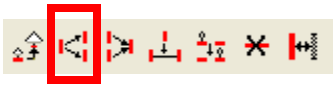
Additional Information
PSS®E Program Operation Manual, Section 5.16, Joining Buses *PSS®E Application Program Interface (API)*, Section 1.114, JOIN

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7.4 Splitting Buses

Activity SPLT

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Changing > Split buses (SPLT)…

The bus sectionalizing activity SPLT enables the user to split a bus into two buses connected by a branch.

The **[Split Buses]** dialog (Figure 7-11) requires identification of the bus to be split and the number of the new bus. The bus name and base voltage to be assigned to the new bus may also be specified. If omitted, the base voltage of the bus being split and a blank name are assigned to the new bus.

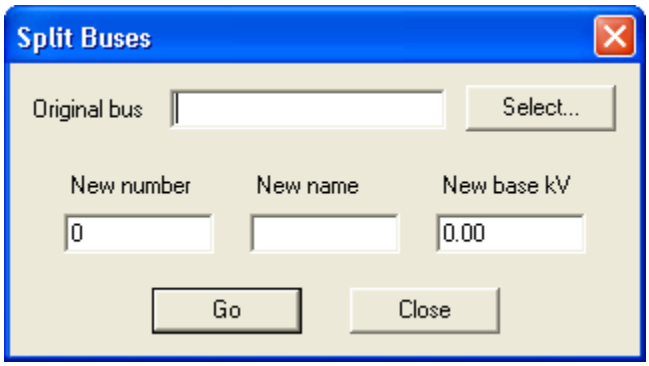


Figure 7-11. Split Buses Dialog

The bus number or extended bus name, as established by the bus input program option setting (refer to Saved Case Specific Option Settings of the *PSS®E Program Operation Manual*), may be entered in the input field. Alternatively, **[Select...]** may be used to open the **[Bus Selection]** dialog (Figure 7-3) where the desired bus may be specified by double-clicking on an entry in the selection list; click **[Filter]** to open a **[Bus Filter]** dialog (Figure 7-4) to apply filtering criteria to limit the number of buses in the selection list.

Click **[Go]** to initiate the splitting process. The new bus is created along with a new jumper branch connecting the original and new buses. The **[Reassign Branches and Equipment]** dialog then opens and any displayed equipment items connected to the bus being split may be specified to be moved to the new bus. Figure 7-12 shows the display if bus 154 were selected to be split in the *savnw.sav* power flow case.

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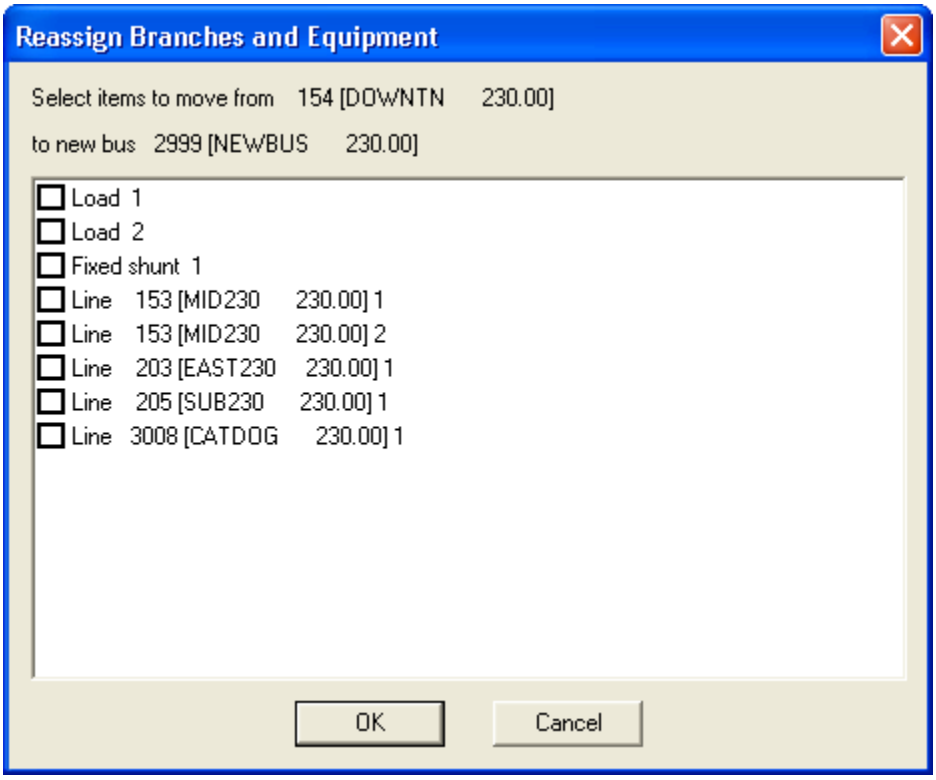


Figure 7-12. Reassign Branches and Equipment Dialog

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Using the *savnw.sav* power flow case, (see Figure 7-13) bus 3003 can be split and circuit 2 from bus 3005 reassigned to the new bus 3020.

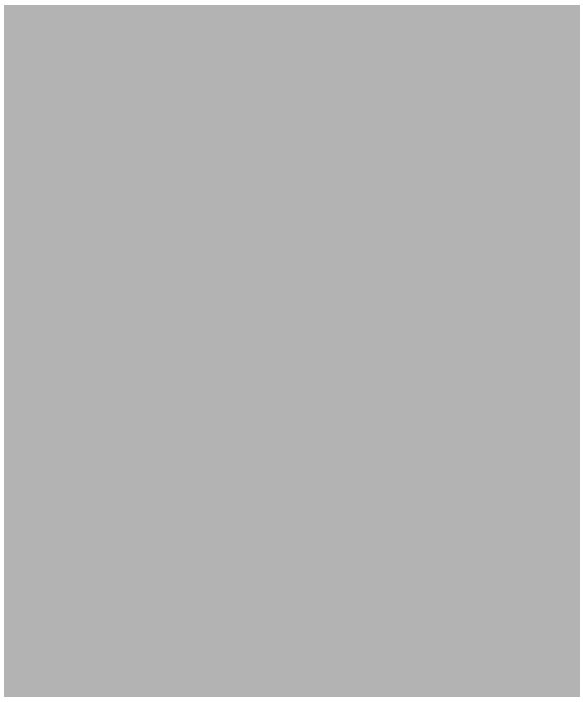


Figure 7-13. Bus 3003 to be Split



Figure 7-14. Bus Selection and Reassignment for Bus Split

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Selection of the bus to be split, and the specification of the number, name and base voltage to be assigned to the new bus, is done in **[Split Buses]** (see Figure 7-14). After clicking **[Go]**, **[Reassign Branches and Equipment]** displays the elements that can be selected to be moved to the new bus 3020. In the example, the option to move circuit 2 from bus 3005 to the new bus has been selected. Clicking **[OK]** completes the process and generates a summary of the action in the **Progress** tab (Figure 7-15).

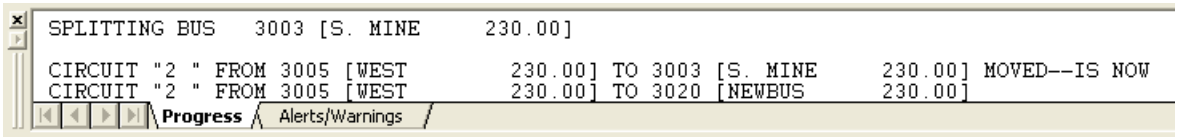


Figure 7-15. Example of Split Buses Output

The result of the splitting process can be seen in **[Diagram]** (see Figure 7-16). The diagram shows the new bus, 3020, the new branch from the new bus to the original bus and the new routing of circuit 2 from bus 3005 to the new bus.

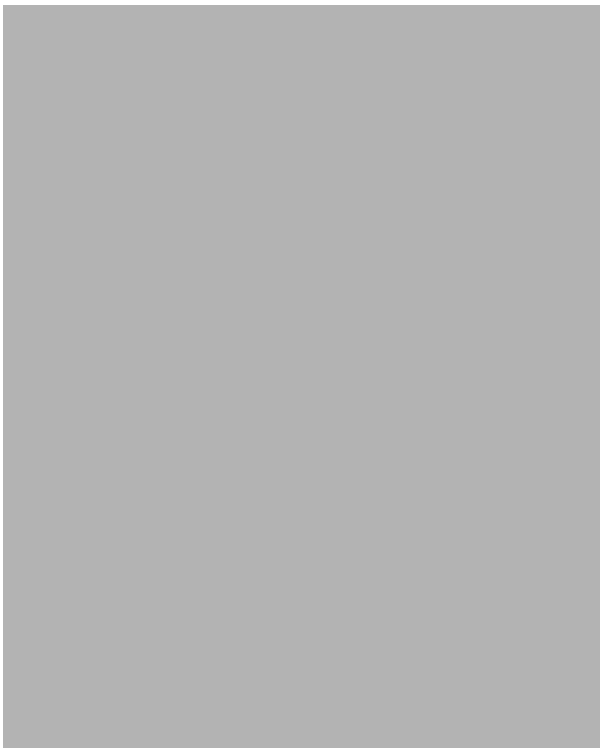


Figure 7-16. Diagram View of New Topology following Bus Split

A one-line Slider file conforming to the original topology will now show circuit 2 out-of-service. A modified Slider file would have to be generated to display the new topology.

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Additional Information
PSS®E Program Operation Manual, Section 5.17, Splitting Buses
PSS®E Application Program Interface (API) Section 1.306, SPLT Section 1.155, MOVE3WND Section 1.156, MOVEBRN Section 1.157, MOVELOAD Section 1.158, MOVELOADS Section 1.159, MOVEMAC Section 1.160, MOVEPLNT Section 1.161, MOVESHUNT Section 1.162, MOVESHUNTS Section 1.163, MOVESWS

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7.5 Tapping a Line

Activity LTAP

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Changing > Tap line (LTAP)...

The line tapping activity LTAP enables the user to introduce a new bus into the working case at a designated location along a specified ac branch. Any non-transformer branch may be tapped with activity LTAP.

The **[Tap Line]** dialog (Figure 7-17) requires the user to specify the line to be tapped, the location of the new bus, and the identifying number, name, and base voltage of the new bus.

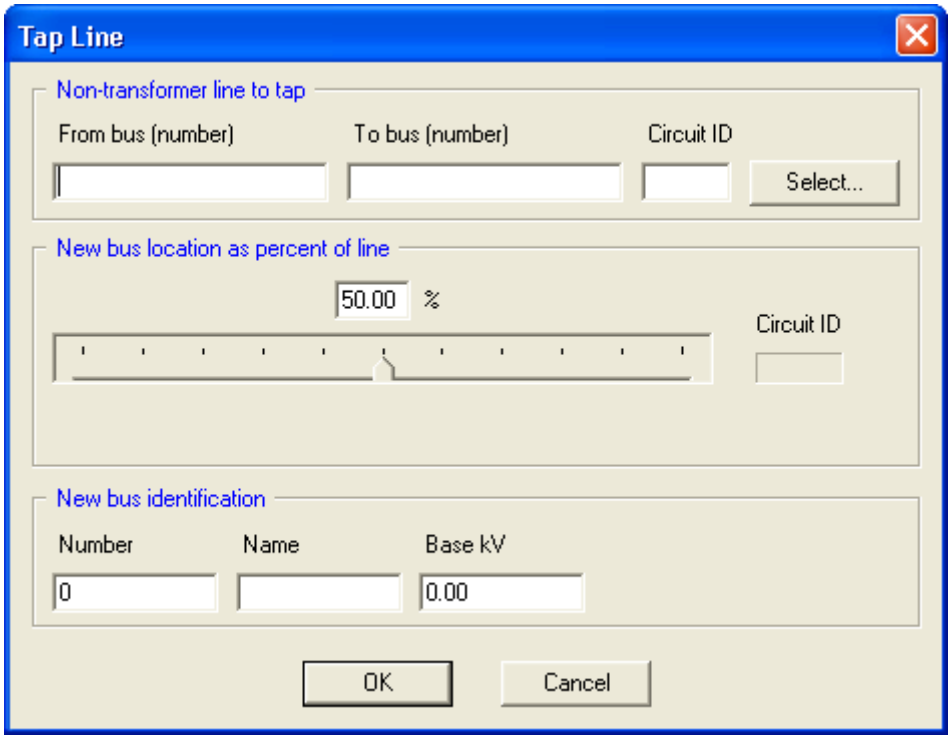


Figure 7-17. Tap Line Dialog

As an example, the branch from bus 3003 to 3005 circuit 1 in the *savnw.sav* power flow case will be tapped at a point 40% of the distance from bus 3003. A new bus will be created with the number 3020, a name NEWBUS and a base voltage of 230 kV. Figure 7-18 shows the original topology as **[Diagram]**.

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Figure 7-18. Line Selected to be Tapped

Click **[Select]** in **[Tap Line]** to open the **[Branch Selection]** dialog (see Figure 7-19). Specifying a bus in the *From bus* list will create a list of the to buses and circiut identifiers of the non-transformer branches connected to that bus. Then specify an entry in the *To bus* list. If required, the **[Filter]** can be used to reduce the listed buses to those in a specified subsystem.

In the example (Figure 7-19), the branch from bus 3003 to bus 3005, circuit 1, has been selected. Clicking **[OK]** will return to **[Tap Line]**.

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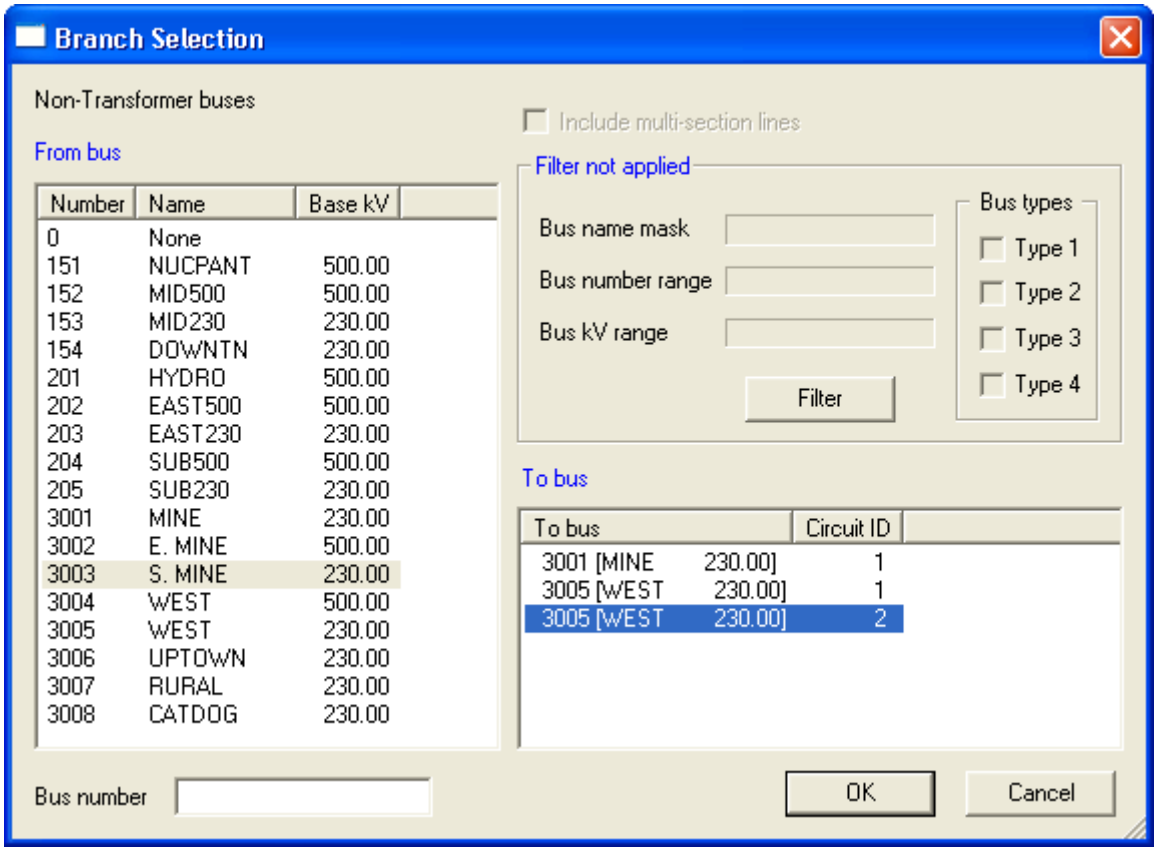


Figure 7-19. Specification of a Branch for the Tap Line

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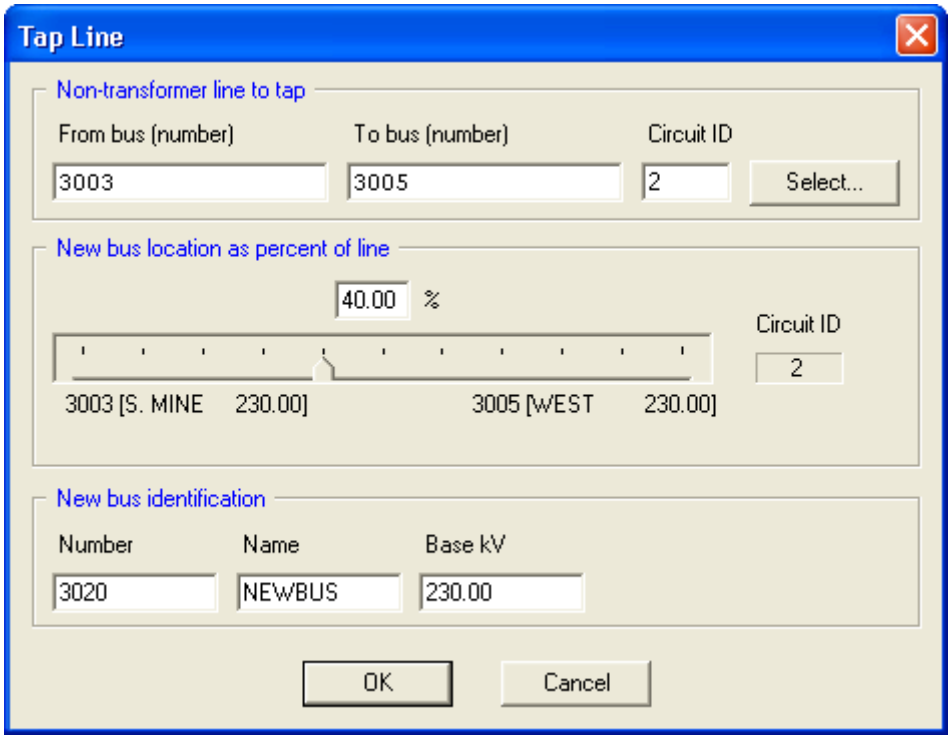


Figure 7-20. Final Stage in Tap Line Activity

In **[Tap Line]**, the tap position, 40% of the distance from bus 3003 and the new bus name, number and base voltage are specified as shown in Figure 7-20. Clicking **[OK]** completes the process and generates a summary of the action at the **Progress** device.

At this point, any Slider diagram established for the original network topology will indicate that the tapped line is out-of-service. The one-line Slider diagram will need to be modified to show the new topology with the new bus. Figure 7-21 shows a modified **[Diagram]** of the new topology around the new bus.

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Figure 7-21. New Topology on Tapped Line

Additional Information
PSS®E Program Operation Manual, Section 5.18, Tapping a Line
PSS®E Application Program Interface (API), Section 1.124, LTAP

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7.6 Changing Equipment Identifiers

Activity MBID

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

PSS®Express permits the user to change the alphanumeric identifiers assigned to specified machines, loads, fixed bus shunts, ac branches, multi-section line groupings, dc lines, FACTS devices, and interarea transfers in the working case. In the GUI, the primary means of changing individual data items is the **[Spreadsheet]** (refer to Spreadsheet View).

	From Area Number/Name	To Area Number/Name	Id	Transaction MW	Transfer MW to PDES
	1 FLAPCO	2 LIGHTCO	A	70.000	Apply
	1 FLAPCO	2 LIGHTCO	B	30.000	Apply
	1 FLAPCO	5 WORLD	A	100.000	Apply
	1 FLAPCO	5 WORLD	B	50.000	Apply
*					Apply

Figure 7-22. Inter-area Transfer Spreadsheet, Identifier

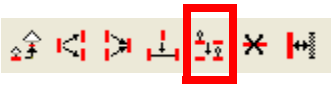
Additional Information
PSS®E Program Operation Manual, Section 5.19, Changing Equipment Identifiers
PSS®E Application Program Interface (API)
Section 1.138, MBID2DC
Section 1.139, MBID3WND
Section 1.140, MBIDATRN
Section 1.141, MBIDBRN
Section 1.142, MBIDFACTS
Section 1.147, MBIDLOAD
Section 1.148, MBIDMAC
Section 1.149, MBIDMDC
Section 1.150, MBIDMSL
Section 1.151, MBIDSHUNT
Section 1.152, MBIDVSC

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7.7 Moving Equipment

Activity MOVE

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Changing > Move network elements (MOVE)…

The equipment transfer activity MOVE allows the user to move specified fixed shunts, switched shunts, loads, machines, and plants from one bus to another. It also provides for connecting the far end of specified branches and one winding of a three-winding transformer to different to buses.

The **[Move Network Elements]** dialog (Figure 7-23) provides tabs for each of the types of equipment that may be moved by activity MOVE.

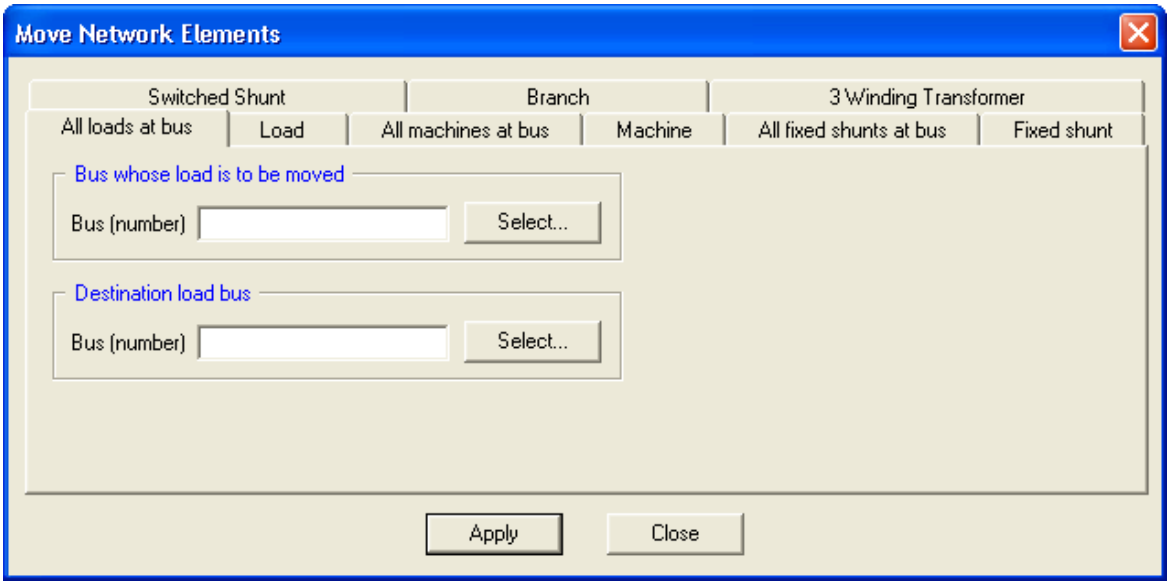


Figure 7-23. Move Network Elements Dialog

Select the equipment category tab, and then specify the individual equipment to be moved and the bus to which the equipment item is to be transferred. Clicking **[Apply]** completes the transfer.

Using the *savnw.sav* power flow case, the 500 kV line from bus 151, which terminates at bus 201 will be moved to terminate at bus 202. The original topology is shown in Figure 7-24.

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Figure 7-24. Original Topology before Moving Branch 151 - 201

From the Branch tab on **[Move Network Elements]** click **[Select...]** to open the **[Branch Selection]** dialog (Figure 7-27) and highlight the desired branch. **[Filter]** may be used to limit the number of entries in the From bus list. Clicking **[OK]** returns to **[Move Network Elements]** where the new to bus can be specified.

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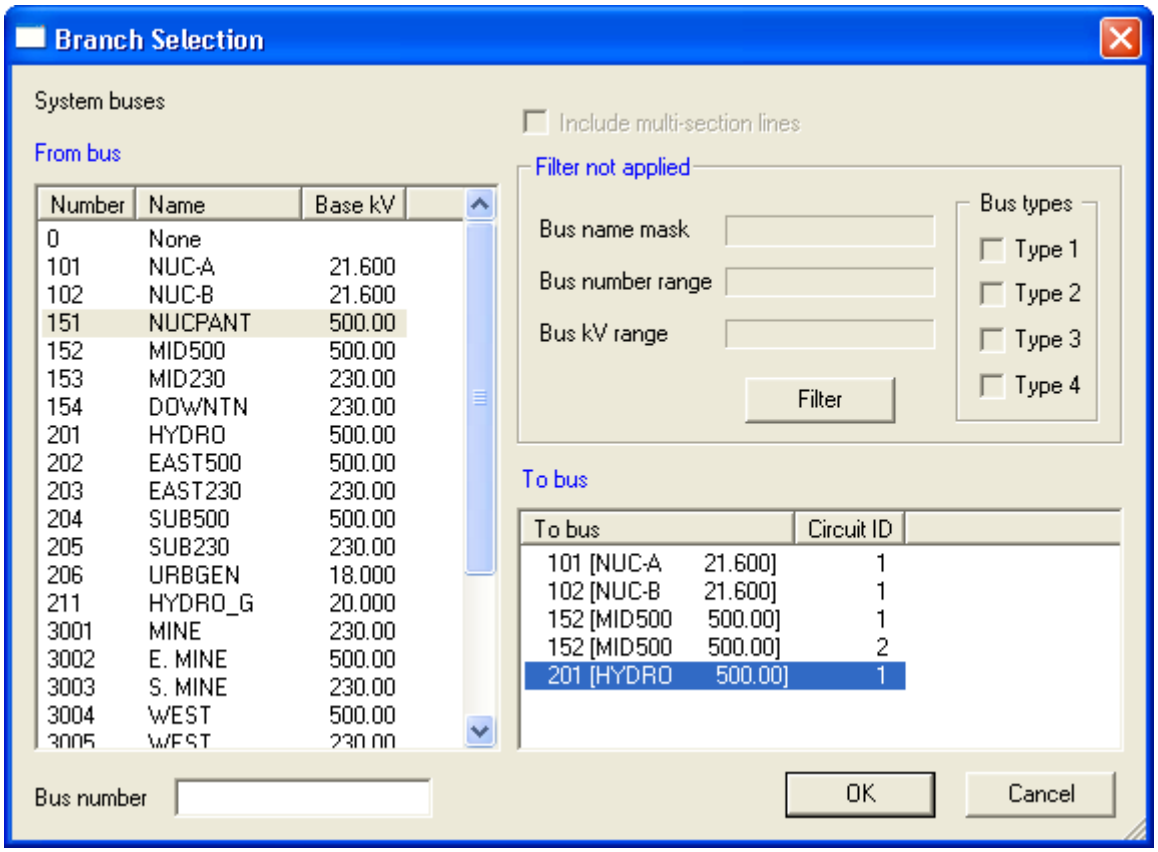


Figure 7-25. Branch Selection Dialog

From the Destination (to bus) area of the Branch tab on **[Move Network Elements]**, the bus number or extended bus name, as established by the bus input program option setting (refer to Saved Case Specific Option Settings of the *PSS®E Program Operation Manual*), may be entered in the bus input field. Alternatively, **[Select...]** may be used to open the **[Bus Selection]** dialog (Figure 7-3) where the desired bus may be specified by double-clicking on an entry in the selection list; click **[Filter]** to open a **[Bus Filter]** dialog (Figure 7-4) to apply filtering criteria to limit the number of buses in the selection list. Clicking **[OK]** in **[Bus Selection]** returns to **[Move Network Elements]** where all fields are now populated (Figure 7-27).

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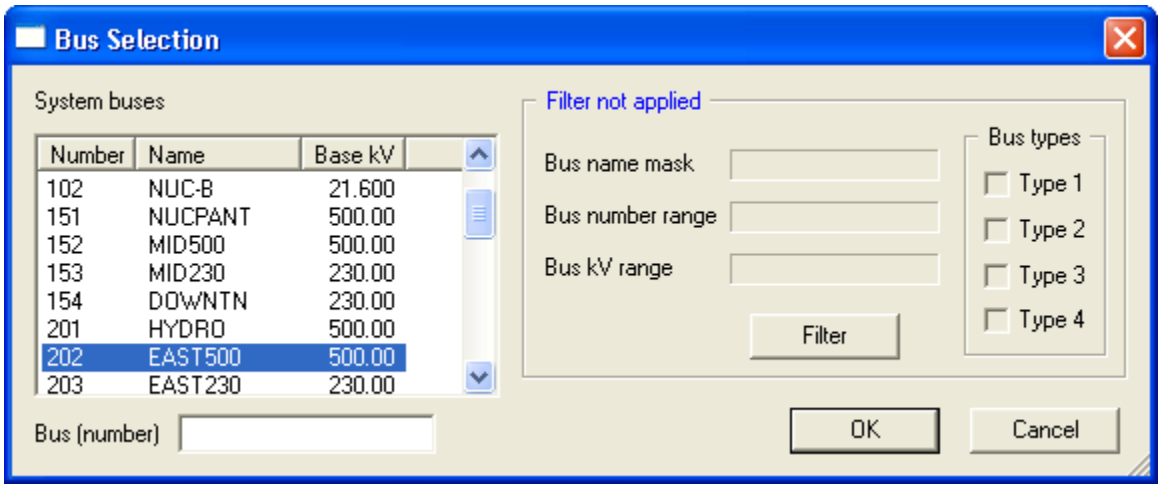


Figure 7-26. Bus Selection Dialog

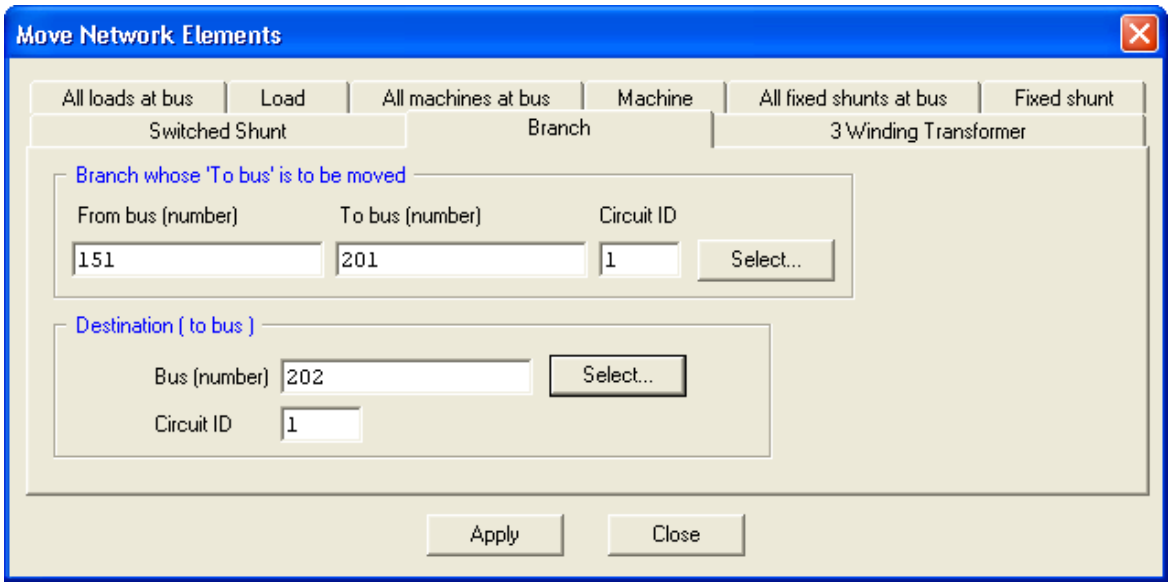


Figure 7-27. Example of Moving a Branch

Click **[Apply]** to complete the move. To end activity MOVE, click **[Close]**.

When the moving process is complete, the original one-line Slider file will not be able to display the new topology. The branch that was moved (in the case of this example, the line from bus 151 to bus 201) will be indicated as out-of-service. The Slider file will need to be modified to display the new line termination. Figure 7-28 shows the new display with the new line location and the indicated line out-of-service from the original topology. The non-existent branch can be deleted from the Slider file. It will no longer be in the network data and will not appear in **[Spreadsheet]** or **[Network Tree]**.

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Figure 7-28. New Topology Following Branch Move

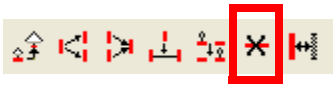
Additional Information
PSS®E Program Operation Manual, Section 5.20, Moving Equipment *PSS®E Application Program Interface (API)* Section 1.155, MOVE3WND Section 1.156, MOVEBRN Section 1.157, MOVELOAD Section 1.158, MOVELOADS Section 1.159, MOVEMAC Section 1.160, MOVEPLNT Section 1.161, MOVESHUNT Section 1.162, MOVESHUNTS Section 1.163, MOVESWS See also: Section 3.6.9, Moving Diagram Elements

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7.8 Removing Buses and Connected Equipment

Activity PURG/EXTR

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Changing > Delete network elements (PURG/EXTR)…

PSS®Express facilitates the users' ability to delete equipment items from the working case. In the GUI, the primary means of deleting individual equipment items is the **[Spreadsheet]** (refer to Chapter 2 Spreadsheet View). In addition, the following may be removed from a specified subsystem:

- subsystem buses and all equipment connected to them.
- all outaged items of a selected equipment category in a subsystem.

The **[Delete Buses and Outaged Network Elements]** dialog (Figure 7-29) enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Delete Buses and Outaged Network Elements]** dialog

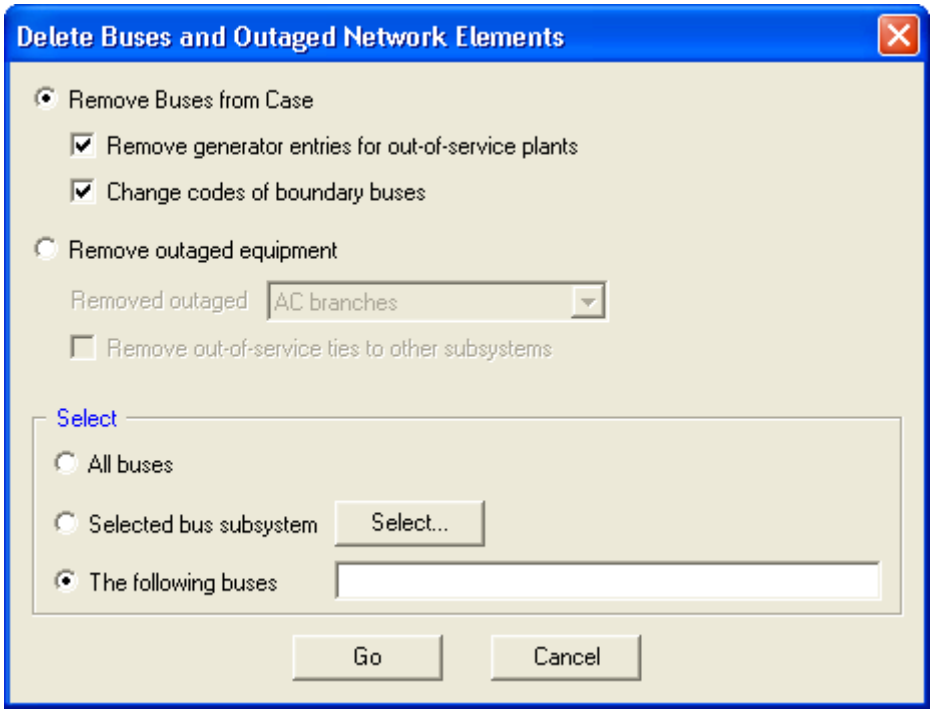


Figure 7-29. Deleting Network Elements Dialog

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7.8.1 Removing Buses

The bus removal activity EXTR removes subsystem buses and all equipment connected to them from the working case.

It is possible to see buses and their associated lines and equipment that have been deleted in **[Diagram]**. The unbound items (items that exist in the diagram but not in the network data because of the deletion) are shown in color specified in **[Diagram Properties]**. Figure 7-30 shows a partial view of the buses deleted from the FLAPCO area of the *savnw.sav* power flow case.



Figure 7-30. Example of Deleted Buses and the Diagram Properties Options

7.8.2 Removing Outaged Equipment

In the GUI, the equipment removal activity PURG deletes all outaged items of a specified equipment category that are contained within a specified subsystem of the working case.

The data categories that can be processed for deletion are available in a pull-down list in **[Delete Buses and Outaged Network Elements]**. In addition, a checkbox selects the option to *Remove out-of-service ties to other subsystems*.

Figure 7-31 shows the output following deletion of three ac lines in the *savnw.sav* case. These branches were taken out-of-service before attempting to delete them.

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Figure 7-31. Example of Removing Outaged Equipment Output

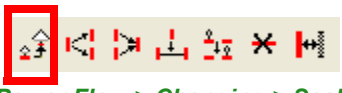
Additional Information
PSS®E Program Operation Manual, Section 5.14, Removing Specified Buses and Connected Equipment Section 5.15, Deleting Equipment *PSS®E Application Program Interface (API)* Section 1.80, EXTR Section 1.211, PURG See also: Section 3.6.3, Editing Functions Section 3.6.4, Binding Data

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7.9 Scaling Loads, Generators, and/or Shunts

Activity SCAL

Requirements / Prerequisites
The working case must contain a validly specified power flow case.
If generation is to be scaled, and a Type 3 bus is in the subsystem being scaled, the working case must be solved to an acceptable mismatch level.



Power Flow > Changing > Scale generation, load, shunt (SCAL)

The load, generation and shunt scaling activity SCAL enables the user to uniformly increase or decrease any or all of the following quantities for a specified grouping of loads, fixed shunts, and machines:

- Load active power.
- Load reactive power.
- Active component of fixed bus shunt admittance.
- Positive reactive component of fixed bus shunt admittance (capacitors).
- Negative reactive component of fixed bus shunt admittance (reactors).
- Generator active power output (positive generation).
- Motor active power output (negative generation).

The **[Scale Powerflow Data]** dialog (Figure 7-32) enables the user to designate for scaling either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Scale Powerflow Data]** dialog.

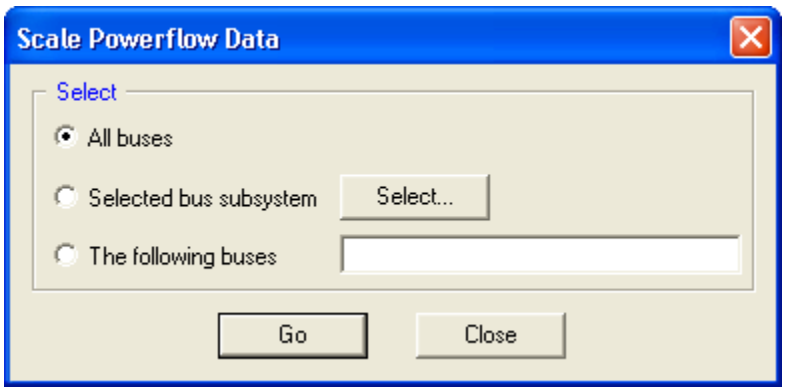


Figure 7-32. Scale Powerflow Data Dialog to Specify Subsystem

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Click **[Go]** to open a second **[Scale Power Flow Data]** dialog (Figure 7-33) where scaling targets for the data categories to be scaled may be specified.

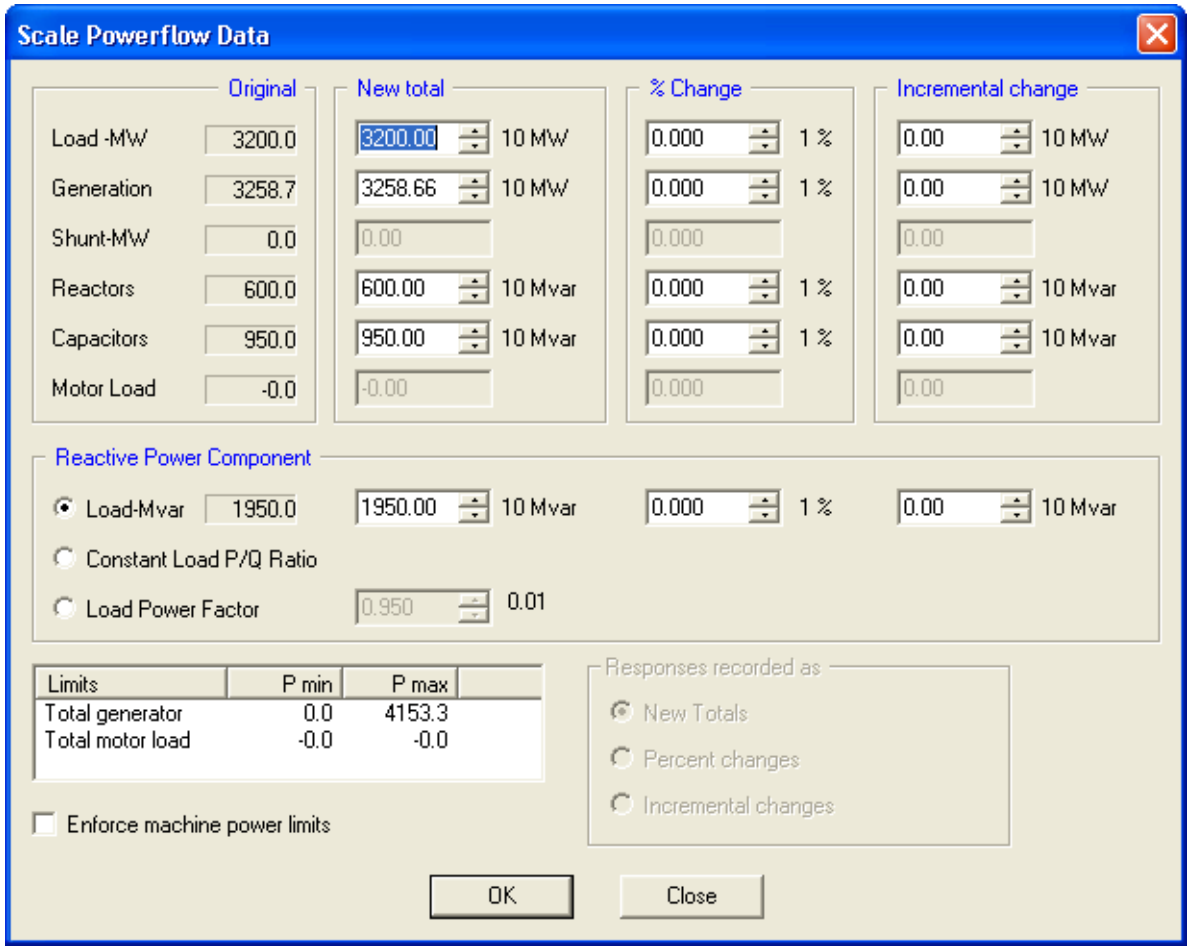


Figure 7-33. Scale Powerflow Data Dialog to Enter Scaling Targets

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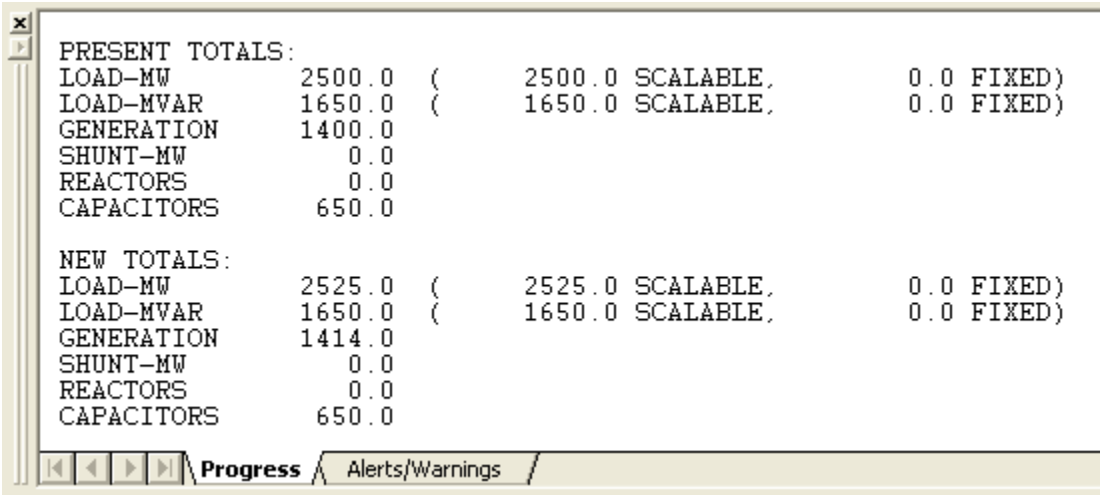


Figure 7-34. Example of Scaling Output

Additional Information
PSS®E ProgramProgram Operation Manual, Section 5.10, Scaling Loads, Generators, and/or Shunts *PSS®E Application Program Interface (API)*, Section 1.280, SCAL

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7.10 Changing Adjustment Enable Flags of Transformers

Activity TFLG

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

Power Flow > Changing > Transformer adjustment flags (TFLG)…

The transformer adjustment enable flag setting activity TFLG allows the user to either enable or disable the adjustment status of all automatically adjustable transformer windings contained in the subsystem specified by the user.

The **[Transformer Adjustment Flags]** dialog (Figure 7-35) enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Transformer Adjustment Flags]** dialog.

The setting to be assigned to the adjustment enable flags of automatically adjustable transformer windings contained in the specified subsystem is specified via the *Allow automatic adjustment* checkbox.



Figure 7-35. Transformer Adjustment Flags Dialog

Additional Information
PSS®E Program Operation Manual, Section 5.31, Changing Adjustment Enable Flags of Transformers *PSS®E Application Program Interface (API)*, Section 1.316, TFLG

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7.11 Importing an ECDI Data File

Requirements / Prerequisites
Economic Dispatch Data File (*.ecd)

File > Import > ECDI File

The file selection window lists only *Files of type: Economic Dispatch Data file (*.ecd)*. Highlight the desired file and click **[Open]**. The economic dispatch data is read into the working case.

Additional Information
PSS®E Program Operation Manual, Section 5.32.1, Economic Dispatch Data File Contents

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7.12 Changing Area Assignments

Activity ARNM

Activity LDAR

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

Power Flow > Renumbering Areas / Owners / Zones …

The area renumbering activity ARNM reassigns buses and/or loads in the working case from their original areas to a designated area. All buses and/or loads in the working case may be reassigned to a designated area; more typically, buses and/or loads in a specified subsystem of the working case are assigned to a designated area.

To change area assignments, select the *Area assignments* tab of the **[Renumbering Areas / Owners / Zones]** dialog (Figure 7-36).

The **[Renumbering Areas / Owners / Zones]** dialog enables the user to designate for area renumbering either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Renumbering Areas / Owners / Zones]** dialog

Use the *Reassign* radio button to indicate if only buses, only loads, or both buses and loads in the designated subsystem are to have their area assignments changed.

Select the area to which subsystem buses and/or loads are to be assigned by highlighting an entry in either the *Used areas* scrolled list or the *Unused areas* scrolled list.

Click **[OK]** or **[Apply]** to implement the area reassignments; a summary is printed at the **Progress** device. When **[OK]** is used, the **[Renumbering Areas / Owners / Zones]** dialog is dismissed; when **[Apply]** is used, the dialog remains and additional renumbering may be implemented.

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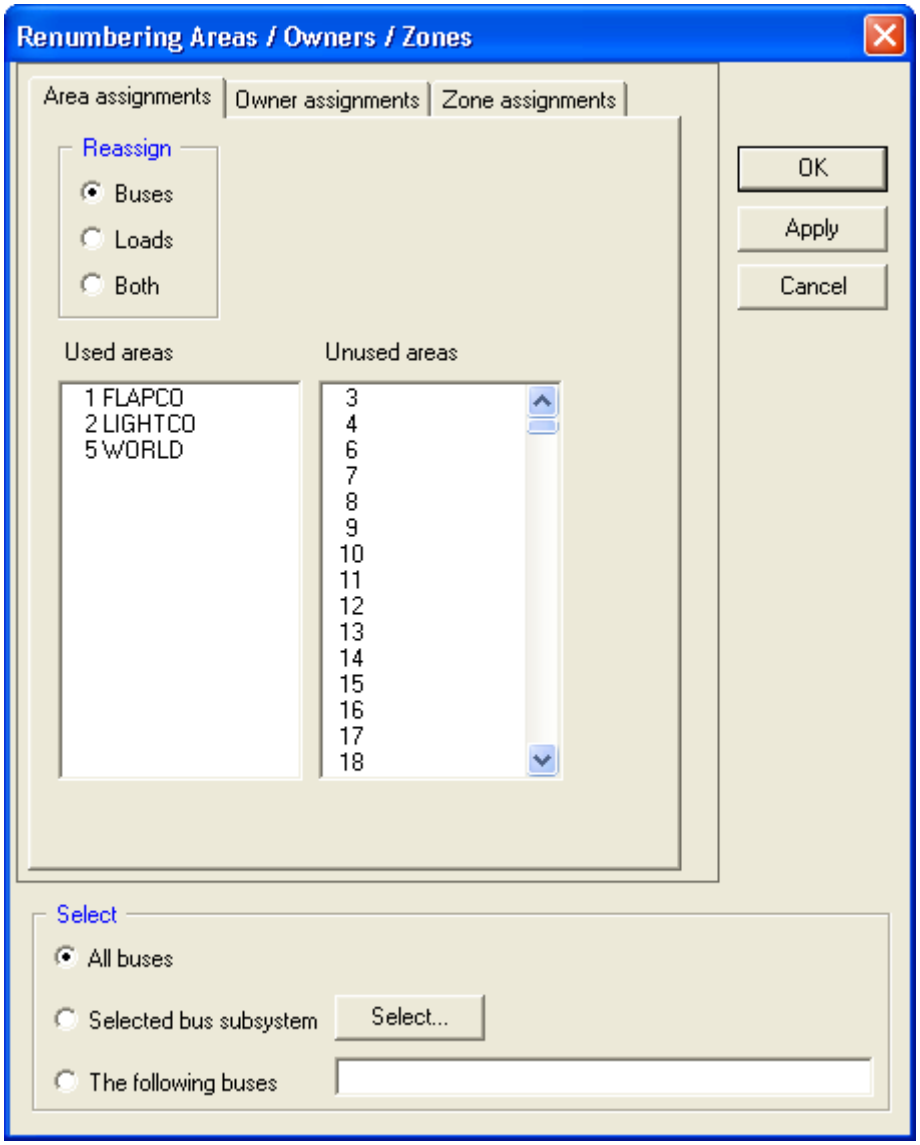


Figure 7-36. Renumbering Areas/Owners/Zones Dialog: Area Assignments Tab

Additional Information
PSS®E Program Operation Manual: Section 5.22, Changing Area Assignments Areas, Zones and Owners

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7.13 Changing Owner Assignments

Activity OWNM

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

Power Flow > Renumbering Areas / Owners / Zones ...

The owner renumbering activity OWNM reassigns buses, loads, machines, branches, FACTS devices and/or VSC dc lines in the working case from their original owners to a designated owner. All elements of the selected equipment types in the working case may be reassigned to a designated owner; more typically, all elements of the selected equipment types in a specified subsystem of the working case are assigned to a designated owner.

To change owner assignments, select the *Owner assignments* tab of the **[Renumbering Areas / Owners / Zones]** dialog (Figure 7-37).

The **[Renumbering Areas / Owners / Zones]** dialog enables the user to designate for owner renumbering either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Renumbering Areas / Owners / Zones]** dialog.

Use the *Reassign* check boxes to indicate one or more of the following equipment types for which elements in the designated subsystem are to have their owner assignments changed:

- Buses
- Loads
- Machines
- Branches (non-transformers branches and transformers)
- FACTS devices
- VSC dc lines

When one or more of branches, FACTS devices, and VSC dc lines have been selected, the *Branch reassignment options* pull-down list is used to select for owner reassignment of series elements either:

- Subsystem branches only
- Subsystem tie lines only
- Subsystem branches and tie lines

Select the owner to which subsystem elements are to be assigned by highlighting an entry in either the *Used owners* scrolled list or the *Unused owners* scrolled list.

Click **[OK]** or **[Apply]** to implement the owner reassignments; a summary is printed at the **Progress** device. When **[OK]** is used, the **[Renumbering Areas / Owners / Zones]** dialog is dismissed; when **[Apply]** is used, the dialog remains and additional renumbering may be implemented.

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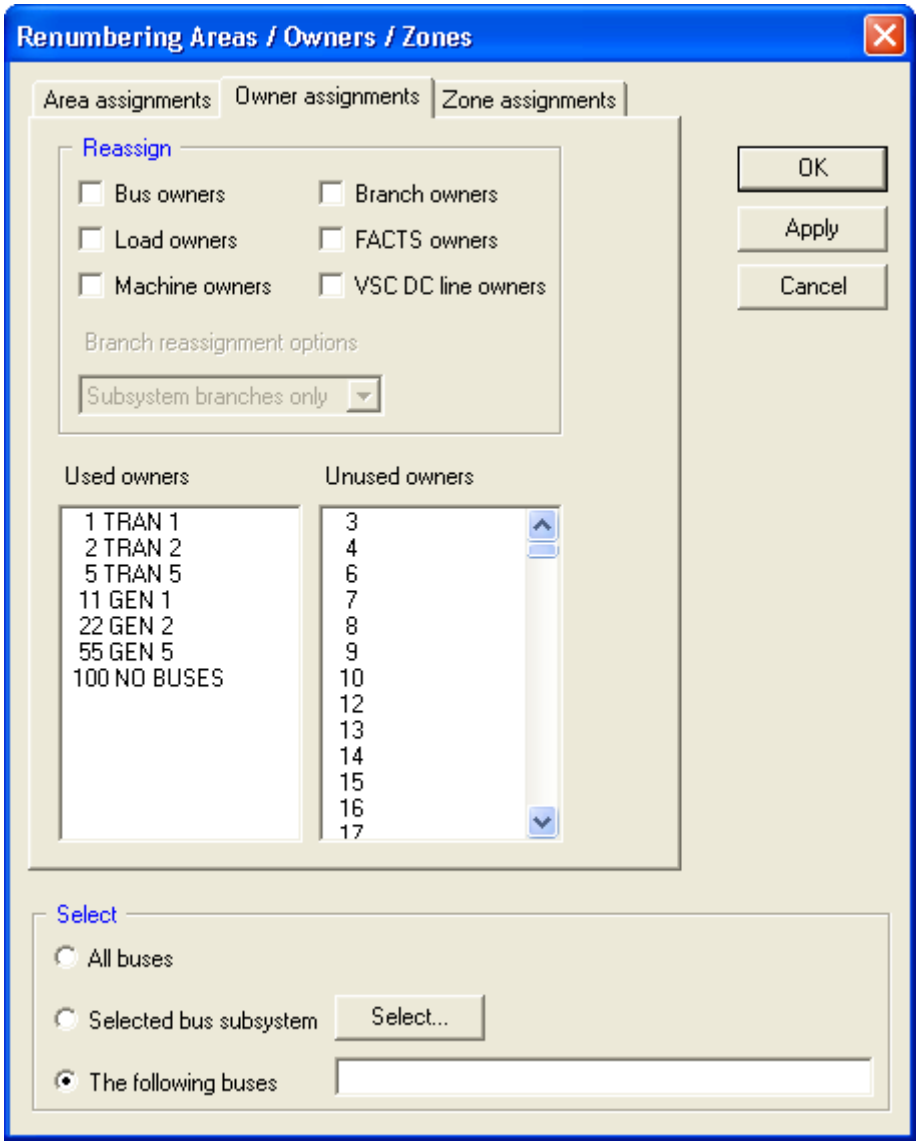


Figure 7-37. Renumbering Areas/Owners/Zones Dialog: Owner Assignments Tab

Additional Information
PSS®E Program Operation Manual: Section 5.23, Changing Owner Assignments Areas, Zones and Owners

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7.14 Changing Zone Assignments

Activity ZONM

Activity LDZO

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

Power Flow > Renumbering Areas / Owners / Zones ...

The zone renumbering activity ZONM reassigns buses and/or loads in the working case from their original zones to a designated zone. All buses and/or loads in the working case may be reassigned to a designated zone; more typically, buses and/or loads in a specified subsystem of the working case are assigned to a designated zone.

To change zone assignments, select the *Zone assignments* tab of the **[Renumbering Areas / Owners / Zones]** dialog (Figure 7-38).

The **[Renumbering Areas / Owners / Zones]** dialog enables the user to designate for zone renumbering either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Renumbering Areas / Owners / Zones]** dialog

Use the *Reassign* radio button to indicate if only buses, only loads, or both buses and loads in the designated subsystem are to have their zone assignments changed.

Select the zone to which subsystem buses and/or loads are to be assigned by highlighting an entry in either the *Used zones* scrolled list or the *Unused zones* scrolled list.

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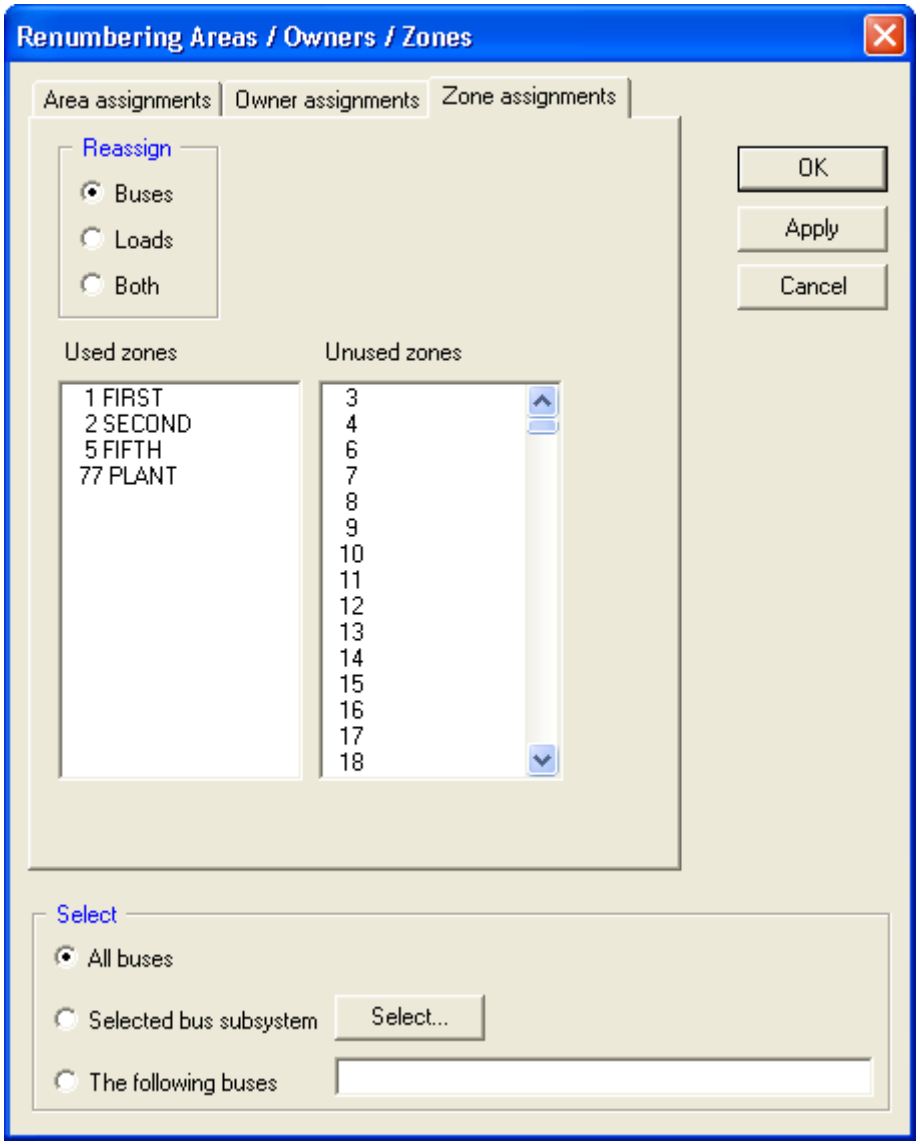


Figure 7-38. Renumbering Areas/Owners/Zones Dialog: Zone Assignments Tab

Click **[OK]** or **[Apply]** to implement the zone reassignments; a summary is printed at the **Progress** device. When **[OK]** is used, the **[Renumbering Areas / Owners / Zones]** dialog is dismissed; when **[Apply]** is used, the dialog remains and additional renumbering may be implemented.

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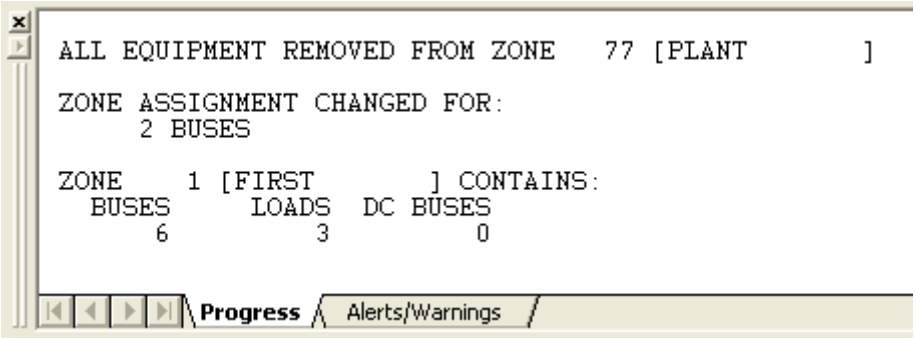


Figure 7-39. Zone Reassignment Output

Additional Information
PSS®E Program Operation Manual: Section 5.24, Changing Zone Assignments Areas, Zones and Owners

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7.15 Renumbering Buses

Activity BSNM

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

The bus renumbering activity BSNM enables the user to change the bus numbers of specified network buses in the working case and retain a tabulation, normally in file form, of the bus number changes made.

Access to the various bus renumbering methods available in activity BSNM is supplied by four dialogs corresponding to four menu entries available in the GUI.

The option for the handling of the output tabulation records produced by activity BSNM is specified in the *Select output destination* area on each BSNM dialog. The following options are available:

- Click the *Data file* radio button on the dialog to preserve the output tabulation records in a Bus Renumbering Translation File. The name of the Bus Renumbering Translation File may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field next to the *Data file* radio button.
- Click the *Report window* radio button on the dialog to write the output tabulation records to the **Report** device.
- Click the *No output* radio button on the dialog to suppress the writing of the output tabulation records.

7.15.1 Renumbering Buses by Bus Number

Power Flow > Renumber Buses > By Bus Number…

The **[Renumber Buses by Bus Number]** dialog (Figure 7-40) is used to renumber buses using the bus number to bus number translation method. See Section 7.15, Renumbering Buses for a description of the *Select output destination* area.

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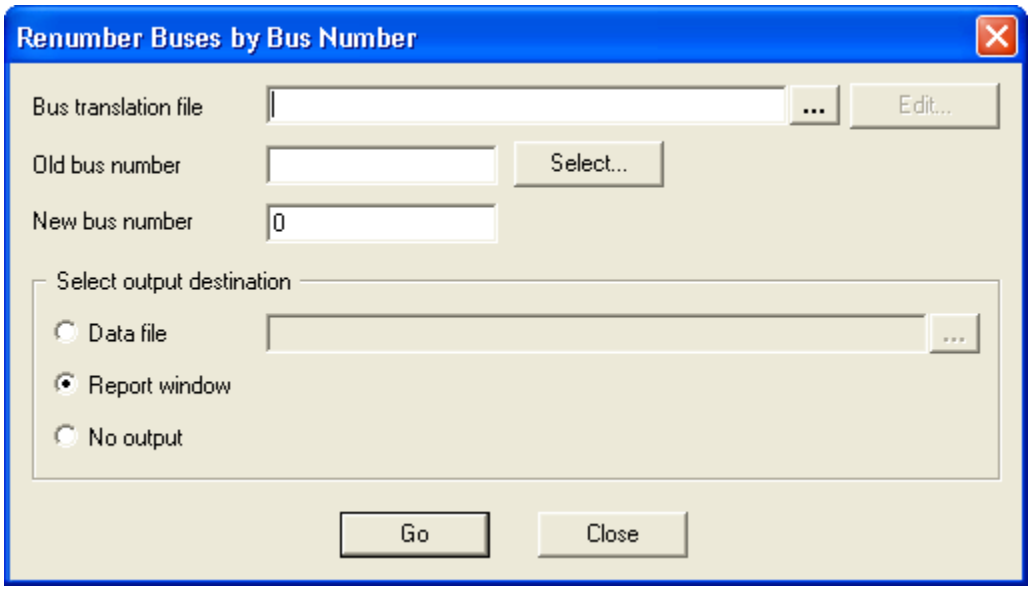


Figure 7-40. Renumber Buses by Bus Number Dialog

7.0.0.1 File Input

If bus number pairs are to be supplied in a Bus Renumbering Translation File, its name is specified in the *Bus translation file* field. The name of the file may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field. Then click **[Go]** to complete the renumbering process and dismiss the dialog.

7.0.0.2 Dialog Input

If bus number pairs are to be specified on the dialog, the old and new bus numbers are entered in the *Old bus number* and *New bus number* fields, respectively.

The old bus number may be entered directly in the *Old bus number* input field. Alternatively, **[Select...]** adjacent to the input field may be used to open the **[Bus Selection]** dialog (Figure 7-3) where the desired bus may be specified by double-clicking on an entry in the selection list; click **[Filter]** to open a **[Bus Filter]** dialog (Figure 7-4) to apply filtering criteria to limit the number of buses in the selection list.

Click **[Go]** to implement the specified bus number change. The dialog remains and another bus number pair may be specified, or click **[Close]** to end activity BSNM.

7.15.2 Renumbering Buses by Name

Power Flow > Renumber Buses > By Bus Name...

The **[Renumber Buses by Bus Name]** dialog (Figure 7-41) is used to renumber buses using the bus name to bus number translation method. See Section 7.15, Renumbering Buses for a description of the *Select output destination* area.

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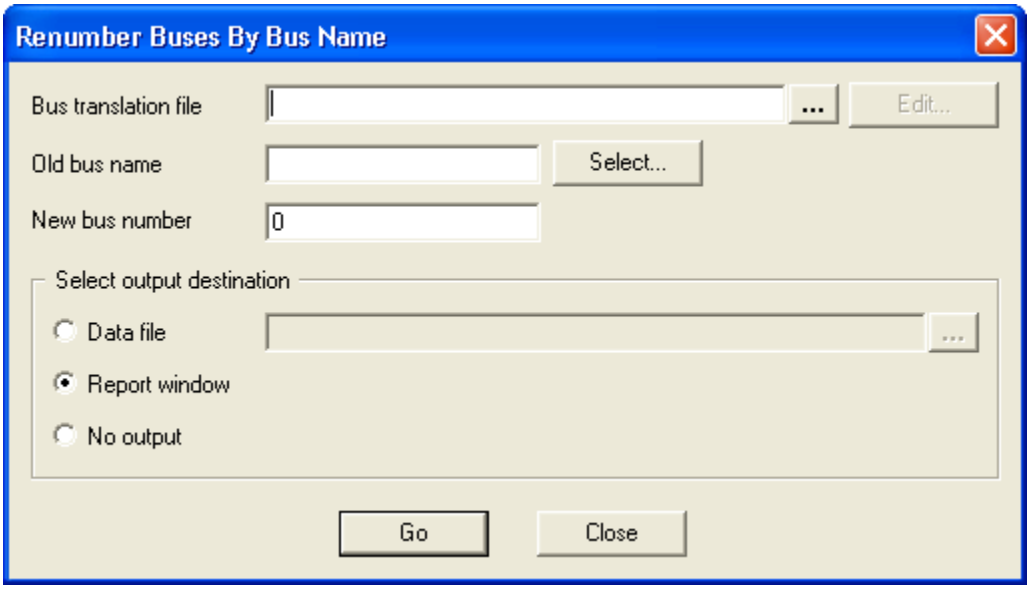


Figure 7-41. Renumber Buses by Bus Name Dialog

7.0.0.3 File Input

If bus name, number pairs are to be supplied in an input data file, its name is specified in the *Bus translation file* field. The name of the file may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field. Then click **[Go]** to complete the renumbering process and dismiss the dialog.

7.0.0.4 Dialog Input

If bus name, number pairs are to be specified on the dialog, the extended bus name and new bus number are entered in the *Old bus name* and *New bus number* fields, respectively.

The old bus name may be entered directly in the *Old bus name* input field. Alternatively, **[Select...]** adjacent to the input field may be used to open the **[Bus Selection]** dialog (Figure 7-3) where the desired bus may be specified by double-clicking on an entry in the selection list; click **[Filter]** to open a **[Bus Filter]** dialog (Figure 7-4) to apply filtering criteria to limit the number of buses in the selection list.

Click **[Go]** to implement the specified bus number change. The dialog remains and another bus name, number pair may be specified, or click **[Close]** to end activity BSNM.

7.15.3 Renumbering Buses by Packing

Power Flow > Renumber Buses > By Packing…

The **[Renumber Buses by Bus Packing]** dialog (Figure 7-42) is used to renumber buses using the bus number packing translation method. See Section 7.15, Renumbering Buses for a description of the *Select output destination* area.

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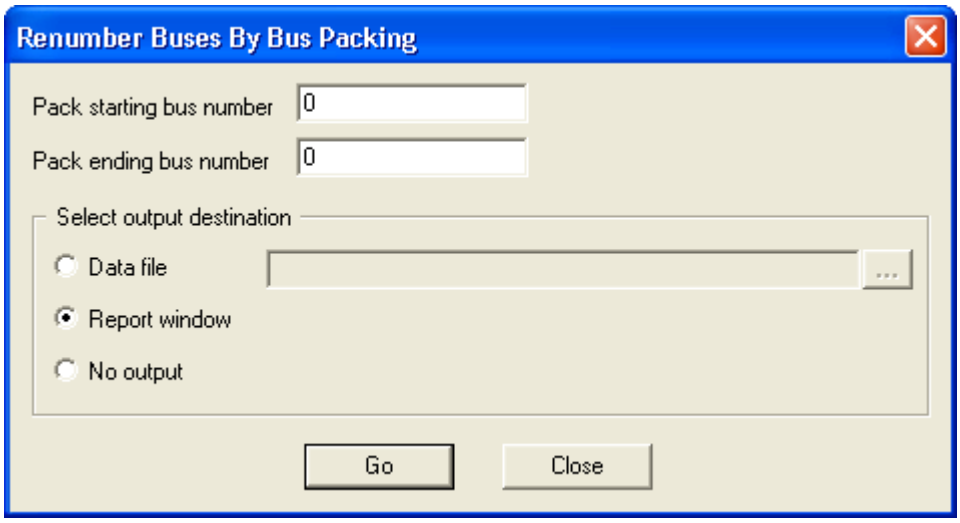


Figure 7-42. Renumber Buses by Bus Packing Dialog

The numbers defining the bus number range are entered in the *Pack starting bus number* and *Pack ending bus number* fields.

Click **[Go]** to implement the packing of bus numbers in the specified range into the low end of the range. The dialog remains and another bus range may be specified, or click **[Close]** to end activity BSNM.

7.15.4 Renumbering Buses By Subsystem

Power Flow > Renumber Buses > By Subsystem…

The **[Renumber Buses by Subsystem]** dialog (Figure 7-43) is used to renumber buses using any of the following bus renumbering methods:

- All buses without area blocking method.
- All buses with area blocking method.
- Subsystem bus number range method.
- Subsystem bus number offset method.

See Section 7.15, Renumbering Buses for a description of the *Select output destination* area.

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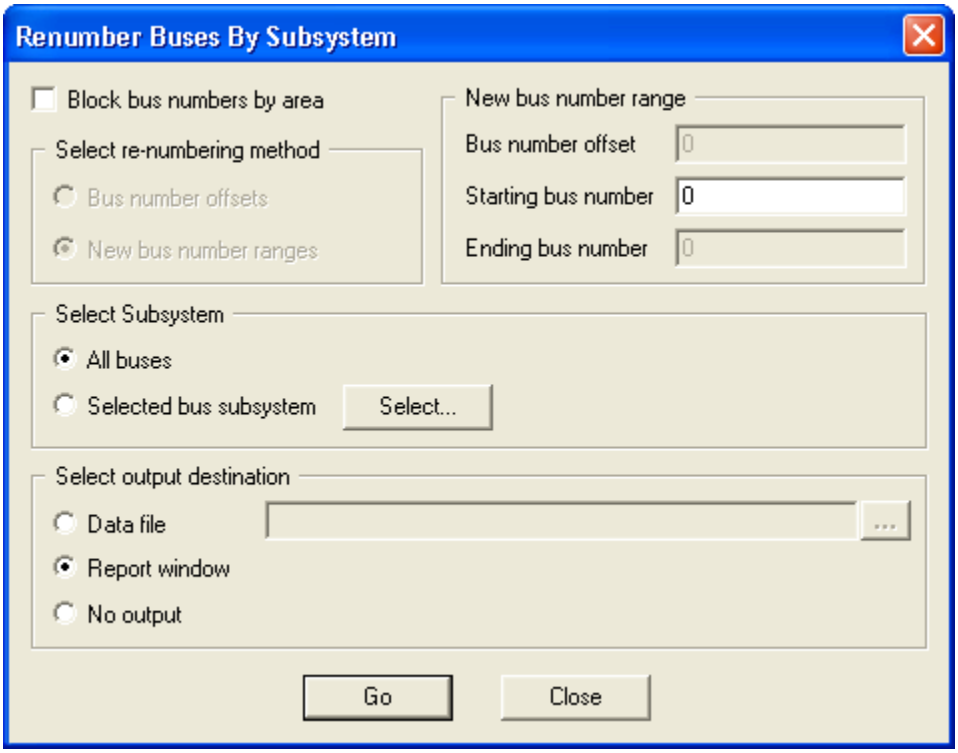


Figure 7-43. Renumber Buses by Subsystem Dialog

7.0.0.5 All Buses Without Area Blocking

To renumber buses using the all buses without area blocking renumbering method, enter the following on the dialog:

- In the *Select Subsystem* area, select the *All buses* radio button.
- Uncheck the *Block bus numbers by area* checkbox.
- In the *New bus number range* area, specify the appropriate value in the *Starting bus number* field.

Then click **[Go]** to complete the renumbering process and dismiss the dialog.

7.0.0.6 All Buses With Area Blocking

To renumber buses using the all buses with area blocking renumbering method, enter the following on the dialog:

- In the *Select Subsystem* area, select the *All buses* radio button.
- Check the *Block bus numbers by area* checkbox.

Then click **[Go]** to open the **[Block Numbers By Area]** dialog (see Figure 7-44).

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Figure 7-44. Example of Block Numbers By Area Dialog

The **[Block Numbers By Area]** dialog contains an entry for each area in the working case that has at least one bus assigned to it. Specify a new bus number range for *each* area on the dialog.Then click **[Go]** to complete the renumbering process and dismiss the dialog.

⚠ No bus number changes are implemented until new number ranges are specified for all areas shown on the dialog. Consequently, unless non-zero bus numbers are entered to indicate the bus number ranges for each area on the dialog, the renumbering process will not be initiated.

7.0.0.7 Subsystem Bus Number Range

To renumber buses in a bus subsystem using the bus number range method, enter the following on the dialog:

- In the *Select Subsystem* area, select the *Selected bus subsystem* radio button.
- Unless the current bus subsystem is to be processed, click **[Select...]** to specify a bus subsystem via a **[Bus Subsystem Selector]**.
- In the *Select re-numbering method* area, select the *New bus number range* radio button.
- In the *New bus number range* area, specify the appropriate values in the *Starting bus number* and *Ending bus number* fields.

Click **[Go]** to implement the specified subsystem renumbering. The dialog remains and another bus subsystem may be renumbered, or click **[Close]** to end activity BSNM.

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7.0.0.8 Subsystem Bus Number Offset

To renumber buses in a bus subsystem using the bus number offset method, enter the following on the dialog:

- In the *Select Subsystem* area, select the *Selected bus subsystem* radio button.
- Unless the current bus subsystem is to be processed, click **[Select...]** to specify a bus subsystem via a **[Bus Subsystem Selector]**.
- In the *Select re-numbering method* area, select the *Bus number offset* radio button.
- In the *New bus number range* area, specify the appropriate value in the *Bus number offset* field.

Click **[Go]** to implement the specified subsystem renumbering. The dialog remains and another bus subsystem may be renumbered, or click **[Close]** to end activity BSNM.

Additional Information
PSS®E Program Operation Manual, Section 5.46, Bus Renumbering *PSS®E Application Program Interface (API)*, Section 1.45, BSNM

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7.16 Renumbering Buses in Auxiliary Data Files

Activity RNFI

Requirements / Prerequisites
The working case must contain a validly specified power flow case.
A Bus Renumbering Translation File must be available.

File > Renumber buses in auxiliary files (RNFI)…

The auxiliary data input file bus renumbering activity RNFI reflects changes in bus numbering in auxiliary data input files. It is used primarily in conjunction with activity BSNM to coordinate bus renumbering implemented in a Saved Case with auxiliary data input files associated with the Saved Case and read by other PSS®Express activities.

The **[Renumber Buses in Auxiliary Files]** dialog (Figure 7-45) is available only when using **[Spreadsheet]**.

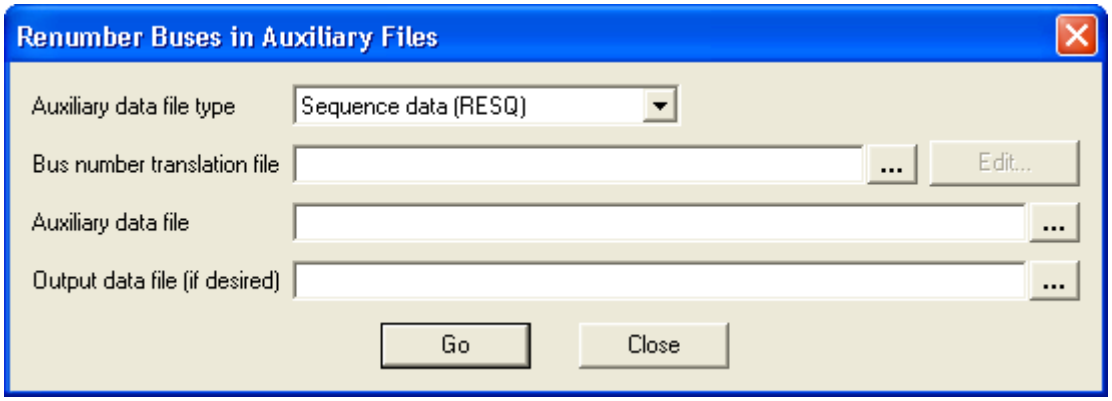


Figure 7-45. Renumber Buses in Auxiliary Files Dialog

First, select the type of file to be processed from the *Auxiliary data file type* pull-down list. An abbreviation for the activity that requires the file type appears in parentheses, as follows:

- Sequence data (RESQ)
- Dynamics data (DYRE)
- Machine impedance data (MCRE)
- Dispatch data (ECDI)
- Dispatch data (INLF)
- Breaker duty data (BKDY)
- Fault specification data (BKDY)
- Fault control data (ASCC)
- Capability curve data (GCAP)
- Load throwover data (ACCC)
- Fault specification data (ANSI)
- Subsystem description data (ACCC)
- Monitored element data (ACCC)
- Contingency description data (ACCC)
- Tripping data (ACCC)

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Then specify the files to be used by activity RNFI.

- Bus Number Translation file (*.trn)
- Auxiliary Data file (files corresponding to *Auxiliary data file type*)
- Output data file (files corresponding to *Auxiliary data file type*)

For each of these files, its name may be typed directly in its input field, or you can click **[...]** next to the input field to specify the name of the file from a file selector dialog.

Additional Information
PSS®E Program Operation Manual, Section 5.21, Renumbering Buses in Auxiliary Files Section 5.46, Bus Renumbering *PSS®E Application Program Interface (API)*, Section 1.273, RNFI

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Chapter 8

Data Reports

8.1 Working Case Data Reporting



Power Flow > List Data…

PSS®Express facilitates the listing of all data in the working case for viewing, problem checking, or case documentation.

Through the **[List Data]** dialog (Figure 8-1), the following tabulations of working case data can be produced:

- Categories of power flow data in tabular form (activity LIST).
- Power flow data grouped together on a bus-by-bus basis (activity EXAM).
- Categories of optimal power flow data in tabular form (activity LSTO).
- Outaged network elements (activity OUTS).
- Fixed and/or switched bus shunts (activity SHNT).
- Extended bus names in alphabetic order (activity ALPH).

Click the radio button on the **[List Data]** dialog corresponding to the desired reporting function. If the selected function has provision for reporting options, the corresponding controls on the dialog are enabled.

The **[List Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[List Data]** dialog.

Reports are routed to the **Report** device.

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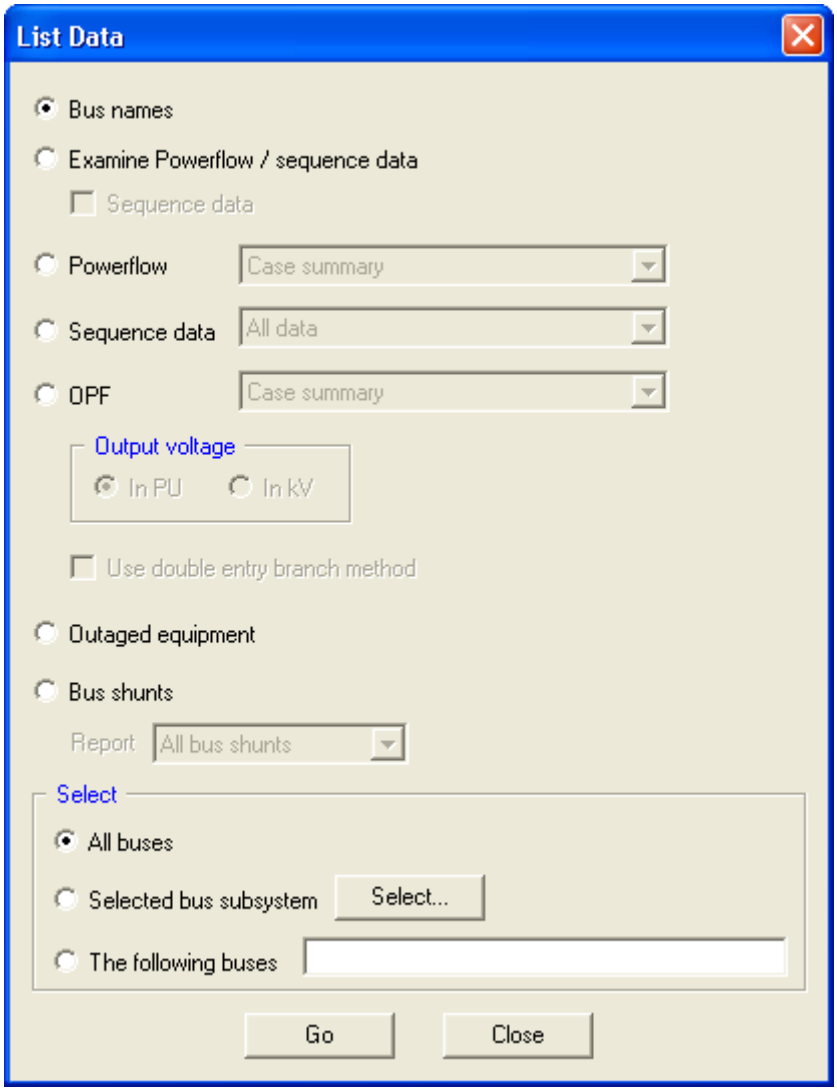


Figure 8-1. List Data Dialog

8.1.1 Power Flow Data

Activity LIST

Requirements / Prerequisites
The working case must contain a non-null case.



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Power Flow > List Data…

The data listing activity LIST tabulates the power flow working case in a form suitable for problem data documentation. The report generated by activity LIST is separated into several categories of data.

Activity LIST is accessible from the *Powerflow* radio button of the **[List Data]** dialog (Figure 8-1). The data category to be tabulated is selected from the drop down list next to the *Powerflow* radio button on the dialog. The *Output voltage* radio buttons provide for the tabulation of ac voltages in either per unit or kilovolts. The *Use double entry branch method* check box is used to select either single or double entry format for branch related category reports when either the *All data* or *Branches* report is specified in the drop down list.

The **[List Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[List Data]** dialog.

Additional Information
PSS®E Program Operation Manual, Section 5.37, Displaying Power Flow Data *PSS®E Application Program Interface (API)*, Section 1.119, LIST

8.1.2 Listing Buses in Alphabetical Order

Activity ALPH

Requirements / Prerequisites
Validly specified power flow case.



Power Flow > List Data…

The bus alphabetic listing activity ALPH prints an alphabetically sorted table of all buses in a specified subsystem of the working case.

Activity ALPH is accessible from the *Bus names* radio button of the **[List Data]** dialog (Figure 8-1).

The **[List Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[List Data]** dialog.

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Additional Information
PSS®E Program Operation Manual, Section 5.39, Listing Buses in Alphabetical Order *PSS®E Application Program Interface (API)*, Section 1.26, ALPH

8.1.3 Listing Buses and their Connected Equipment

Activity EXAM

Requirements / Prerequisites
The working case must contain a non-null case.



Power Flow > List Data…

The data examination activity EXAM produces a tabulation of power flow data organized by bus. For each bus tabulated, its bus data is followed by the data associated with each network element that is connected to the bus.

Activity EXAM is accessible from the *Examine Powerflow / sequence data* radio button of the **[List Data]** dialog (Figure 8-1). The *sequence data* check box must be unchecked.

The **[List Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[List Data]** dialog.

Additional Information
PSS®E Program Operation Manual, Section 5.38, Listing Components of a Bus *PSS®E Application Program Interface (API)*, Section 1.74, EXAM

8.1.4 Outaged Equipment

Activity OUTS

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > List Data…

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The outaged equipment reporting activity OUTS tabulates those components in the working case that are removed from service.

Activity OUTS is accessible from the *Outaged equipment* radio button of the **[List Data]** dialog (Figure 8-1).

The **[List Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[List Data]** dialog.

Additional Information
PSS®E Program Operation Manual, Section 5.42, Listing Outaged Equipment *PSS®E Application Program Interface (API)*, Section 1.188, OUTS

8.1.5 Listing Bus Shunts

Activity SHNT

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > List Data…

The bus shunt summary activity SHNT tabulates fixed and/or switched bus shunts contained in the working case.

Activity SHNT is accessible from the *Bus shunts* radio button of the **[List Data]** dialog (Figure 8-1). The type(s) of bus shunts to be tabulated is selected from the *Report* drop down list below the *Bus shunts* radio button on the dialog.

The **[List Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[List Data]** dialog.

Additional Information
PSS®E Program Operation Manual, Section 5.41, Listing Bus Shunts *PSS®E Application Program Interface (API)*, Section 1.296, SHNT

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8.2 Listing File Information

File > File information (SIZE/SHOW/BUSN)…

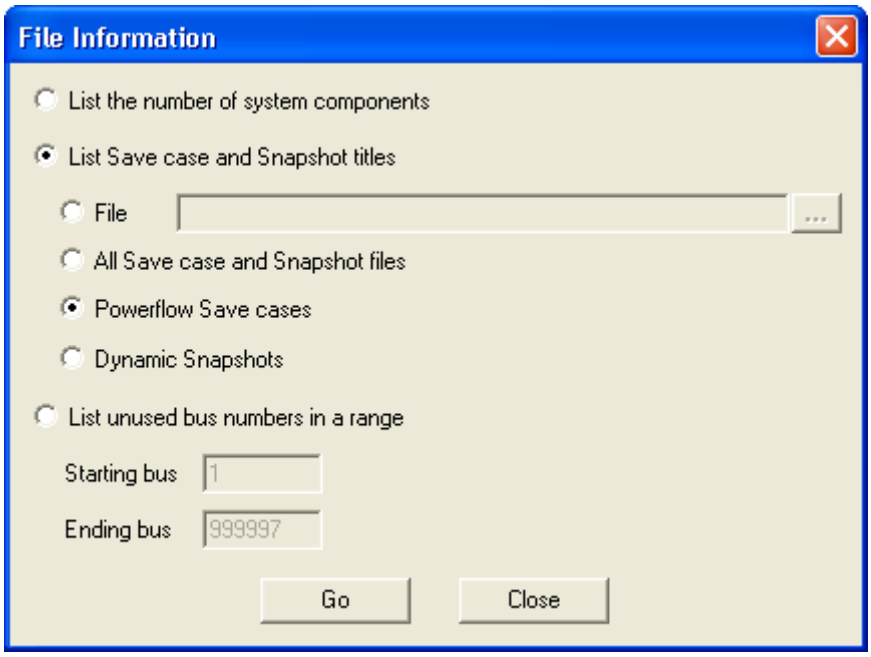


Figure 8-2. File Information Dialog

Three reports are available from the **[File Information]** dialog:

- System Components
- Saved Case and Snapshot Filenames
- Unused Bus Numbers

8.2.1 System Components

Activity SIZE

Requirements / Prerequisites
The working case must contain a non-null case..

The case size summary activity SIZE tabulates the number of components in the working case, along with the maximum number permitted at the current size level of PSS®Express working memory.

Activity SIZE is accessible from the *List the number of system components* radio button of the **[File Information]** dialog.

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Additional Information
PSS®E Program Operation Manual, Section 5.34.3, Editing the Two-Line Case Title or the Long Title *PSS®E Application Program Interface (API)*, Section 1.302, SIZE

8.2.2 Saved Case and Snapshot Filenames

Activity SHOW

Selecting the *List Save case and Snapshot titles* option produces a report of filenames in the current working directory. The user may specify whether to open the **[Select file to display title]** dialog, only Powerflow Save cases, only Dynamics Snapshot files, or All Save cases and Snapshot files.

Specific information on this option may be found at Section 6.2, Listing Saved Case Filenames and Section 20.7, Listing Snapshot Filenames.

Additional Information
PSS®E Program Operation Manual, Section 18.14, Listing Saved Case or Snapshot Files

8.2.3 Unused Bus Numbers

Activity BUSN

Requirements / Prerequisites
Validly specified power flow case.

The unused bus number summary activity BUSN tabulates those numbers, from within a user specified bus number range, which are not assigned to buses in the working case.

Activity BUSN is accessible from the *List unused bus numbers in a range* radio button of the **[File Information]** dialog. The bus number range is specified in the *Starting bus* and *Ending bus* input fields on the dialog.

Additional Information
PSS®E Program Operation Manual, Section 5.36, Listing Unused Bus Numbers *PSS®E Application Program Interface (API)*, Section 1.49, BUSN

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Chapter 9

Case Comparison

9.1 Comparing Power Flow Case Totals

Activity CMPR

Requirements / Prerequisites
Validly specified power flow case. If losses, mismatches, or interchange are to be compared, both cases should be solved to an acceptable mismatch level. Saved Case file (*.sav)

File > Compare…

The **[Compare Case Totals]** dialog (Figure 9-1) creates a report comparing working case totals by area, owner, or zone with a specified saved case. Click **[…]** to open the selection window for the required *Saved Case file* (*.sav). Click **[Go]** to perform the activity. The comparison is routed to the **Report** tab by default (Figure 9-2).

You will need to click **[Close]** to end this activity.

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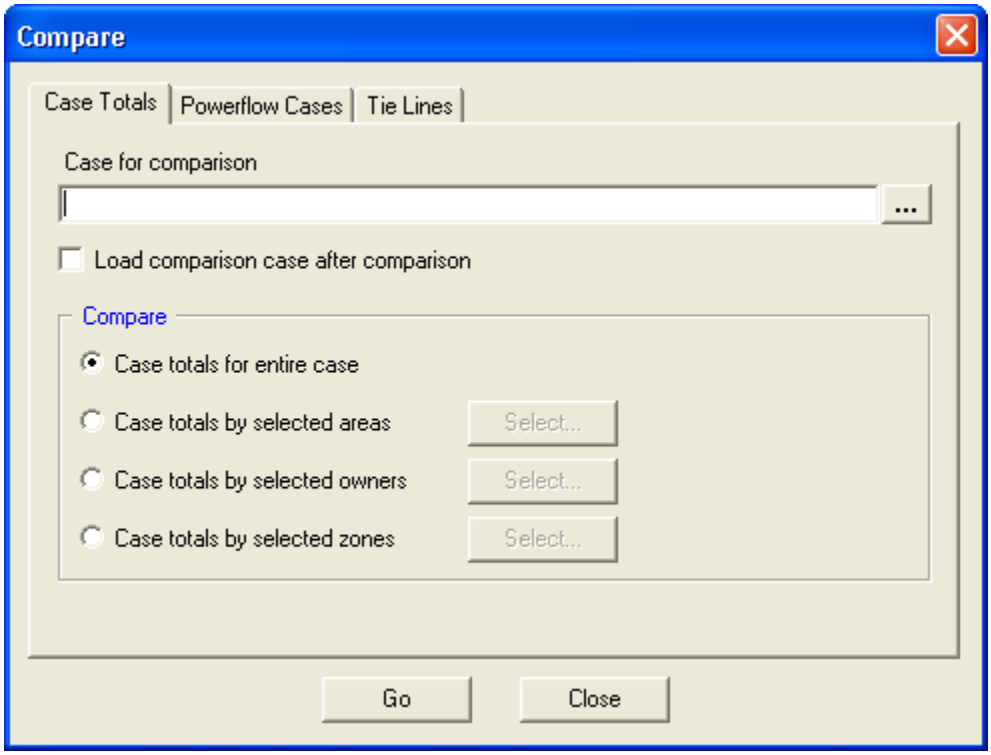
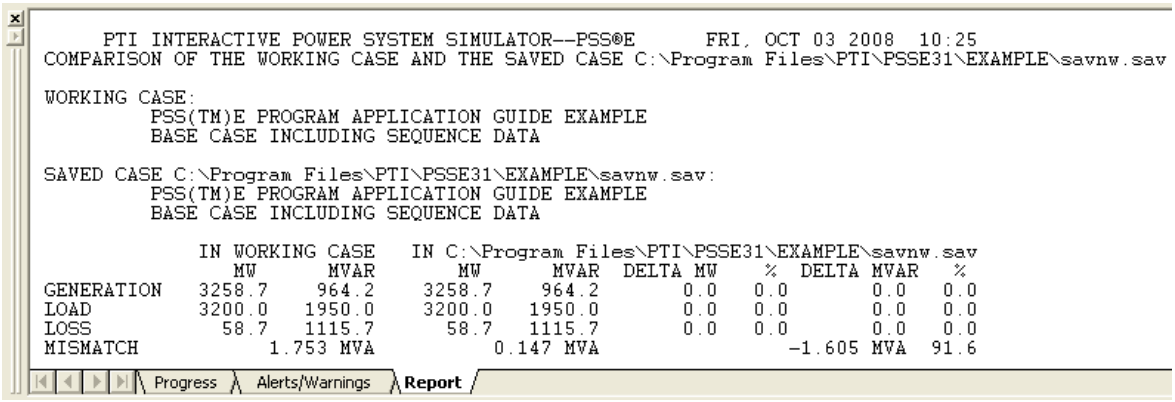


Figure 9-1. Compare Case Totals Dialog



```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      FRI, OCT 03 2008  10:25
COMPARISON OF THE WORKING CASE AND THE SAVED CASE C:\Program Files\PTI\PSSE31\EXAMPLE\savnw.sav

WORKING CASE:
         PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
         BASE CASE INCLUDING SEQUENCE DATA

SAVED CASE C:\Program Files\PTI\PSSE31\EXAMPLE\savnw.sav:
         PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
         BASE CASE INCLUDING SEQUENCE DATA

              IN WORKING CASE    IN C:\Program Files\PTI\PSSE31\EXAMPLE\savnw.sav
                 MW      MVAR       MW      MVAR  DELTA MW    %  DELTA MVAR   %
GENERATION    3258.7    964.2    3258.7    964.2       0.0   0.0       0.0   0.0
LOAD          3200.0   1950.0    3200.0   1950.0       0.0   0.0       0.0   0.0
LOSS            58.7   1115.7      58.7   1115.7       0.0   0.0       0.0   0.0
MISMATCH           1.753 MVA          0.147 MVA                -1.605 MVA  91.6
```

Progress | Alerts/Warnings | Report

Figure 9-2. Example of Compare Case Totals Report

Additional Information
PSS®E Program Operation Manual, Section 11.26, Comparing Power Flow Case Totals

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9.2 Comparing Power Flow Cases

Activity DIFF

Requirements / Prerequisites
Validly specified power flow case. If bus voltages, line flows, or line losses are to be compared, both cases should be solved to an acceptable mismatch level. Saved Case file (*.sav) - - - Managing Case Titles

File > Compare...

The **[Compare Powerflow Cases]** dialog (Figure 9-3) creates a report comparing a specified bus subsystem quantity in the working case with a saved case. Click **[...]** to open the selection window for the required *Saved Case file* (*.sav).

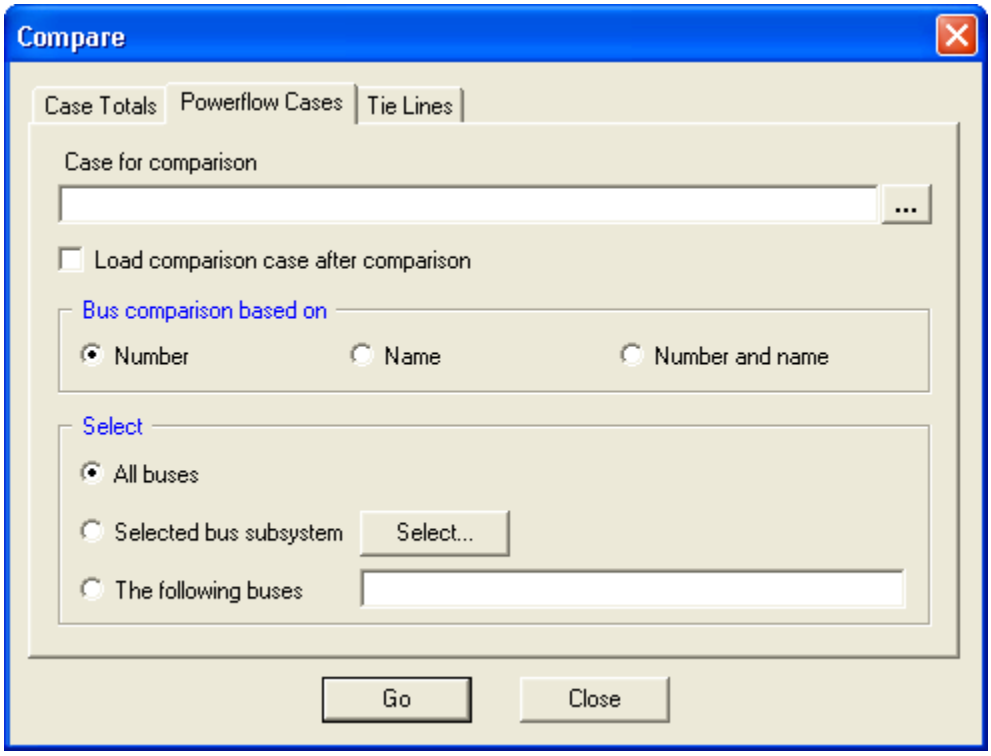


Figure 9-3. Compare Power Flow Cases Dialog

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Click **[Go]** to open the **[Select Powerflow Comparison Options]** dialog (Figure 9-4).

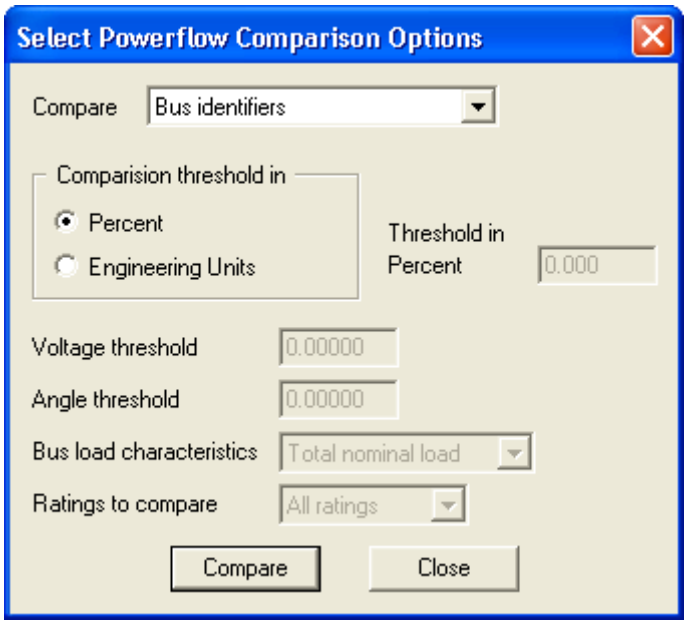


Figure 9-4. Select Powerflow Comparison Options Dialog

The dialog activates comparison threshold values, load characteristics, or ratings appropriate to the specification of one of the following quantities:

- Bus identifiers
- Bus type codes
- Machine status
- Generator MW
- Generator MW or MVAR
- Bus loads
- Bus shunts
- Switched shunts
- Voltage
- Voltage and angle
- MBASE & ZSORCE
- MBASE & ZPOS
- MBASE & ZNEG
- MBASE & ZZERO
- Negative sequence bus shunts
- Zero sequence bus shunts
- Branch status
- Line R, X, B
- Line shunts
- Line ratings
- Metered end
- Transformers
- Flows MW or MVAR (from bus)
- Flows MW or MVAR (from & to)
- Line MW or MVAR losses
- Zero sequence R, X, B
- Zero sequence line shunts
- Connection codes
- Zero sequence mutuals
- Multi-section lines
- Multi-section line metered end
- Bus load status
- Line lengths
- Generator MVAR
- Flows MW (from bus)
- Flows MVAR (from bus)
- Flows MW (from & to)
- Flows MVAR (from & to)
- Line MW losses
- Line MVAR losses
- Fixed shunt status
- Switched shunt status

Bus load characteristics that may be compared include one of the following quantities:

- Total nominal load
- Constant MVA
- Constant current
- Constant admittance

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Click **[Compare]** to perform the activity. The comparison is routed to the **Report** tab by default (Figure 9-5).

You will need to click **[Close]** to end this activity.

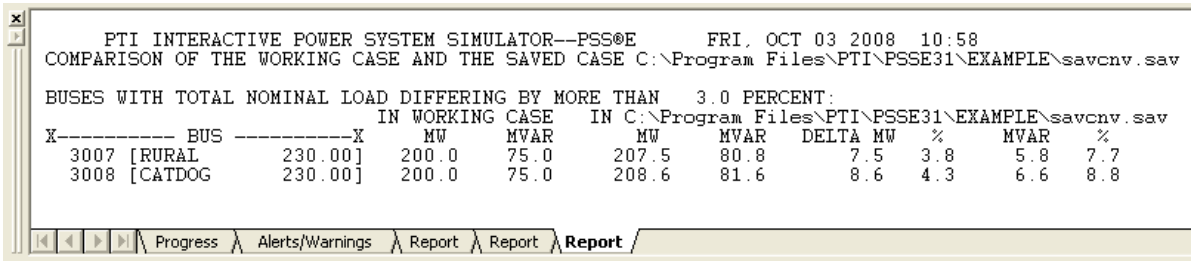


Figure 9-5. Example of Compare Cases Report

Additional Information
PSS®E Program Operation Manual, Section 11.26, Comparing Power Flow Case Totals

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9.3 Comparing AC Tie Branches

Activity DFTI

Requirements / Prerequisites
Validly specified power flow case. If line flows or line losses are to be compared, both cases should be solved to an acceptable mismatch level. Saved Case file (*.sav) - - - Managing Case Titles

File > Compare…

The **[Compare Tie Lines]** dialog (Figure 9-6) creates a report comparing a specified tie line quantity in the working case with a specified saved case. Click **[…]** to open the selection window for the required *Saved Case file* (*.sav).

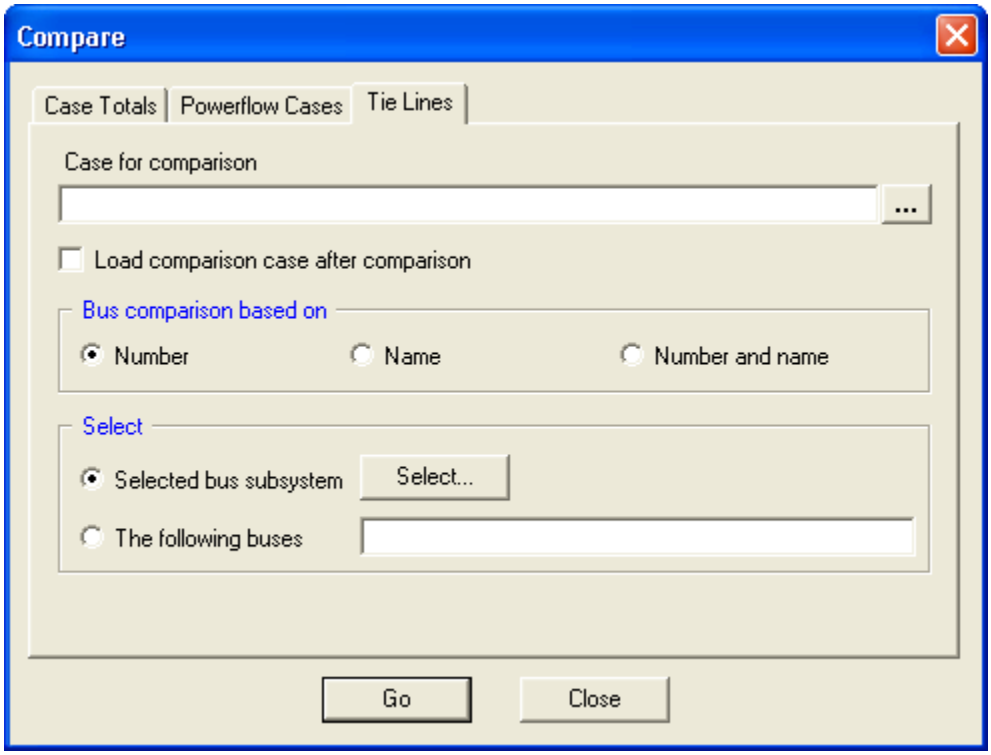


Figure 9-6. Compare Tie Lines Dialog

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Click **[Go]** to open the **[Select Tie Line Comparison Options]** dialog (Figure 9-7).

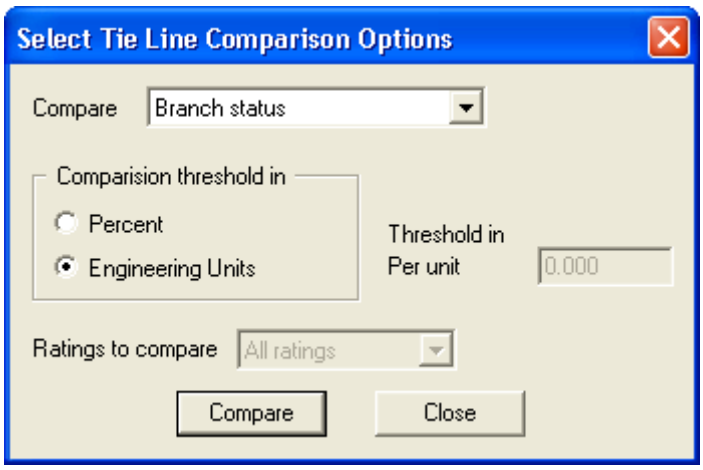


Figure 9-7. Select Tie Line Comparison Options Dialog

The dialog activates comparison threshold values or ratings appropriate to the specification of one of the following quantities:

- Branch status
- Line R, X, B
- Line shunts
- Line ratings
- Metered end
- Line length
- Transformers
- Flows MW or MVAR (from bus)
- Flows MW or MVAR (from & to)
- Line MW or MVAR losses
- Zero sequence R, X, B
- Zero sequence line shunts
- Connection codes
- Flow MW (from bus)
- Flow MVAR (from bus)
- Flow MW (from & to)
- Flow MVAR (from & to)
- Line MW losses
- Line MVAR losses

Click **[Compare]** to perform the activity. The comparison is routed to the **Report** tab by default

You will need to click **[Close]** to end this activity.

Additional Information
PSS®E Program Operation Manual, Section 11.29, Comparing AC Tie Branches

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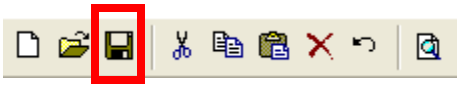
Chapter 10

Creating Power Flow Data Files

10.1 Creating a Saved Case File

Activity SAVE

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



File > Save...

The case saving activity SAVE stores the working case into a user-specified Saved Case file in a compressed format.

Activity SAVE is accessible from the *Case Data* tab of the **[Save Network Data]** dialog (Figure 10-1). The Saved Case File may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field.

Click **[OK]** to save the working case in the designated file.

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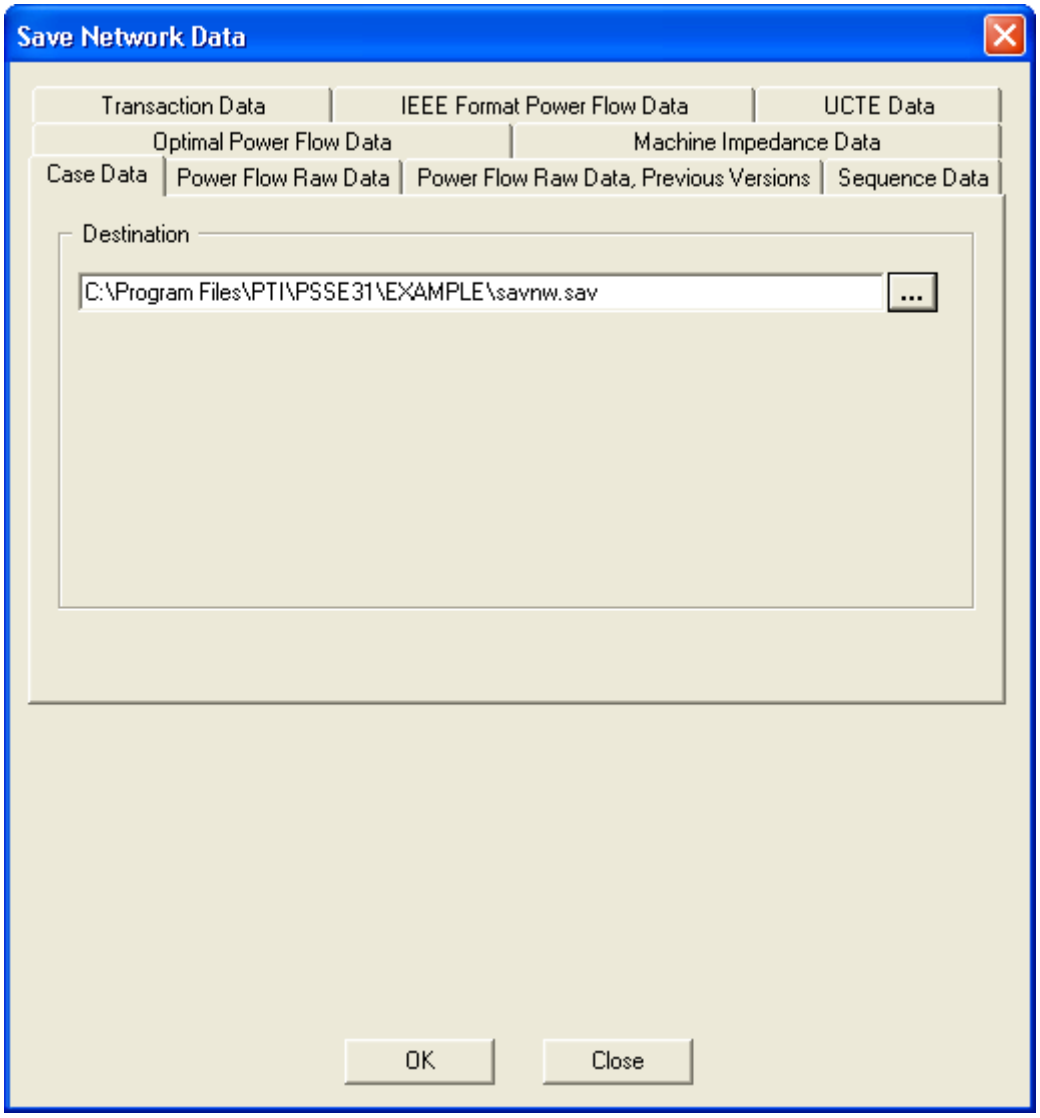


Figure 10-1. Save Network Data Dialog, Case Data

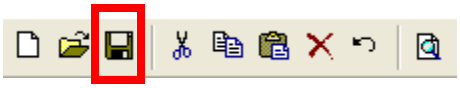
Additional Information
PSS®E Program Operation Manual, Section 5.45, Creating a Saved Case File *PSS®E Application Program Interface (API)*, Section 1.279, SAVE

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10.2 Saving the Working Case in an IEEE Common Format File

Activity RWCM

Requirements / Prerequisites
The working case must contain a validly specified power flow case solved to an acceptable mismatch level.



File > Save...

The IEEE Common Format output activity RWCM writes the working case as a file in IEEE common tape format data records.

Activity RWCM is accessible from the *IEEE Format Power Flow Data* tab of the **[Save Network Data]** dialog (Figure 10-2).

To write the data records to a file, select the *Data file* radio button in the *Destination* area of the dialog. The name of the file may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field.

To write the data records to the **Report** device, select the *Report window* radio button in the *Destination* area of the dialog.

To ignore dc lines and FACTS devices in the data records, uncheck the *Add power to loads* checkbox; to represent their power injections as load, check the *Add power to loads* checkbox.

Click **[OK]** to save the working case in the designated file in the IEEE format.

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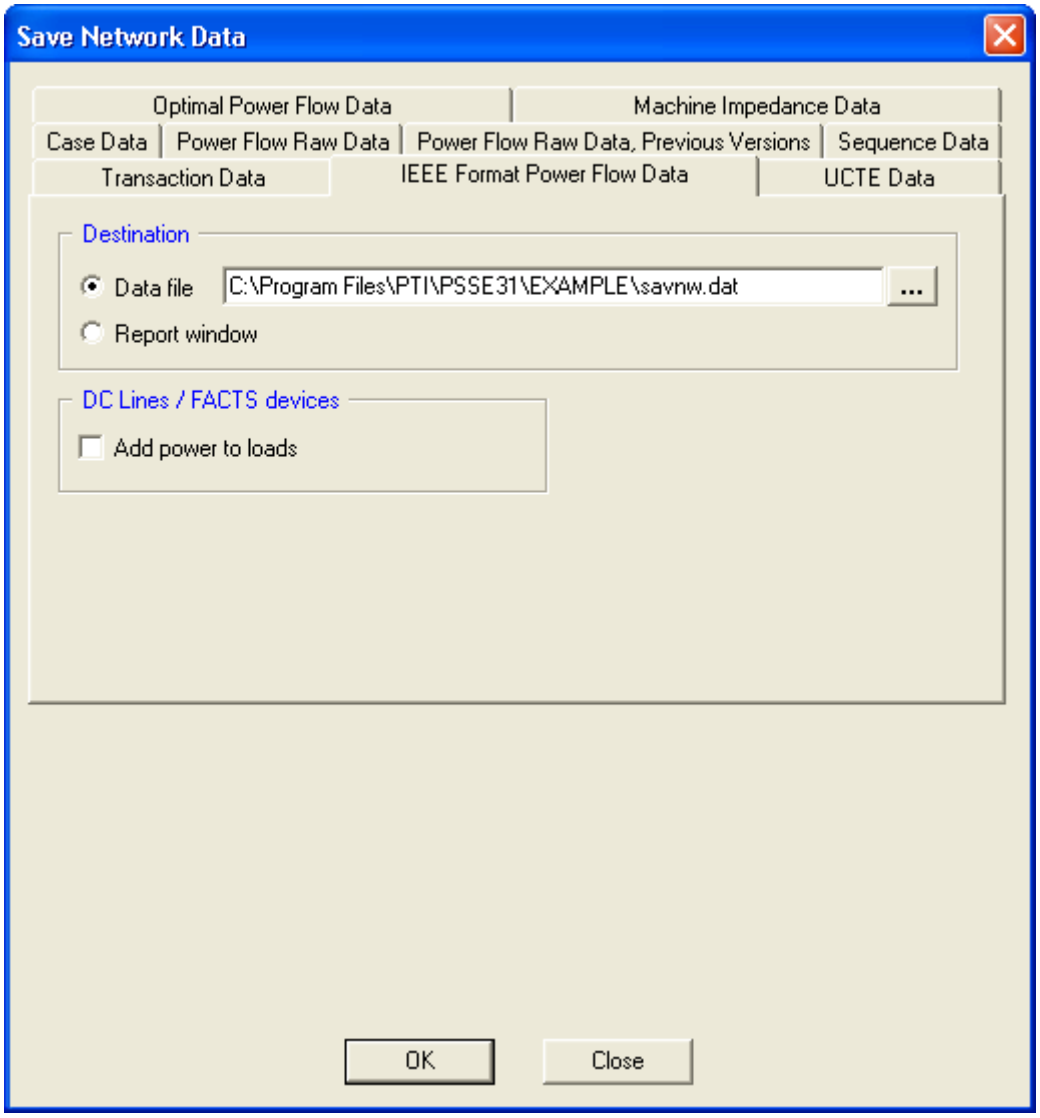


Figure 10-2. Save Network Data Dialog, IEEE Common Format

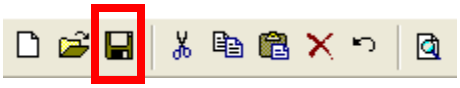
Additional Information
PSS®E Program Operation Manual, Section 5.47, Saving the Working Case in an IEEE Common Format File *PSS®E Application Program Interface (API)*, Section 1.274, RWCM

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10.3 Saving Machine Impedance Data

Activity RWMA

Requirements / Prerequisites
The working case must contain a validly specified power flow case.
If generator reactive powers are to be used in calculating the reactive power split fractions, the case must be solved to an acceptable mismatch level.
Machine impedance data (MBASE, ZSORCE, XTRAN, and GENTAP) must be correctly specified for those machines to be processed.



File > Save...

The machine impedance data output activity RWMA writes out machine parametric data from the working case in the form of a Machine Impedance Data File.

Activity RWMA is accessible from the *Machine Impedance Data* tab of the **[Save Network Data]** dialog (Figure 10-3).

To write the data records to a file, select the *Data file* radio button in the *Destination* area of the dialog. The name of the file may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field.

To write the data records to the **Report** device, select the *Report window* radio button in the *Destination* area of the dialog.

To exclude data data records for out-of-service machines, uncheck the *Include records for out-of-service machines* checkbox; to include them, check the *Include records for out-of-service machines* checkbox.

Designate the quantities to be used in calculating the active and reactive power split fractions by selecting the appropriate entries from the two drop-down lists in the *Machine Impedance* area of the dialog.

The **[Save Network Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Save Network Data]** dialog.

Click **[OK]** to save the machine parametric data in Machine Impedance Data File format.

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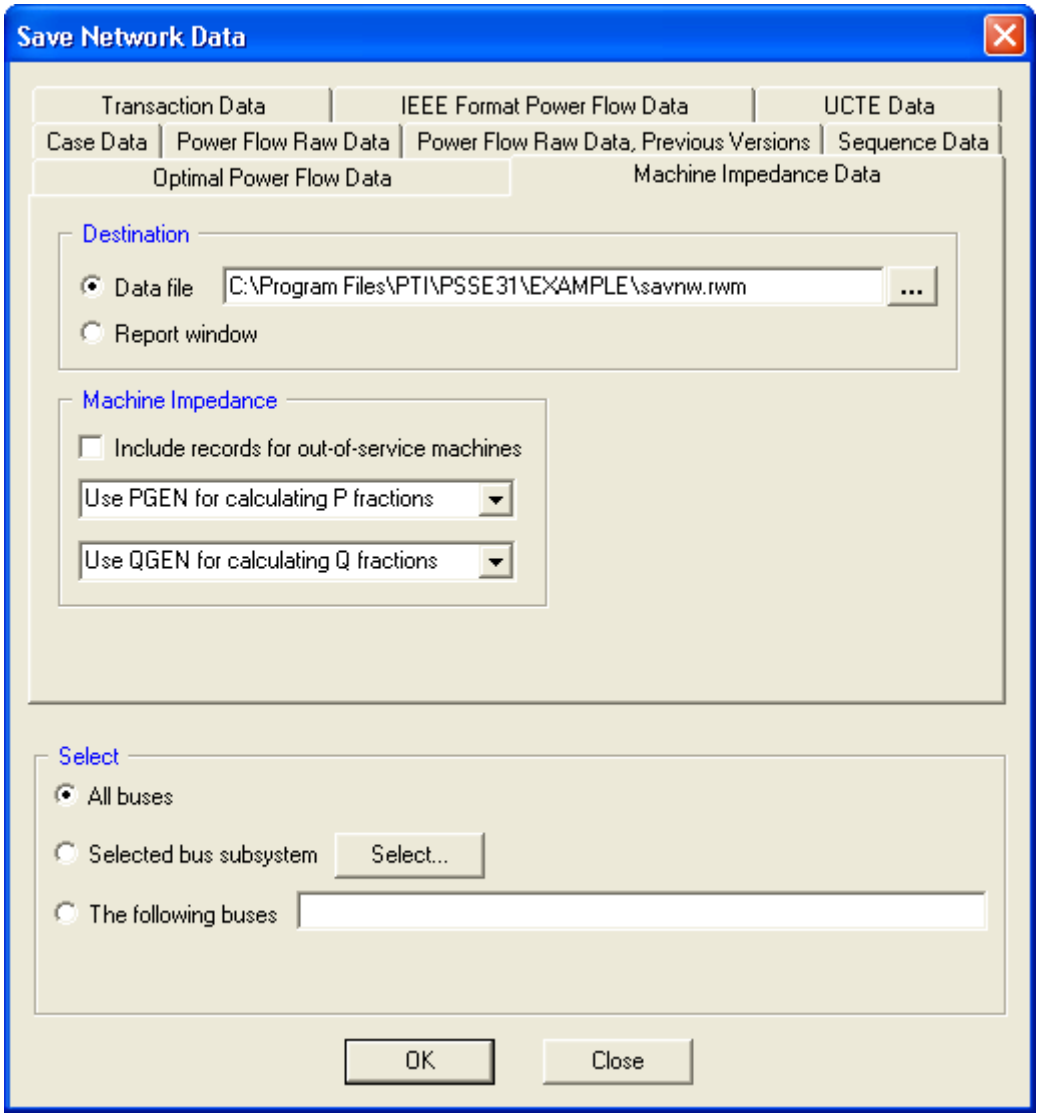


Figure 10-3. Save Network Data Dialog, Machine Impedance Data

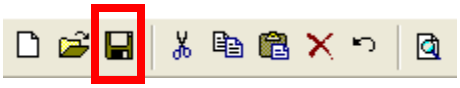
Additional Information
PSS®E Program Operation Manual, Section 5.48, Saving Machine Impedance Data *PSS®E Application Program Interface (API)*, Section 1.275, RWMA

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10.4 Creating a Power Flow Raw Data File

Activity RAWD

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



File > Save...

The Raw Data File output activity RAWD writes the working case in the form of a Power Flow Raw Data File. That is, when directed to a file, the data records written by activity RAWD are in a format suitable for input to activity READ.

Activity RAWD is accessible from the *Power Flow Raw Data* tab of the **[Save Network Data]** dialog (Figure 10-4).

To write the data records to a file, select the *Data file* radio button in the *Destination* area of the dialog. The name of the file may be specified either via a file selector dialog by clicking **[...]** next to the input field, or by entering its name directly in the input field.

To write the data records to the **Report** device, select the *Report window* radio button in the *Destination* area of the dialog.

The user must designate the intended use of the Power Flow Raw Data File by selecting one of the following entries from the *Configure RAW file to* drop-down list:

- *Initialize working case*: suitable for use by activity READ with IC on the first data record set to 0 (default).
- *Add to working case*: suitable for use by activity READ with IC on the first data record set to 1.
- *Use with RDCH*: suitable for use by activity RDCH (i.e., the three case identification data records are omitted).

The **[Save Network Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Save Network Data]** dialog. If the *All area ties* radio button is selected, RAWD writes data records only for branches between areas.

Options to be used in writing Power Flow Raw Data records are specified via checkboxes and a drop-down list in the *Include* area of the dialog. The checkboxes provide options to include network components: isolated buses, out of service branches, subsystem data, subsystem tie lines, and/or bus names. The drop-down list selects of one of the following load options:

- Include all loads at subsystem buses
- Include subsystem loads at all buses
- Include all loads at subsystem buses and subsystem loads at non-subsystem buses

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Click **[OK]** to save the power flow data in Power Flow Raw Data File format.

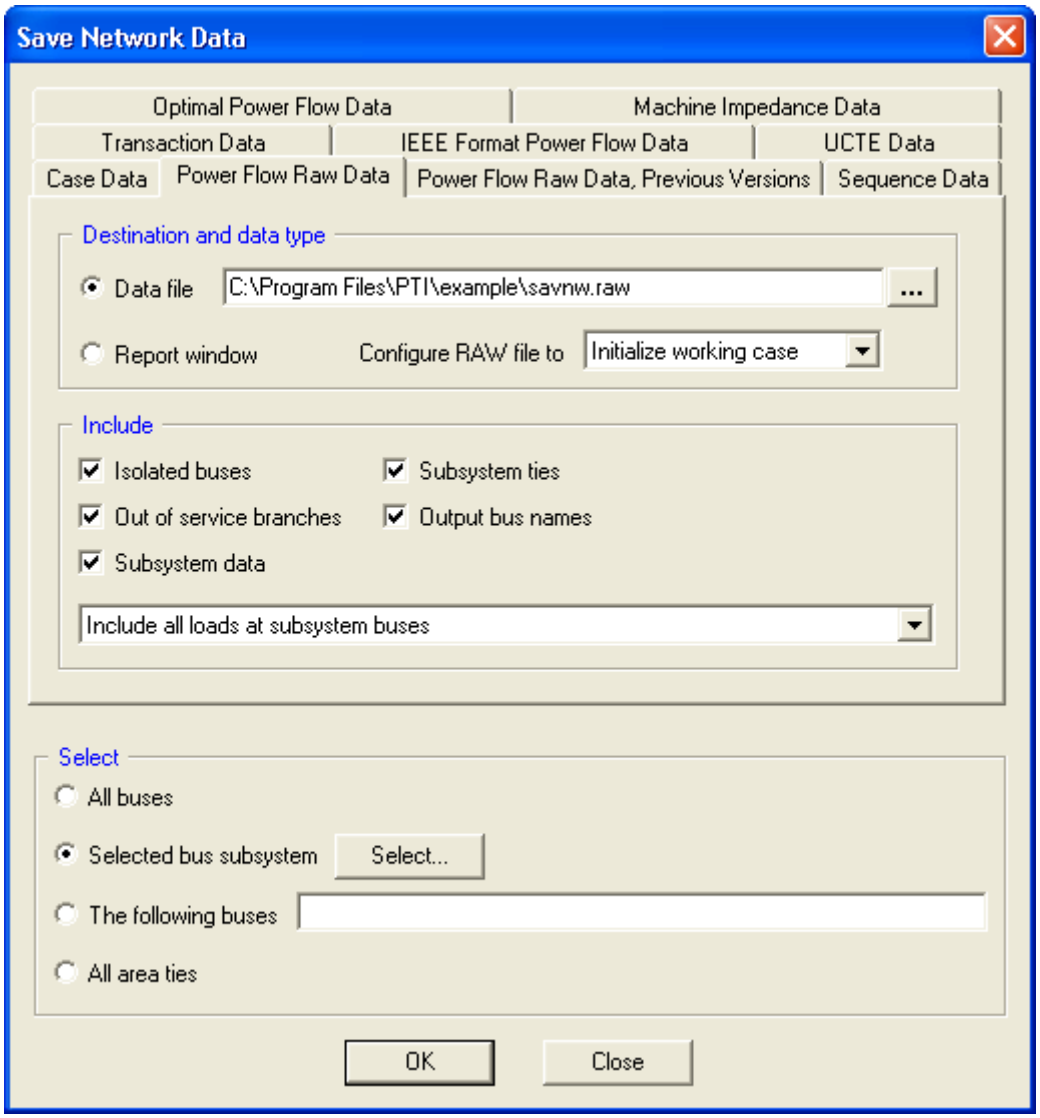


Figure 10-4. Save Network Data Dialog, Power Flow Raw Data

Additional Information
PSS®E Program Operation Manual: Section 5.49, Creating a Power Flow Raw Data File Section 5.2.1, Power Flow Raw Data File Contents *PSS®E Application Program Interface (API)*, Section 1.254, RAWD_2

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Chapter 11

Power Flow Solutions

The Power Flow menu provides access to most of the PSS®Express steady-state analyses. Included are the power flow solutions for both ac and dc network analysis, data access and listing, network, dispatch, load and topology manipulations. Methods are available for checking network conditions and exporting results. Analysis methods provided in the power flow menu include the following:

- Decoupled Newton-Raphson Power Flow Solution (Section 11.2.1)
- Fully-Coupled Newton-Raphson Power Flow Solution (Section 11.2.2)
- Fixed Slope Decoupled Newton-Raphson Power Flow Solution (Section 11.2.3)
- Newton-Raphson Power Flow Solution with Inertial Governot Dispatch (Section 11.2.4)
- Gauss-Seidel Power Flow Solution (Section 11.2.5)
- AC Contingency Analysis (Section)

Short-circuit analysis (Chapter 15), the Optimal Power Flow (Chapter 19), transmission access analysis (Chapter 18), and dynamic simulation and disturbances (Chapter 21) methods are available on separate menus.

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11.1 Specifying Solution Parameters

11.1.1 Boundary Conditions



Power Flow > Solution > Parameters…

The constant power characteristic holds the load power constant as long as the bus voltage exceeds the value specified by the modifiable parameter PQBRAK. The **[Solution Parameters]** dialog **General** tab provides access to this parameter. Notification of the change is routed to the **Progress** tab.

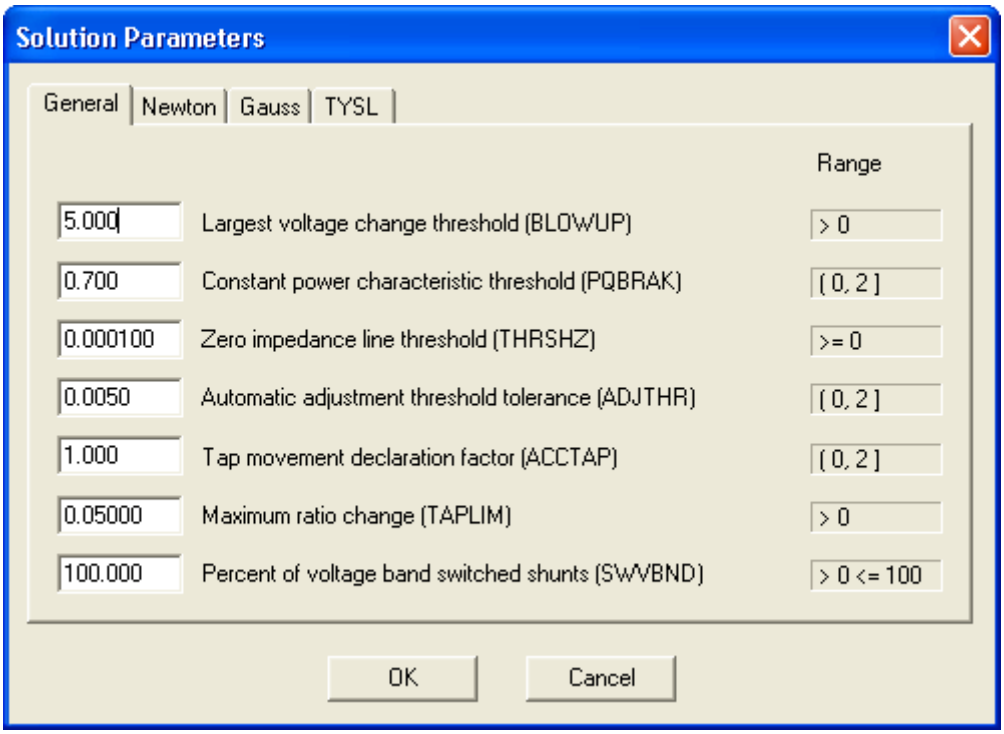


Figure 11-1. Power Flow Solution Parameters Dialog

Additional Information
PSS®E Program Operation Manual, Section 6.3.19, Characteristics of Activity SOLV

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11.1.2 Gauss-Seidel Controls



Power Flow > Solution > Parameters...

On the **Gauss** tab of **[Solution Parameters]**, changes can be made to the acceleration factors, ACCP/ACCQ, and the TOL and ITMX parameters. The BLOWUP parameter is available on the **General** tab.

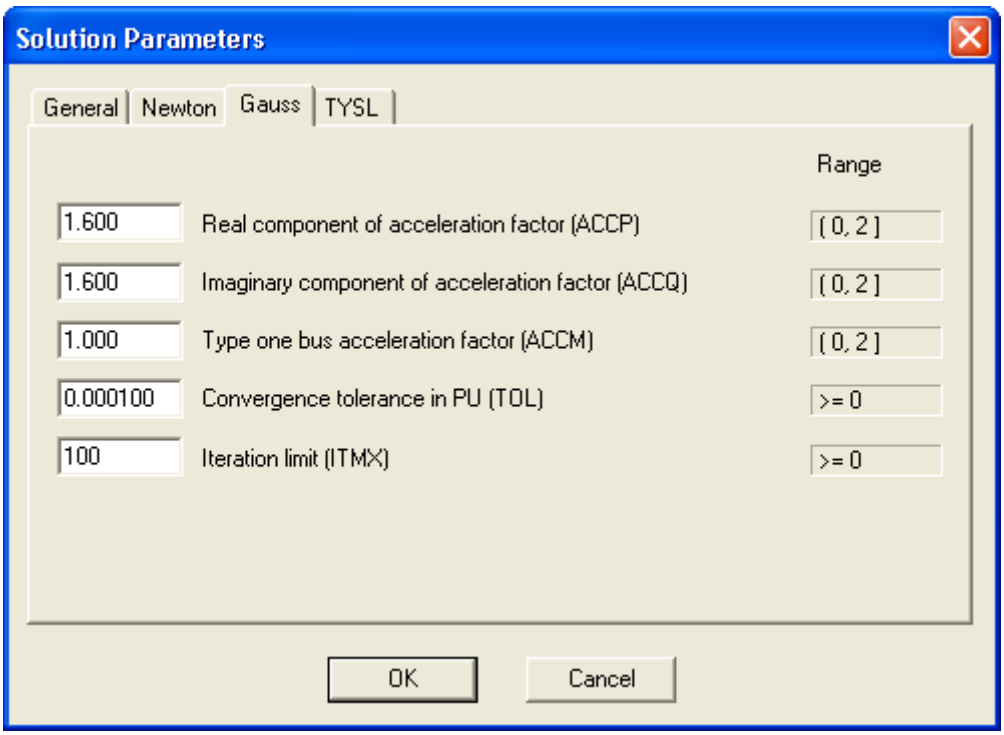
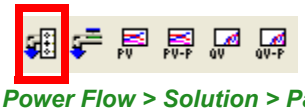


Figure 11-2. Power Flow Solution Parameters Dialog, Gauss

Additional Information
PSS®E Program Operation Manual, Load Data

11.1.3 Newton-Raphson Solution Controls



Power Flow > Solution > Parameters...

On the **Newton** tab of **[Solution Parameters]**, changes can be made to the the following:

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- ACCN - the acceleration factor
- TOL - the largest mismatch in MW and Mvar
- ITMXN - the maximum number of iterations
- DVLIM - the largest change in bus voltage
- NDVFCT - the non-divergent improvement factor
- VCTOLQ - the controlled bus Q mismatch convergence tolerance
- VCTOLV - the controlled bus voltage error convergence tolerance

The BLOWUP parameter is available on the **General** tab.

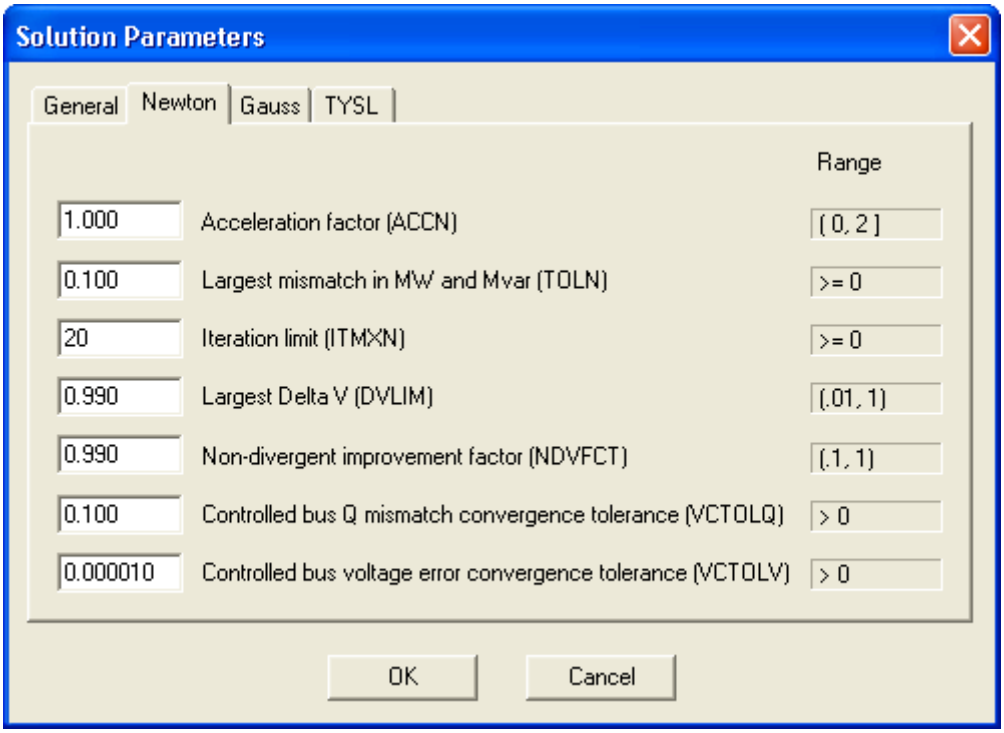


Figure 11-3. Power Flow Solution Parameters Dialog, Newton

Additional Information
PSS®E Program Operation Manual, Section 6.5.1, Characteristics of Activity FNSL

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11.2 Running Power Flow Solutions

11.2.1 Decoupled Newton-Raphson Power Flow Solution

Activity NSOL

Requirements / Prerequisites
Validly specified power flow case with voltages that are a reasonable estimate of a solution.



Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)...

The three Newton-Raphson iterative algorithms solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. The **[Loadflow Solutions]** dialog (Figure 11-4) provides options for all three Newton-Raphson solution methods.



The toolbar button opens **[Loadflow Solutions]** or automatically executes the previously used solution method without opening the dialog. To re-open the dialog for each analysis, enable the option *Show this window when using the Solve toolbar button*.

Newton-Raphson controls are modified using **[Solution Parameters]**. See Section 11.1.3.

Automatic adjustments that allow maneuvering the system to meet specified voltage, branch flow, and area net interchange schedules are enabled or disabled using **[Program Settings]**. See Changing Program Settings. The user must specify *Solution options* for *Tap adjustment*, *Switched shunt adjustments* and *Area interchange control*; these options override automatic adjustments.

Checkboxes to enable/disable *Solution options* for *Flat start*, *Adjust phase shift*, and *Adjust DC taps* are also available. *VAR limits* may also be specified to control the number of iterations during which generator reactive power limits will first be applied.

Click **[Solve]** to process the working case. Solution output is routed to the **Progress** tab (Figure 11-5).

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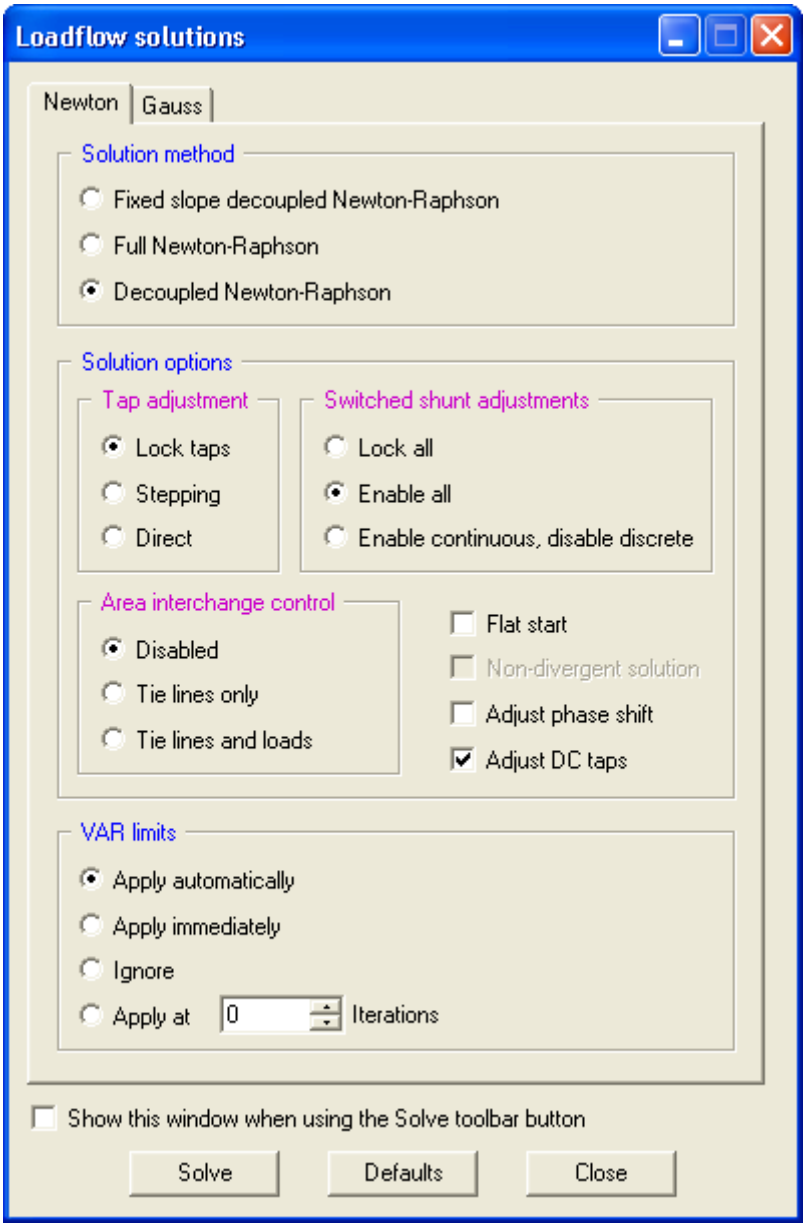


Figure 11-4. Loadflow Solutions Dialog, Newton-Raphson, Decoupled

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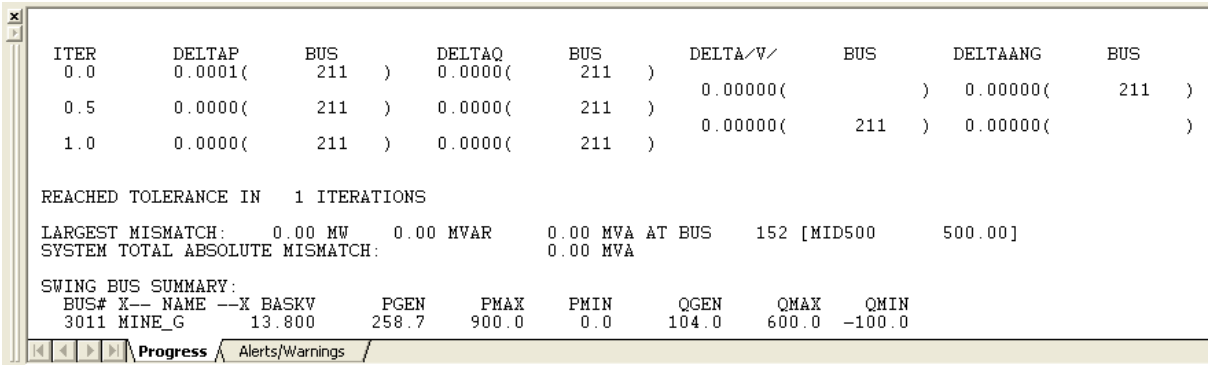


Figure 11-5. Example of Newton-Raphson Decoupled Solution Output

Additional Information
PSS®E Program Operation Manual, Section 6.6, Applying Decoupled Newton-Raphson Power Flow Solution

11.2.2 Fully-Coupled Newton-Raphson Power Flow Solution

Activity FNSL

Requirements / Prerequisites
Validly specified power flow case with voltages that are a reasonable estimate of a solution.



Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)…

The three Newton-Raphson iterative algorithms solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. The **[Loadflow Solutions]** dialog (Figure 11-6) provides options for all three Newton-Raphson solution methods.



The toolbar button opens **[Loadflow Solutions]** or automatically executes the previously used solution method without opening the dialog. To re-open the dialog for each analysis, enable the option *Show this window when using the Solve toolbar button*.

Newton-Raphson controls are modified using **[Solution Parameters]**. See Section 11.1.3.

Automatic adjustments that allow maneuvering the system to meet specified voltage, branch flow, and area net interchange schedules are enabled or disabled using **[Program Settings]**. See Changing Program Settings. The user must specify *Solution options* for *Tap adjustment*, *Switched shunt adjustments* and *Area interchange control*; these options override automatic adjustments.

Activity FNSL requires the prior execution of activity ORDR. If the need for a new bus ordering is detected, activity ORDR is automatically executed before beginning the voltage change calculation.

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Checkboxes to enable/disable *Solution options* for *Flat start*, *Non-divergent solution, Adjust phase shift*, and *Adjust DC taps* are also available. *VAR limits* may also be specified to control the number of iterations during which generator reactive power limits will first be applied.

Click **[Solve]** to process the working case. Solution output is routed to the **Progress** tab (Figure 11-7).

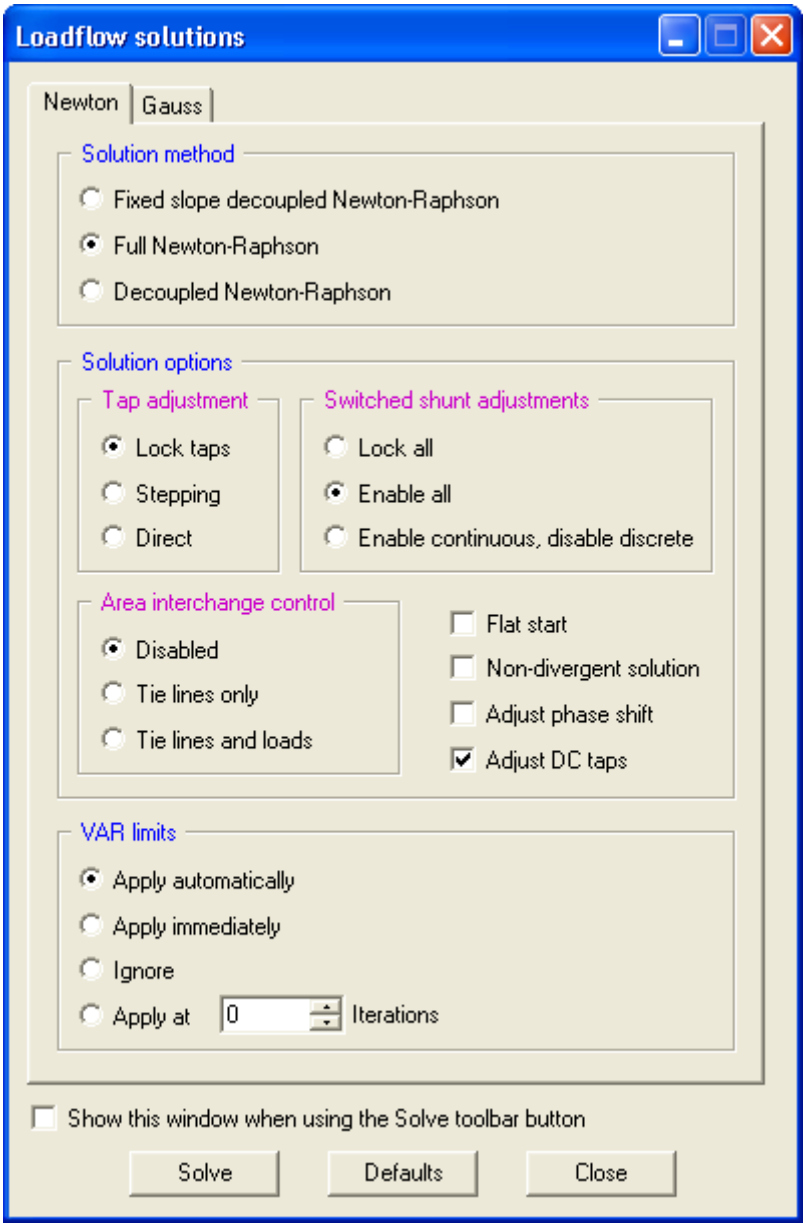


Figure 11-6. Loadflow Solutions Dialog, Newton-Raphson, Fully-Coupled

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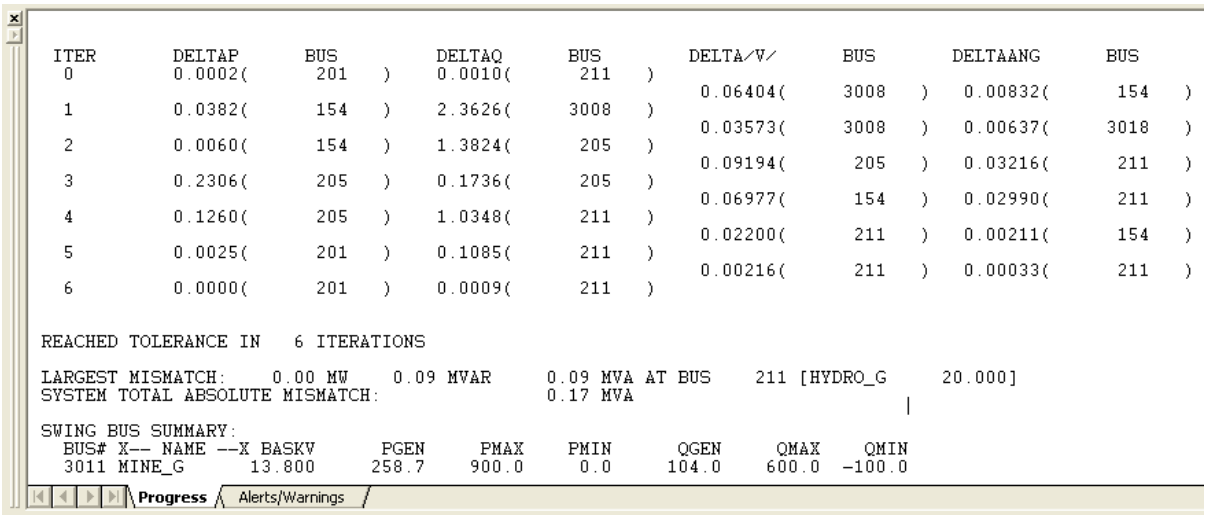


Figure 11-7. Example of Newton-Raphson Fully-Coupled Solution Output

Additional Information
PSS®E Program Operation Manual, Section 6.5, Applying Fully-Coupled Newton-Raphson Power Flow Solution

11.2.3 Fixed Slope Decoupled Newton-Raphson Power Flow Solution

Activity FDNS

Requirements / Prerequisites
Validly specified power flow case with voltages that are a reasonable estimate of a solution.



Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)…

The three Newton-Raphson iterative algorithms solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. The **[Loadflow Solutions]** dialog (Figure 11-8) provides options for all three Newton-Raphson solution methods.

The toolbar button opens **[Loadflow Solutions]** or automatically executes the previously used solution method without opening the dialog. To re-open the dialog for each analysis, enable the option *Show this window when using the Solve toolbar button*.

Newton-Raphson controls are modified using **[Solution Parameters]**. See Section 11.1.3.

Automatic adjustments that allow maneuvering the system to meet specified voltage, branch flow, and area net interchange schedules are enabled or disabled using **[Program Settings]**. See

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Changing Program Settings. The user must specify *Solution options* for *Tap adjustment*, *Switched shunt adjustments* and *Area interchange control*; these options override automatic adjustments.

Checkboxes to enable/disable *Solution options* for *Flat start*, *Non-divergent solution, Adjust phase shift*, and *Adjust DC taps* are also available. *VAR limits* may also be specified to control the number of iterations during which generator reactive power limits will first be applied.

Click **[Solve]** to process the working case. Solution output is routed to the **Progress** tab (Figure 11-9).

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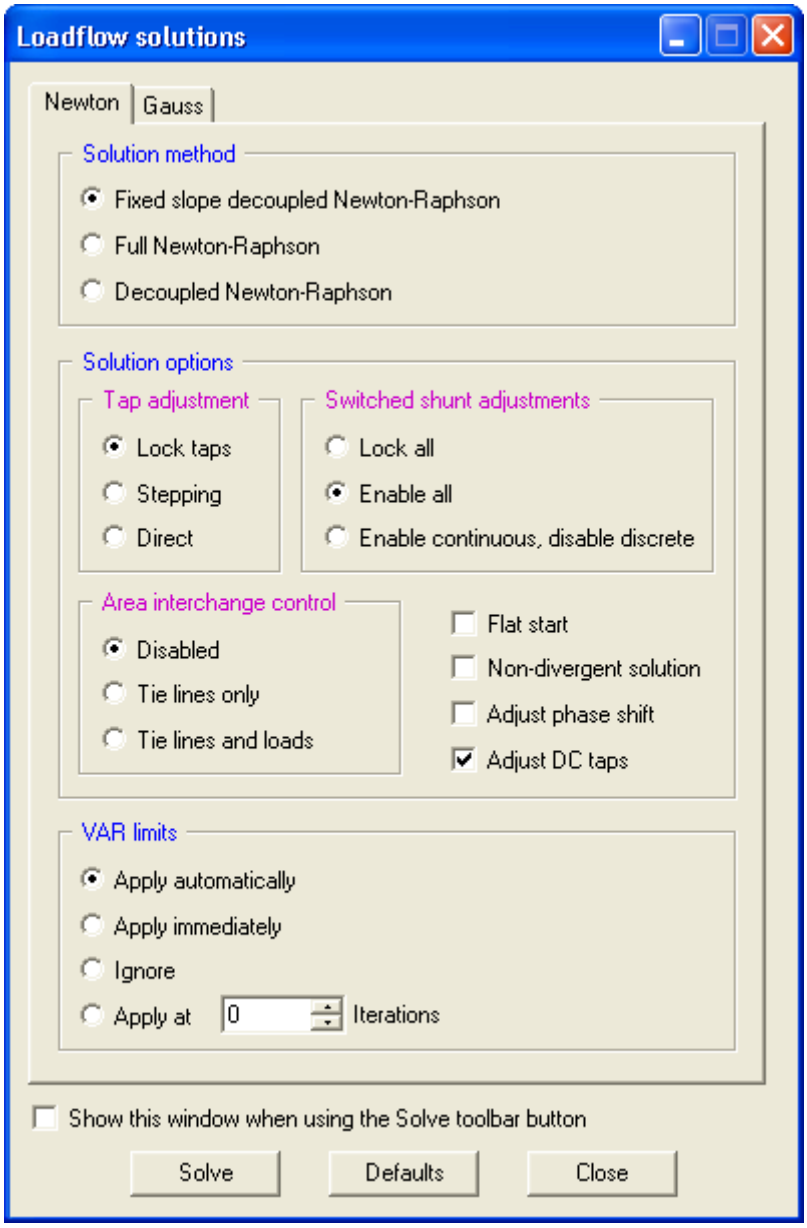


Figure 11-8. Loadflow Solutions Dialog, Newton-Raphson, Fixed Slope Decoupled

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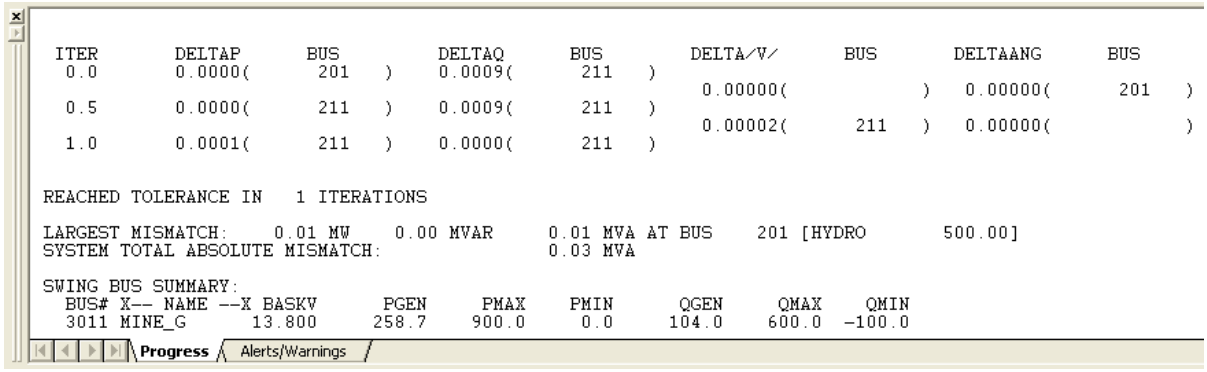


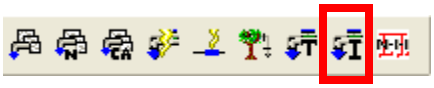
Figure 11-9. Example of Newton-Raphson Fixed Slope Decoupled Solution Output

Additional Information
PSS®E Program Operation Manual, Section 6.7, Applying Fixed Slope Decoupled Newton-Raphson Power Flow Solution

11.2.4 Newton-Raphson Power Flow Solution with Inertial / Governor Dispatch

Activity INLF

Requirements / Prerequisites
Validly specified power flow case, solved in the pre-event condition.
No prior power flow solution activities in working case.
Unit Inertia and Governor Data File (*.inl) containing machine data for all in-service machines.



Power Flow > Solution > N-R solution with inertial / governor dispatch (INLF)…

The **[N-R Solution with Inertial / Governor Redispatch]** dialog requires specification of *Solution options*, *Switched shunt adjustments*, *VAR limits*, and the *Solution type* (*Governor* or *Inertial*). The *Governor* solution requires specification of *Plimits* from among the following options:

- Pre PSS®Express Rev-29 approach
- 0.0 to 1.0 for both types
- Working case values for both types

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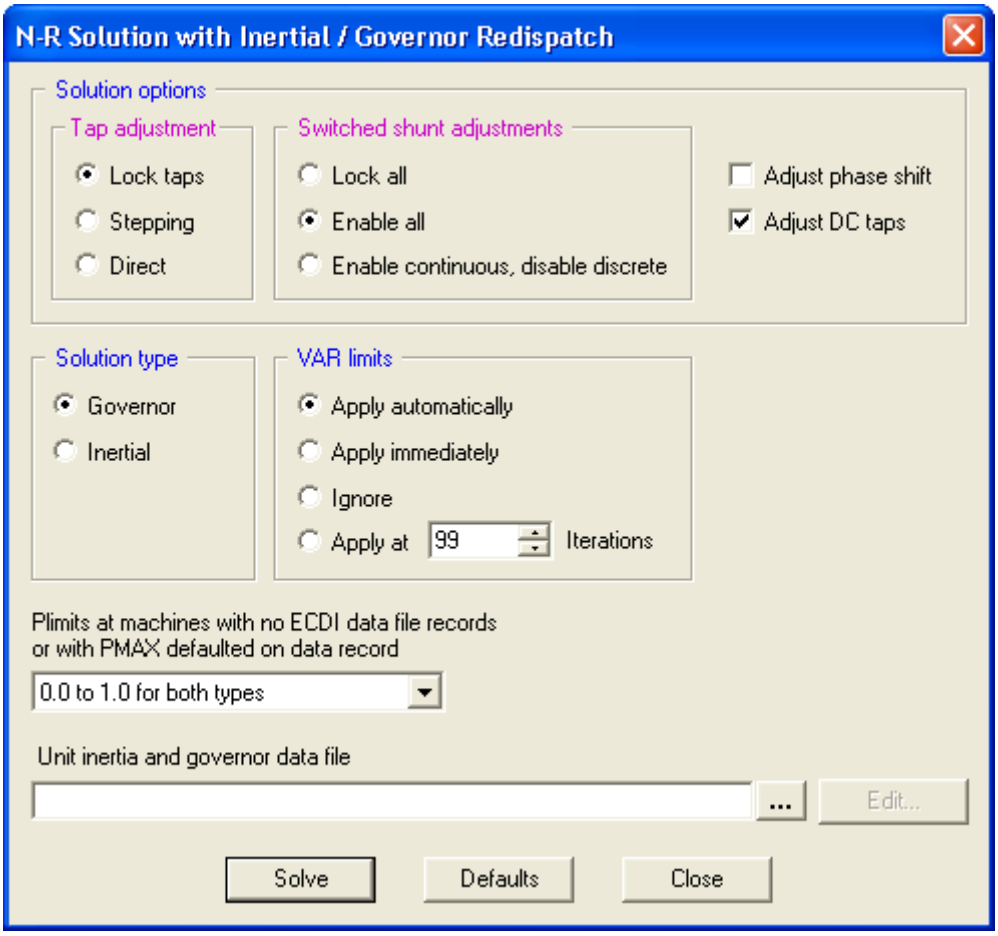


Figure 11-10. N-R Solution with Inertial / Governor Redispatch Dialog

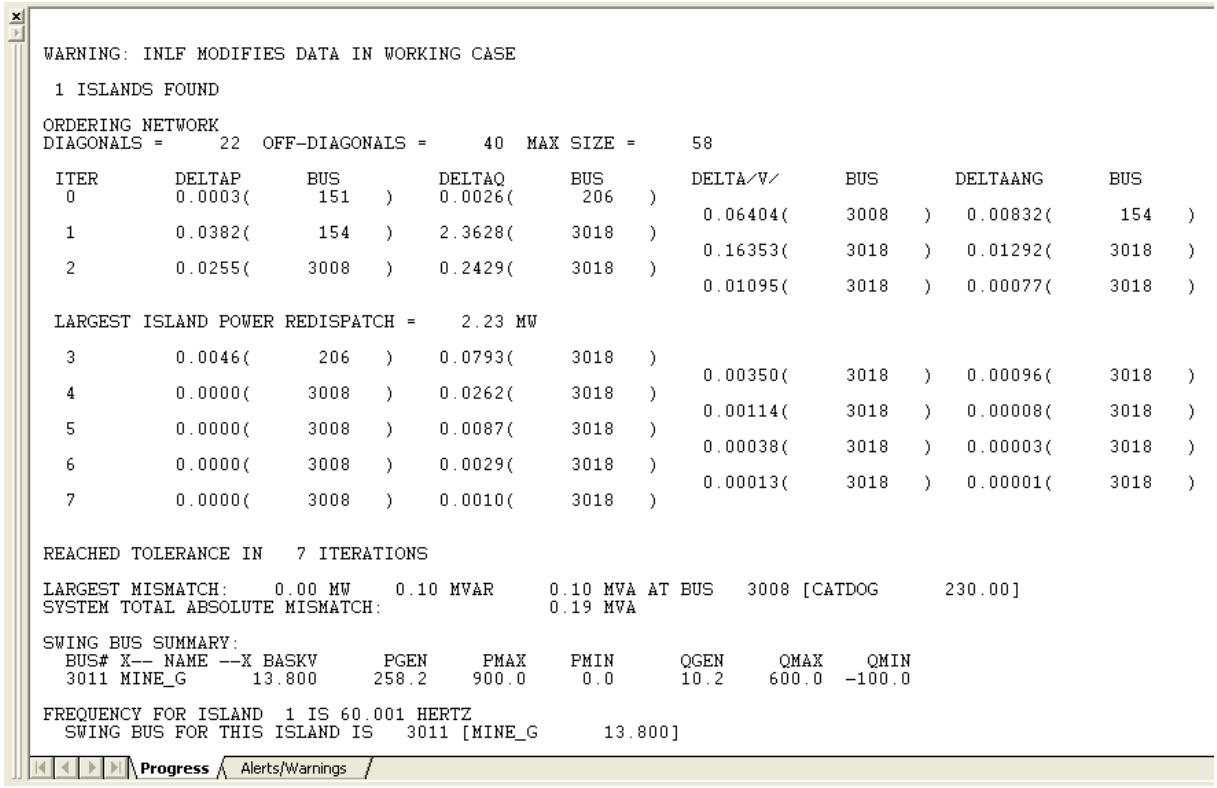
Click **[…]** to open the selection window to save data in a *Unit Inertia and Governor Data file* (*.inl), which can be a new file or a previously-built file to be over-written.

Area interchange control and the non-divergent solution option are always disabled in this solution.

Click **[Solve]** to process the working case. Solution output is routed to the **Progress** tab (Figure 11-9).

This activity modifies the data in the working case. If you wish to protect the original data, you must save the redispatch data with a different filename.

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```
WARNING: INLF MODIFIES DATA IN WORKING CASE

 1 ISLANDS FOUND

ORDERING NETWORK
DIAGONALS =     22  OFF-DIAGONALS =     40  MAX SIZE =     58

 ITER       DELTAP      BUS         DELTAQ      BUS         DELTA/V/      BUS       DELTAANG      BUS
  0         0.0003(      151   )    0.0026(      206   )
                                                            0.06404(      3008   )   0.00832(      154   )
  1         0.0382(      154   )    2.3628(      3018   )
                                                            0.16353(      3018   )   0.01292(      3018   )
  2         0.0255(      3008   )   0.2429(      3018   )
                                                            0.01095(      3018   )   0.00077(      3018   )

 LARGEST ISLAND POWER REDISPATCH =    2.23 MW

  3         0.0046(      206   )    0.0793(      3018   )
                                                            0.00350(      3018   )   0.00096(      3018   )
  4         0.0000(      3008   )   0.0262(      3018   )
                                                            0.00114(      3018   )   0.00008(      3018   )
  5         0.0000(      3008   )   0.0087(      3018   )
                                                            0.00038(      3018   )   0.00003(      3018   )
  6         0.0000(      3008   )   0.0029(      3018   )
                                                            0.00013(      3018   )   0.00001(      3018   )
  7         0.0000(      3008   )   0.0010(      3018   )

REACHED TOLERANCE IN   7 ITERATIONS

LARGEST MISMATCH:    0.00 MW    0.10 MVAR     0.10 MVA AT BUS   3008 [CATDOG      230.00]
SYSTEM TOTAL ABSOLUTE MISMATCH:               0.19 MVA

SWING BUS SUMMARY:
  BUS# X-- NAME --X BASKV      PGEN     PMAX    PMIN      QGEN     QMAX    QMIN
  3011 MINE_G      13.800     258.2    900.0     0.0      10.2    600.0  -100.0

FREQUENCY FOR ISLAND  1 IS 60.001 HERTZ
  SWING BUS FOR THIS ISLAND IS   3011 [MINE_G      13.800]
```

Progress | Alerts/Warnings

Figure 11-11. Example of Newton-Raphson Solution with Governor Redispatch

Additional Information
PSS®E Program Operation Manual, Section 6.8, Applying Newton-Raphson Power Flow Solution with Inertial / Governor Dispatch Section 6.8.1, Inertia and Governor Response Data File Contents

11.2.5 Gauss-Seidel Power Flow Solution

Activity SOLV

Requirements / Prerequisites
Validly specified power flow case with voltages that are a reasonable estimate of a solution. No series capacitors.



Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)…

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The Gauss-Seidel iterative algorithms solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. The **[Loadflow Solutions]** dialog (Figure 11-12) provides options for either the Gauss-Seidel or Modified Gauss-Seidel solutions.


The toolbar button opens **[Loadflow Solutions]** or automatically executes the previously used solution method without opening the dialog. To re-open the dialog for each analysis, enable the option *Show this window when using the Solve toolbar button*.

Gauss-Seidel controls are modified using **[Solution Parameters]**. See Section 11.1.2.

Automatic adjustments that allow maneuvering the system to meet specified voltage, branch flow, and area net interchange schedules are enabled or disabled using **[Program Settings]**. See Changing Program Settings. The user must specify *Solution options* for *Switched shunt adjustments* and *Area interchange control*; these options override automatic adjustments.

Checkboxes to enable/disable *Solution options* for *Flat start*, *Ignore Var limits*, *Adjust DC taps*, and *Adjust taps* are also available.

Click **[Solve]** to process the working case. Solution output is routed to the **Progress** tab (Figure 11-13).

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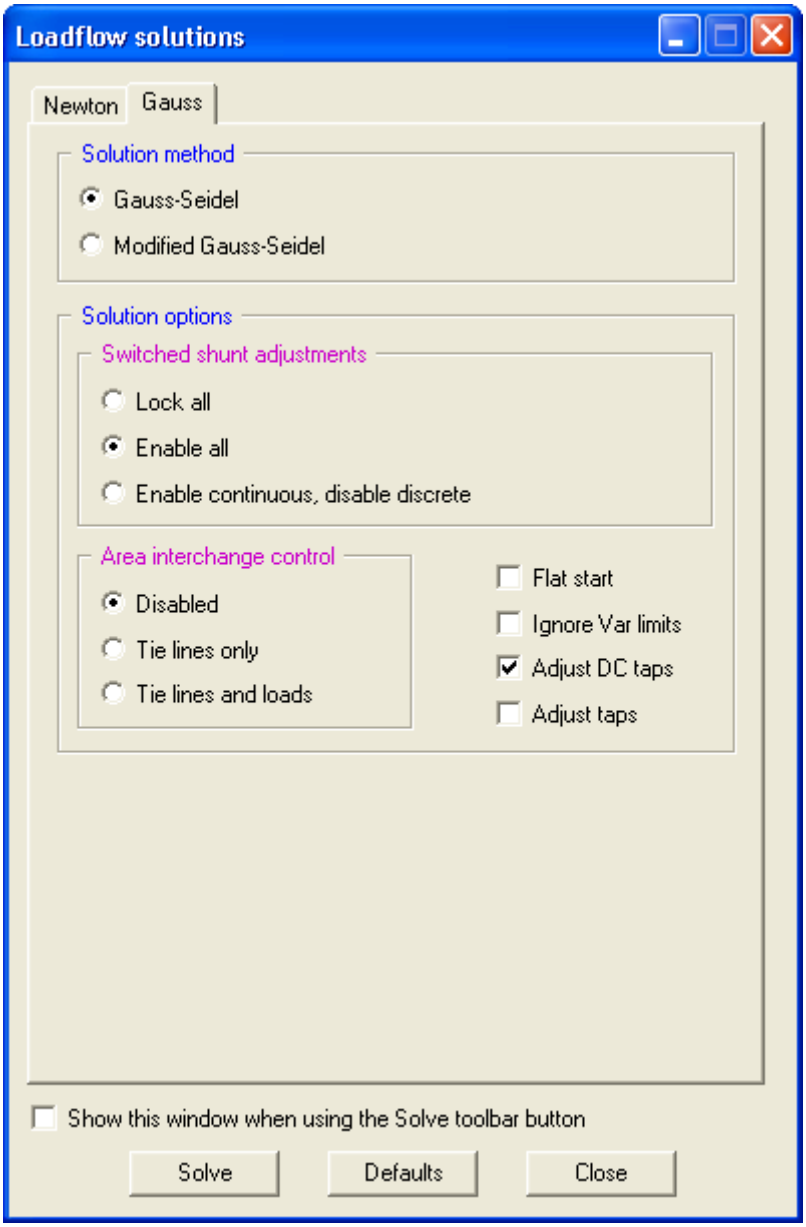


Figure 11-12. Loadflow Solutions Dialog, Gauss-Seidel

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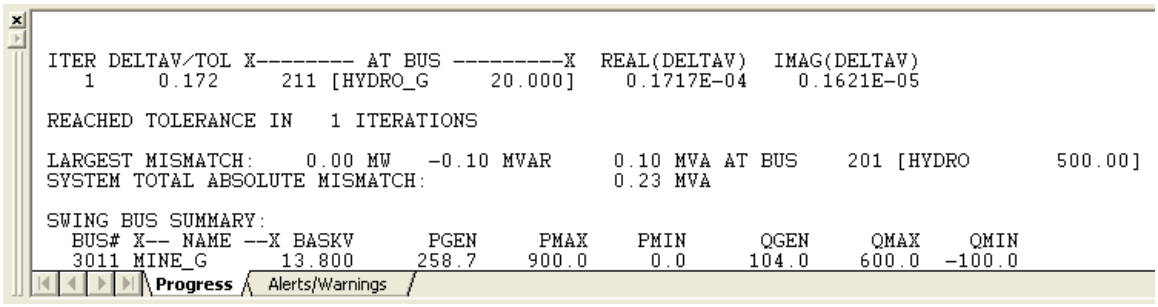


Figure 11-13. Example of Gauss-Seidel Solution Output

Additional Information
PSS®E Program Operation Manual, Section 6.3, Applying Gauss-Seidel Power Flow Solution

11.2.6 Modified Gauss-Seidel Power Flow Solution

Activity MSLV

Requirements / Prerequisites
Validly specified power flow case with voltages that are a reasonable estimate of a solution.



Power Flow > Solution > Solve (NSOL/FNSL/FDNS/SOLV/MSLV)...

The Gauss-Seidel iterative algorithms solve for the bus voltages needed to satisfy the bus boundary conditions contained in the working case. The **[Loadflow Solutions]** dialog (Figure 11-14) provides options for either the Gauss-Seidel or Modified Gauss-Seidel solutions.

The toolbar button opens **[Loadflow Solutions]** or automatically executes the previously used solution method without opening the dialog. To re-open the dialog for each analysis, enable the option *Show this window when using the Solve toolbar button*.

Gauss-Seidel controls are modified using **[Solution Parameters]**. See Section 11.1.2.

Automatic adjustments that allow maneuvering the system to meet specified voltage, branch flow, and area net interchange schedules are enabled or disabled using **[Program Settings]**. See Changing Program Settings. The user must specify *Solution options* for *Switched shunt adjustments* and *Area interchange control*; these options override automatic adjustments.

Checkboxes to enable/disable *Solution options* for *Flat start*, *Ignore Var limits*, *Adjust DC taps*, and *Adjust taps* are also available.

Click **[Solve]** to process the working case. Solution output is routed to the **Progress** tab.

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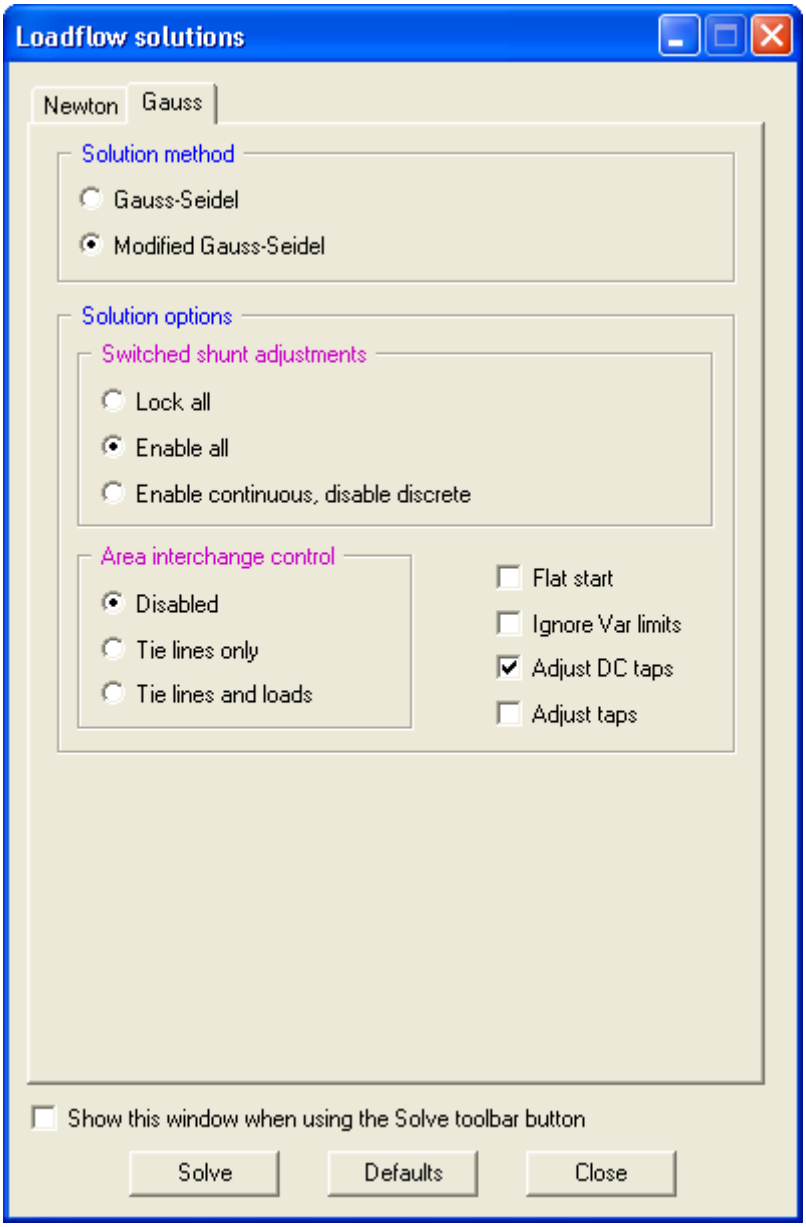


Figure 11-14. Loadflow Solutions Dialog, Modified Gauss-Seidel

Additional Information
PSS®E Program Operation Manual, Section 6.4, Applying Modified Gauss-Seidel Power Flow Solution

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Chapter 12

Power Flow Data Verification

PSS®Express provides a variety of means by which data can be validated in order to identify suspect parameters, conflicting voltage controls, unacceptable tap controls, and isolated buses or inadvertent islands. The data checking is accompanied by the generation of reports at the **Report** device for examination.



Data checking does not validate the state of a power flow solution.

12.1 Checking Branch Parameters

Activity BRCH

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

Power Flow > Check Data > Branch parameters (BRCH)…

The branch parameter checking activity BRCH tabulates those branches where impedances or other characteristics are such that they may be detrimental to the rate of convergence of one or more of the power flow solution activities.

In the **[Check Branch Parameters]** dialog (Figure 12-1), one or more data checks may be selected. Seven of the checks use limit or threshold values. The threshold values shown on the dialog may be used, or you may specify different values that might be more appropriate for the working case under examination. Threshold values are specified either by typing them directly in the data value fields or by using the up and down arrow buttons next to each input field.

The check box for the final check, missing zero sequence impedance, is disabled if sequence data is not present in the working case.

The **[Check Branch Parameters]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Check Branch Parameters]** dialog

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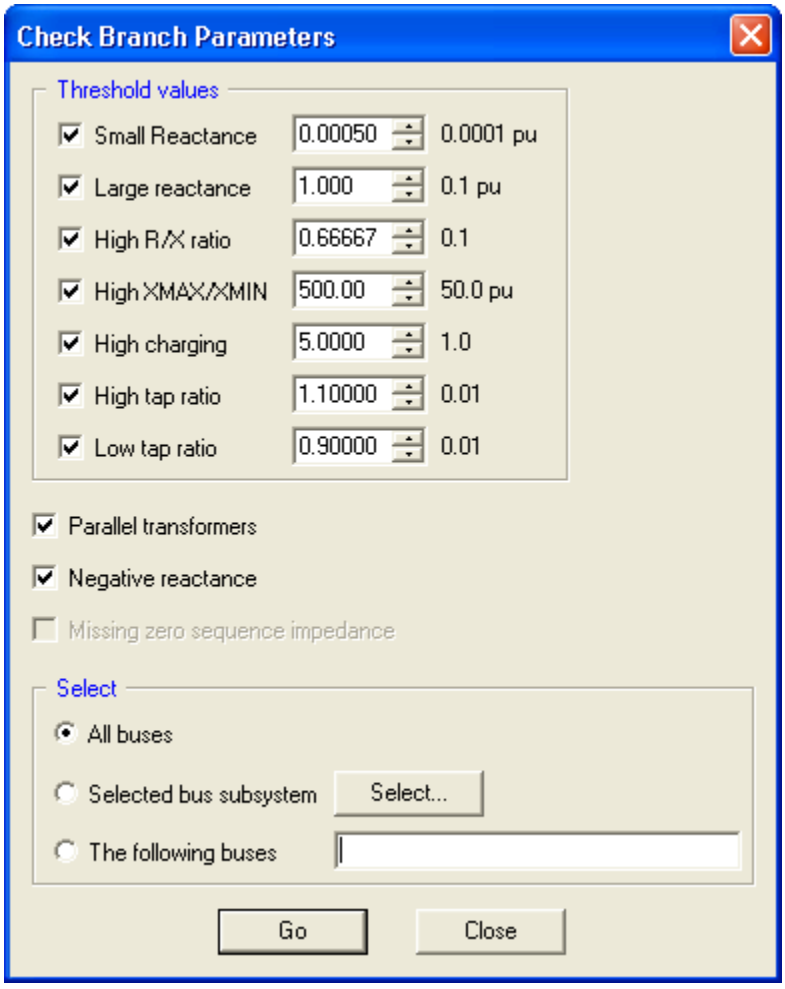


Figure 12-1. Check Branch Parameters Dialog

Additional Information
PSS®E Program Operation Manual, Section 5.27, Checking Branch Parameters *PSS®E Application Program Interface (API)*, Section 1.43, BRCH

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12.2 Checking/Changing Controlled Bus Scheduled Voltage

Activity CNTB

Requirements / Prerequisites
The working case must contain a validly specified power flow case solved to an acceptable mismatch level.

Power Flow > Check Data > Check/Change controlled bus scheduled voltages (CNTB)…

The bus scheduled voltage checking activity CNTB tabulates the voltage setpoints and desired voltage bands of voltage controlling equipment in the working case, and, optionally, allows the user to specify new scheduled voltages. It also performs certain checks on voltage controlling buses that are not themselves voltage controlled buses. It may be instructed to process all such buses, or only those with suspect or conflicting voltage schedules or other errors.

In the **[Check / Change Controlled Bus Scheduled Voltages]** dialog (Figure 12-2), three control options are specified:

- Process either all voltage controlled and controlling buses in the subsystem, or only those with apparent conflicting voltage objective data or other errors:
- Either operate in an interactive mode, in which new voltage schedules may be specified, or in a reporting mode:
- Either include or omit Type 4 controlled buses and out-of-service or disabled controlling equipment in its checking and reporting.

The **[Check / Change Controlled Bus Scheduled Voltages]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Check / Change Controlled Bus Scheduled Voltages]** dialog.

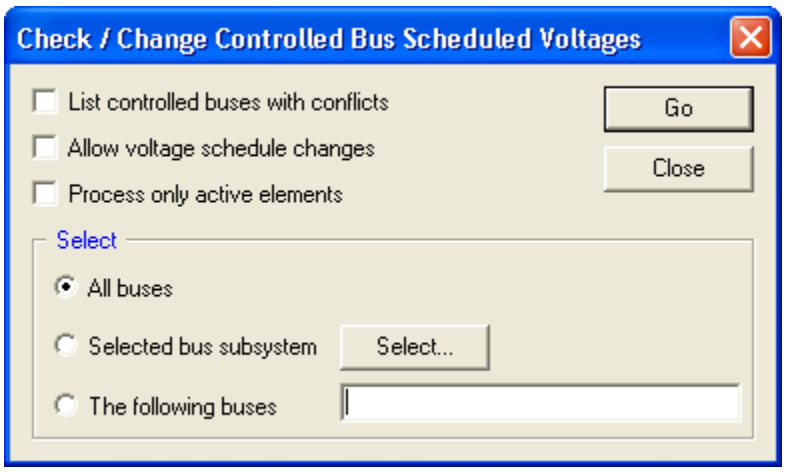


Figure 12-2. Check/Change Controlled Bus Scheduled Voltages Dialog

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If the voltage schedule changing option was specified, as each voltage controlled bus is processed, the user has the option of entering a new scheduled voltage (Figure 12-3).

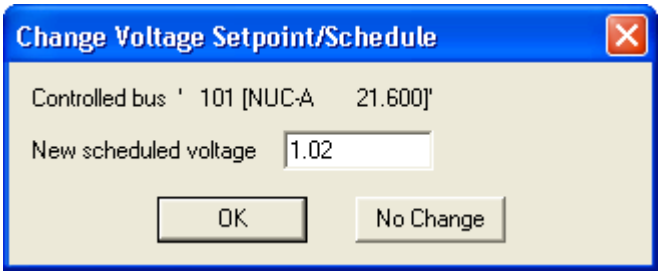


Figure 12-3. Change Voltage Setpoint Dialog

Any voltage controlling transformers with load drop compensation that are controlling voltage at the controlled bus being processed are listed. A new voltage band may be specified for any or all of such transformers (Figure 12-4).

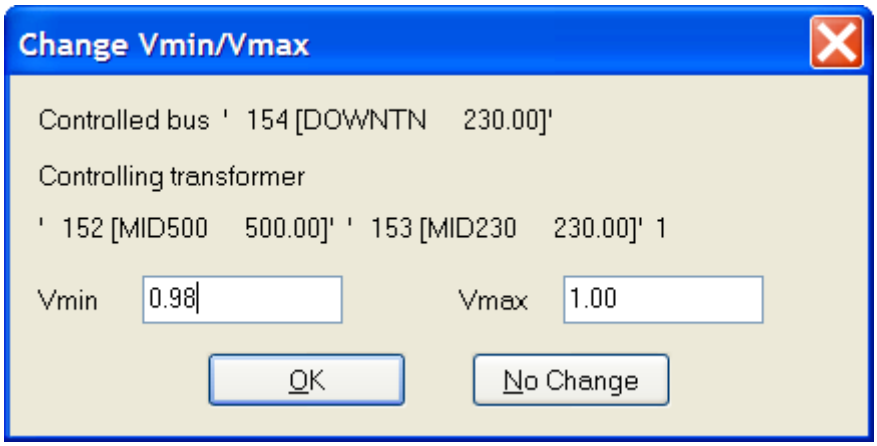


Figure 12-4. Change Vmin/Vmax Dialog

Additional Information
PSS®E Program Operation Manual, Section 5.29, Checking/Changing Controlled Bus Scheduled Voltage *PSS®E Application Program Interface (API)*, Section 1.57, CNTB

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12.3 Checking/Changing Transformer Adjustment Data

Activity TPCH

Requirements / Prerequisites
The working case must contain a validly specified power flow case.

Power Flow > Check Data > Check / Change transformer adjustment data (TPCH)…

The controlling transformer parameter checking activity TPCH performs several checks on the adjustment data associated with voltage and flow controlling transformers.

In the **[Check / Change Transformer Adjustment Data]** dialog (Figure 12-5), data checks are selected through check boxes. Those checks that use a threshold value have a value field next to the test description; you may override any default threshold by typing the desired value in its field.

The **[Check / Change Transformer Adjustment Data]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Check / Change Transformer Adjustment Data]** dialog.

If any transformers are listed as failing a check, the user has the option of modifying the data of all transformers tabulated (Figure 12-6).

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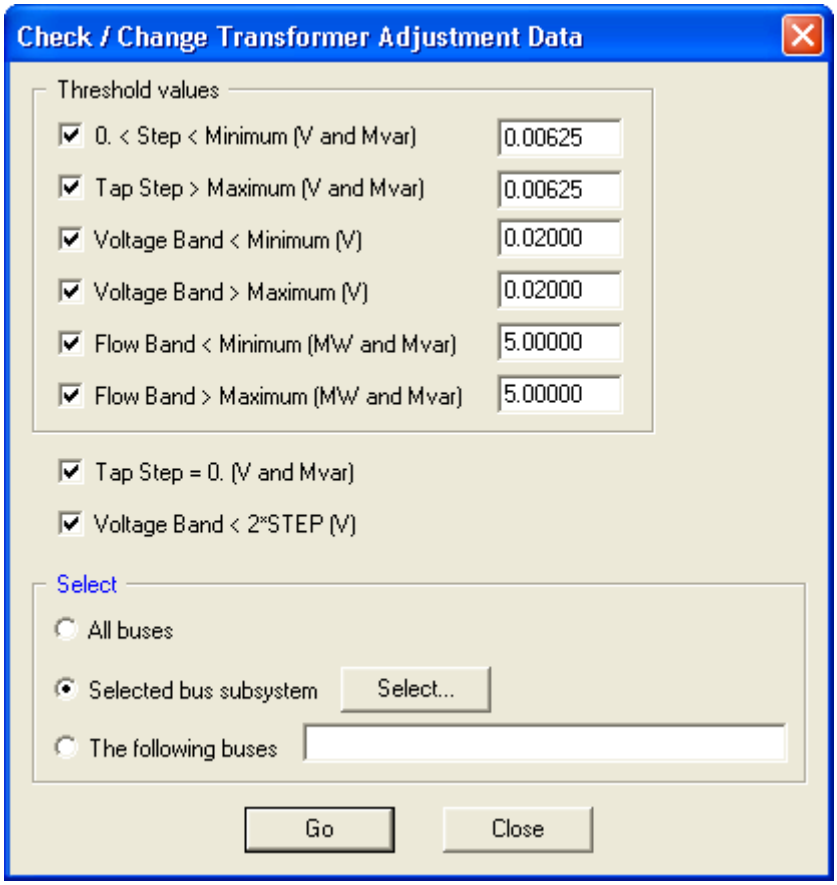


Figure 12-5. Check/Change Transformer Adjustment Data Dialog

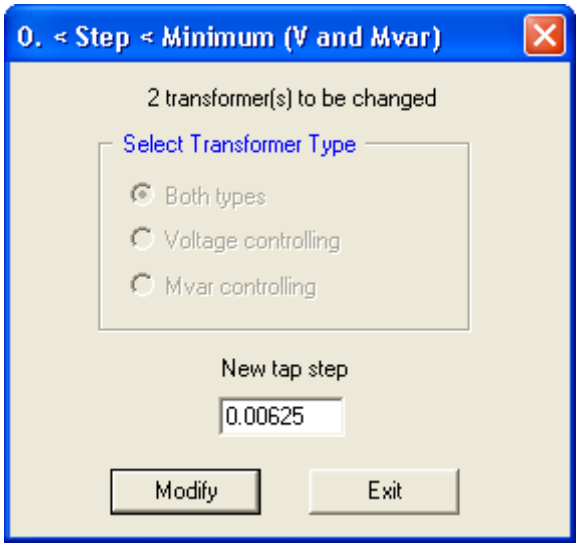


Figure 12-6. Change Transformer Tap Step Dialog

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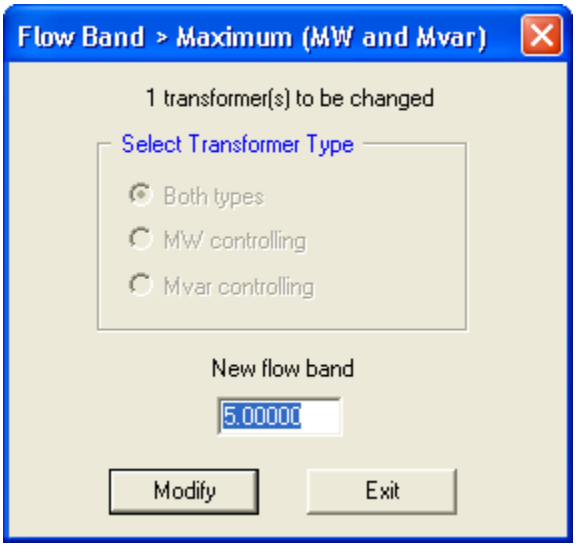


Figure 12-7. Change Transformer Flow Band Dialog

Additional Information
PSS®E Program Operation Manual, Section 5.30, Checking/Changing Transformer Adjustment Data *PSS®E Application Program Interface (API)*, Section 1.322, TPCH

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12.4 Check for Islands Without a Swing Bus

Activity TREE

Requirements / Prerequisites
The working case must contain a validly specified power flow case.



Power Flow > Check Data > Buses not in swing tree (TREE)

The network continuity checking activity TREE enables the user to identify buses not connected back to a Type 3 (swing) bus through the in-service ac network. It also tabulates in-service branches connected to Type 4 (disconnected) buses. Each swingless island may optionally be disconnected.

For each swingless island is detected, a dialog is brought up which allows the user to select the next action of activity TREE:

- Click **[Yes]** to disconnect this island, and then check for another swingless island.
- Click **[No]** to leave this island unchanged and check for another swingless island.
- Click **[Cancel]** to leave this island unchanged and exit activity TREE.



Figure 12-8. Example Dialog of Activity TREE

Additional Information
PSS®E Program Operation Manual, Section 5.28, Check for Islands Without a Swing Bus *PSS®E Application Program Interface (API)*, Section 1.326, TREE

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Chapter 13

Power Flow Reports

13.1 Area/Zone-Based Reports

13.1.1 Area-to-Area Interchange

Activity INTA

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / zone based reports…

The **[Area/Zone Based Reports]** dialog (Figure 13-1) provides options for interchange reports and tie line loading reports, of all subsystems or selected subsystems, by area or zone.

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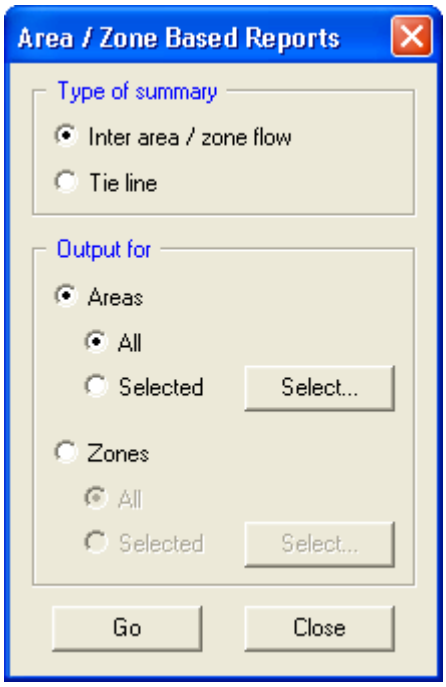


Figure 13-1. Area/Zone Based Reports Dialog, Area

Click **[Go]** to produce the report (Figure 13-2), which is displayed in the **Report** tab by default.

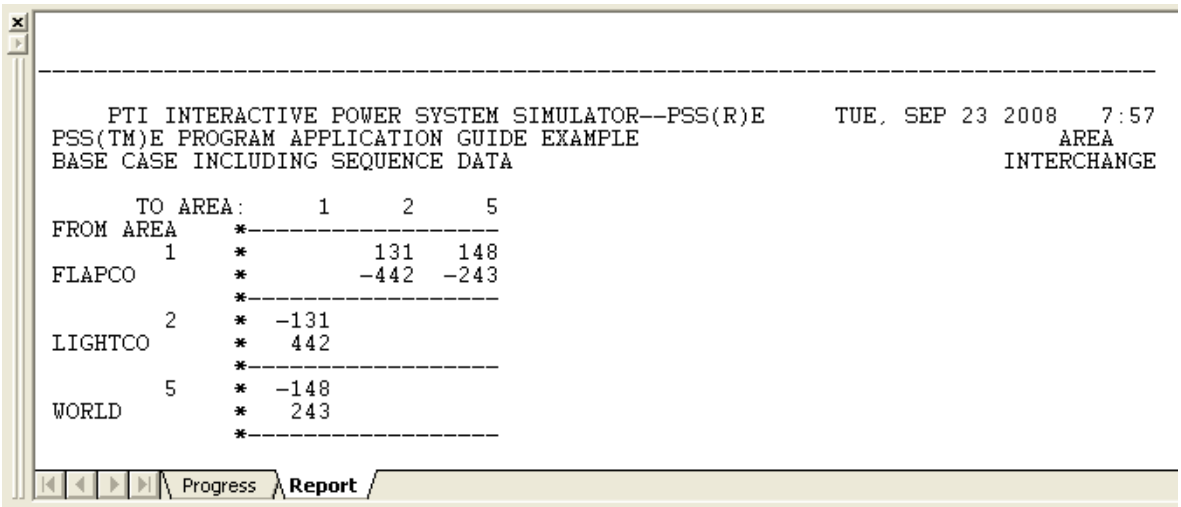


Figure 13-2. Example of Area Interchange Report

Additional Information
PSS®E Program Operation Manual, Section 11.11, Summarizing Area-to-Area Interchange

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13.1.2 Zone-to-Zone Interchange

Activity INTZ

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / zone based reports…

The **[Area/Zone Based Reports]** dialog (Figure 13-3) provides options for interchange reports and tie line loading reports, of all subsystems or selected subsystems, by area or zone.

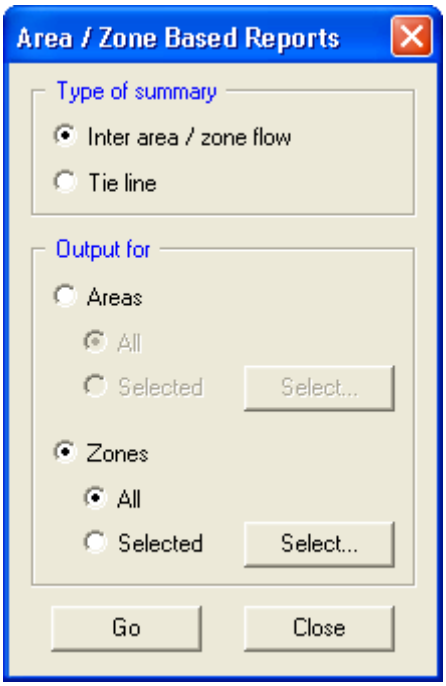


Figure 13-3. Area/Zone Based Reports Dialog, Zone

Click **[Go]** to produce the report (Figure 13-4), which is displayed in the **Report** tab by default.

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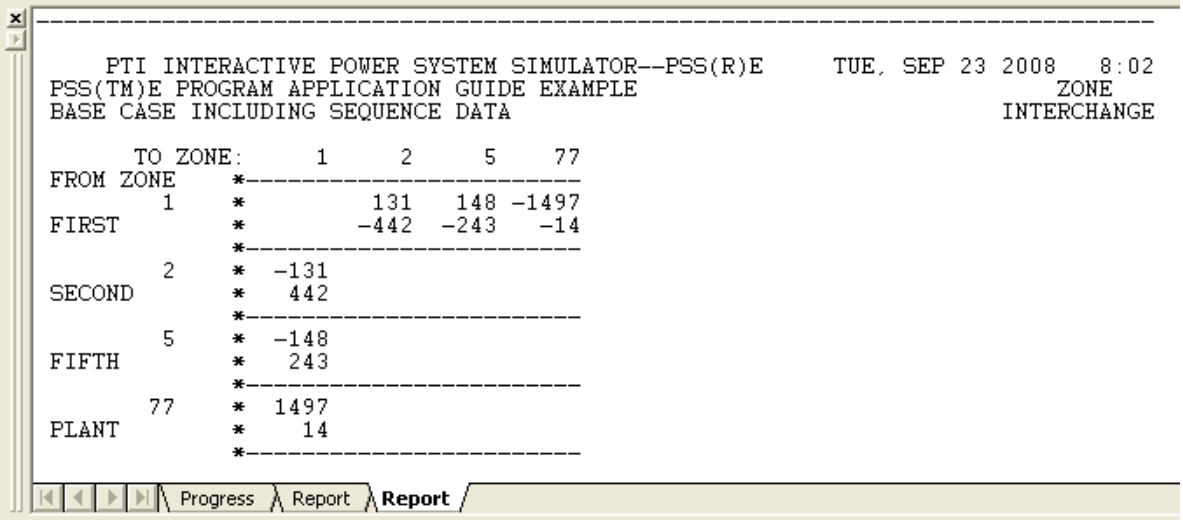


Figure 13-4. Example of Zone Interchange Report

Additional Information
PSS®E Program Operation Manual, Section 11.12, Summarizing Zone-to-Zone Interchange

13.1.3 Loadings on Tie Lines from Interchange Areas

Activity TIES

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / zone based reports…

The **[Area/Zone Based Reports]** dialog (Figure 13-5) provides options for interchange reports and tie line loading reports, of all subsystems or selected subsystems, by area or zone.

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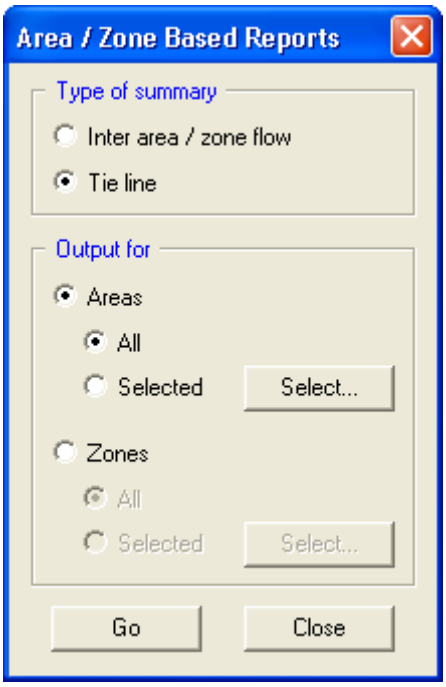
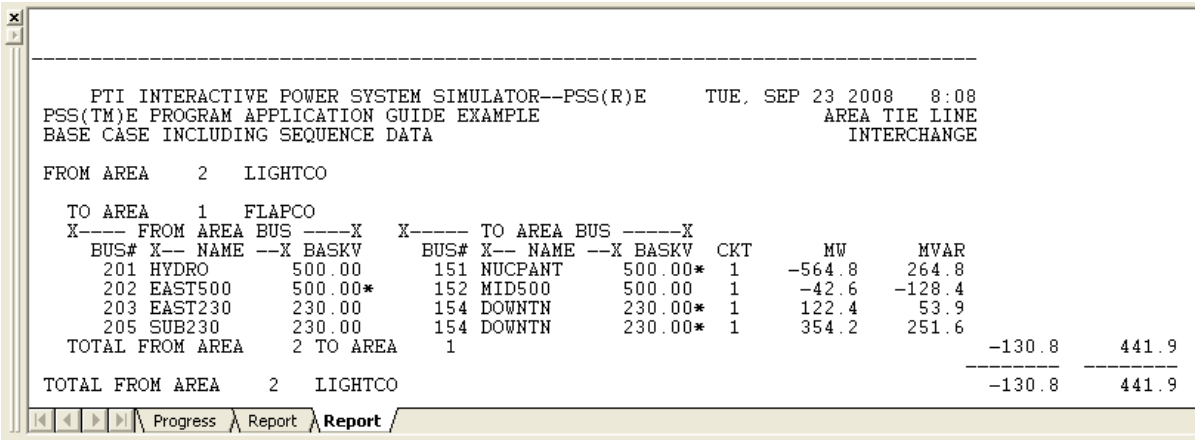


Figure 13-5. Area/Zone Based Reports Dialog, Area Tie Line Loadings

Click **[Go]** to produce the report (Figure 13-9), which is displayed in the **Report** tab by default.



```
    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   8:08
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                      AREA TIE LINE
BASE CASE INCLUDING SEQUENCE DATA                                 INTERCHANGE

FROM AREA     2    LIGHTCO

  TO AREA     1    FLAPCO
  X---- FROM AREA BUS ----X    X----- TO AREA BUS -----X
    BUS# X-- NAME --X BASKV      BUS# X-- NAME --X BASKV  CKT      MW       MVAR
     201 HYDRO        500.00       151 NUCPANT      500.00*  1    -564.8     264.8
     202 EAST500      500.00*      152 MID500       500.00   1     -42.6    -128.4
     203 EAST230      230.00       154 DOWNTN       230.00*  1     122.4      53.9
     205 SUB230       230.00       154 DOWNTN       230.00*  1     354.2     251.6
  TOTAL FROM AREA     2 TO AREA     1                                       -130.8      441.9
                                                                          --------    --------
TOTAL FROM AREA     2    LIGHTCO                                           -130.8      441.9
```

Progress | Report | Report

Figure 13-6. Example of Area Tie Line Loadings Report

Additional Information
PSS®E Program Operation Manual, Section 11.13, Summarizing Loadings on Ties from Interchange Areas

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13.1.4 Loadings on Tie Lines from Zones

Activity TIEZ

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / zone based reports…

The **[Area/Zone Based Reports]** dialog (Figure 13-5) provides options for interchange reports and tie line loading reports, of all subsystems or selected subsystems, by area or zone.

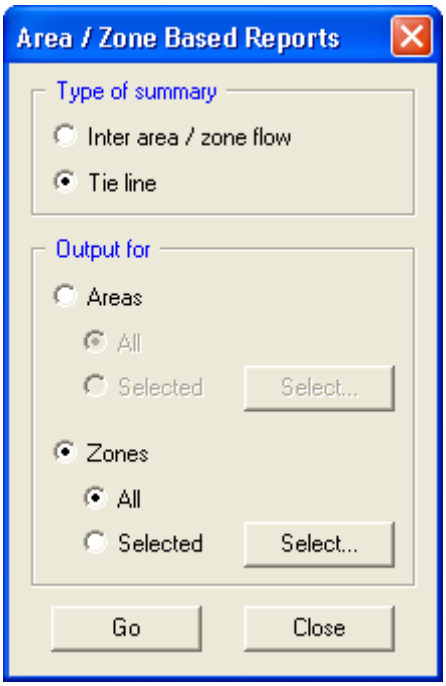
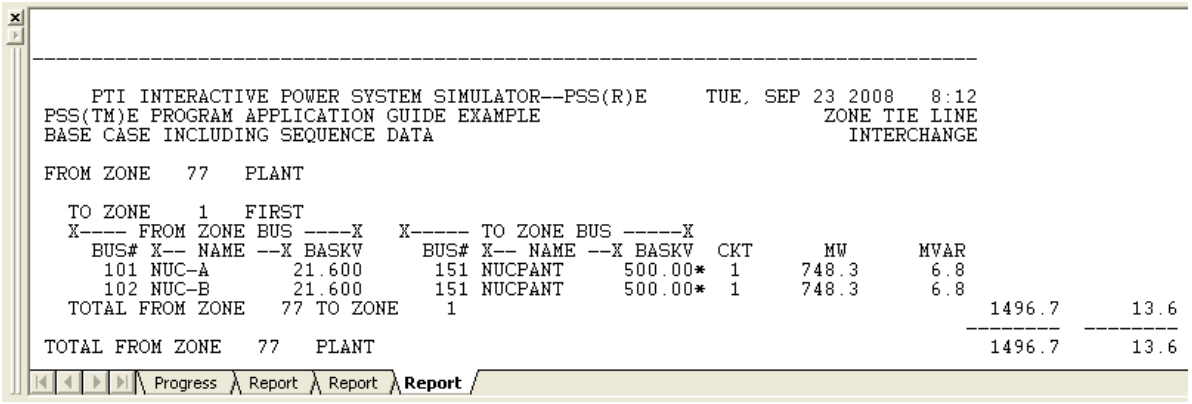


Figure 13-7. Area/Zone Based Reports Dialog, Zone Tie Line Loadings

Click **[Go]** to produce the report (Figure 13-9), which is displayed in the **Report** tab by default.

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```
--------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   8:12
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                      ZONE TIE LINE
BASE CASE INCLUDING SEQUENCE DATA                                 INTERCHANGE

FROM ZONE    77    PLANT

  TO ZONE     1    FIRST
  X---- FROM ZONE BUS ----X   X----- TO ZONE BUS -----X
    BUS# X-- NAME --X BASKV     BUS# X-- NAME --X BASKV  CKT      MW       MVAR
     101 NUC-A       21.600      151 NUCPANT     500.00*  1     748.3      6.8
     102 NUC-B       21.600      151 NUCPANT     500.00*  1     748.3      6.8
  TOTAL FROM ZONE    77 TO ZONE     1                                    1496.7      13.6
                                                                         --------  --------
TOTAL FROM ZONE    77    PLANT                                           1496.7      13.6
```

Progress | Report | Report | **Report**

Figure 13-8. Example of Zone Tie Line Loadings Report

Additional Information
PSS®E Program Operation Manual, Section 11.14, Summarizing Loadings on Ties from Zones

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13.2 Area/Owner/Zone Totals

13.2.1 Area Totals

Activity AREA

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / owner / zone totals…

The **[Area/Owner/Zone Totals]** dialog (Figure 13-9) provides options for subsystem totals, of all subsystems or selected subsystems, by area, owner, or zone. The area report may also include zone subtotals.

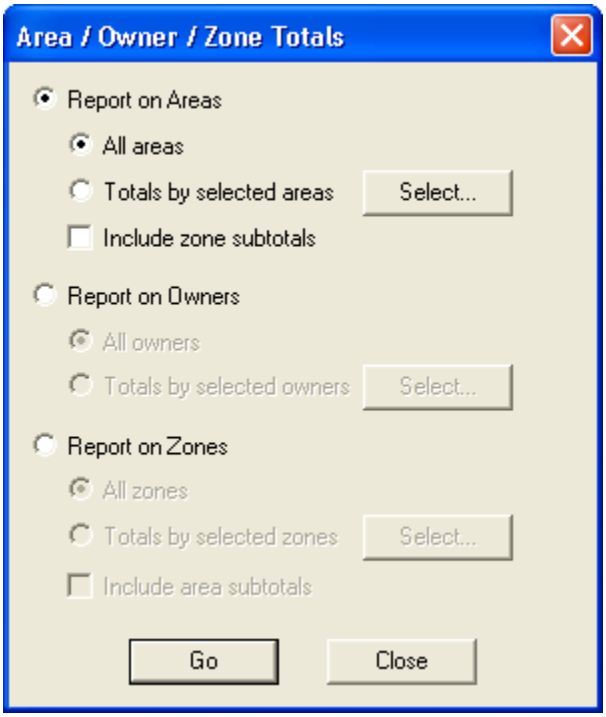
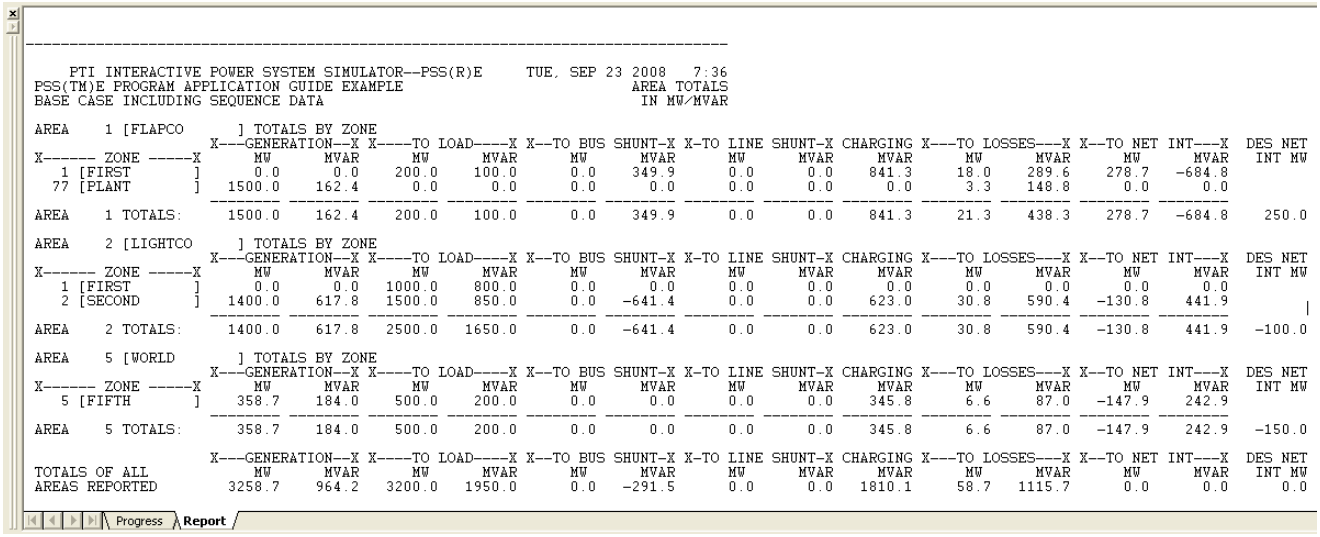


Figure 13-9. Area/Owner/Zone Totals Dialog, Area

Click **[Go]** to produce the report (Figure 13-10), which is displayed in the **Report** tab by default.

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```
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    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   7:36
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                      AREA TOTALS
BASE CASE INCLUDING SEQUENCE DATA                               IN MW/MVAR

AREA    1 [FLAPCO      ] TOTALS BY ZONE
                      X---GENERATION---X X----TO LOAD----X X--TO BUS SHUNT-X X-TO LINE SHUNT-X CHARGING X---TO LOSSES---X X--TO NET INT---X  DES NET
X------ ZONE -----X       MW      MVAR       MW      MVAR       MW      MVAR       MW      MVAR      MVAR       MW      MVAR       MW      MVAR   INT MW
   1 [FIRST       ]       0.0       0.0    200.0    100.0       0.0    349.9       0.0       0.0     841.3     18.0     289.6     278.7   -684.8
  77 [PLANT       ]    1500.0     162.4       0.0       0.0       0.0       0.0       0.0       0.0       0.0       3.3     148.8       0.0       0.0
                      -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- --------
AREA    1 TOTALS:        1500.0     162.4    200.0    100.0       0.0    349.9       0.0       0.0     841.3     21.3     438.3     278.7   -684.8    250.0

AREA    2 [LIGHTCO     ] TOTALS BY ZONE
                      X---GENERATION---X X----TO LOAD----X X--TO BUS SHUNT-X X-TO LINE SHUNT-X CHARGING X---TO LOSSES---X X--TO NET INT---X  DES NET
X------ ZONE -----X       MW      MVAR       MW      MVAR       MW      MVAR       MW      MVAR      MVAR       MW      MVAR       MW      MVAR   INT MW
   1 [FIRST       ]       0.0       0.0   1000.0    800.0       0.0       0.0       0.0       0.0       0.0       0.0       0.0       0.0       0.0
   2 [SECOND      ]    1400.0     617.8    500.0    850.0       0.0   -641.4       0.0       0.0     623.0      30.8     590.4   -130.8     441.9
                      -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- --------
AREA    2 TOTALS:        1400.0     617.8   2500.0   1650.0       0.0   -641.4       0.0       0.0     623.0      30.8     590.4   -130.8     441.9   -100.0

AREA    5 [WORLD       ] TOTALS BY ZONE
                      X---GENERATION---X X----TO LOAD----X X--TO BUS SHUNT-X X-TO LINE SHUNT-X CHARGING X---TO LOSSES---X X--TO NET INT---X  DES NET
X------ ZONE -----X       MW      MVAR       MW      MVAR       MW      MVAR       MW      MVAR      MVAR       MW      MVAR       MW      MVAR   INT MW
   5 [FIFTH       ]     358.7     184.0    500.0    200.0       0.0       0.0       0.0       0.0     345.8       6.6      87.0   -147.9     242.9
                      -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- -------- --------
AREA    5 TOTALS:         358.7     184.0    500.0    200.0       0.0       0.0       0.0       0.0     345.8       6.6      87.0   -147.9     242.9   -150.0

                      X---GENERATION---X X----TO LOAD----X X--TO BUS SHUNT-X X-TO LINE SHUNT-X CHARGING X---TO LOSSES---X X--TO NET INT---X  DES NET
TOTALS OF ALL             MW      MVAR       MW      MVAR       MW      MVAR       MW      MVAR      MVAR       MW      MVAR       MW      MVAR   INT MW
AREAS REPORTED         3258.7     964.2   3200.0   1950.0       0.0   -291.5       0.0       0.0    1810.1      58.7    1115.7       0.0       0.0     0.0
```

Progress | Report

Figure 13-10. Example of Area Totals Report with Zone Subtotals

Additional Information
PSS®E Program Operation Manual, Section 11.7, Summarizing Area Totals

13.2.2 Owner Totals

Activity OWNR

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / owner / zone totals…

The **[Area/Owner/Zone Totals]** dialog (Figure 13-11) provides options for subsystem totals, of all subsystems or selected subsystems, by area, owner, or zone:

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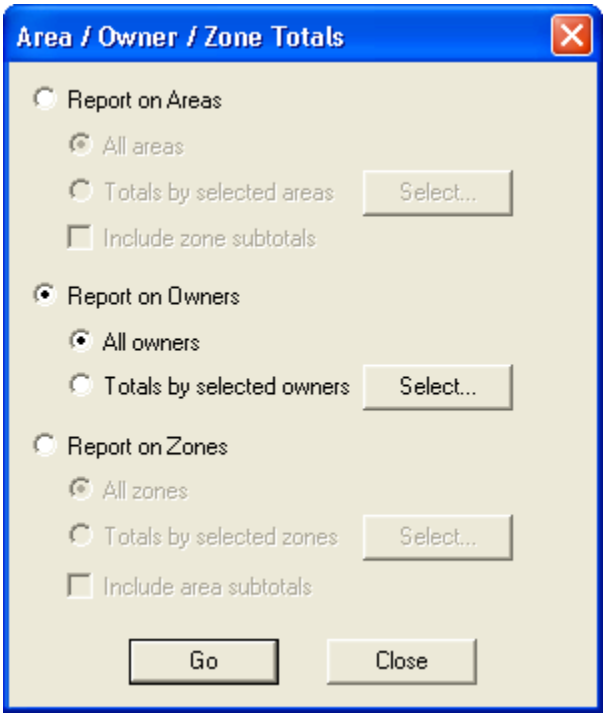
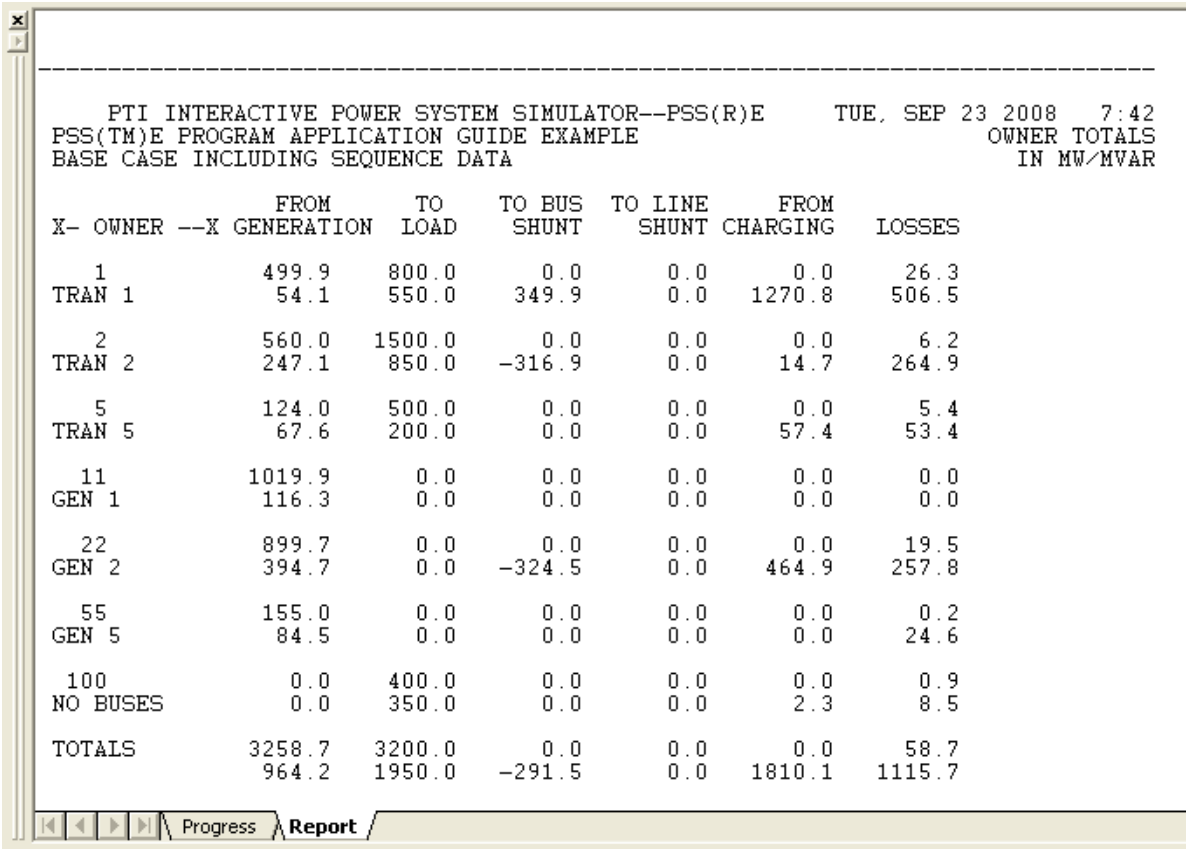


Figure 13-11. Area/Owner/Zone Totals Dialog, Owner

Click **[Go]** to produce the report (Figure 13-12), which is displayed in the **Report** tab by default.

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------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   7:42
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                        OWNER TOTALS
BASE CASE INCLUDING SEQUENCE DATA                                   IN MW/MVAR

                  FROM      TO     TO BUS  TO LINE     FROM
X- OWNER --X GENERATION   LOAD     SHUNT    SHUNT CHARGING    LOSSES

   1            499.9    800.0       0.0      0.0      0.0      26.3
TRAN 1           54.1    550.0     349.9      0.0   1270.8     506.5

   2            560.0   1500.0       0.0      0.0      0.0       6.2
TRAN 2          247.1    850.0    -316.9      0.0     14.7     264.9

   5            124.0    500.0       0.0      0.0      0.0       5.4
TRAN 5           67.6    200.0       0.0      0.0     57.4      53.4

  11           1019.9      0.0       0.0      0.0      0.0       0.0
GEN 1           116.3      0.0       0.0      0.0      0.0       0.0

  22            899.7      0.0       0.0      0.0      0.0      19.5
GEN 2           394.7      0.0    -324.5      0.0    464.9     257.8

  55            155.0      0.0       0.0      0.0      0.0       0.2
GEN 5            84.5      0.0       0.0      0.0      0.0      24.6

 100              0.0    400.0       0.0      0.0      0.0       0.9
NO BUSES          0.0    350.0       0.0      0.0      2.3       8.5

TOTALS         3258.7   3200.0       0.0      0.0      0.0      58.7
                964.2   1950.0    -291.5      0.0   1810.1    1115.7
```

Progress | Report

Figure 13-12. Example of Owner Totals Report

Additional Information
PSS®E Program Operation Manual, Section 11.8, Summarizing Owner Totals

13.2.3 Zone Totals

Activity ZONE

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Area / owner / zone totals…

The **[Area/Owner/Zone Totals]** dialog (Figure 13-11) provides options for subsystem totals, of all subsystems or selected subsystems, by area, owner, or zone. The zone report may also include area subtotals.

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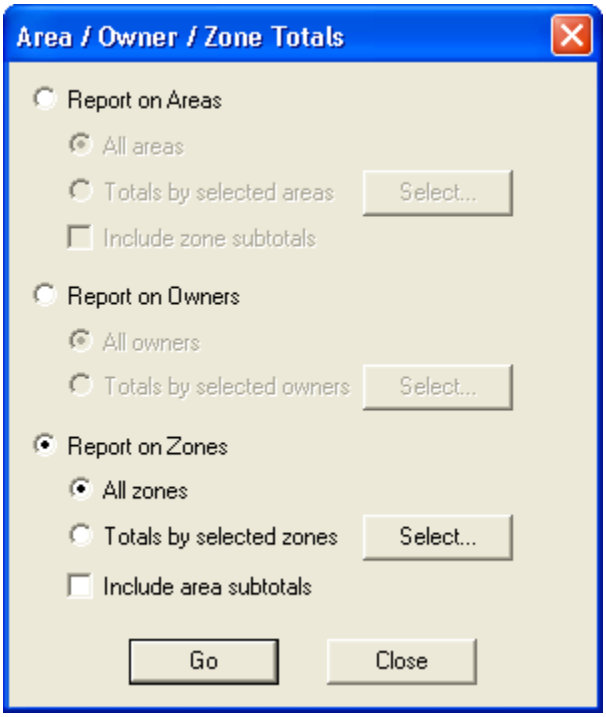
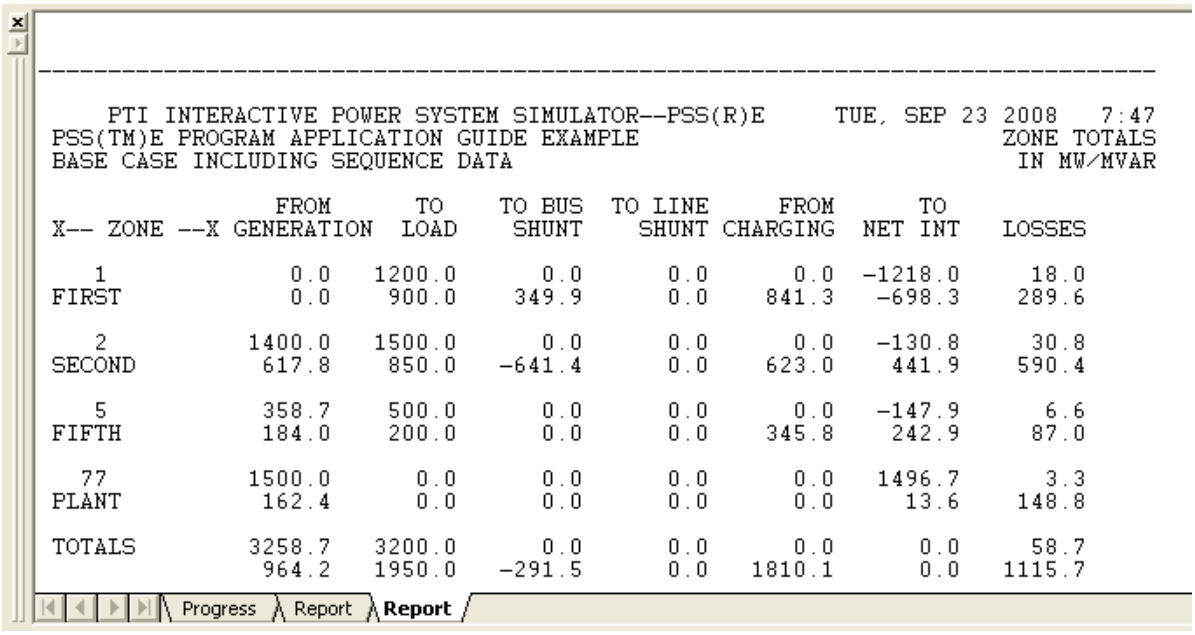


Figure 13-13. Area/Owner/Zone Totals Dialog, Zone

Click **[Go]** to produce the report (Figure 13-12), which is displayed in the **Report** tab by default.



```
------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   7:47
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                          ZONE TOTALS
BASE CASE INCLUDING SEQUENCE DATA                                    IN MW/MVAR

                FROM       TO     TO BUS  TO LINE     FROM       TO
X-- ZONE --X GENERATION   LOAD     SHUNT    SHUNT CHARGING  NET INT   LOSSES

   1             0.0   1200.0       0.0      0.0      0.0  -1218.0     18.0
FIRST            0.0    900.0     349.9      0.0    841.3   -698.3    289.6

   2          1400.0   1500.0       0.0      0.0      0.0   -130.8     30.8
SECOND         617.8    850.0    -641.4      0.0    623.0    441.9    590.4

   5           358.7    500.0       0.0      0.0      0.0   -147.9      6.6
FIFTH          184.0    200.0       0.0      0.0    345.8    242.9     87.0

  77          1500.0      0.0       0.0      0.0      0.0   1496.7      3.3
PLANT          162.4      0.0       0.0      0.0      0.0     13.6    148.8

TOTALS        3258.7   3200.0       0.0      0.0      0.0      0.0     58.7
               964.2   1950.0    -291.5      0.0   1810.1      0.0   1115.7
```

Progress | Report | Report

Figure 13-14. Example of Zone Totals Report

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Additional Information
PSS®E Program Operation Manual, Section 11.9, Summarizing Zone Totals

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13.3 Bus-Based Reports

Through the **[Bus Based Reports]** dialog (Figure 13-15), several power flow solution reports may be obtained.

The **[Bus Based Reports]** dialog enables the user to designate for processing either the entire working case or a specified bus subsystem (refer to Section 4.8, Subsystem Selection of the *PSS®E Program Operation Manual)*. A bus subsystem may be specified either via a **[Bus Subsystem Selector]** dialog by clicking **[Select...]**, or by entering buses directly in the **[Bus Based Reports]** dialog.

Click **[Go]** to produce the report . The report is written to the **Report** device.

13.3.1 Producing a Standard Power Flow Solution Report

Activity POUT

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Bus based reports…

In the **[Bus Based Reports]** dialog (Figure 13-15), select *Powerflow output* with the *Wide format output* check box unchecked. Click **[Go]** to produce the report (Figure 13-16).

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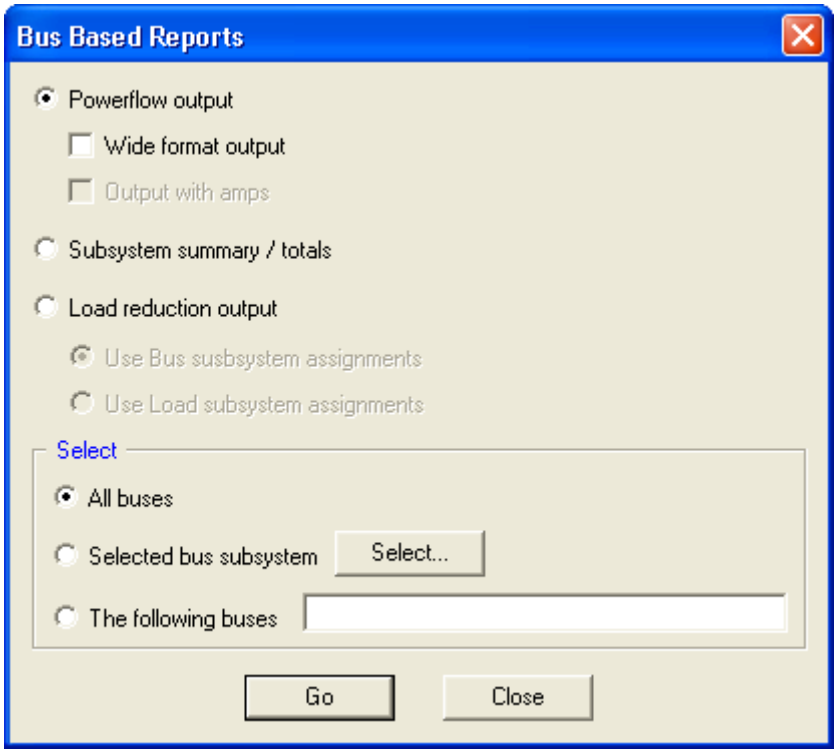
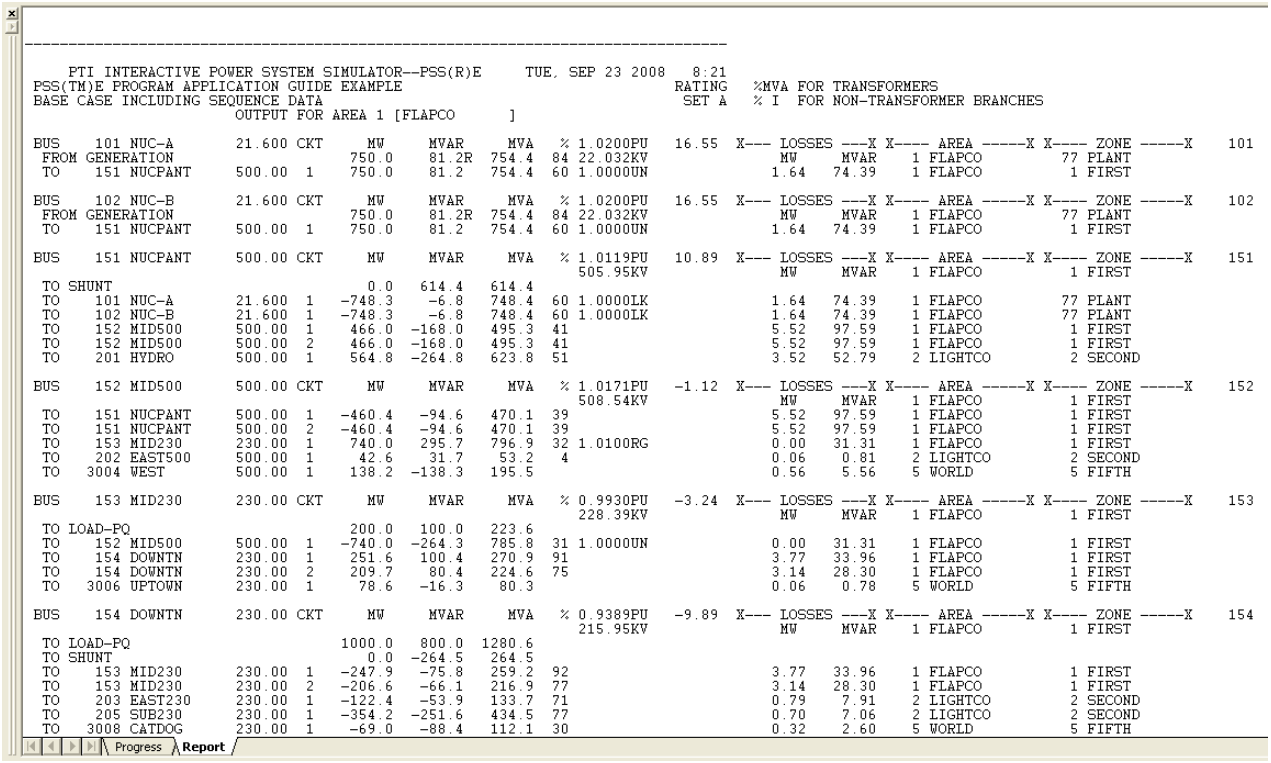


Figure 13-15. Bus Based Reports Dialog, Standard Format

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------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   8:21
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                          RATING   %MVA FOR TRANSFORMERS
BASE CASE INCLUDING SEQUENCE DATA                                    SET A   % I  FOR NON-TRANSFORMER BRANCHES
                  OUTPUT FOR AREA 1 [FLAPCO      ]

BUS    101 NUC-A       21.600 CKT     MW     MVAR     MVA   % 1.0200PU   16.55  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    101
 FROM GENERATION                    750.0    81.2R  754.4  84 22.032KV              MW     MVAR     1 FLAPCO          77 PLANT
 TO    151 NUCPANT     500.00  1    750.0    81.2   754.4  60 1.0000UN            1.64   74.39     1 FLAPCO           1 FIRST

BUS    102 NUC-B       21.600 CKT     MW     MVAR     MVA   % 1.0200PU   16.55  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    102
 FROM GENERATION                    750.0    81.2R  754.4  84 22.032KV              MW     MVAR     1 FLAPCO          77 PLANT
 TO    151 NUCPANT     500.00  1    750.0    81.2   754.4  60 1.0000UN            1.64   74.39     1 FLAPCO           1 FIRST

BUS    151 NUCPANT     500.00 CKT     MW     MVAR     MVA   % 1.0119PU   10.89  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    151
                                                              505.95KV              MW     MVAR     1 FLAPCO           1 FIRST
 TO SHUNT                             0.0    614.4   614.4
 TO    101 NUC-A       21.600  1   -748.3     -6.8   748.4  60 1.0000LK            1.64   74.39     1 FLAPCO          77 PLANT
 TO    102 NUC-B       21.600  1   -748.3     -6.8   748.4  60 1.0000LK            1.64   74.39     1 FLAPCO          77 PLANT
 TO    152 MID500      500.00  1    466.0   -168.0   495.3  41                     5.52   97.59     1 FLAPCO           1 FIRST
 TO    152 MID500      500.00  2    466.0   -168.0   495.3  41                     5.52   97.59     1 FLAPCO           1 FIRST
 TO    201 HYDRO       500.00  1    564.8   -264.8   623.8  51                     3.52   52.79     2 LIGHTCO          2 SECOND

BUS    152 MID500      500.00 CKT     MW     MVAR     MVA   % 1.0171PU   -1.12  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    152
                                                              508.54KV              MW     MVAR     1 FLAPCO           1 FIRST
 TO    151 NUCPANT     500.00  1   -460.4    -94.6   470.1  39                     5.52   97.59     1 FLAPCO           1 FIRST
 TO    151 NUCPANT     500.00  2   -460.4    -94.6   470.1  39                     5.52   97.59     1 FLAPCO           1 FIRST
 TO    153 MID230      230.00  1    740.0    295.7   796.9  32 1.0100RG            0.00   31.31     1 FLAPCO           1 FIRST
 TO    202 EAST500     500.00  1     42.6     31.7    53.2   4                     0.06    0.81     2 LIGHTCO          2 SECOND
 TO   3004 WEST        500.00  1    138.2   -138.3   195.5                         0.56    5.56     5 WORLD            5 FIFTH

BUS    153 MID230      230.00 CKT     MW     MVAR     MVA   % 0.9930PU   -3.24  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    153
                                                              228.39KV              MW     MVAR     1 FLAPCO           1 FIRST
 TO LOAD-PQ                         200.0    100.0   223.6
 TO    152 MID500      500.00  1   -740.0   -264.3   785.8  31 1.0000UN            0.00   31.31     1 FLAPCO           1 FIRST
 TO    154 DOWNTN      230.00  1    251.6    100.4   270.9  91                     3.77   33.96     1 FLAPCO           1 FIRST
 TO    154 DOWNTN      230.00  2    209.7     80.4   224.6  75                     3.14   28.30     1 FLAPCO           1 FIRST
 TO   3006 UPTOWN      230.00  1     78.6    -16.3    80.3                         0.06    0.78     5 WORLD            5 FIFTH

BUS    154 DOWNTN      230.00 CKT     MW     MVAR     MVA   % 0.9389PU   -9.89  X--- LOSSES ---X X---- AREA -----X X---- ZONE -----X    154
                                                              215.95KV              MW     MVAR     1 FLAPCO           1 FIRST
 TO LOAD-PQ                        1000.0    800.0  1280.6
 TO SHUNT                             0.0   -264.5   264.5
 TO    153 MID230      230.00  1   -247.9    -75.8   259.2  92                     3.77   33.96     1 FLAPCO           1 FIRST
 TO    153 MID230      230.00  2   -206.6    -66.1   216.9  77                     3.14   28.30     1 FLAPCO           1 FIRST
 TO    203 EAST230     230.00  1   -122.4    -53.9   133.7  71                     0.79    7.91     2 LIGHTCO          2 SECOND
 TO    205 SUB230      230.00  1   -354.2   -251.6   434.5  77                     0.70    7.06     2 LIGHTCO          2 SECOND
 TO   3008 CATDOG      230.00  1    -69.0    -88.4   112.1  30                     0.32    2.60     5 WORLD            5 FIFTH

 Progress  Report
```

Figure 13-16. Example of Standard Format Power Flow Solution Report

Additional Information
PSS®E Program Operation Manual, Section 11.2, Producing a Standard Power Flow Solution Report

13.3.2 Wide-Format Power Flow Solution Report

Activity LOUT

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Bus based reports…

In the **[Bus Based Reports]** dialog (Figure 13-17), select *Powerflow output* with the *Wide format output* check box checked and the *Output with amps* check box unchecked. Click **[Go]** to produce the report (Figure 13-18).

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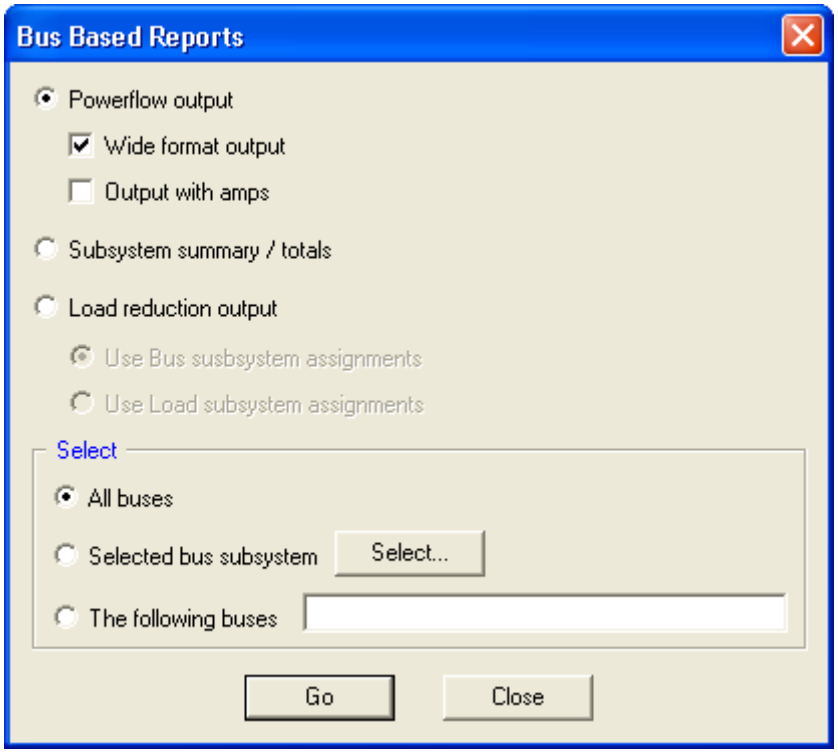
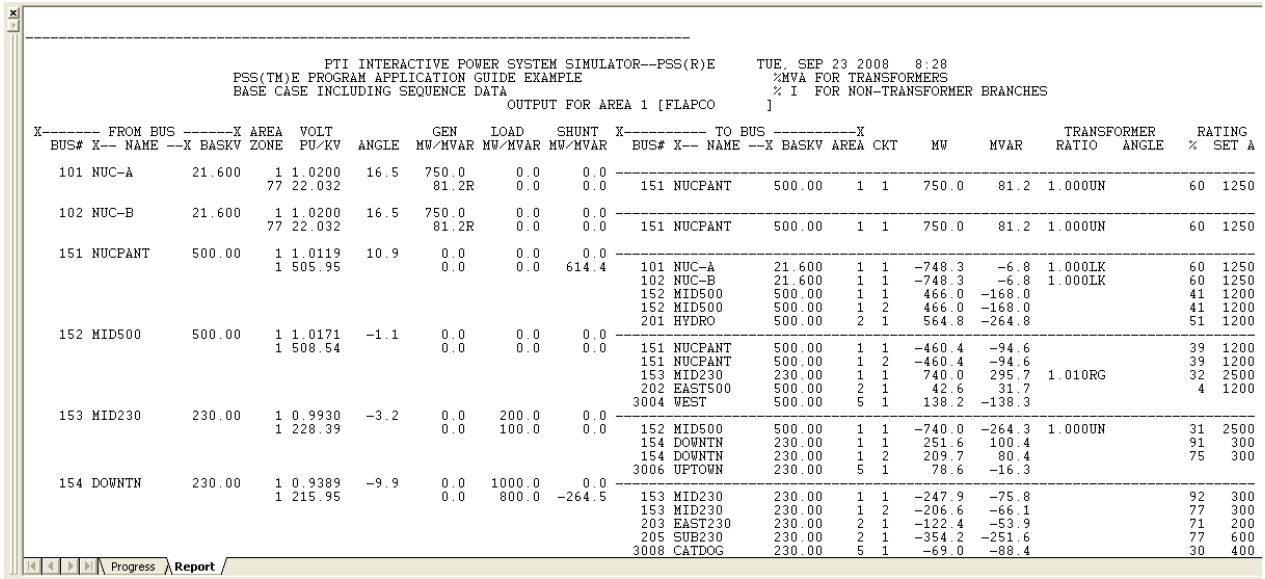


Figure 13-17. Bus Based Reports Dialog, Wide Format



```
                                    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   8:28
                          PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                     %MVA FOR TRANSFORMERS
                          BASE CASE INCLUDING SEQUENCE DATA                              % I  FOR NON-TRANSFORMER BRANCHES
                                                        OUTPUT FOR AREA 1 [FLAPCO      ]

X------- FROM BUS ------X AREA  VOLT            GEN    LOAD    SHUNT  X---------- TO BUS ----------X                          TRANSFORMER     RATING
  BUS# X-- NAME --X BASKV ZONE  PU/KV  ANGLE  MW/MVAR MW/MVAR MW/MVAR   BUS# X-- NAME --X BASKV AREA CKT    MW     MVAR    RATIO   ANGLE   %  SET A

   101 NUC-A       21.600    1 1.0200   16.5   750.0     0.0     0.0 ------------------------------------------------------------------------------
                            77 22.032           81.2R    0.0     0.0    151 NUCPANT     500.00    1  1    750.0    81.2  1.000UN           60  1250

   102 NUC-B       21.600    1 1.0200   16.5   750.0     0.0     0.0 ------------------------------------------------------------------------------
                            77 22.032           81.2R    0.0     0.0    151 NUCPANT     500.00    1  1    750.0    81.2  1.000UN           60  1250

   151 NUCPANT     500.00    1 1.0119   10.9     0.0     0.0     0.0 ------------------------------------------------------------------------------
                             1 505.95            0.0     0.0   614.4    101 NUC-A       21.600    1  1   -748.3    -6.8  1.000LK           60  1250
                                                                       102 NUC-B       21.600    1  1   -748.3    -6.8  1.000LK           60  1250
                                                                       152 MID500      500.00    1  1    466.0  -168.0                     41  1200
                                                                       152 MID500      500.00    1  2    466.0  -168.0                     41  1200
                                                                       201 HYDRO       500.00    2  1    564.8  -264.8                     51  1200

   152 MID500      500.00    1 1.0171   -1.1     0.0     0.0     0.0 ------------------------------------------------------------------------------
                             1 508.54            0.0     0.0     0.0    151 NUCPANT     500.00    1  1   -460.4   -94.6                     39  1200
                                                                       151 NUCPANT     500.00    1  2   -460.4   -94.6                     39  1200
                                                                       153 MID230      230.00    1  1    740.0   295.7  1.010RG            32  2500
                                                                       202 EAST500     500.00    2  1     42.6    31.7                      4  1200
                                                                      3004 WEST        500.00    5  1    138.2  -138.3

   153 MID230      230.00    1 0.9930   -3.2     0.0   200.0     0.0 ------------------------------------------------------------------------------
                             1 228.39            0.0   100.0     0.0    152 MID500      500.00    1  1   -740.0  -264.3  1.000UN           31  2500
                                                                       154 DOWNTN      230.00    1  1    251.6   100.4                     91   300
                                                                       154 DOWNTN      230.00    1  2    209.7    80.4                     75   300
                                                                      3006 UPTOWN      230.00    5  1     78.6   -16.3

   154 DOWNTN      230.00    1 0.9389   -9.9     0.0  1000.0     0.0 ------------------------------------------------------------------------------
                             1 215.95            0.0   800.0  -264.5    153 MID230      230.00    1  1   -247.9   -75.8                     92   300
                                                                       153 MID230      230.00    1  2   -206.6   -66.1                     77   300
                                                                       203 EAST230     230.00    2  1   -122.4   -53.9                     71   200
                                                                       205 SUB230      230.00    2  1   -354.2  -251.6                     77   600
                                                                      3008 CATDOG      230.00    5  1    -69.0   -88.4                     30   400

 Progress  Report
```

Figure 13-18. Example of Wide Format Power Flow Solution Report

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Additional Information
PSS®E Program Operation Manual, Section 11.3, Producing a Wide-Format Power Flow Solution Report

13.3.3 Wide-Format Power Flow Solution Report with Branch Current Loadings

Activity LAMP

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Bus based reports…

In the **[Bus Based Reports]** dialog (Figure 13-19), select *Powerflow output* with the *Wide format output* and *Output with amps* check boxes both checked. Click **[Go]** to produce the report (Figure 13-20).

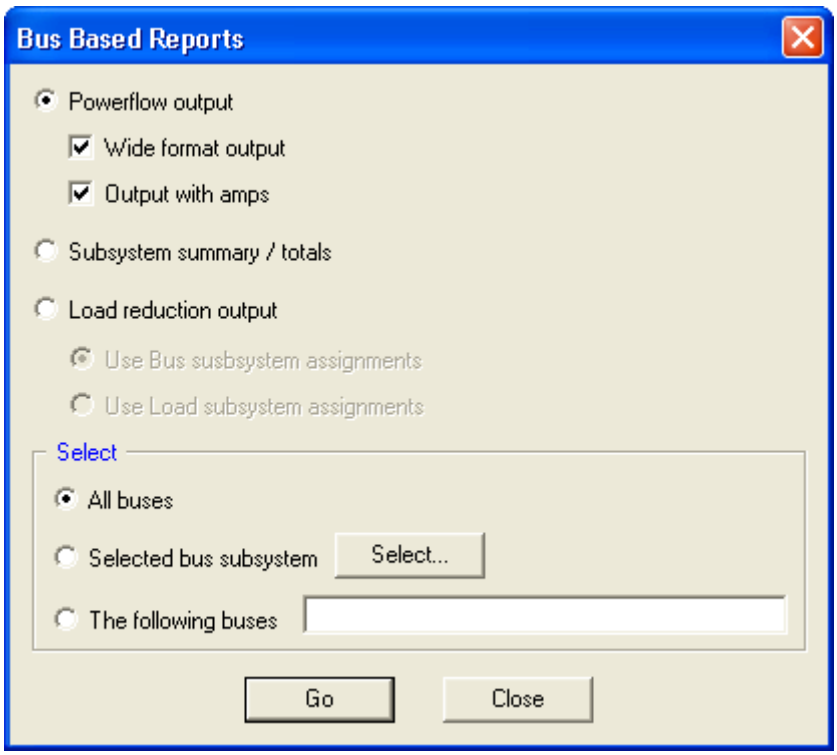
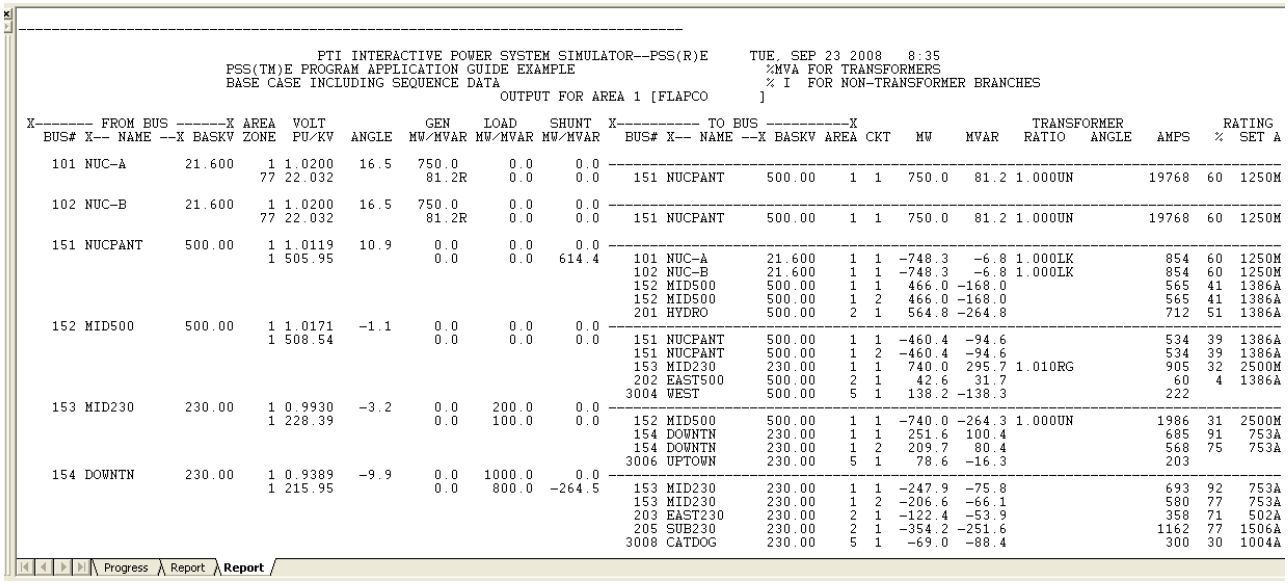


Figure 13-19. Bus Based Reports Dialog, Wide Format, with Branch Current Loadings

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PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E TUE, SEP 23 2008 8:35
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE %MVA FOR TRANSFORMERS
BASE CASE INCLUDING SEQUENCE DATA % I FOR NON-TRANSFORMER BRANCHES
OUTPUT FOR AREA 1 [FLAPCO]

FROM BUS BUS#	NAME	BASKV	AREA ZONE	VOLT PU/KV	ANGLE	GEN MW/MVAR	LOAD MW/MVAR	SHUNT MW/MVAR	TO BUS BUS#	NAME	BASKV	AREA	CKT	MW	MVAR	TRANSFORMER RATIO	ANGLE	AMPS	RATING %	SET A
101	NUC-A	21.600	1	1.0200	16.5	750.0	0.0	0.0												
			77	22.032		81.2R	0.0	0.0	151	NUCPANT	500.00	1	1	750.0	81.2	1.000UN		19768	60	1250M
102	NUC-B	21.600	1	1.0200	16.5	750.0	0.0	0.0												
			77	22.032		81.2R	0.0	0.0	151	NUCPANT	500.00	1	1	750.0	81.2	1.000UN		19768	60	1250M
151	NUCPANT	500.00	1	1.0119	10.9	0.0	0.0	0.0												
			1	505.95		0.0	0.0	614.4	101	NUC-A	21.600	1	1	-748.3	-6.8	1.000LK		854	60	1250M
									102	NUC-B	21.600	1	1	-748.3	-6.8	1.000LK		854	60	1250M
									152	MID500	500.00	1	1	466.0	-168.0			565	41	1386A
									152	MID500	500.00	1	2	466.0	-168.0			565	41	1386A
									201	HYDRO	500.00	2	1	564.8	-264.8			712	51	1386A
152	MID500	500.00	1	1.0171	-1.1	0.0	0.0	0.0												
			1	508.54		0.0	0.0	0.0	151	NUCPANT	500.00	1	1	-460.4	-94.6			534	39	1386A
									151	NUCPANT	500.00	1	2	-460.4	-94.6			534	39	1386A
									153	MID230	230.00	1	1	740.0	295.7	1.010RG		905	32	2500M
									202	EAST500	500.00	2	1	42.6	31.7			60	4	1386A
									3004	WEST	500.00	5	1	138.2	-138.3			222		
153	MID230	230.00	1	0.9930	-3.2	0.0	200.0	0.0												
			1	228.39		0.0	100.0	0.0	152	MID500	500.00	1	1	-740.0	-264.3	1.000UN		1986	31	2500M
									154	DOWNTN	230.00	1	1	251.6	100.4			685	91	753A
									154	DOWNTN	230.00	1	2	209.7	80.4			568	75	753A
									3006	UPTOWN	230.00	5	1	78.6	-16.3			203		
154	DOWNTN	230.00	1	0.9389	-9.9	0.0	1000.0	0.0												
			1	215.95		0.0	800.0	-264.5	153	MID230	230.00	1	1	-247.9	-75.8			693	92	753A
									153	MID230	230.00	1	2	-206.6	-66.1			580	77	753A
									203	EAST230	230.00	2	1	-122.4	-53.9			358	71	502A
									205	SUB230	230.00	2	1	-354.2	-251.6			1162	77	1506A
									3008	CATDOG	230.00	5	1	-69.0	-88.4			300	30	1004A

Progress | Report | **Report**

Figure 13-20. Example of Wide Format Power Flow Solution Report with Branch Current Loadings

Additional Information
PSS®E Program Operation Manual, Section 11.4, Producing a Wide-Format Power Flow Solution Report in Amps

13.3.4 Summary of Subsystem Conditions

Activity SUBS

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Bus based reports…

In the **[Bus Based Reports]** dialog (Figure 13-21), select *Subsystem summary / totals*. Click **[Go]** to produce the report (Figure 13-22).

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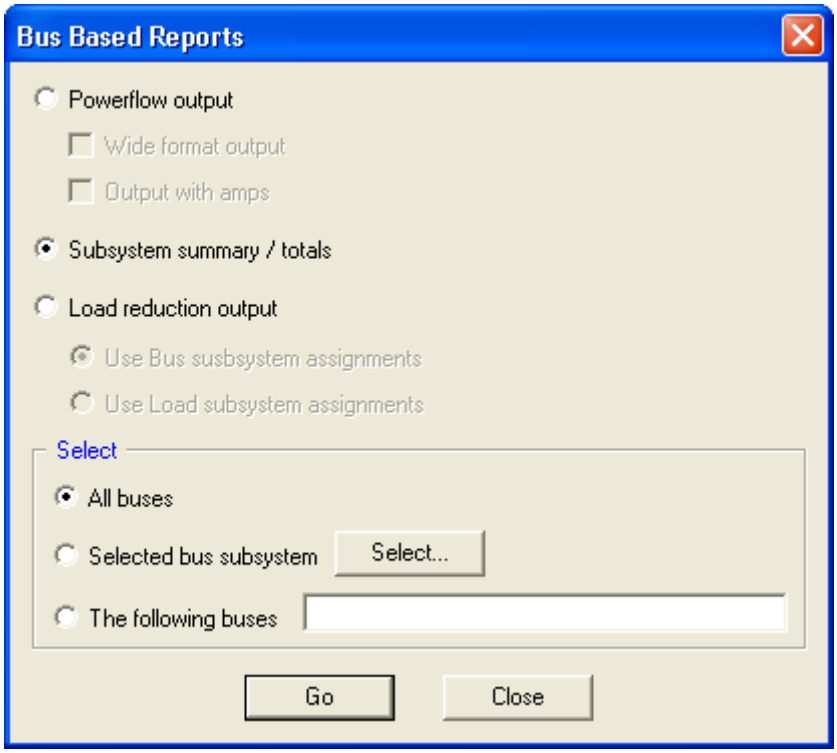
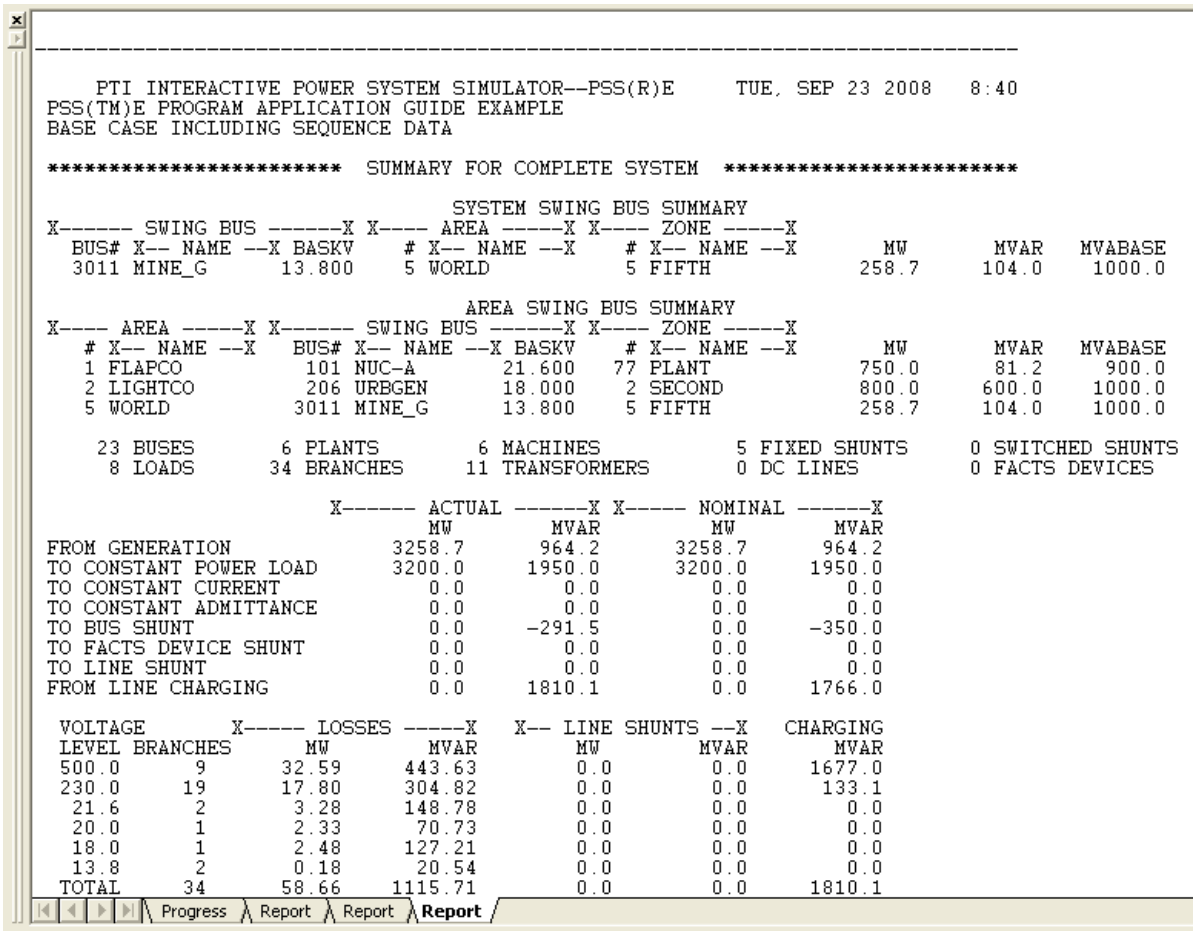


Figure 13-21. Bus Based Reports Dialog, Subsystem Summary

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```
------------------------------------------------------------------------------

    PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E     TUE, SEP 23 2008   8:40
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

************************  SUMMARY FOR COMPLETE SYSTEM  ************************

                                  SYSTEM SWING BUS SUMMARY
X------ SWING BUS ------X X---- AREA -----X X---- ZONE -----X
  BUS# X-- NAME --X BASKV     # X-- NAME --X     # X-- NAME --X       MW       MVAR   MVABASE
  3011 MINE_G      13.800     5 WORLD           5 FIFTH           258.7      104.0    1000.0

                                  AREA SWING BUS SUMMARY
X---- AREA -----X X------ SWING BUS ------X X---- ZONE -----X
   # X-- NAME --X   BUS# X-- NAME --X BASKV     # X-- NAME --X       MW       MVAR   MVABASE
   1 FLAPCO          101 NUC-A       21.600    77 PLANT           750.0      81.2     900.0
   2 LIGHTCO         206 URBGEN      18.000     2 SECOND          800.0     600.0    1000.0
   5 WORLD          3011 MINE_G      13.800     5 FIFTH           258.7      104.0    1000.0

    23 BUSES       6 PLANTS        6 MACHINES          5 FIXED SHUNTS     0 SWITCHED SHUNTS
     8 LOADS      34 BRANCHES     11 TRANSFORMERS      0 DC LINES         0 FACTS DEVICES

                        X------ ACTUAL ------X X----- NOMINAL ------X
                                  MW        MVAR         MW        MVAR
FROM GENERATION                3258.7      964.2       3258.7      964.2
TO CONSTANT POWER LOAD         3200.0     1950.0       3200.0     1950.0
TO CONSTANT CURRENT               0.0        0.0          0.0        0.0
TO CONSTANT ADMITTANCE            0.0        0.0          0.0        0.0
TO BUS SHUNT                      0.0     -291.5          0.0     -350.0
TO FACTS DEVICE SHUNT             0.0        0.0          0.0        0.0
TO LINE SHUNT                     0.0        0.0          0.0        0.0
FROM LINE CHARGING                0.0     1810.1          0.0     1766.0

 VOLTAGE       X----- LOSSES -----X   X-- LINE SHUNTS --X   CHARGING
 LEVEL BRANCHES       MW        MVAR        MW        MVAR       MVAR
 500.0      9      32.59      443.63       0.0         0.0     1677.0
 230.0     19      17.80      304.82       0.0         0.0      133.1
  21.6      2       3.28      148.78       0.0         0.0        0.0
  20.0      1       2.33       70.73       0.0         0.0        0.0
  18.0      1       2.48      127.21       0.0         0.0        0.0
  13.8      2       0.18       20.54       0.0         0.0        0.0
 TOTAL     34      58.66     1115.71       0.0         0.0     1810.1
```

Progress | Report | Report | **Report**

Figure 13-22. Example of Subsystem Summary Report

Additional Information
PSS®E Program Operation Manual, Section 11.5, Summarizing Subsystem Conditions

13.3.5 Summarizing Load Reduction

Activity LODR

Requirements / Prerequisites
The working case must contain a validly specified power flow case solved to an acceptable mismatch tolerance.



Power Flow > Reports > Bus based reports…

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The load reduction reporting activity LODR summarizes the reduction in:

- Nominal constant MVA load caused when voltage at a load bus falls below PQBRAK.
- Nominal constant current load caused when voltage at a load bus falls below 0.5 pu.

In the **[Bus Based Reports]** dialog (Figure 13-23), select *Load reduction output*. If subsystem selection by area, owner and/or zone had been specified, select the desired option for defining subsystem loads. Click **[Go]** to produce the report.

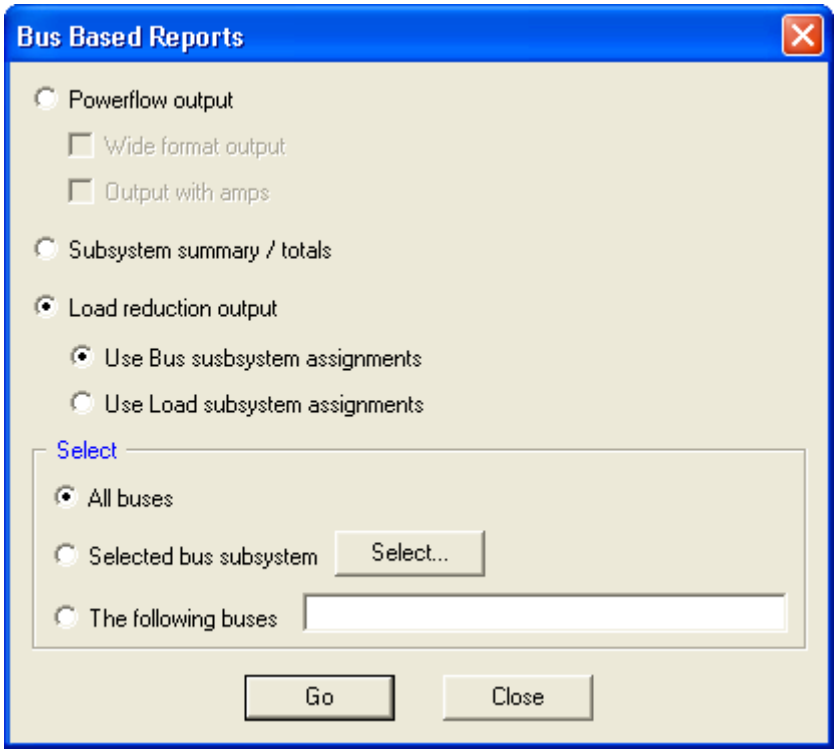


Figure 13-23. Bus Based Reports Dialog, Load Reduction Output

Additional Information
PSS®E Program Operation Manual, Section 11.6, Summarizing Load Reduction Section 11.1.1, Boundary Conditions

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13.4 Limit Checking Reports



Power Flow > Reports > Limit checking reports…

The **[Limit Checking Reports]** dialog provides tabs to produce all PSS®Express limit checking reports:

- Machine Reactive Capability
- Generator Bus Limits
- Machine Terminal Limits
- Transmission Line Overloads
- Transformer Overloads
- Branch Overloads
- Transmission Line Overloads
- Transformer Overloads
- Branch Current Ratings
- Voltage Controlled Buses
- Controlling Transformers
- Out-of-limit Bus Voltage

Note that each report can be restricted by Area, Owner, Zone, Base kV and Bus, using the *Select* option of the dialog.

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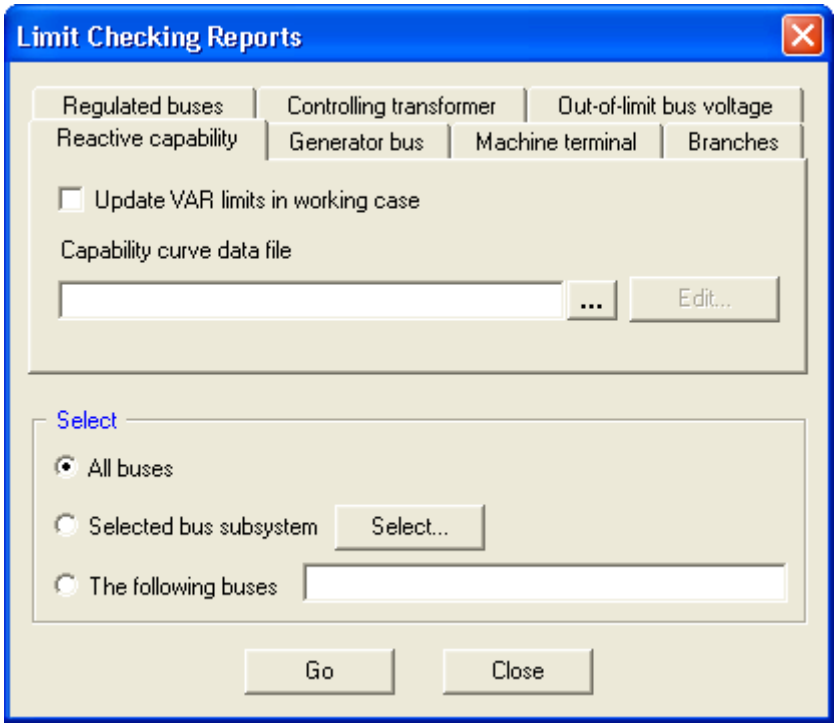


Figure 13-24. Limit Checking Reports Dialog

13.4.1 Machine Reactive Capability

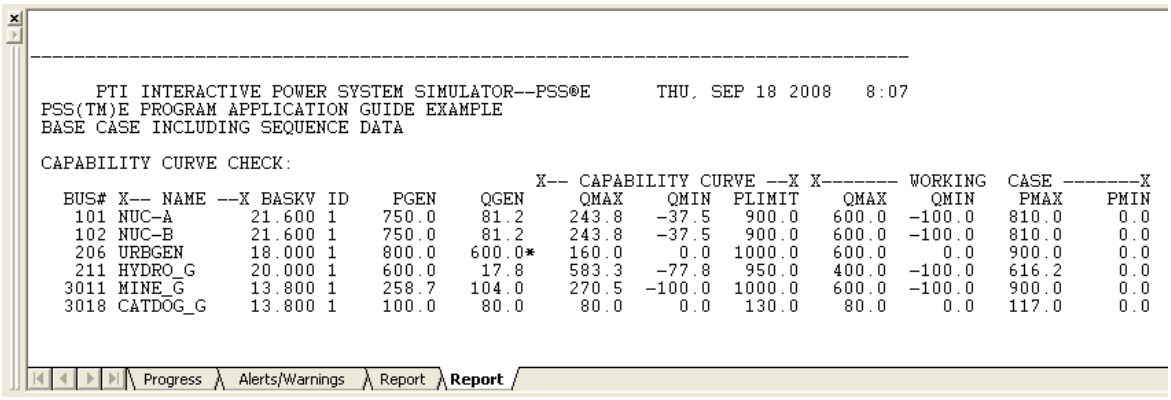
Activity GCAP

Requirements / Prerequisites
Validly specified power flow case. Machine Capability Data File (*.gcp)

Power Flow > Reports > Limit checking reports…

Clicking the *Reactive capability* tab in **[Limit Checking Reports]** will generate a report that tabulates machine loading and limit data. There is a checkbox option to update the machine reactive power limits in the power flow case. The *PSS®E Program Operation Manual* (Section 11.21.1) provides details of the data records for the required Machine Capability Data file (*.gcp).

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      THU, SEP 18 2008   8:07
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

CAPABILITY CURVE CHECK:
                                                X-- CAPABILITY CURVE --X X------- WORKING  CASE --------X
  BUS# X-- NAME --X BASKV ID     PGEN     QGEN      QMAX     QMIN  PLIMIT     QMAX     QMIN     PMAX     PMIN
   101 NUC-A       21.600 1     750.0     81.2     243.8   -37.5   900.0    600.0  -100.0    810.0      0.0
   102 NUC-B       21.600 1     750.0     81.2     243.8   -37.5   900.0    600.0  -100.0    810.0      0.0
   206 URBGEN      18.000 1     800.0    600.0*    160.0     0.0  1000.0    600.0     0.0    900.0      0.0
   211 HYDRO_G     20.000 1     600.0     17.8     583.3   -77.8   950.0    400.0  -100.0    616.2      0.0
  3011 MINE_G      13.800 1     258.7    104.0     270.5  -100.0  1000.0    600.0  -100.0    900.0      0.0
  3018 CATDOG_G    13.800 1     100.0     80.0      80.0     0.0   130.0     80.0     0.0    117.0      0.0

 Progress  Alerts/Warnings  Report  Report
```

Figure 13-25. Example of Reactive Capability Report

Additional Information
PSS®E Program Operation Manual, Section 11.21, Producing a Machine Reactive Capability Report Section 11.21.1, Machine Capability Curve Data File Contents

13.4.2 Generator Bus Limits

Activity GENS

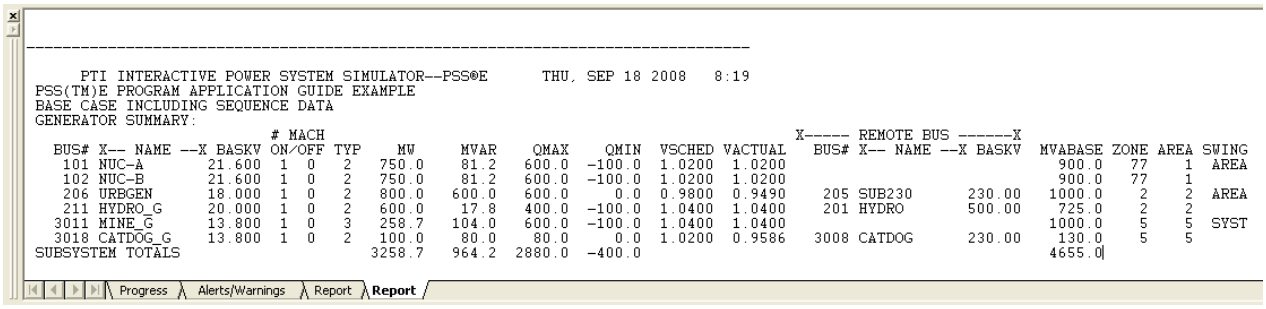
Requirements / Prerequisites
Validly specified power flow case. Plant outputs and voltages are valid only if the working case is solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Generator bus* tab in **[Limit Checking Reports]** will generate a report that tabulates the loading and voltage conditions at generator buses. The following reports can be selected:

- Var limited plants with unequal var limits
- All var limited plants
- On-line plants
- All plants

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```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      THU, SEP 18 2008   8:19
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA
GENERATOR SUMMARY:
                    # MACH                                                    X------ REMOTE BUS ------X
  BUS# X-- NAME --X BASKV ON/OFF TYP    MW     MVAR    QMAX    QMIN  VSCHED VACTUAL   BUS# X-- NAME --X BASKV  MVABASE ZONE AREA SWING
   101 NUC-A       21.600  1   0   2   750.0    81.2   600.0  -100.0 1.0200  1.0200                            900.0   77    1  AREA
   102 NUC-B       21.600  1   0   2   750.0    81.2   600.0  -100.0 1.0200  1.0200                            900.0   77    1
   206 URBGEN      18.000  1   0   2   800.0   600.0   600.0     0.0 0.9800  0.9490    205 SUB230      230.00  1000.0    2    2  AREA
   211 HYDRO_G     20.000  1   0   2   600.0    17.8   400.0  -100.0 1.0400  1.0400    201 HYDRO       500.00   725.0    2    2
  3011 MINE_G      13.800  1   0   3   258.7   104.0   600.0  -100.0 1.0400  1.0400                            1000.0    5    5  SYST
  3018 CATDOG_G    13.800  1   0   2   100.0    80.0    80.0     0.0 1.0200  0.9586   3008 CATDOG      230.00   130.0    5    5
SUBSYSTEM TOTALS                     3258.7   964.2  2880.0  -400.0                                             4655.0
```

Progress | Alerts/Warnings | Report | **Report**

Figure 13-26. Example of Generator Bus Report, On-line Plants

Additional Information
PSS®E Program Operation Manual, Section 11.22, Producing a Generator Bus Limits Report

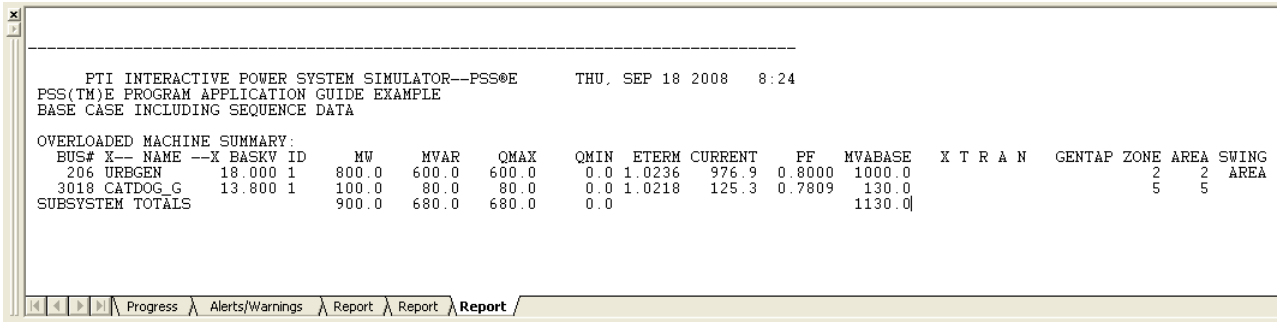
13.4.3 Machine Terminal Limits

Activity GEOL

Requirements / Prerequisites
Validly specified power flow case. Machine outputs and voltages are valid only if the working case is solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Machine terminal* tab in **[Limit Checking Reports]** will generate a report that tabulates machine loading and voltage conditions at the generator terminals of on-line machines at type two and three buses in the working case. The report may include all on-line machines or only those that are overloaded.



```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      THU, SEP 18 2008   8:24
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

OVERLOADED MACHINE SUMMARY:
  BUS# X-- NAME --X BASKV ID     MW     MVAR    QMAX    QMIN  ETERM CURRENT    PF   MVABASE   X T R A N   GENTAP ZONE AREA SWING
   206 URBGEN      18.000 1    800.0   600.0   600.0     0.0 1.0236   976.9  0.8000  1000.0                      2    2  AREA
  3018 CATDOG_G    13.800 1    100.0    80.0    80.0     0.0 1.0218   125.3  0.7809   130.0                      5    5
SUBSYSTEM TOTALS              900.0   680.0   680.0     0.0                         1130.0
```

Progress | Alerts/Warnings | Report | Report | **Report**

Figure 13-27. Example of Machine Terminal Limits Report, Overloaded Machines Only

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International.

Additional Information
PSS®E Program Operation Manual, Section 11.23, Producing a Machine Terminal Limits Report

13.4.4 Branch Overloads

Activity RAT3

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Branches* tab in **[Limit Checking Reports]** and then selecting the *both transformer and non-transformer branches* option from the pull-down list will generate a report that tabulates branch overloads. Options are available to specify the *Rating Set* and *Percent Loading* limit.

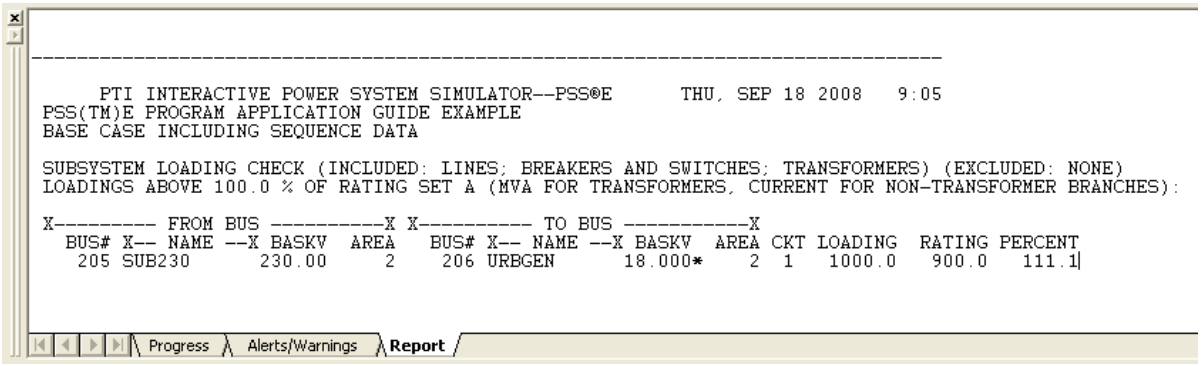


Figure 13-28. Example of Branch Overload Report, Default Settings

Additional Information
PSS®E Program Operation Manual, Section 11.16, Producing a Branch Overload Checking Report

13.4.5 Transmission Line Overloads

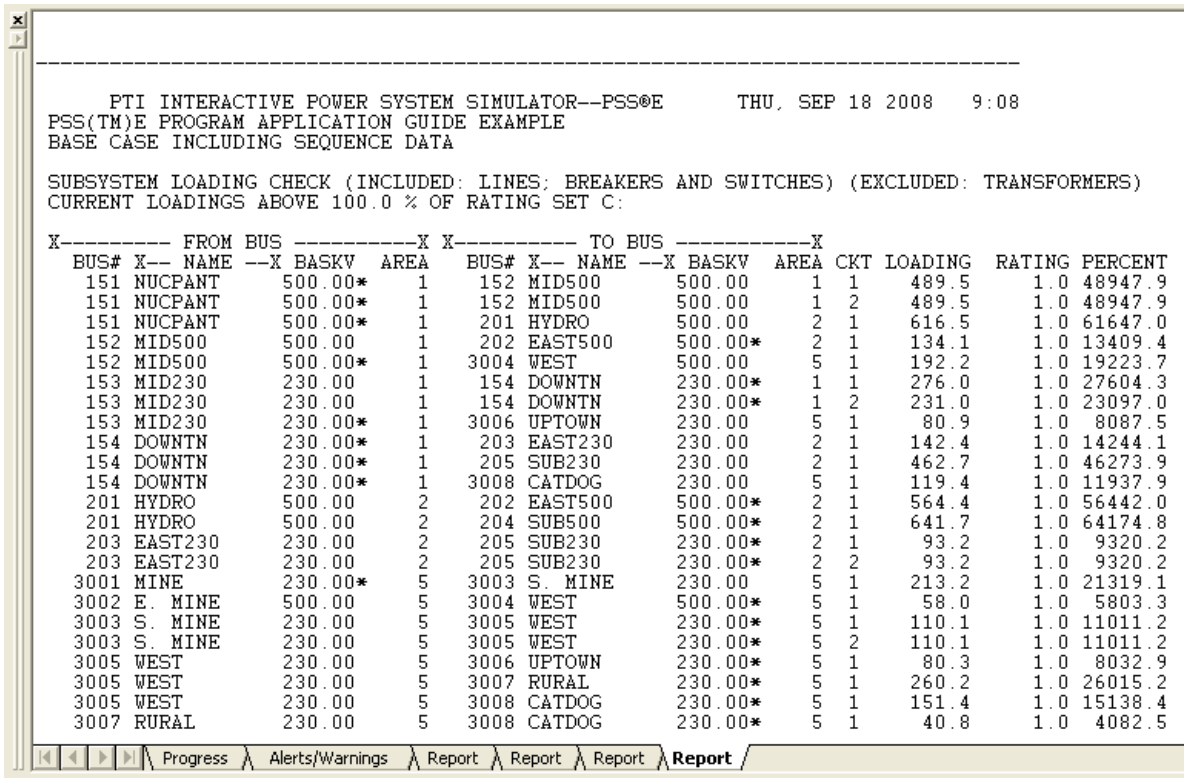
Activity OLTL

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

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Clicking the *Branches* tab in **[Limit Checking Reports]** and then selecting the *only non-transformer branches* option from the pull-down list will generate a report that tabulates transmission line overloads. Options are available to specify the *Rating Set* and *Percent Loading* limit.



PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E THU, SEP 18 2008 9:08
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES) (EXCLUDED: TRANSFORMERS)
CURRENT LOADINGS ABOVE 100.0 % OF RATING SET C:

FROM BUS				TO BUS						
BUS#	NAME	BASKV	AREA	BUS#	NAME	BASKV	AREA	CKT	LOADING	RATING	PERCENT
151	NUCPANT	500.00*	1	152	MID500	500.00	1	1	489.5	1.0	48947.9
151	NUCPANT	500.00*	1	152	MID500	500.00	1	2	489.5	1.0	48947.9
151	NUCPANT	500.00*	1	201	HYDRO	500.00	2	1	616.5	1.0	61647.0
152	MID500	500.00	1	202	EAST500	500.00*	2	1	134.1	1.0	13409.4
152	MID500	500.00*	1	3004	WEST	500.00	5	1	192.2	1.0	19223.7
153	MID230	230.00	1	154	DOWNTN	230.00*	1	1	276.0	1.0	27604.3
153	MID230	230.00	1	154	DOWNTN	230.00*	1	2	231.0	1.0	23097.0
153	MID230	230.00*	1	3006	UPTOWN	230.00	5	1	80.9	1.0	8087.5
154	DOWNTN	230.00*	1	203	EAST230	230.00	2	1	142.4	1.0	14244.1
154	DOWNTN	230.00*	1	205	SUB230	230.00	2	1	462.7	1.0	46273.9
154	DOWNTN	230.00*	1	3008	CATDOG	230.00	5	1	119.4	1.0	11937.9
201	HYDRO	500.00	2	202	EAST500	500.00*	2	1	564.4	1.0	56442.0
201	HYDRO	500.00	2	204	SUB500	500.00*	2	1	641.7	1.0	64174.8
203	EAST230	230.00	2	205	SUB230	230.00*	2	1	93.2	1.0	9320.2
203	EAST230	230.00	2	205	SUB230	230.00*	2	2	93.2	1.0	9320.2
3001	MINE	230.00*	5	3003	S. MINE	230.00	5	1	213.2	1.0	21319.1
3002	E. MINE	500.00	5	3004	WEST	500.00*	5	1	58.0	1.0	5803.3
3003	S. MINE	230.00	5	3005	WEST	230.00*	5	1	110.1	1.0	11011.2
3003	S. MINE	230.00	5	3005	WEST	230.00*	5	2	110.1	1.0	11011.2
3005	WEST	230.00	5	3006	UPTOWN	230.00*	5	1	80.3	1.0	8032.9
3005	WEST	230.00	5	3007	RURAL	230.00*	5	1	260.2	1.0	26015.2
3005	WEST	230.00	5	3008	CATDOG	230.00*	5	1	151.4	1.0	15138.4
3007	RURAL	230.00	5	3008	CATDOG	230.00*	5	1	40.8	1.0	4082.5

Progress | Alerts/Warnings | Report | Report | Report | Report

Figure 13-29. Example of Transmission Line Overload Report, Rate C

Additional Information
PSS®E Program Operation Manual, Section 11.18, Producing a Transmission Line Overload Checking Report

13.4.6 Transformer Overloads

Activity OLTR

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Branches* tab in **[Limit Checking Reports]** and then selecting the *only transformers* option from the pull-down list will generate a report that tabulates transformer overloads. Options are available to specify the *Rating Set* and *Percent Loading* limit.

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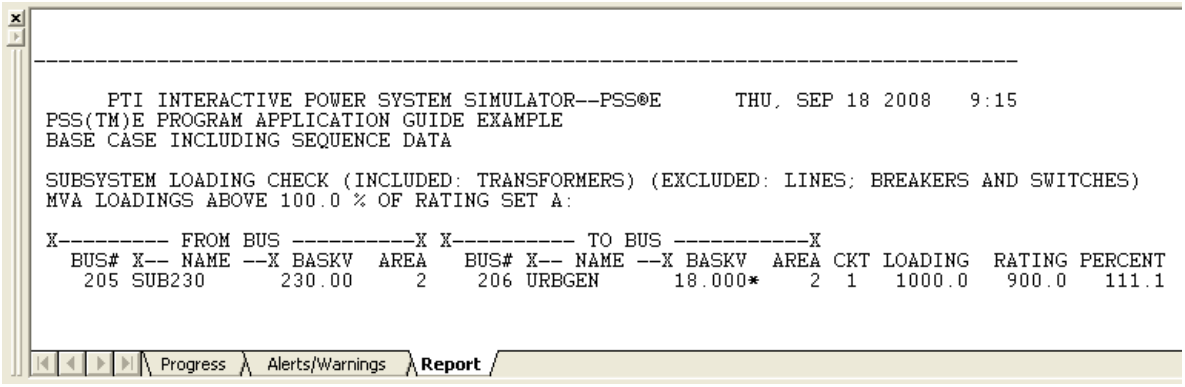


Figure 13-30. Example of Transformer Overload Report, Default Options

Additional Information
PSS®E Program Operation Manual, Section 11.17, Producing a Transformer Overload Checking Report

13.4.7 Branch Current Ratings

Activity RATE

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

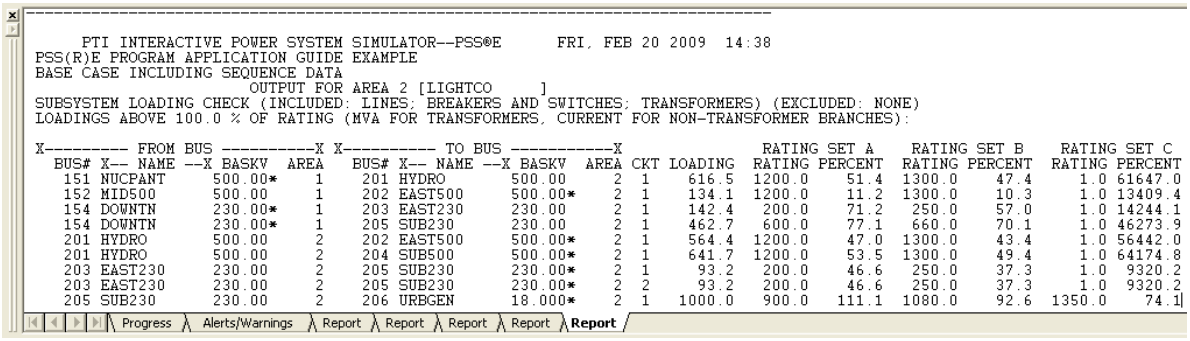
Clicking the *Branches* tab in **[Limit Checking Reports]** provides options to include any of the following network components in the output:

- Include non-transformer branches that are not breakers or switches
- Include transformer branches
- Include branches that are breakers or switches.

Options are available to specify the *Rating Set* and *Percent Loading* limit.

Selecting all three network component options with the all ratings option will generate a report that tabulates branch current ratings.

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PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E FRI, FEB 20 2009 14:38
PSS(R)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA
OUTPUT FOR AREA 2 [LIGHTCO]
SUBSYSTEM LOADING CHECK (INCLUDED: LINES; BREAKERS AND SWITCHES; TRANSFORMERS) (EXCLUDED: NONE)
LOADINGS ABOVE 100.0 % OF RATING (MVA FOR TRANSFORMERS, CURRENT FOR NON-TRANSFORMER BRANCHES):

FROM BUS				TO BUS					RATING SET A		RATING SET B		RATING SET C	
BUS#	X-- NAME	--X BASKV	AREA	BUS#	X-- NAME	--X BASKV	AREA	CKT	LOADING	RATING	PERCENT	RATING	PERCENT	RATING	PERCENT
151	NUCPANT	500.00*	1	201	HYDRO	500.00	2	1	616.5	1200.0	51.4	1300.0	47.4	1.0	61647.0
152	MID500	500.00	1	202	EAST500	500.00*	2	1	134.1	1200.0	11.2	1300.0	10.3	1.0	13409.4
154	DOWNTN	230.00*	1	203	EAST230	230.00	2	1	142.4	200.0	71.2	250.0	57.0	1.0	14244.1
154	DOWNTN	230.00*	1	205	SUB230	230.00	2	1	462.7	600.0	77.1	660.0	70.1	1.0	46273.9
201	HYDRO	500.00	2	202	EAST500	500.00*	2	1	564.4	1200.0	47.0	1300.0	43.4	1.0	56442.0
201	HYDRO	500.00	2	204	SUB500	500.00*	2	1	641.7	1200.0	53.5	1300.0	49.4	1.0	64174.8
203	EAST230	230.00	2	205	SUB230	230.00*	2	1	93.2	200.0	46.6	250.0	37.3	1.0	9320.2
203	EAST230	230.00	2	205	SUB230	230.00*	2	2	93.2	200.0	46.6	250.0	37.3	1.0	9320.2
205	SUB230	230.00	2	206	URBGEN	18.000*	2	1	1000.0	900.0	111.1	1080.0	92.6	1350.0	74.1

Progress | Alerts/Warnings | Report | Report | Report | Report | Report

Figure 13-31. Example of Branch Current Ratings Report, Default Option

Additional Information
PSS®E Program Operation Manual, Section 11.19, Producing a Branch Current Ratings Report

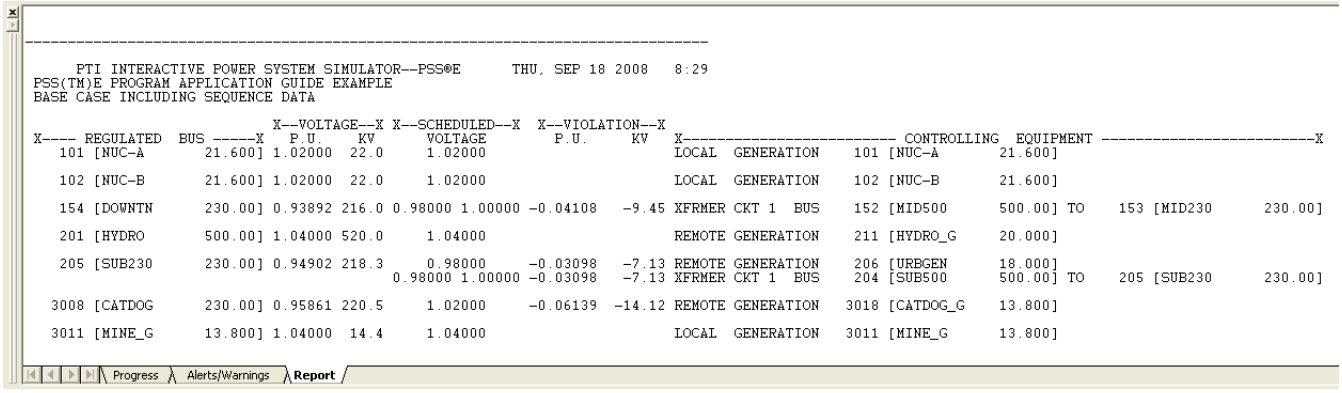
13.4.8 Voltage Controlled Buses

Activity REGB

Requirements / Prerequisites
Validly specified power flow case. Bus voltages are valid only if the working case is solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Regulated buses* tab in **[Limit Checking Reports]** will generate a report that tabulates those buses where voltages are controlled by generation, switched shunts, voltage controlling transformers, FACTS devices, and/or VSC dc line converters.



PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E THU, SEP 18 2008 8:29
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE
BASE CASE INCLUDING SEQUENCE DATA

REGULATED BUS	X--VOLTAGE--X P.U.	KV	X--SCHEDULED--X VOLTAGE	P.U.	X--VIOLATION--X P.U.	KV	CONTROLLING EQUIPMENT
101 [NUC-A 21.600]	1.02000	22.0	1.02000				LOCAL GENERATION 101 [NUC-A 21.600]
102 [NUC-B 21.600]	1.02000	22.0	1.02000				LOCAL GENERATION 102 [NUC-B 21.600]
154 [DOWNTN 230.00]	0.93892	216.0	0.98000 1.00000	-0.04108	-9.45		XFRMER CKT 1 BUS 152 [MID500 500.00] TO 153 [MID230 230.00]
201 [HYDRO 500.00]	1.04000	520.0	1.04000				REMOTE GENERATION 211 [HYDRO_G 20.000]
205 [SUB230 230.00]	0.94902	218.3	0.98000	-0.03098	-7.13		REMOTE GENERATION 206 [URBGEN 18.000]
			0.98000 1.00000	-0.03098	-7.13		XFRMER CKT 1 BUS 204 [SUB500 500.00] TO 205 [SUB230 230.00]
3008 [CATDOG 230.00]	0.95861	220.5	1.02000	-0.06139	-14.12		REMOTE GENERATION 3018 [CATDOG_G 13.800]
3011 [MINE_G 13.800]	1.04000	14.4	1.04000				LOCAL GENERATION 3011 [MINE_G 13.800]

Progress | Alerts/Warnings | Report

Figure 13-32. Example of Regulated Buses Report

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International.

Additional Information
PSS®E Program Operation Manual, Section 11.24, Producing a Regulated Bus Report

13.4.9 Controlling Transformers

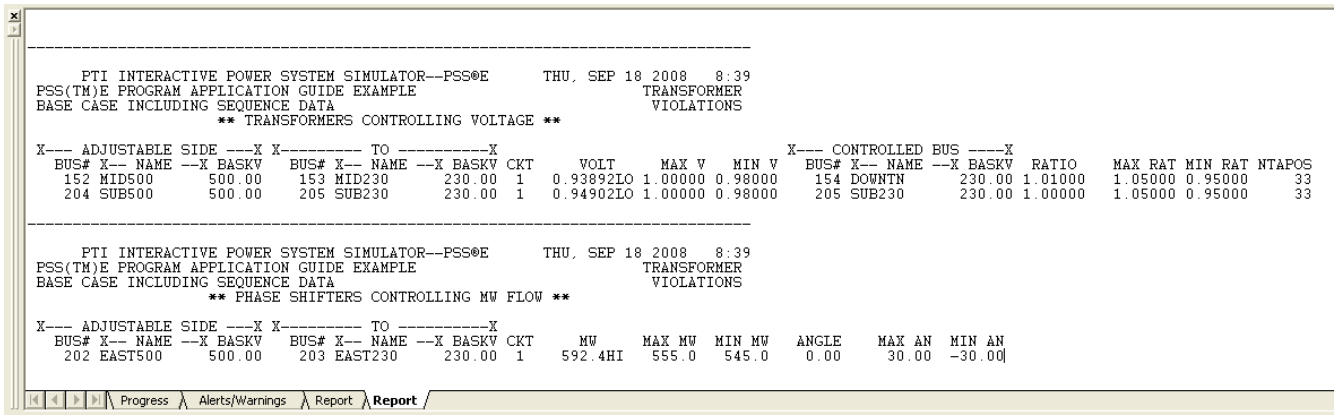
Activity TLST

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Controlling transformer* tab in **[Limit Checking Reports]** will generate a report that tabulates those transformers in the power flow case where off-nominal turns ratio or phase-shift angle may be adjusted by power flow solution activities.

The user has the option of listing *All controlling transformers* or of restricting the report to *Controlling transformer violations*, those transformers where the controlled quantity is outside of its specified band. The *Output with voltage reversed* option produces a report reversing the voltage option specified (pu or kV) in the Program Settings Dialog.



```
------------------------------------------------------------------------------

     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      THU, SEP 18 2008   8:39
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                      TRANSFORMER
BASE CASE INCLUDING SEQUENCE DATA                                VIOLATIONS
                  ** TRANSFORMERS CONTROLLING VOLTAGE **

X--- ADJUSTABLE SIDE ---X X--------- TO ----------X                          X--- CONTROLLED BUS ----X
  BUS# X-- NAME --X BASKV   BUS# X-- NAME --X BASKV CKT     VOLT     MAX V   MIN V   BUS# X-- NAME --X BASKV  RATIO    MAX RAT MIN RAT NTAPOS
   152 MID500      500.00    153 MID230      230.00  1   0.93892LO 1.00000 0.98000    154 DOWNTN      230.00 1.01000   1.05000 0.95000      33
   204 SUB500      500.00    205 SUB230      230.00  1   0.94902LO 1.00000 0.98000    205 SUB230      230.00 1.00000   1.05000 0.95000      33

------------------------------------------------------------------------------

     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      THU, SEP 18 2008   8:39
PSS(TM)E PROGRAM APPLICATION GUIDE EXAMPLE                      TRANSFORMER
BASE CASE INCLUDING SEQUENCE DATA                                VIOLATIONS
                  ** PHASE SHIFTERS CONTROLLING MW FLOW **

X--- ADJUSTABLE SIDE ---X X--------- TO ----------X
  BUS# X-- NAME --X BASKV   BUS# X-- NAME --X BASKV CKT     MW     MAX MW  MIN MW   ANGLE    MAX AN  MIN AN
   202 EAST500     500.00    203 EAST230     230.00  1    592.4HI   555.0   545.0    0.00     30.00  -30.00

 Progress | Alerts/Warnings | Report | Report
```

Figure 13-33. Example of Controlling Transformer Report, Violations Only

Additional Information
PSS®E Program Operation Manual, Section 11.25, Producing a Controlling Transformer Report

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13.4.10 Out-of-limit Bus Voltage

Activity VCHK

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance.

Power Flow > Reports > Limit checking reports…

Clicking the *Out-of-limit bus voltage* tab in **[Limit Checking Reports]** will generate a report that tabulates those buses where voltage magnitude is outside the specified range. The example (Figure 13-34) is based on the default limits, which can be changed by the user in .01 pu increments.

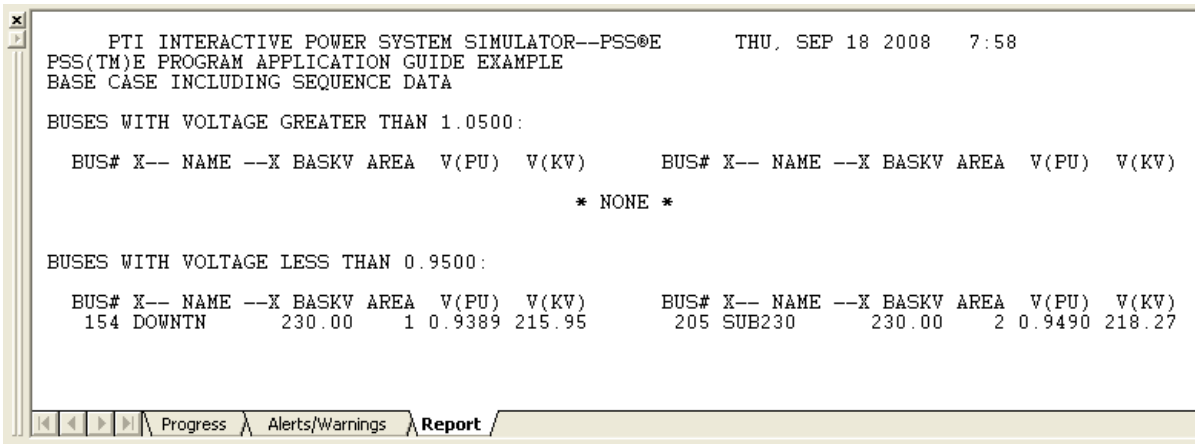


Figure 13-34. Example of Out-of-Limit Bus Voltage Report

Additional Information
PSS®E Program Operation Manual, Section 10.20, Producing an Out-of-Limits Bus Voltage Report

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13.5 Reporting DC Network Conditions

Activity MTDC

Requirements / Prerequisites
Validly specified power flow case solved to an acceptable mismatch tolerance. Unblocked multi-terminal dc lines must be present in the working case.

Power Flow > Reports > Multi-terminal DC line solution output (MTDC)

The multi-terminal dc line power flow output activity MTDC prints power flow solution results at all dc buses of all unblocked multi-terminal dc lines in the working case.

Activity MTDC displays a popup message *No unblocked multi-terminal dc lines in this case* and terminates if these network conditions are not found. Click **[OK]** to remove the message.

There are no user-defined options. The report is displayed in the **Report** tab by default (Figure 13-35).

```
     PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS®E      WED, OCT 15 2008  14:35
PSS(TM)E SAMPLE CASE                                        MULTI-TERMINAL DC
ALL DATA CATEGORIES WITH SEQUENCE DATA                            "1          "

       BUS# X-- NAME --X BASKV AREA ZONE CKT     MW      AMPS      KV
FROM      1 DC1           (DC)     4    4                        534.60
  TO    401 COGEN-1     500.00     4    9      -321.0   -600.4
  TO      5 DC5           (DC)     4    4  1    321.0    600.4

       BUS# X-- NAME --X BASKV AREA ZONE CKT     MW      AMPS      KV
FROM      2 DC2           (DC)     2    2                        500.00
  TO    212 INVERT1     230.00     2    7       296.4    592.8
  TO      5 DC5           (DC)     4    4  1   -296.4   -592.8

       BUS# X-- NAME --X BASKV AREA ZONE CKT     MW      AMPS      KV
FROM      3 DC3           (DC)     4    4                        534.60
  TO    402 COGEN-2     500.00     6    9      -321.0   -600.4
  TO      5 DC5           (DC)     4    4  1    321.0    600.4

       BUS# X-- NAME --X BASKV AREA ZONE CKT     MW      AMPS      KV
FROM      4 DC4           (DC)     2    2                        499.55
  TO    213 INVERT2     230.00     2    7       303.8    608.1
  TO      5 DC5           (DC)     4    4  1   -303.8   -608.1

       BUS# X-- NAME --X BASKV AREA ZONE CKT     MW      AMPS      KV
FROM      5 DC5           (DC)     4    4                        517.19
  TO      1 DC1           (DC)     4    4  1   -310.5   -600.4
  TO      2 DC2           (DC)     2    2  1    306.6    592.8
  TO      3 DC3           (DC)     4    4  1   -310.5   -600.4
  TO      4 DC4           (DC)     2    2  1    314.5    608.1

 Progress | Alerts/Warnings | Report
```

Figure 13-35. Example of DC Network Conditions Report

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13.6 Restricting the Output of Large Reports

13.6.1 By Area

Subsystem > Area…

The **[Area Subsystem Selector]** dialog is similar to the area tab in the **[Bus Subsystem Selector]** (see Figure 2-9). The buttons respond the same way, but the *Memorize* and *Recall* options are not available. This dialog is used to limit report output and does not affect the **[Spreadsheet]** display.

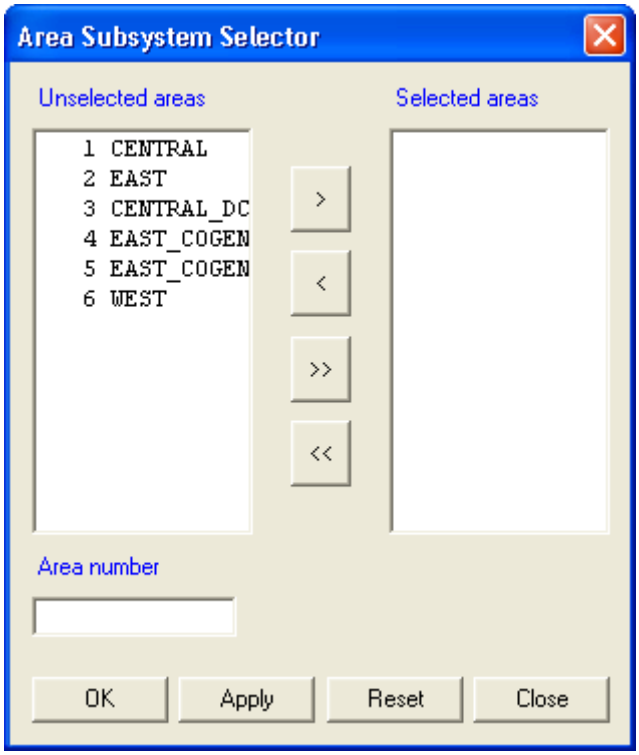


Figure 13-36. Area Subsystem Selector Dialog

13.6.2 By Owner

Subsystem > Owner…

The **[Owner Subsystem Selector]** dialog is similar to the owner tab in the **[Bus Subsystem Selector]** (see Figure 2-9). The buttons respond the same way, but the *Memorize* and *Recall* options are not available. This dialog is used to limit report output and does not affect the **[Spreadsheet]** display.

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International.

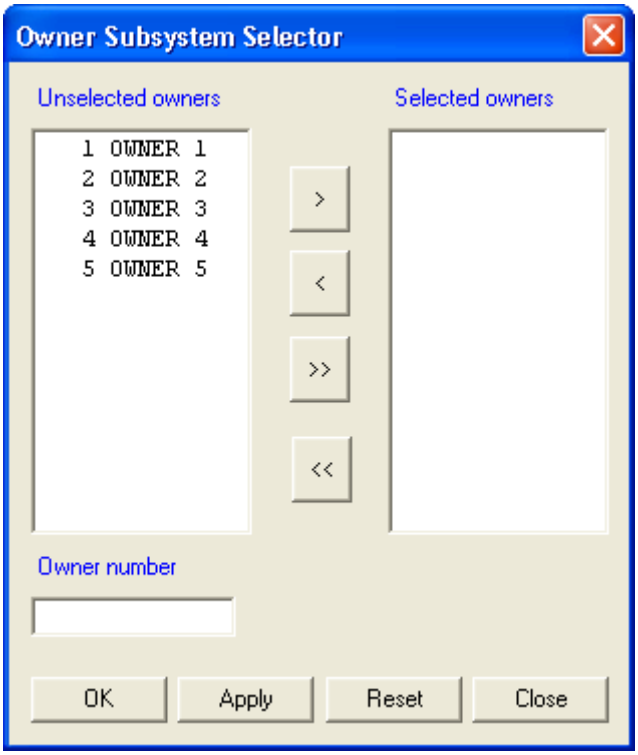


Figure 13-37. Owner Subsystem Selector Dialog

13.6.3 By Zone

Subsystem > Zone…

The **[Zone Subsystem Selector]** dialog is similar to the zone tab in the **[Bus Subsystem Selector]** (see Figure 2-9). The buttons respond the same way, but the *Memorize* and *Recall* options are not available. This dialog is used to limit report output and does not affect the **[Spreadsheet]** display.

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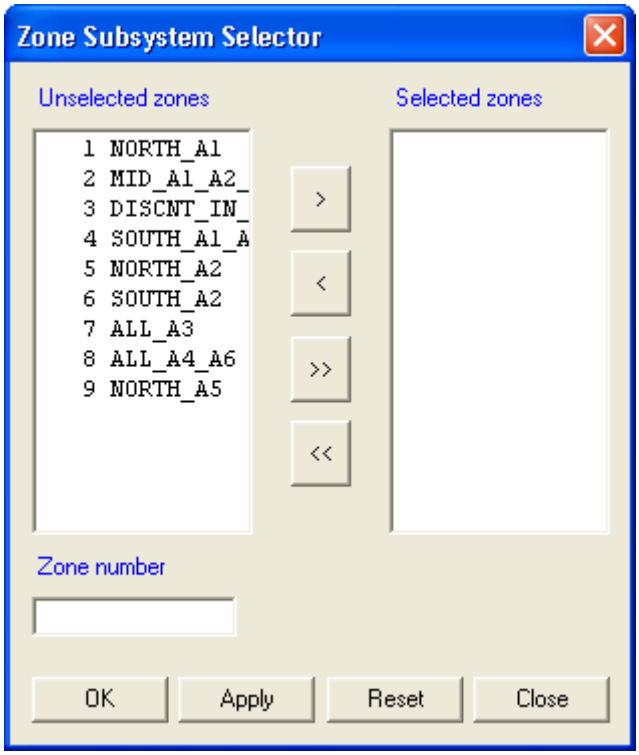


Figure 13-38. Zone Subsystem Selector Dialog

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13.7 Exporting Power Flow Results to Microsoft Excel®



Power Flow > Reports > Export ACCC, PV/QV results to Excel…

OR

Start > Programs > PSSE > Export results to Excel…

The **[Export PSS®Express Data/Result to Excel]** dialog (Figure 13-40) requires specification of the Contingency for export:

- ACCC Analysis Results
- PV Solution Results
- QV Solution Results

If *ACCC Analysis Results* is selected, additional options are available. Click **[Options]** to open **[Specify psseexcel.accc Options]** (Figure 13-39). The *Solution Type* (*Contingency*, *Corrective Action*, or *Tripping*), *Rating*, and filters for *Bus* and *System Mismatch* may be specified.

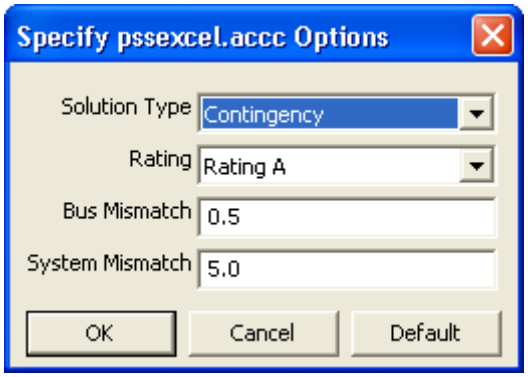


Figure 13-39. Specify psseexcel.accc Options Dialog

Click **[OK]** to return to **[Export PSS®Express Data/Result to Excel]**. Click **[…]** to open selection windows for the file to be exported and the Excel output file, which can be a new file with the default name of the *.acc file, a designated filename, or a previously-built file to be over-written.

Checkboxes are provided to select the desired *Quantities to Export* and contingencies.

Click **[Export]** to complete the action. The Excel® Workbook will open in a new window.

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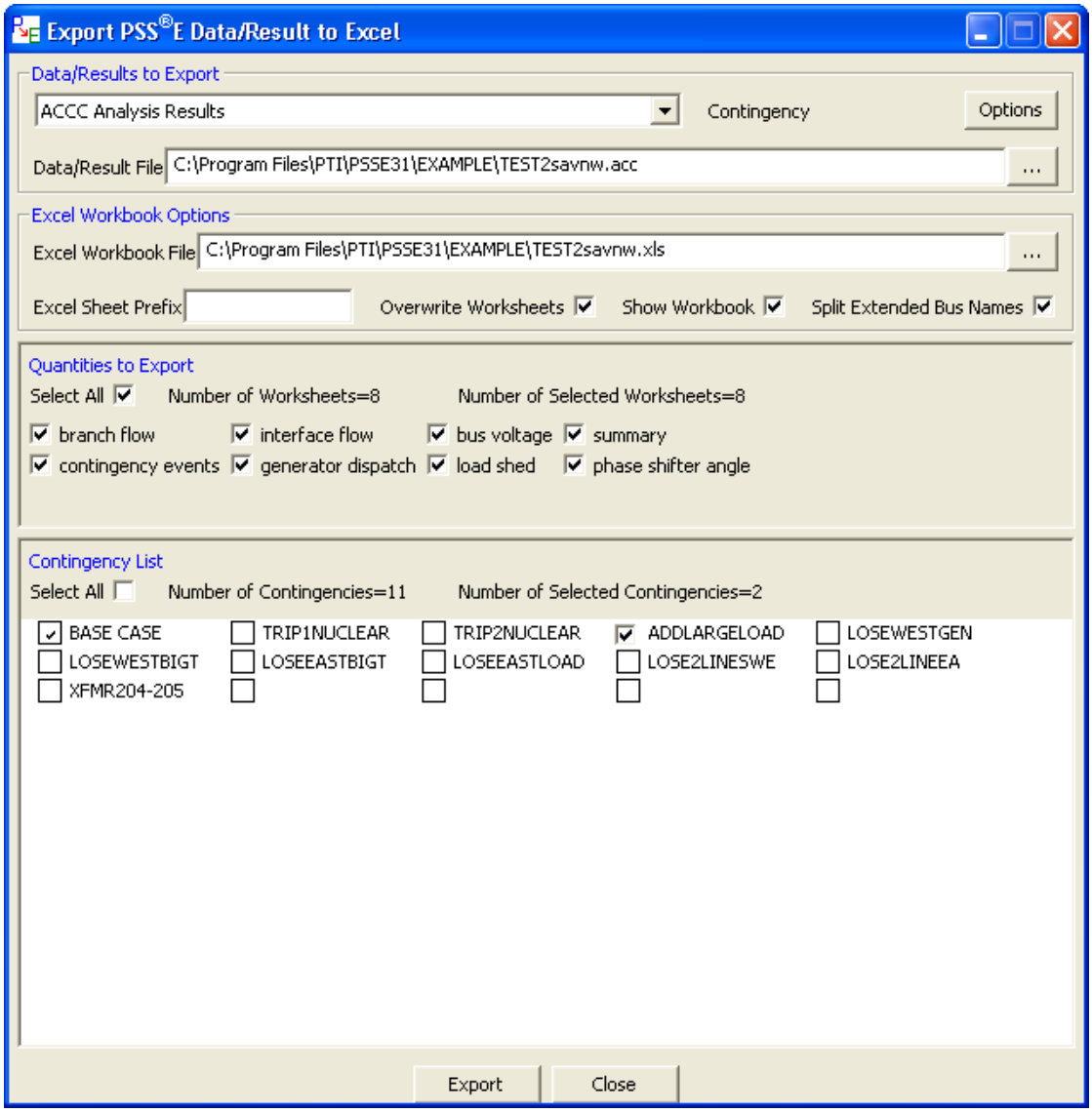


Figure 13-40. Export Data/Result to Excel Dialog

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International.

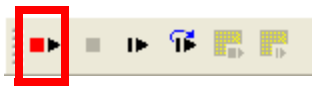
SIEMENS

Chapter 14

Program Automation

14.1 Reproducing User Actions

A sequence of PSS®Express actions for batch commands may be recorded in a Python script (*.py, *.pyc, or *.pyw) or Response file (*.idv). See Section 14.2 for details on specific types of files.



I/O Control > Start recording…

A file selector window opens for specification of the type of file to record and the file name.



I/O Control > Stop recording

When done recording, select ***I/O Control > Stop recording*** to close the file.

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14.2 Creating Program Automation Files

14.2.1 Response File

Activity IDEV



I/O Control > Start recording…

An experienced user can create a response file using a text editor. The simplest and most robust way, however, is to select the ***I/O Control > Start recording*** option. Note that this is equivalent to executing activity ECHO in previous versions of PSS®Express.

Having selected a file, the user executes the required sequence of activities or operations using the menu and toolbar functions in the interface. The resulting response file will contain, in Response File form, a series of commands reflecting the user's sequence of activities. The user can to tailor this basic Response File for subsequent runs by editing the created response file by changing some of the filenames and bus numbers specifying faulted nodes, switched branches, and so on.

As an example it can be assumed that, using the *savnw.sav* power flow case, the user wishes to open one circuit between buses 151 and 152, solve the case, and then display the power flow results for bus 151.

Using the interface:

- Right-click the branch in **[Diagram]** and select *Switch* from the pop-up menu.
- Employ the ***Power Flow > Solution > Solve*** option or the *Solve* toolbar button.
- Select the *Bus Based Output* toolbar button and, subsequently, bus number 151.

If this series of operations were recorded, a response file would be constructed (see Figure 14-1).



Figure 14-1. Created Response File

It can be seen that the Response file contains PSS®Express batch (BAT_) commands. Consequently, the manual creation of a Response File, which could be done with the text editor, requires an intimate familiarity with the PSS®Express batch commands. These commands are described in the *PSS®E API Manual*.

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Additional Information
PSS®E Program Operation Manual, Section 16.12, Running a Response File

14.2.2 Python and IPLAN Files

The process described in the previous section for creating a Response file can also be used to create a Python file (**.py*) by selecting this file type instead of the Response file type (**.idv*). The file type is selected. The program records the sequence of events selected by the user in the same manner.

Alternatively the user can create a Python file using the Python programming language. This will facilitate a more sophisticated control of PSS®Express by enabling looping and decision making based on results of analyses. Standard concepts such as looping and subroutine calling are available. For details on the Python programming language refer to a Python reference book or http://www.python.org.

It is not possible to construct an IPLAN file using the PSS®Express recording facility. The file must be created manually and compiled. It is very beneficial to be familiar with at least one programming language and the standard programming concepts such as assignment statements, looping and subroutine calls.

For details on the IPLAN programming language, the IPLAN language interface to the PSS®Express working case and dynamics data, and the IPLAN compiler, refer to the *IPLAN Program Manual*.

Additional Information
PSS®E Program Operation Manual, Section 16.4, Python Programs

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14.3 Executing Automation Files

14.3.1 Run Automation File



I/O Control > Run program Automation file…

Existing Python scripts (*.py), Response files (*.idv), IPLAN programs (*.irf) and PSAS/PSEB (*.psa/*.pse) command files may be executed from within the user interface. PSS®Express will open a file selector window and prompt for the appropriate filename to execute. Figure 14-2 shows the dialog for selecting an Automation File:



Figure 14-2. Select Program Automation File Dialog

Only five file types are shown. This is because command line input and batch commands can be assembled in the same file, so the **Response File** file type (*.idv) is used for either (or both). When

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the selected file is opened, the processes packaged in the Response file or programmed into the IPLAN or Python file will be initiated.

A **[Terminal Read]** dialog (Figure 14-3) allows specification of data output to a file. Clicking **[OK]** without specifying a filename sends the data to a **Report** tab.

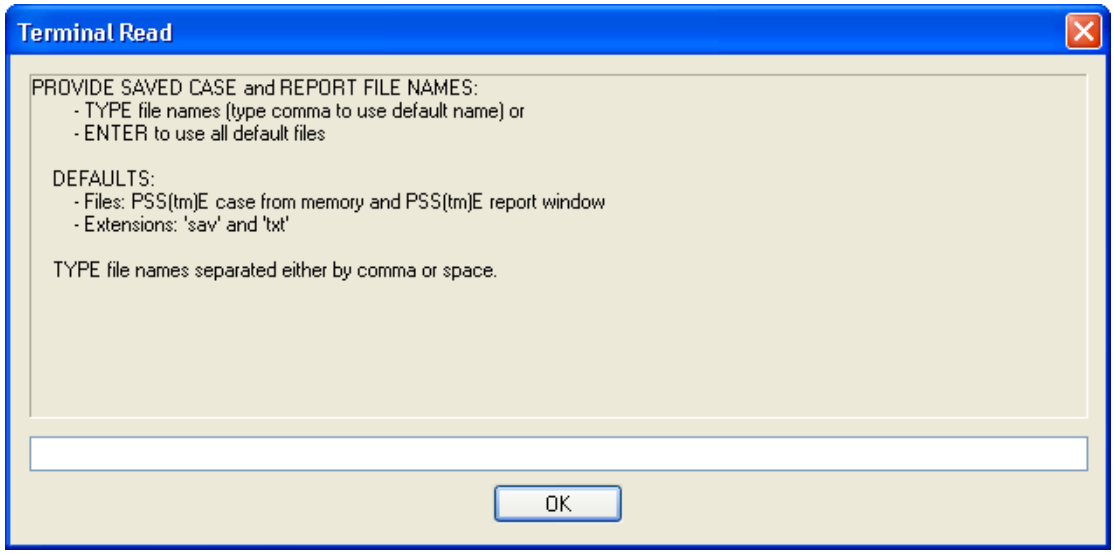


Figure 14-3. Terminal Read Dialog for Automated Activity

14.3.2 IPLAN Program File

Activity EXEC

Requirements / Prerequisites
IPLAN Executable Program File generated by IPLAN language compiler program.

Select the desired IPLAN File (*.irf) from **[Select Program Automation File to Run]**.

For details on the IPLAN programming language, the IPLAN language interface to the PSS®Express working case and dynamics data, and the IPLAN compiler, refer to the *IPLAN Program Manual*.

Additional Information
PSS®E Program Operation Manual, Section 16.16, Launching an IPLAN Program File

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14.3.3 PSEB Command File, Power Flow Calculation

Activity PSEB

Requirements / Prerequisites
PSEB Command file (*.pse)

Select the desired PSEB Command File (*.pse) from **[Select Program Automation File to Run]**.

Additional Information
PSS®E Program Operation Manual, Section 16.14, Building a Response File, Power Flow Calculation

14.3.4 PSAS Command File, Dynamic Simulation

Activity PSAS

Requirements / Prerequisites
PSAS Command file (*.psa)

Select the desired PSAS Command File (*.psa) from **[Select Program Automation File to Run]**.

Additional Information
PSS®E Program Operation Manual, Section 16.15, Building a Response File, Dynamic Simulation

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14.3.5 Argument Passing

These actions allow arguments to be passed to a response file. Highlight the desired file on **[Select Program Automation File to Run]** (Figure 14-2). Click **[Arguments]**. The **[Automation Arguments]** dialog (Figure 14-4) allows a single line of arguments to be input.

Enter the argument string in the text box provided, and click **[OK]**. Click **[Open]** to run the automation file. The response file will be executed within the PSS®Express user interface with results being displayed appropriately to **[Spreadsheet]** and **Report** and **Progress** tabs.

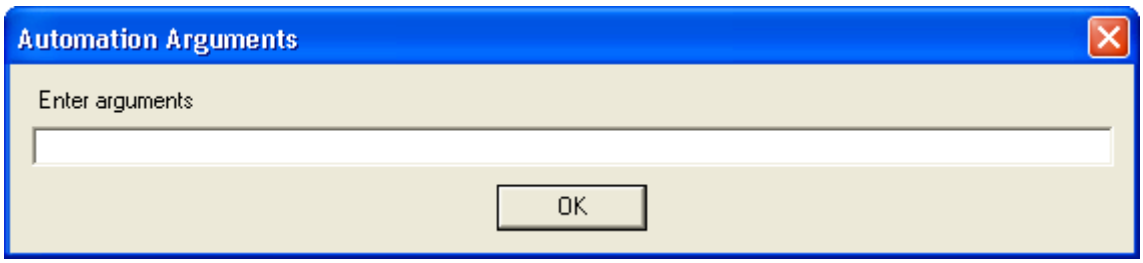
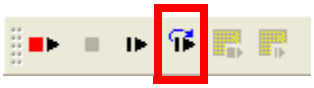


Figure 14-4. Automation Arguments Dialog

Additional Information
PSS®E Program Operation Manual, Section 16.7.1, Arguments in Python Files Section 16.7.2, Arguments in Response Files Section 16.7.3, Arguments in IPLAN Files

14.3.6 Re-Run Last Automation File

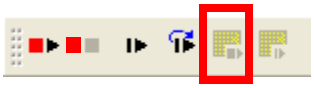


I/O Control > Re-run program Automation file…

When the re-run option is selected, the last-run automation file will run again without reselecting it from a file selector window. The **[Terminal Read]** dialog (Figure 14-3) allows specification of data output to a file. Clicking **[OK]** without specifying a filename sends the data to a **Report** tab.

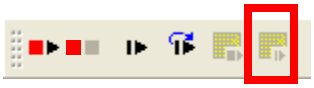
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14.4 Editing an Automation File



I/O Control > Edit last recorded program Automation file

If a text editor has been defined in the **[Preferences]** dialog (see Changing Program Preferences), selecting this option will display the last Automation file recorded by the program. If this is selected, the text editor will display the file as it is created. You may have to refresh the view in the text editor to see the changes in the file.



I/O Control > Edit last run program Automation file

If a text editor has been defined in the **[Preferences]** dialog, selecting this option will display the last Automation file run by the program. If this is selected, the text editor will display the last Automation file run by the program. Selecting this item will re-run the last selected program automation file without forcing you to re-select it.

Additional Information
PSS®E Program Operation Manual, Section 16.13, Recording User Actions

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14.5 Inserting Comments into the Progress Output

Activity TEXT

Misc > Insert text into the Progress stream (TEXT)

During the operation of PSS®Express, a variety of information is sent to the **Progress** tab. This function provides a mechanism for inserting descriptive comments. Comments can be directed to any output device selected for recording the work session (see OPEN). A comment or message is entered directly into the input field in the dialog.

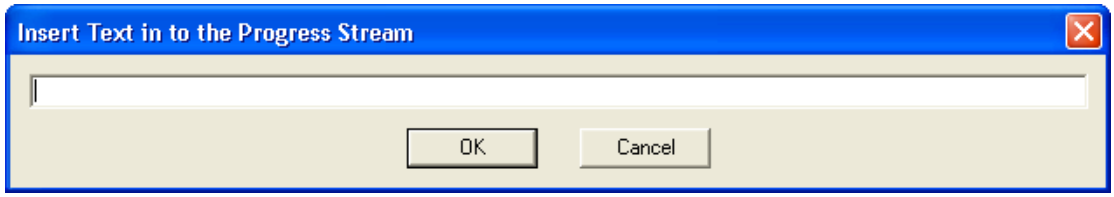


Figure 14-5. Insert Text into the Progress Stream Dialog

Additional Information
PSS®E Program Operation Manual, Section 18.16, Inserting Comments into the Progress Output

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Chapter 15

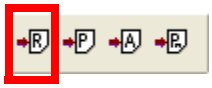
Output Controls

15.1 Directing Solution Output

Output from an interactive session in PSS®Express is typically directed to [Output] view. Since lists and reports can be lengthy, the user may direct the output to alternative devices.

15.1.1 Global Report Destination

Activity OPEN



I/O Control > Direct Report output (OPEN)…

By default, reports are sent to the **Report** tab in [Output] view. Options are available for redirecting reports to a file, printer, or to the current progress device. The option also exists for suppressing all reporting output. Report and progress destinations can be the same.

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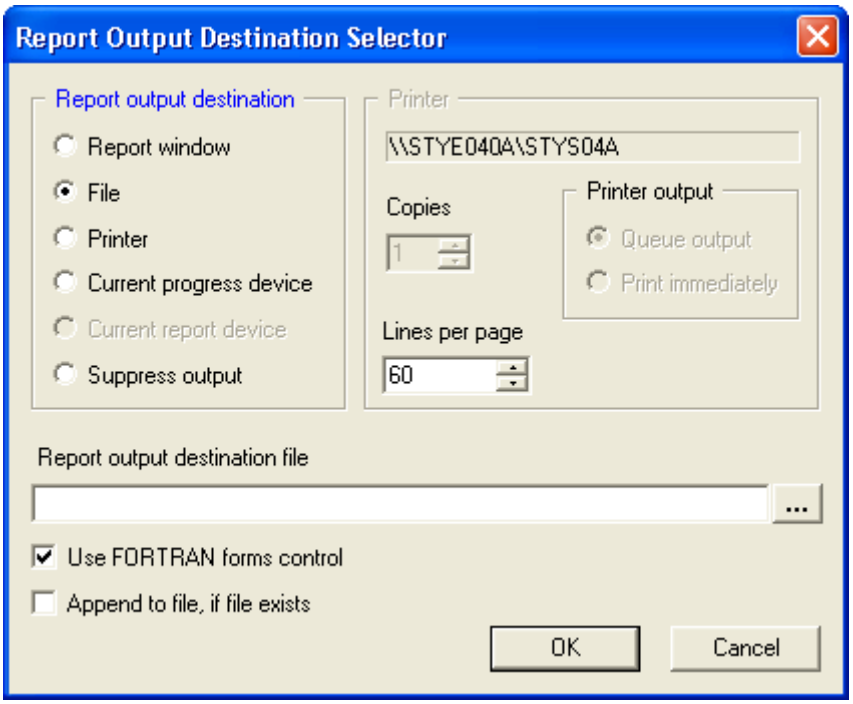


Figure 15-1. Report Output Destination Selector

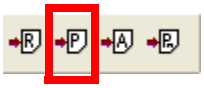
If the Printer option is selected, the user further specifies the printer and printing options. If file output is selected, the user must enter a filename in the *Report output destination file* field. Fortran forms control can be enabled if either file or printer output is specified.

The *Append to file, if file exists* option may be selected if file output is selected. If enabled, then any new report output will be appended to the specified file instead of overwriting it.

Additional Information
PSS®E Program Operation Manual, Section 18.7, Selecting Report Output Destination

15.1.2 Progress Destination

Activity PDEV



I/O Control > Direct Progress output (PDEV)…

This option can be used to direct **Progress** output to an alternative device. Specification for the **[Progress Output Destination Selector]** dialog is essentially the same as that described for directing **Report** output. The only difference is that the default output destination is the **Progress** tab.

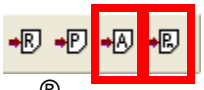
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Additional Information
PSS®E Program Operation Manual, Section 18.6, Selecting Progress Output Destination *PSS®E Program Operation Manual*, Section 4.4, Virtual Output Devices

15.1.3 Alerts and Prompts Destination

Activity ODEV

Requirements / Prerequisites
None, although in practice, activity ODEV would be executed only from a Response File or an IPLAN program. Response File (*.idv)



PSS®Express generated alerts and prompts are sent to **Alerts/Warnings** tab in **[Output]** view. The ***I/O Control > Direct Alert output...*** and ***I/O Control > Direct Prompt output...*** options direct these outputs to an alternate destination or suppress output. The dialogs are similar to those used for redirecting **Progress** and **Report** output.

Additional Information
PSS®E Program Operation Manual, Section 18.4, Selecting Prompt Output Destination *PSS®E Program Operation Manual*, Section 4.4, Virtual Output Devices

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15.2 Setting the Path for Use with & Filenames

Activity PATH

I/O Control > Set path for use with "&" filenames (PATH)…

This function allows you to specify a directory path name that can later be represented by an ampersand (e.g., &myfile). This shorthand method for specifying a path name can be used by PSS®Express file accessing activities. A filename prefixed by this ampersand will be obtained from the directory specified in the path setting.

Select the directory path to be accessed by the & short-hand method by clicking any file in the desired directory (see Figure 15-2).

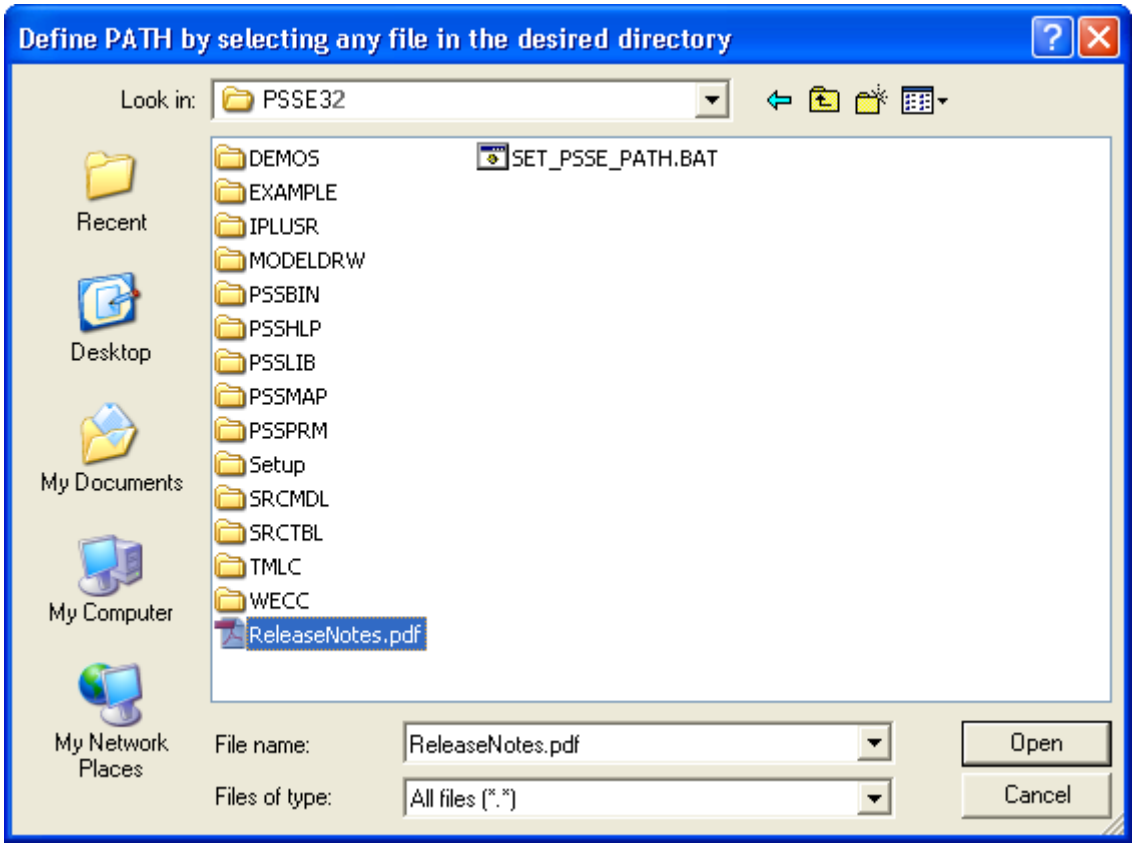


Figure 15-2. Define PATH Dialog

Additional Information
PSS®E Program Operation Manual, Section 18.9, Setting the Path for Use with & Filenames
PSS®E Program Operation Manual, Section 2.4, PSS®E Data Files

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Chapter 16

Scenarios

PSS®Express studies are made up of a myriad number of files. Keeping track of these files has always been the responsibility of the PSS®Express user. A scenario file collects in one place all the files that are used in the study and can be used to automatically open files and perform automated operations upon open and close. A new facility allows the scenario editor to collect all the files contained in the scenario file into a zip archive file. This zip file, which contains all the files identified as being part of the study, can easily be exchanged and the whole study setup recreated on another users machine. When a scenario file is open, accessing files not currently in the scenario will automatically add them to the scenario.

16.1 Creating a Scenario File



File > Scenarios > New Scenario…

The use of any of the Scenario functionality requires the creation of a Scenario file. The location of the Scenario file is important, typically the directory in which most of the files to be used in the study are located. Scenario files are created with, and identified by, a *.pssxml suffix.

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16.2 Building a Scenario



File > Scenarios > Edit Scenario…

When a scenario file is open, accessing files not currently in the scenario will automatically add them. You may also build your scenario interactively by using the scenario editor.

The **[Scenario Editor]** dialog (Figure 16-1) provides a template where files may be added or deleted for a scenario. A dialog to change the root path to the location of the scenario file is also available. Double-clicking a column heading will sort them alphabetically.

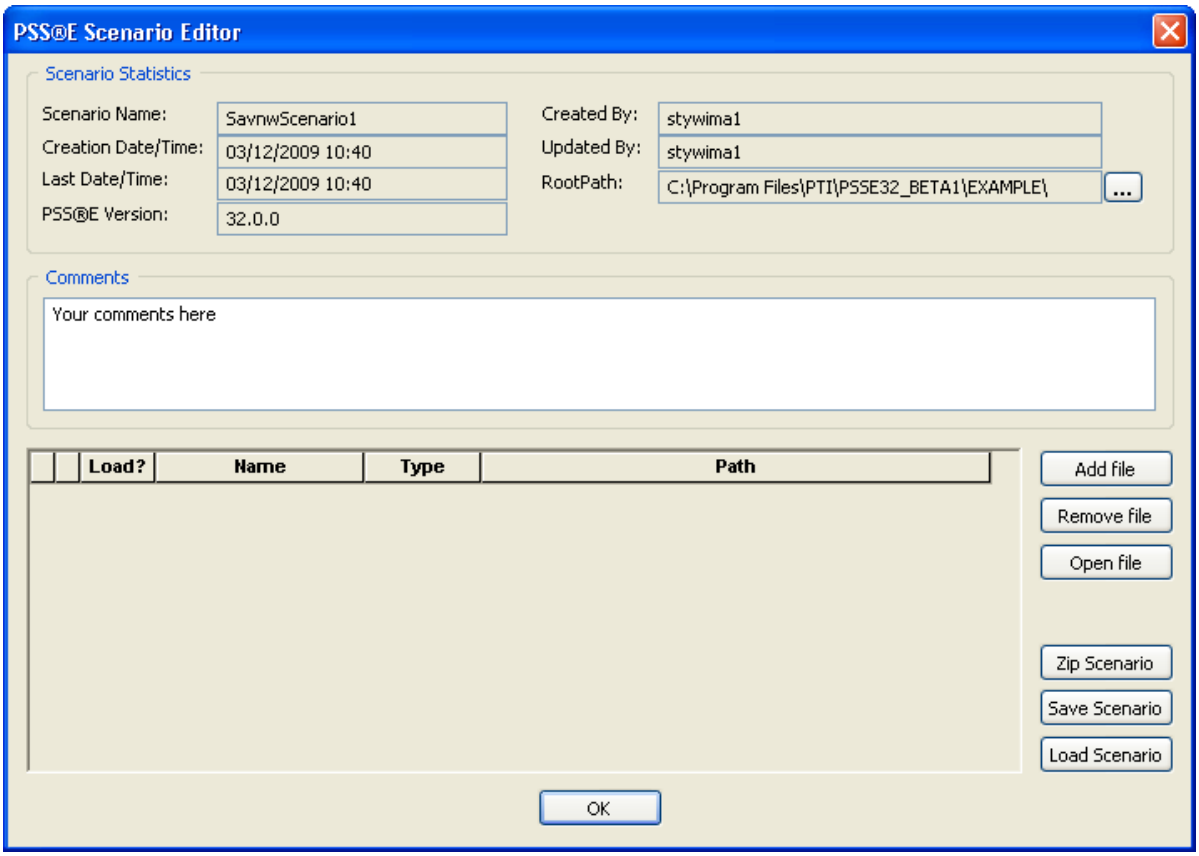


Figure 16-1. Scenario Editor Dialog

Click **[Add file]** to open the file addition dialog (Figure 16-2).

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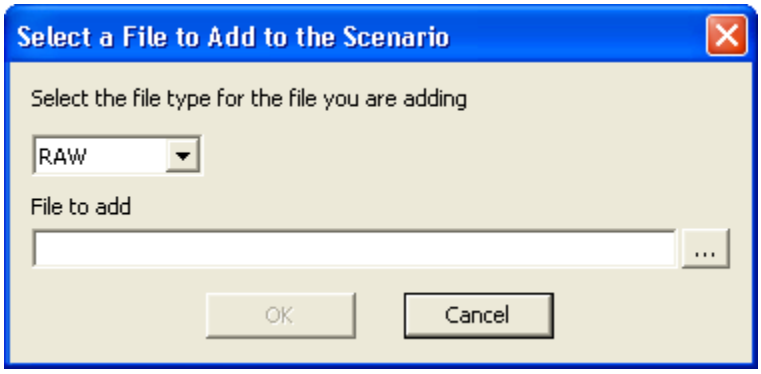


Figure 16-2. Select a File to Add to the Scenario Dialog

The user must specify the file type in order to open the appropriate directory of available files. The following file types may be added to a scenario:

RAW	Power Flow Raw Data file	DFX	Distribution Factor Data file
RAW_OPT	Power Flow Data file, Options	ACC	Contingency Solution Output file
SAV	Saved Case file	SBS	Bus Subsystem file
DYR	Dynamics Model Raw Data file	PYT	Python file
DYR_ADD	Add Dynamics Model Data	IDV	Response file
SNP	Dynamics Snapshot Data file	IRF	Compiled IPLAN Program file
SLD	Slider Binary file	PSE	PSEB Command file
OUT	Channel Output file	PSA	PSAS Command file
SEQ	Sequence Data file	AUTO_OPEN	Python, Response, IPLAN, PSEB or PSAS file
OPF	Optimal Power Flow Data file		
SUB	Subsystem Description file	AUTO_CLOS	Python, Response, IPLAN, PSEB or PSAS file
MON	Monitored Element Description file		
CON	Contingency Description Data file	EVS	Event Study file (*.evs)
		UCT	UCTE Data file (*.uct)

Clicking **[...]** opens the selection window. Highlight the file and click **[Open]** to return to the scenario dialog, and click **[OK]** to add the file to the scenario. Certain file types, such as RAW, SAV, DYR and SNP can only have one instance in a Scenario file. If one already exists, adding a new one replaces the existing file. Other files types, such as SLD, can have multiple instances in the Scenario File.

The checkbox in the **[Load?]** column must be selected to enable loading a file when the scenario is opened.

After all desired files have been added and enabled for loading, click **[Save Scenario]**.

If you want to use the edited scenario at this time, click **[Load Scenario]**.

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You will need to click **[OK]** to end this activity.

All files in the scenario can be compressed into a zip archive for transfer to another machine by clicking **[Zip Scenario]** to open the file compression dialog.

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16.3 Opening an Existing Scenario File



File > Scenarios > Open Scenario…

Opening a scenario loads all enabled files in the selected Scenario file (*.pssxml) into the working case. If another case is active, you will be prompted to save it and any associated Snapshot files.

Any file in the scenario identified as an AUTO_OPEN file will be executed when the scenario file is loaded if the **[Load?]** item is checked. This allows automatic execution of any Automation file that performs study pre-processing.

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16.4 Closing a Scenario File



File > Scenarios > Close Scenario…

Closing the Scenario file will close/unload all files specified in the Scenario file.

Any file in the scenario identified as an AUTO_CLOS file will be executed when the scenario file is closed if the **[Load?]** item is checked. This allows automatic execution of any Automation file that performs study post-processing.

Additional Information
PSS®E Program Operation Manual, Chapter 23, Scenarios

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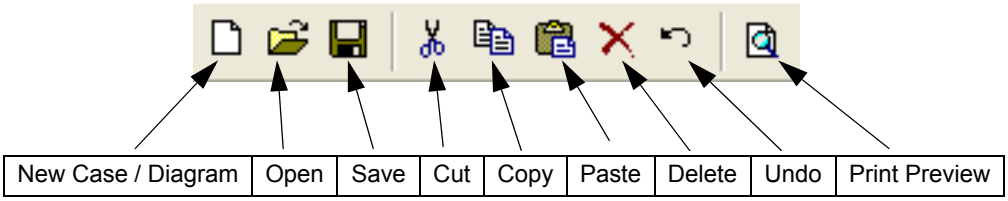
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Appendix A

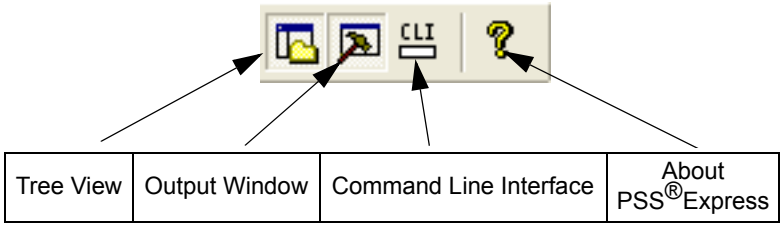
Summary of Toolbar Selections

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A.1 File

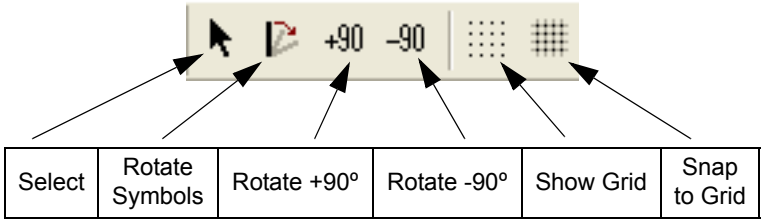


New Case / Diagram	Create a new case or diagram.
Open	Open an existing file.
Save	Save the active documents.
Cut	Cut the selection and put it on the clipboard.
Copy	Copy the selection and put it on the clipboard.
Paste	Insert clipboard contents.
Delete	Delete selected items.
Undo	Undo the previous action.
Print Preview	Print preview.



Tree View	Show or hide the network tree.
Output Window	Show or hide **[Output]**.
Command Line Interface	Show or hide **[Command Line]**.
About PSS®Express	Display program information, version number and copyright.

A.2 Diagram

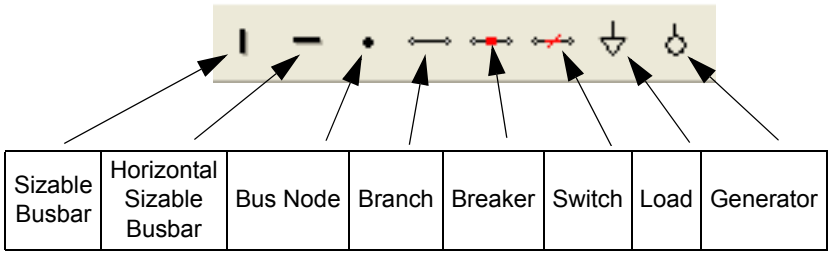


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Select	Select diagram items in **[Diagram]**. Items can be selected using common selection techniques (e.g., dragging a rectangle around several objects, clicking an item and then holding down the **[Ctrl]** key to add more selections to the selection list). The selected items can then be manipulated* in many ways.
Rotation	Rotate diagram items. If the rotation item is selected, and then a diagram item is selected, the cursor changes to a circular arrow. Holding down the left mouse button while dragging the cursor will rotate the selected item around its center.
Rotate +90°	Rotate a selected item positive 90 degrees.
Rotate -90°	Rotate a selected item negative 90 degrees.
Show Grid	Toggle on or off the display of a grid in **[Diagram]**.
Snap to Grid	Toggle on or off the feature that causes the location of any newly created† diagram item to snap to the nearest grid point in **[Diagram]**.

* Manipulation implies that an existing [Diagram] can be modified subsequent to its construction or during its development.

† Creation implies that diagram items, network items or simple annotation items, are selected from the Diagram Toolbar for construction of a new network diagram. This can imply the construction of a new power flow case if the diagram items are bound to network items.

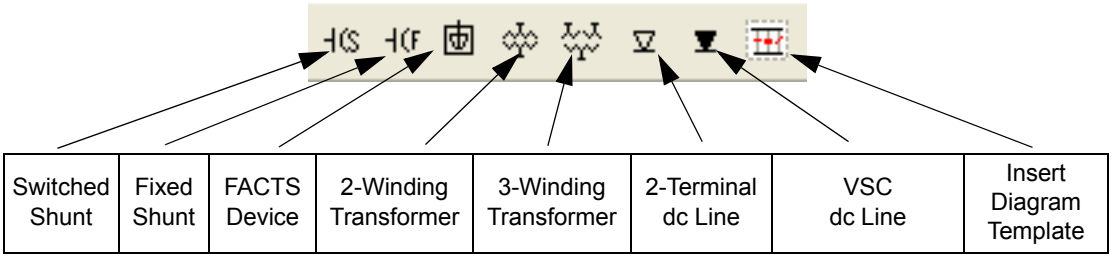


Sizable Busbar	The basic building block of a PSS®Express case and **[Diagram]**. Buses need to exist in a **[Diagram]** before any lines or equipment can be drawn. Buses have a number of discrete ports arranged along both sides of the busbar. When connecting lines and equipment to a bus, the connection point will snap to the nearest port.
Horizontal Sizable Busbar	The same as a regular bus, except that when drawn on the **[Diagram]** it is placed horizontally across the screen instead of vertically.
Bus Node	Select when busbar representation of the bus is not desired. The bus node has a number of ports stacked in the center of the node. When connecting lines or equipment to a bus node, the connection point will snap to the center.
*Branch**	Select to create a line between two buses. When the branch element is selected, the cursor changes to a crosshair. The branch is started by placing the cross-hair on the from bus and clicking. Any number of intermediate kneepoints may then be created by clicking the towards the to bus. Clicking the to bus will complete the creation of the branch.
Breaker	Select to create a line with a breaker between two buses. When the breaker element is selected, the cursor changes to a crosshair. The line is started by placing the cross-hair on the from bus and clicking. Clicking the to bus will complete the creation of the breaker element.

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Switch	Select to create a line with a switch between two buses. When the switch element is selected, the cursor changes to a crosshair. The line is started by placing the cross-hair on the from bus and clicking. Clicking the to bus will complete the creation of the switch element.
Load	Select to create a load on a bus. When the load element is selected, the cursor changes to a crosshair. The load is started by placing the crosshair on the bus and pressing the left mouse button. The mouse is then dragged to where the load symbol is to appear and released.
Generator	Select to create a generator on a bus. When the generator element is selected, the cursor changes to a crosshair. The generator is started by placing the crosshair on the bus and pressing the left mouse button. The mouse is then dragged to where the generator symbol is to appear and released.

* At any point during the creation of a branch, breaker, or switch, the element may be canceled and removed by pressing the **[Esc]** key. The attachment point of an element on a bus may be changed by **[Ctrl]** clicking the attachment point of the element and then moving it to another port on the bus.



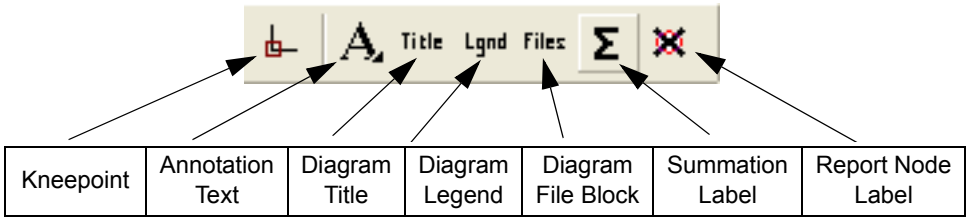
Switched Shunt	Select to create a switched shunt on a bus. When the switched shunt element is selected, the cursor changes to a crosshair. The switched shunt is started by placing the crosshair on the bus and pressing the left mouse button. The mouse is then dragged to where the switched shunt symbol is to appear and released.
Fixed Shunt	Select to create a fixed shunt on a bus. When the fixed shunt item is selected, the cursor changes to a crosshair. The fixed shunt is started by placing the crosshair on the bus and pressing the left mouse button. The mouse is then dragged to where the fixed shunt symbol is to appear and released.
FACTS Device	Select to create a FACTS device between two buses. The FACTS device is started by selecting one or two buses. One bus is selected to create a shunt element FACTS device. Two buses are selected to create a series element FACTS device. When the FACTS device item is selected, the cursor changes to a crosshair. If a single bus was selected, the FACTS device is started by placing the crosshair on the selected bus and pressing the left mouse button. The mouse is then dragged to where the FACTS device is to appear and released. If two buses were selected, the FACTS device is started by placing the crosshair on the sending bus and clicking. Any number of intermediate kneepoints may then be created by clicking the way to the terminal bus. Clicking the terminal bus will complete the creation of the FACTS device. At any point during the creation of the FACTS device, the FACTS device may be canceled and removed by pressing the **[Esc]** key.

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*Two-winding Transformer**	Used to create a two-winding transformer between two buses. The two-winding transformer is started by placing the crosshair on the from bus and clicking. Any number of intermediate kneepoints may then be created by clicking the way to the to bus. Clicking the to bus will complete the creation of the two-winding transformer.
Three-winding Transformer	Select to create a three-winding transformer between three buses. The three-winding transformer is created by first selecting three buses. The three buses will be regarded as the FROM, TO, and last bus in the order they were initially selected. The three-winding transformer item is then selected, the cursor placed in **[Diagram]** at the desired location for the symbol to be placed, and the left mouse button clicked. Any number of intermediate kneepoints may be added to the links between the symbol and the three buses, or the attachment points modified in the manner described above.
Two-terminal dc Line	Select to create a two-terminal dc line between two buses. When the dc line item is selected, the cursor changes to a crosshair. The dc line is started by placing the crosshair on the from bus and clicking. Any number of intermediate kneepoints may then be created by clicking towards the to bus. Clicking the to bus will complete creation of the dc line.
Voltage Source Converter (VSC) dc Line	Select to create a VSC dc line between two buses. When the VSC dc line item is selected, the cursor changes to a crosshair. The VSC dc line is started by placing the crosshair on the from bus and clicking. Any number of intermediate kneepoints may then be created by clicking towards the to bus. Clicking the to bus will complete creation of the VSC dc line.
Insert Diagram Template	Select to open a dialog that displays a list of available diagram templates.

* At any point during the creation of the two-winding transformer, two-terminal dc line, or VSC dc line the two-winding transformer may be canceled and removed by pressing the **[Esc]** key. The attachment point of an element on a bus may be changed by **[Ctrl]** clicking the attachment point of the element and then moving it to another port on the bus.

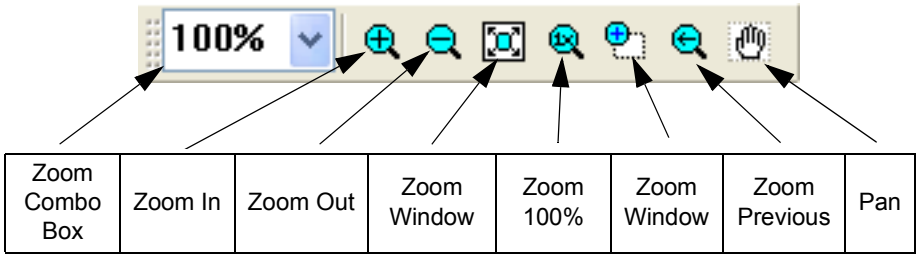
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Kneepoint	Select to put bends, or kneepoints, in a link that connects two diagram items. Links are used to represent lines, the connections between two- and three-winding transformer symbols and buses, and the connections between equipment and buses. When the kneepoint item is selected the cursor changes to a crosshair. Clicking a link will place a red square on the link. This red square can later be dragged with the select item to achieve the desired shape. Kneepoints can be deleted by selecting the kneepoint and pressing the **[Delete]** key.
Annotation Text	Select to place annotation text anywhere on **[Diagram]**. The annotation text item is selected and text can then be placed anywhere in **[Diagram]** by clicking at the location to display the text annotation.
Diagram Title	Select to place a title on **[Diagram]**. The title item displays the two-line diagram title as well as the time and date of the last diagram update on the third line. The time and date is updated on the title item whenever a solution is run and the diagram is open in the application. The title item is selected and titles can then be placed anywhere in **[Diagram]** by clicking the location where the title is to be placed.
Diagram Legend	Select to place a legend on **[Diagram]**. The legend item displays the two-line diagram legend. The legend item is selected and legends can then be placed anywhere in **[Diagram]** by clicking at the location where the legend is to be displayed.
Diagram File Block	Select to place a Diagram File Block on **[Diagram]**. The Diagram File Block contains the current case filename and the current Diagram filename. The files item is selected and file blocks can then be placed anywhere in **[Diagram]** by clicking the location where the file block is to be displayed.
Summation Label	Select to place a summation record on **[Diagram]**. The summation item is selected and summations can then be placed anywhere in **[Diagram]** by clicking. As each summation is placed, the Edit Summation dialog is displayed that allows the setting of the summation records (see Section 3.8.3, Adding a Summation Record).
Report Node Label	Select to place a report node record on **[Diagram]**. The report node item is selected and report nodes can then be placed anywhere in **[Diagram]** by clicking. As each report node is placed, the **[Edit Report Node]** dialog is displayed that allows the setting of the report nodes (see Section 3.8.4, Adding a Report Node).

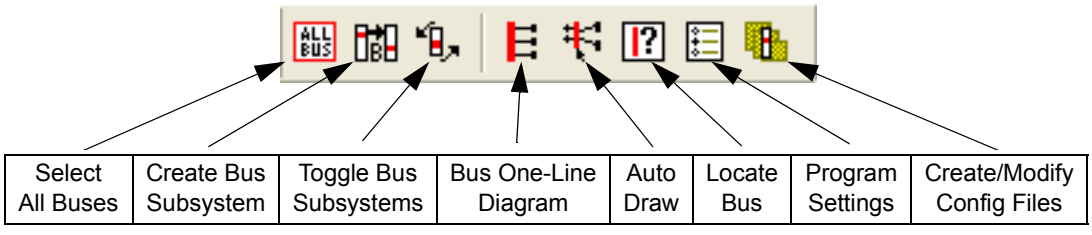
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A.3 Zoom



Zoom Combo Box	Select a predefined magnification to apply to **[Diagram]**. Click the arrow to the right of the combo box for a pull-down list of predefined magnification from which to choose. You can also type in a desired magnification if it is not found in the list. The *Zoom Combo Box* is automatically reflects the current magnification, which can be changed through the use of the other zoom tools.
Zoom in/Zoom out	Increase/decrease magnification of the diagram.
Zoom Extent	Increase/decrease diagram magnification to fill window.
Zoom 100%	Quickly return **[Diagram]** magnification to actual size.
Zoom Window	Increase/decrease magnification of a selected portion of **[Diagram]**. The user selects this item and then drags a rectangle over **[Diagram]** to encompass the part of **[Diagram]** to be zoomed. The selected part of the **[Diagram]** is then zoomed to fit in the extent of **[Diagram]** window.
Zoom Previous	Quickly return the **[Diagram]** magnification to its previous setting.
Pan	Scroll around the **[Diagram]**. Upon selecting this item, the cursor in **[Diagram]** changes to a hand. **[Diagram]** is then panned by holding down the left mouse button and dragging the hand around the window.

A.4 General Task

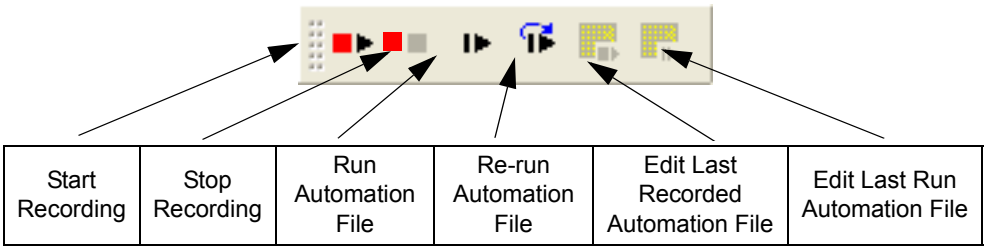


Select All Buses	Select all buses.
Create Bus Subsystem	Create a bus subsystem.
Toggle Bus Subsystems	Toggle between the primary bus subsystem and an alternate.
GOUT/GEXM	Create a graphical power flow bus display (GOUT/GEXM).
Auto Draw	Set interaction mode to automatically draw parts of the network.
Locate Bus	Locate and center the selected bus on the Diagram.

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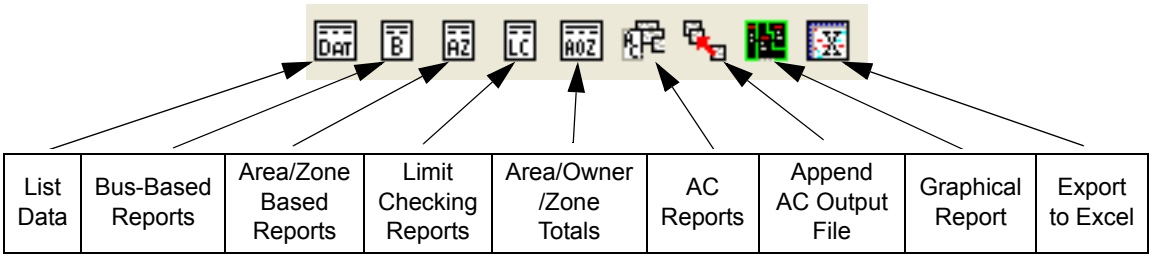
Program Settings	Change general program settings.
Create/Modify Config Files	Create or modify SUB, MON, and CON configuration files.

A.5 Automation



Start Recording	Start recording automation file.
Stop Recording	Stop recording automation file.
Run Automation File	Select and run a program automation file.
Re-run Automation File	Run the last used program automation file.
Edit Last Recorded Automation File	Edit the last automation file recorded.
Edit Last Run Automation File	Edit the last program automation file used.

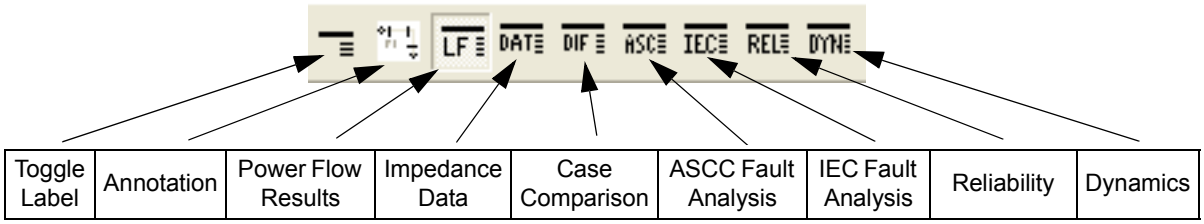
A.6 Reporting



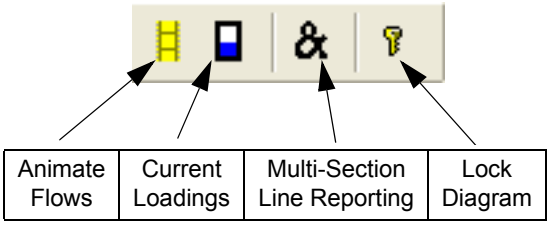
List Data	List Powerflow, sequence and OPF data.
Bus-Based Reports	Generate Bus based reports.
Area/Zone Based Reports	Generate Area/Zone based reports.
Limit Checking Reports	Generate limit checking reports.
Area/Owner /Zone Totals	Generate Area/Owner/Zone total reports.
AC Reports	Generate ac contingency solution output reports.
Append AC Output File	Append to existing ac contingency solution output file.
Graphical Report	Create a graphical report.
Export to Excel	Export report to an Excel spreadsheet.

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A.7 Results (Display on Diagram)

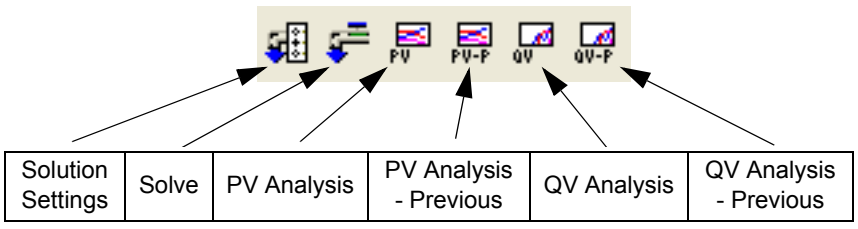


Toggle Labels	Show or hide the display of labels on the active diagram.
Annotation	Edit the diagram annotation on the active diagram.
Power Flow Results	Display power flow results on the active diagram.
Impedance Data	Display power flow impedance data on the active diagram.
Case Comparison	Run graphical case comparison on the active diagram.
ASCC Fault Analysis	Display automatic sequencing fault analysis on the active diagram.
IEC Fault Analysis	Display IEC results on the active diagram.
Reliability	Display reliability results on the active diagram.
Dynamics	Display dynamics results on the active diagram.



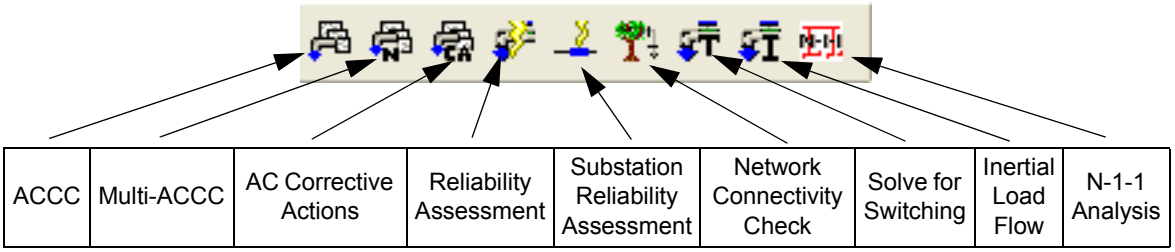
Animate Flows	Animate flows on the active diagram.
Current Loadings	Display current loading bar charts on the active diagram.
Multi-Section Line Reporting	Multisection Line reporting on the active diagram.
Lock Diagram	Lock the diagram.

A.8 Analysis



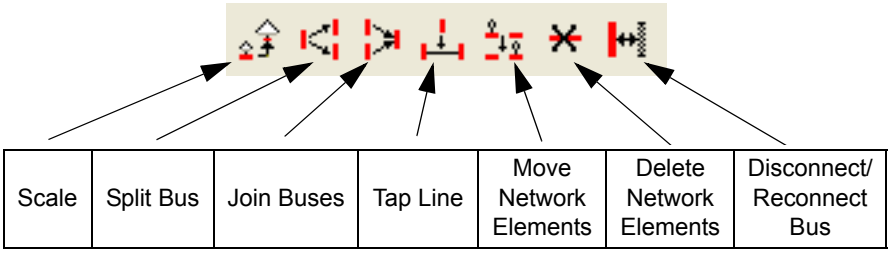
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Solution Settings	Change network solution settings.
Solve	Solve.
PV Analysis	Perform a PV analysis.
PV Analysis - Previous	Perform a PV analysis using previous results.
QV Analysis	Perform a QV analysis.
QV Analysis - Previous	Perform a QV analysis using previous results.



ACCC	Perform ac contingency calculation.
Multi-ACCC	Perform multi-level ac contingency calculation.
AC Corrective Actions	Perform ac corrective actions.
Reliability Assessment	Perform reliability assessment.
Substation Reliability Assessment	Perform substation reliability assessment.
Network Connectivity Check	Perform network connectivity check (TREE).
Solve for Switching	Perform switching study (TYSL).
Inertial Load Flow	Perform Newton-Raphson solution with inertial / governor dispatch (INLF).
N-1-1 Analysis	Perform N-1-1 analysis.

A.9 Topology

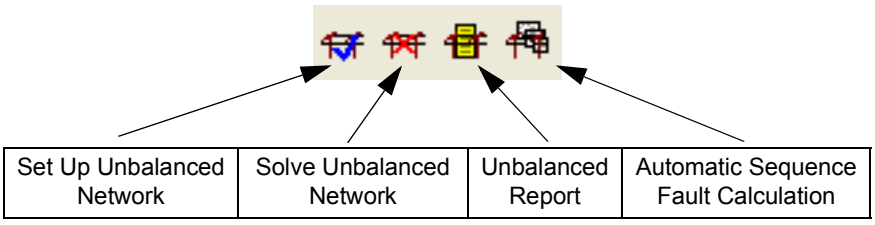


Scale	Scale load, generation, and shunts.
Split Bus	Split a bus into two buses.
Join Buses	Join two buses into one.

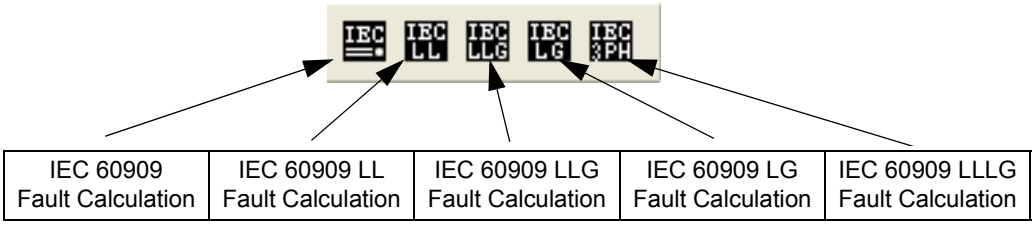
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Tap Line	Insert another bus into a line.
Move Network Elements	Move equipment /lines between buses.
Delete Network Elements	Delete buses/equipment/lines.
Disconnect/Reconnect Bus	Disconnect/Reconnect bus to network.

A.10 Fault Analysis



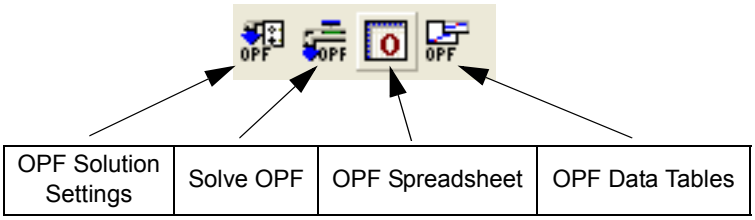
Set Up Unbalanced Network	Setup network for unbalanced solution.
Solve Unbalanced Network	Solve and report network with unbalances.
Unbalanced Report	Unbalanced network report.
Automatic Sequence Fault Calculation	Automatic sequence fault calculation.



IEC 60909 Fault Calculation	IEC 60909 fault calculation.
IEC 60909 LL Fault Calculation	Perform an IEC LL fault calculation.
IEC 60909 LLG Fault Calculation	Perform an IEC LLG fault calculation.
IEC 60909 LG Fault Calculation	Perform an IEC LG fault calculation.
IEC 60909 LLLG Fault Calculation	Perform an IEC LLLG fault calculation.

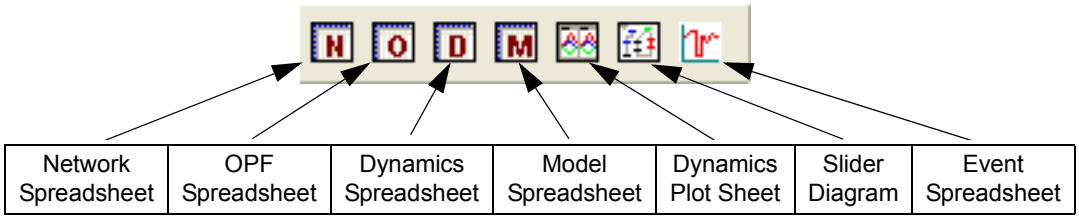
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A.11 Optimal Power Flow



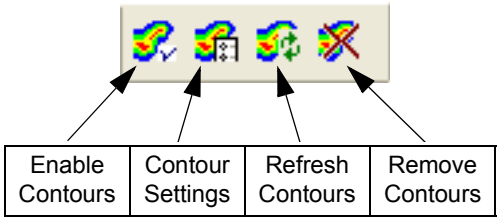
OPF Solution Settings	Change OPF solution settings.
Solve OPF	Solve OPF.
OPF Spreadsheet	Make the OPF spreadsheet active.
OPF Data Tables	Create/modify OPF data tables.

A.12 Activation



Network Spreadsheet	Make the Network spreadsheet active.
OPF Spreadsheet	Make the OPF spreadsheet active.
Dynamics Spreadsheet	Make the Dynamics spreadsheet active.
Model Spreadsheet	Make the Model spreadsheet active.
Dynamics Plot Sheet	Make the Dynamics plot sheet active.
Slider Diagram	Make the Slider Diagram active.
Event Spreadsheet	Make the Event spreadsheet active.

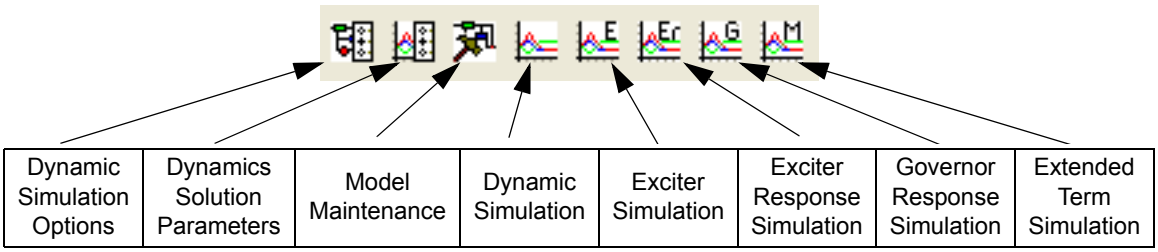
A.13 Contour



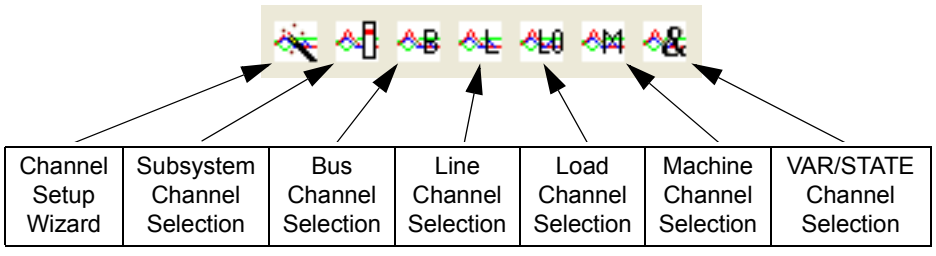
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Enable Contours	Enable color contouring.
Contour Settings	Edit contour settings.
Refresh Contours	Refresh contours.
Remove Contours	Remove contours.

A.14 Dynamics Simulation



Dynamic Simulation Options	Change Dynamic simulation options.
Dynamics Solution Parameters	Change Dynamics solution parameters.
Model Maintenance	Perform network model maintenance functions.
Dynamic Simulation	Initialize and perform a Dynamic simulation.
Exciter Simulation	Initialize and perform an Exciter simulation.
Exciter Response Simulation	Initialize and perform an Exciter response ratio simulation.
Governor Response Simulation	Initialize and perform a Governor Response simulation.
Extended Term Simulation	Initialize and perform an Extended Term simulation.

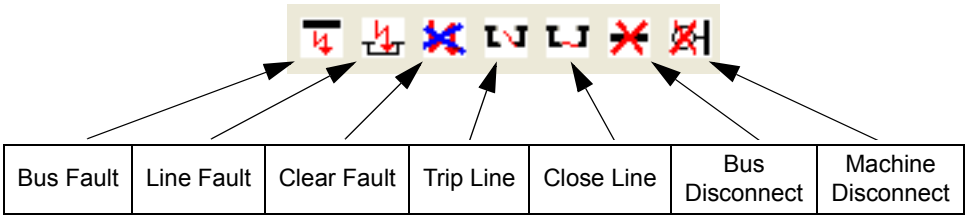


Channel Setup Wizard	Launch Channel Setup Wizard.
Subsystem Channel Selection	Select channels by subsystem.
Bus Channel Selection	Select bus channels.
Line Channel Selection	Select branch and 3 winding transformer channels.
Load Channel Selection	Select load channels.

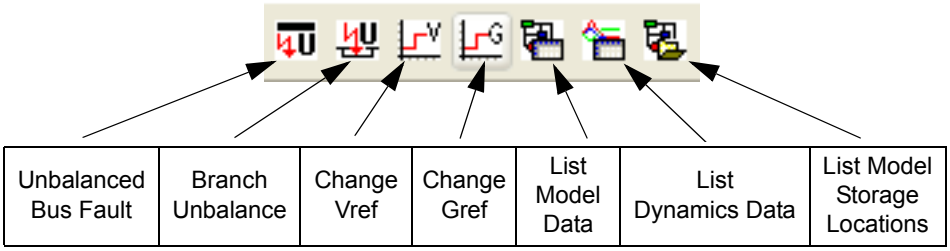
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Machine Channel Selection	Select machine channels.
VAR/STATE Channel Selection	Select VAR/STATE channels.

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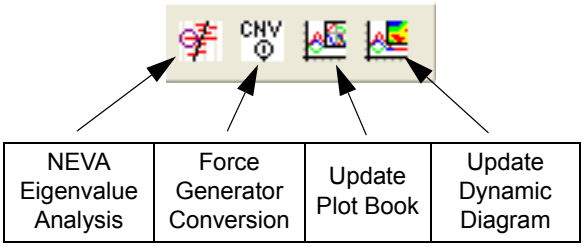


Bus Fault	Apply a bus fault.
Line Fault	Apply a line fault.
Clear Fault	Clear a fault.
Trip Line	Trip a line.
Close Line	Close a line.
Bus Disconnect	Disconnect a bus.
Machine Disconnect	Disconnect a machine.



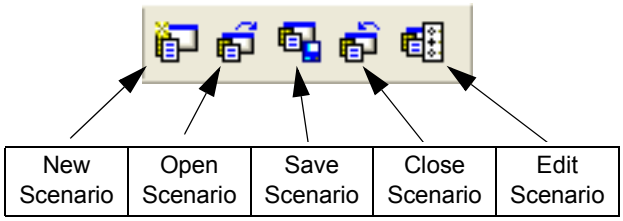
Unbalanced Bus Fault	Calculate and apply an unbalanced bus fault.
Branch Unbalance	Calculate and apply a branch unbalance.
Change Vref	Change or increment Vref.
Change Gref	Change or increment Gref.
List Model Data	List model and data by model type.
List Dynamics Data	List Dynamics variables in network.
List Model Storage Locations	List model storage locations by type and subsystem.

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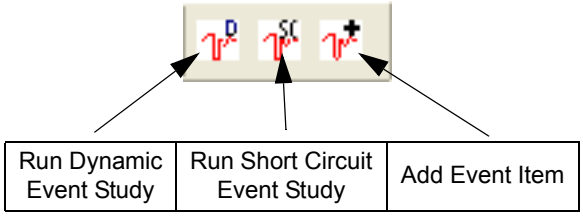
NEVA Eigenvalue Analysis	Launch NEVA Eigenvalue analysis module.
Force Generator Conversion	Force generator conversion when initializing Dynamic Simulation.
Update Plot Book	Update Plot Book during Dynamic Simulation if *Plot every <n> time steps* option is specified.
Update Dynamic Diagram	Update Dynamic Diagram during Dynamic Simulation if *Plot every <n> time steps* option is specified.

A.15 Scenarios



New Scenario	Create new scenario.
Open Scenario	Open existing scenario.
Save Scenario	Save open scenario.
Close Scenario	Close open scenario.
Edit Scenario	Edit existing scenario.

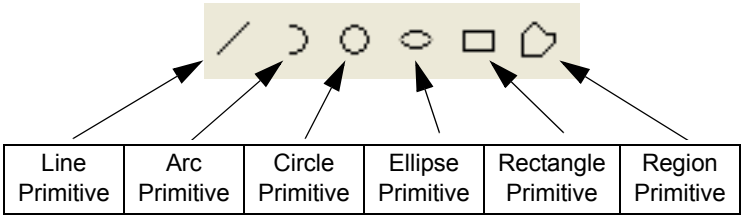
A.16 Event Studies



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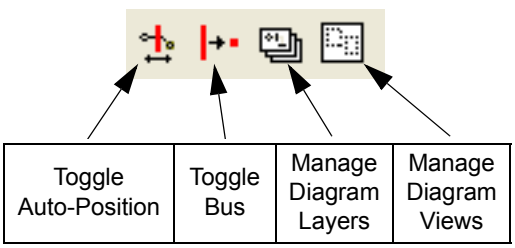
Run Dynamic Event Study	Launch dynamic event study.
Run Short Circuit Event Study	Launch short circuit event study.
Add Event Item	Add event item to existing event study.

A.17 Diagram Primitives



Line Primitive	Select to place annotation lines anywhere on **[Diagram]**. The line primitive item is selected and lines can then be placed anywhere in **[Diagram]** by clicking. Be aware, they are not branches!
Arc Primitive	Select to place annotation arcs anywhere on **[Diagram]**. The arc primitive item is selected and arcs can then be placed anywhere in **[Diagram]** by clicking.
Circle Primitive	Select to place annotation circles anywhere on **[Diagram]**. The circle primitive item is selected and circles can then be placed anywhere in the **[Diagram]** by clicking.
Ellipse Primitive	Select to place annotation ellipses anywhere on **[Diagram]**. The ellipse primitive item is selected and ellipses can then be placed anywhere in **[Diagram]** by clicking.
Rectangle Primitive	Select to place annotation rectangles anywhere on **[Diagram]**. The rectangle primitive item is selected and rectangles can then be placed anywhere in **[Diagram]** by clicking.
Region Primitive	Select to place annotation regions anywhere on **[Diagram]**. The region primitive item is selected and multi-segment polygon regions can then be placed anywhere in **[Diagram]** by clicking the end point for each edge of the polygon. The region is complete when the last point clicked is on the first point.

A.18 Diagram Options



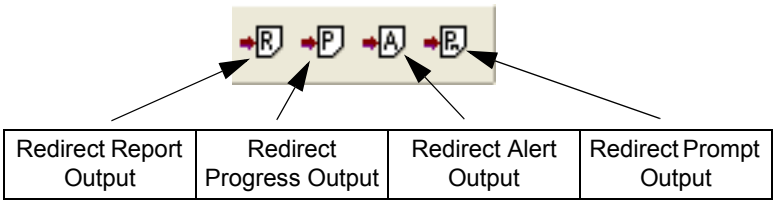
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Toggle Auto-Position	Toggle auto-position for selected elements.
Toggle Bus	Toggle bus symbol orientation.
Manage Diagram Layers	Manage order of **[Diagram]** layers.
Manage Diagram Views	Manage **[Diagram]** saved views.

A.19 Custom Toolbar

Refer to Section 5.3, Creating Custom Toolbars to establish functions for a custom toolbar.

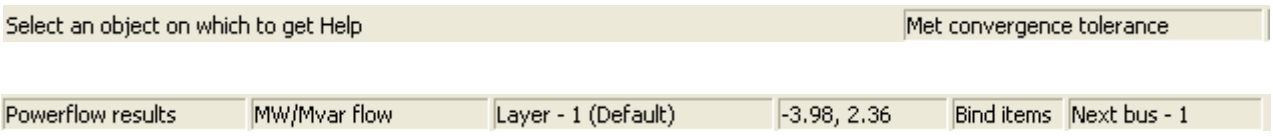
A.20 Redirect Output



Redirect Report Output	Redirect reports to a file, printer, or to the current progress device; suppress all reporting output.
Redirect Progress Output	Redirect progress to a file, printer, or to the current progress device; suppress all progress output.
Redirect Alert Output	Redirect alerts to an alert box, file, printer, or to the current progress or reporting device; suppress all alerts output.
Redirect Prompt Output	Redirect prompts and warnings to a prompt window, file, printer, or to the current progress or reporting device; suppress all prompts output.

A.21 Status Bar

The **Status** bar fields, when in view, indicate various conditions for the user's reference. Interactions with **[Diagram]** cause updates to items in the fields. The fields are listed below.



Select an object on which to get Help	Pops up tooltip on diagram component under cursor.
Met convergence tolerance	Notification that the last activity run met the specified limits.

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Powerflow results	Indicates the type of results currently shown in **[Diagram]**. These include, power flow, impedances, case differences, and short circuit results.
MW/Mvar flow	Indicates units currently in use for the indicated type of results.
4.08, -1.49	The current cursor location in **[Diagram]** is constantly updated in the **Status** bar. The X and Y location of the cursor are displayed in the units specified for **[Diagram]** in **[Diagram Settings]**.
Bind items	The current binding mode for new items created in **[Diagram]** is displayed in the **Status** bar. The field displays either *Bind items*, indicating that new network items added to **[Diagram]** will create new data in the working case, or *No binding*, indicating that no data is added to the working case. Items added when this field displays *No binding* will be displayed in the color red, indicating no corresponding case data exists.
Next bus - 1	The next bus number for new buses created in **[Diagram]** is also displayed in the **Status** bar. Autonumbering is handled with the Power Flow Options menu; see Section 11.1, Specifying Solution Parameters. If autonumbering is on, this field displays the number of the next bus created. If autonumbering is off, this field displays *Autonumber disabled*.

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Appendix B

Network Component Data Records

Following are screen captures of the data records available when PSS®Express-32.0.0 was released. Since data records are a new feature, other network components may be updated using these dialogs, depending upon the version of PSS®Express you are using.

See Section 3.6.1, Using Data Records for a description of their use.

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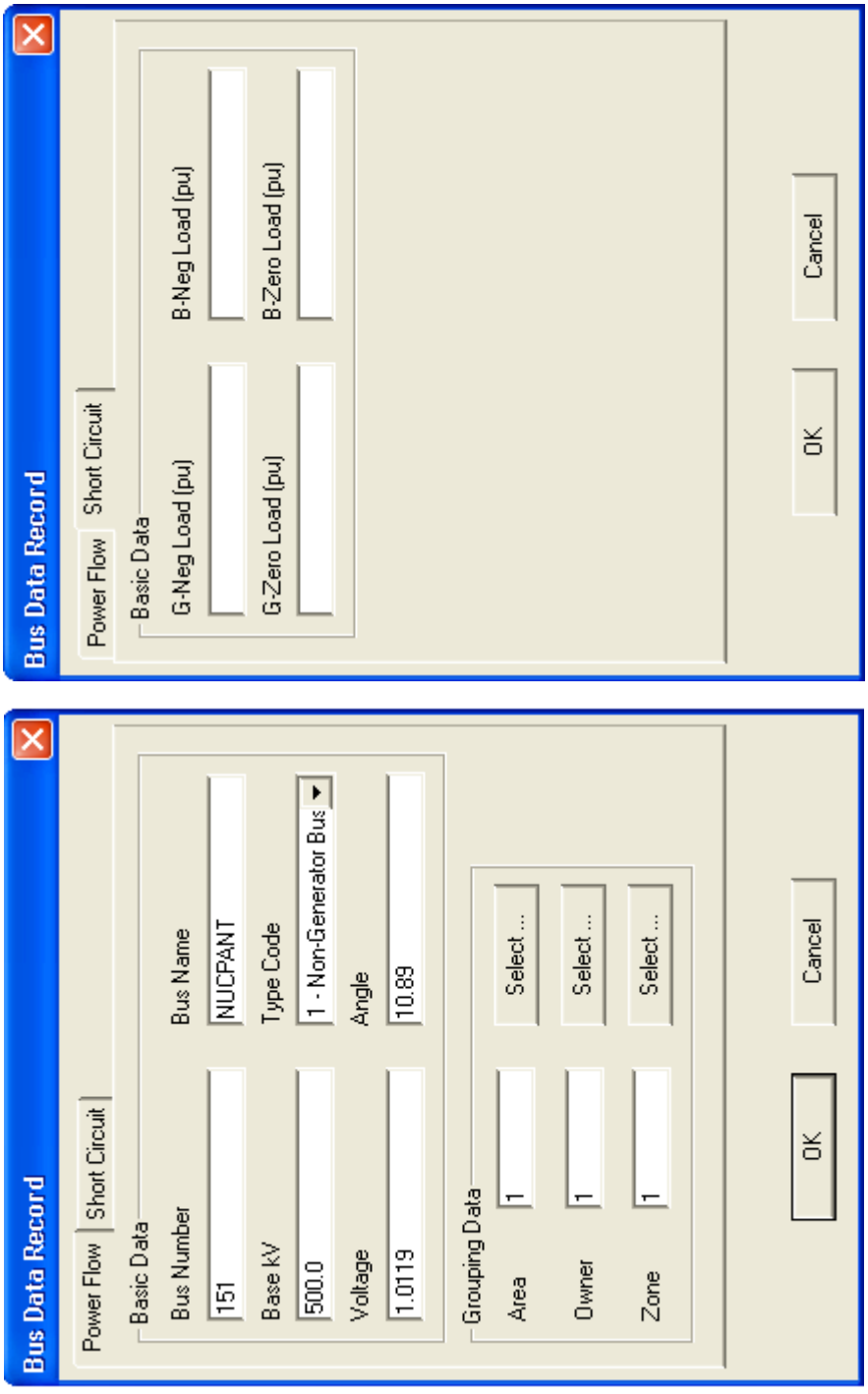


Figure B-1. Bus Data Records

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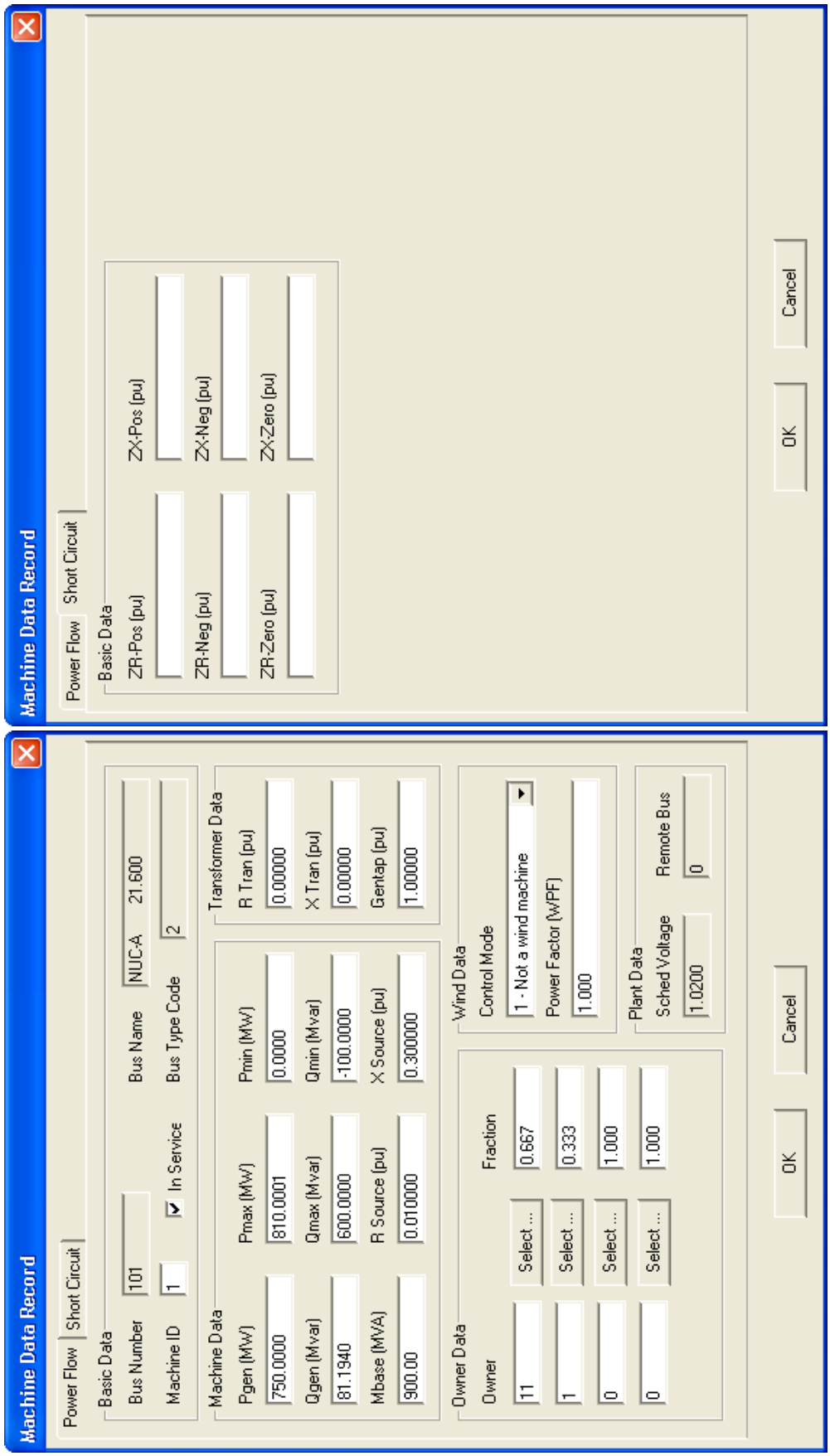


Figure B-2. Machine Data Records

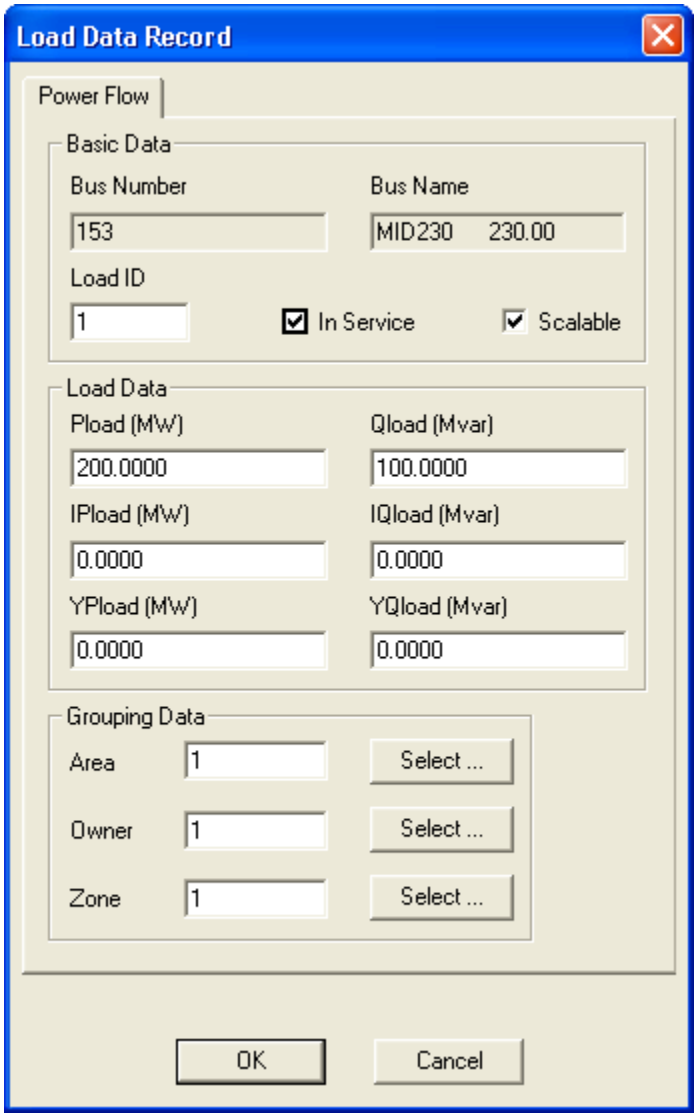


Figure B-3. Load Data Record

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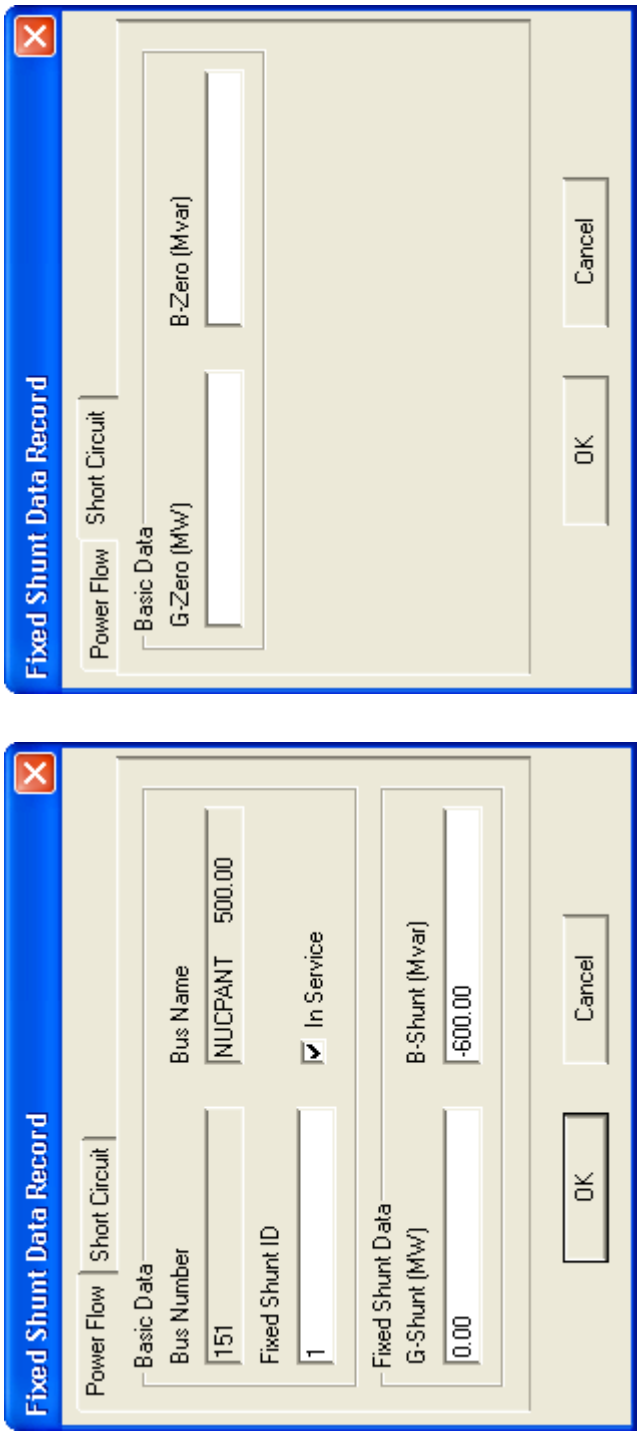


Figure B-4. Fixed Shunt Data Records

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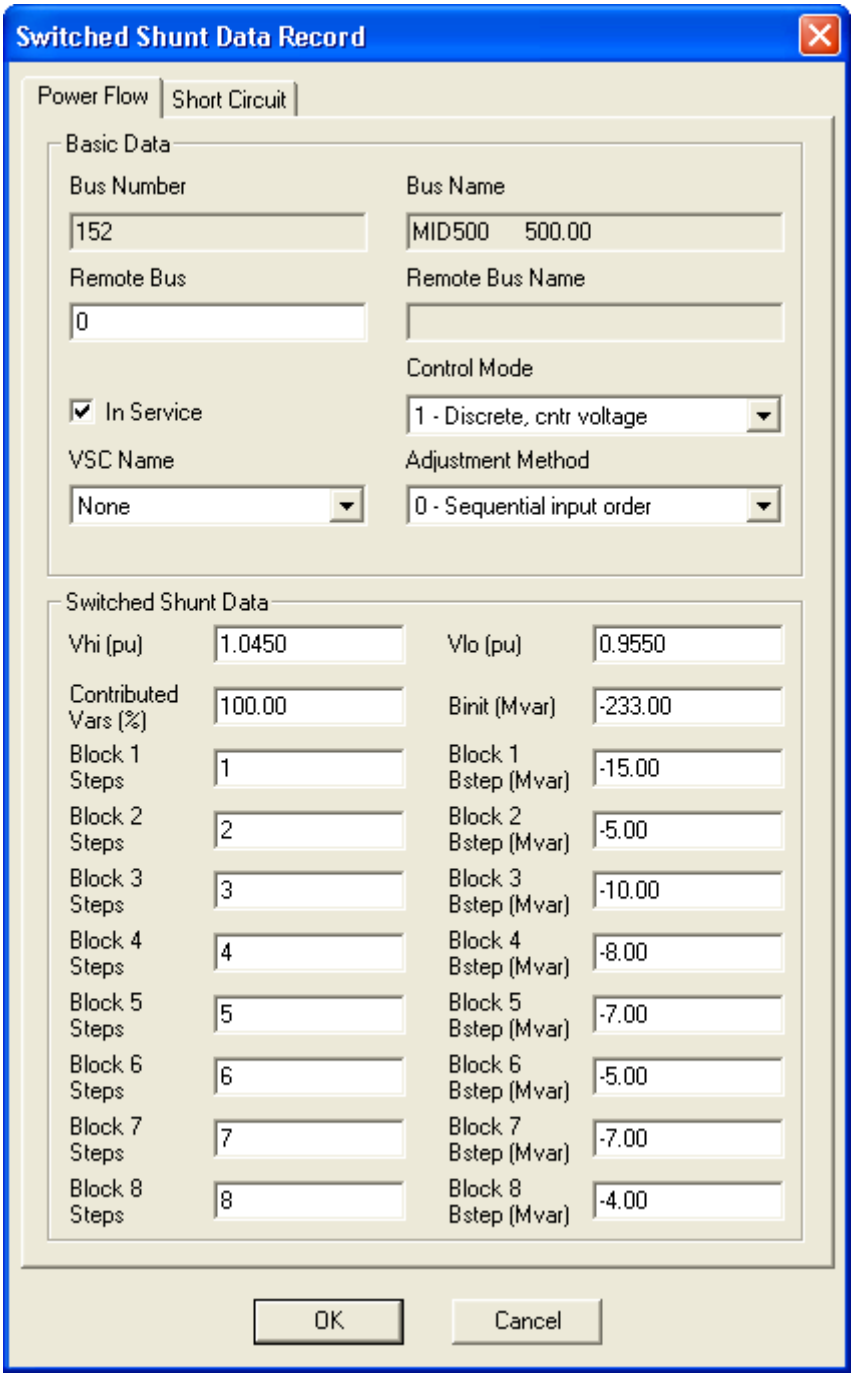


Figure B-5. Switched Shunt Data Record, Power Flow

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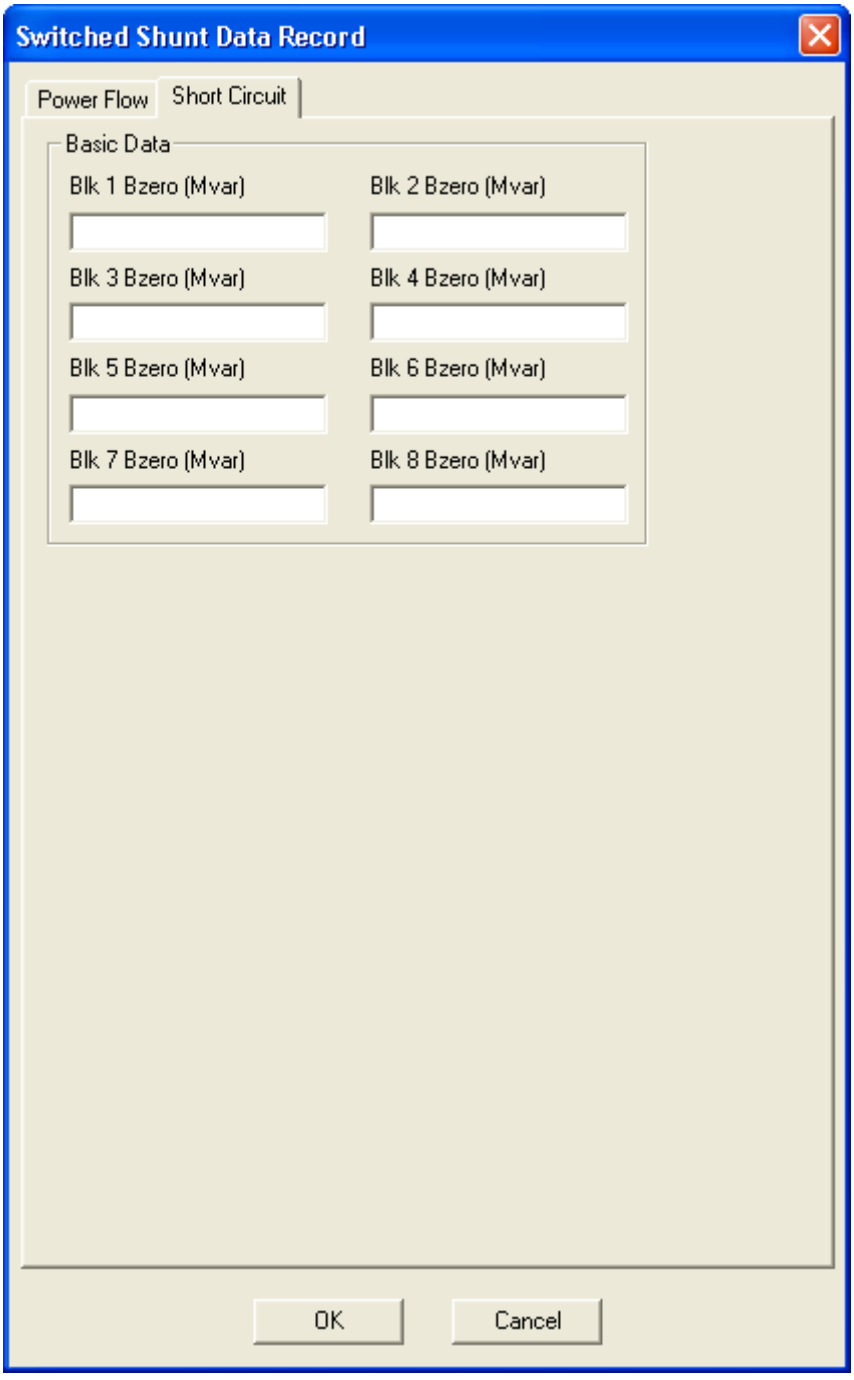


Figure B-6. Switched Shunt Data Record, Short Circuit

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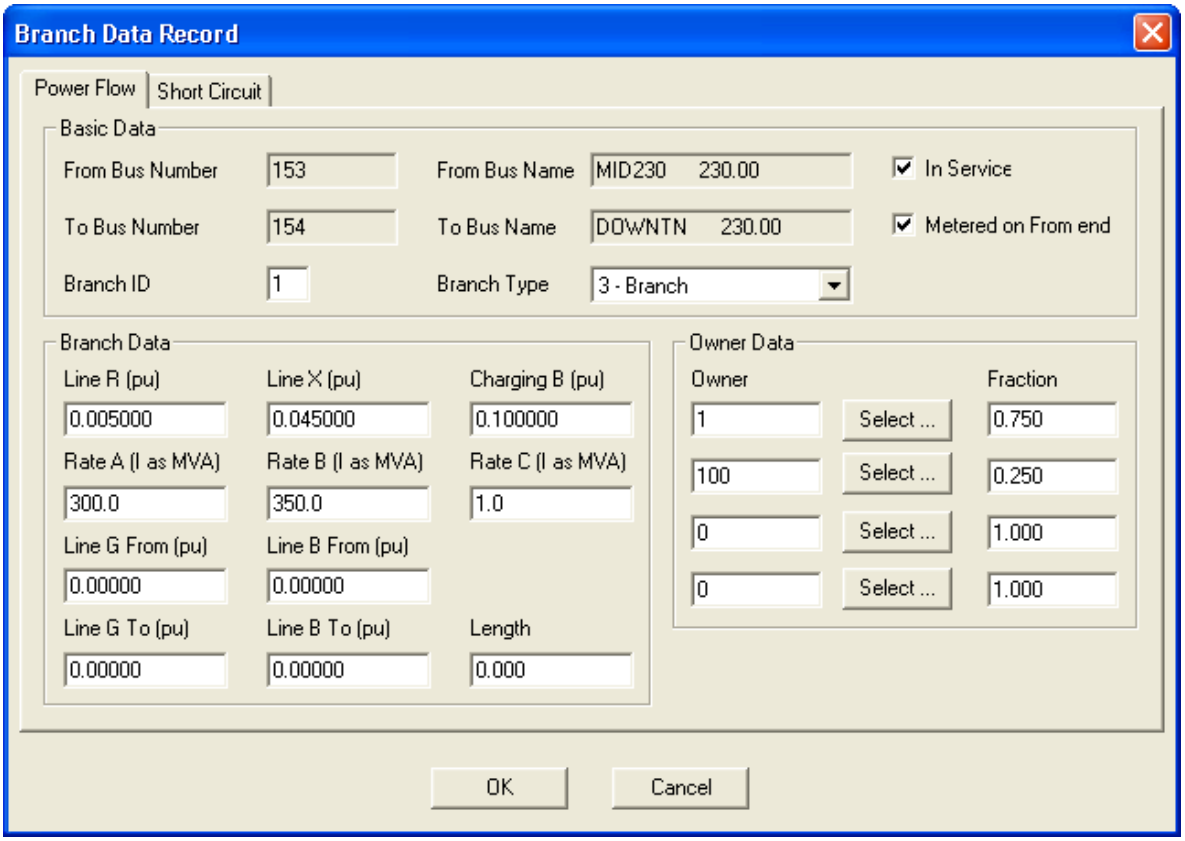


Figure B-7. Branch Data Record, Power Flow

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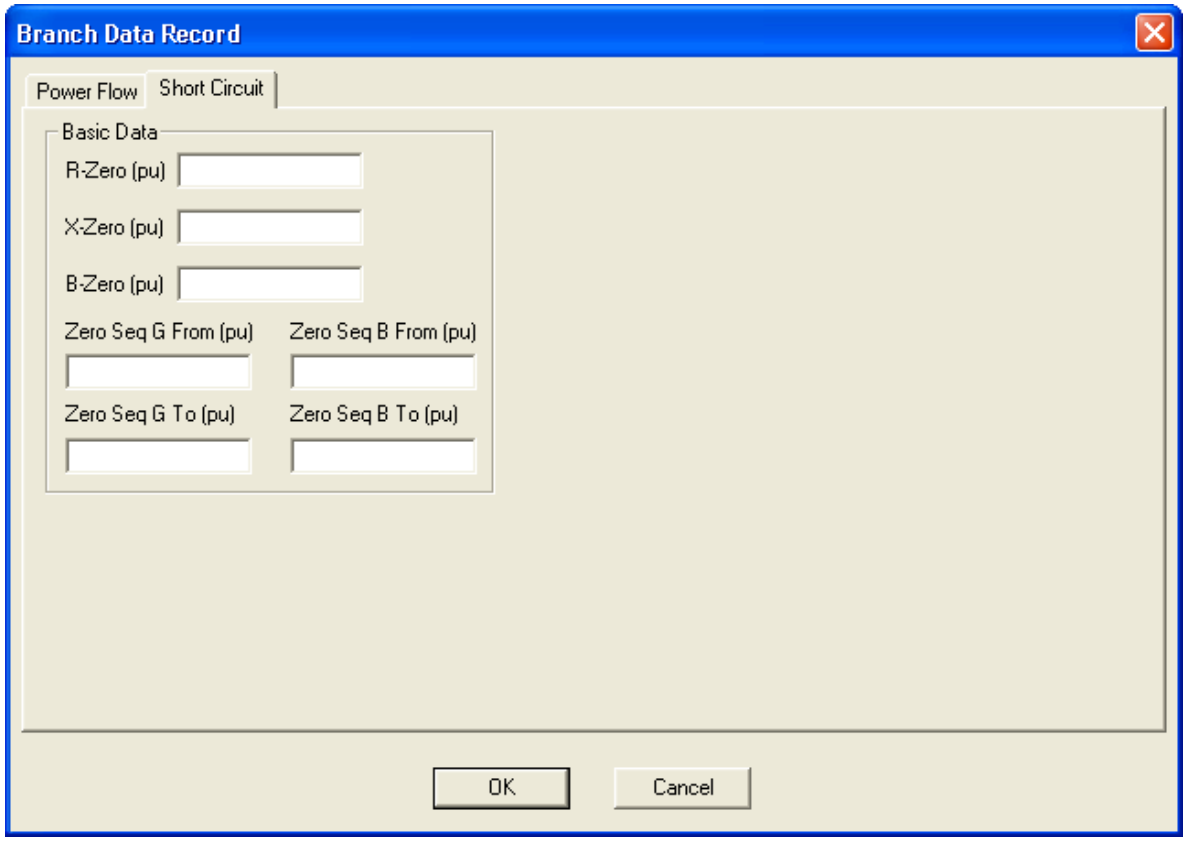


Figure B-8. Branch Data Record, Short Circuit

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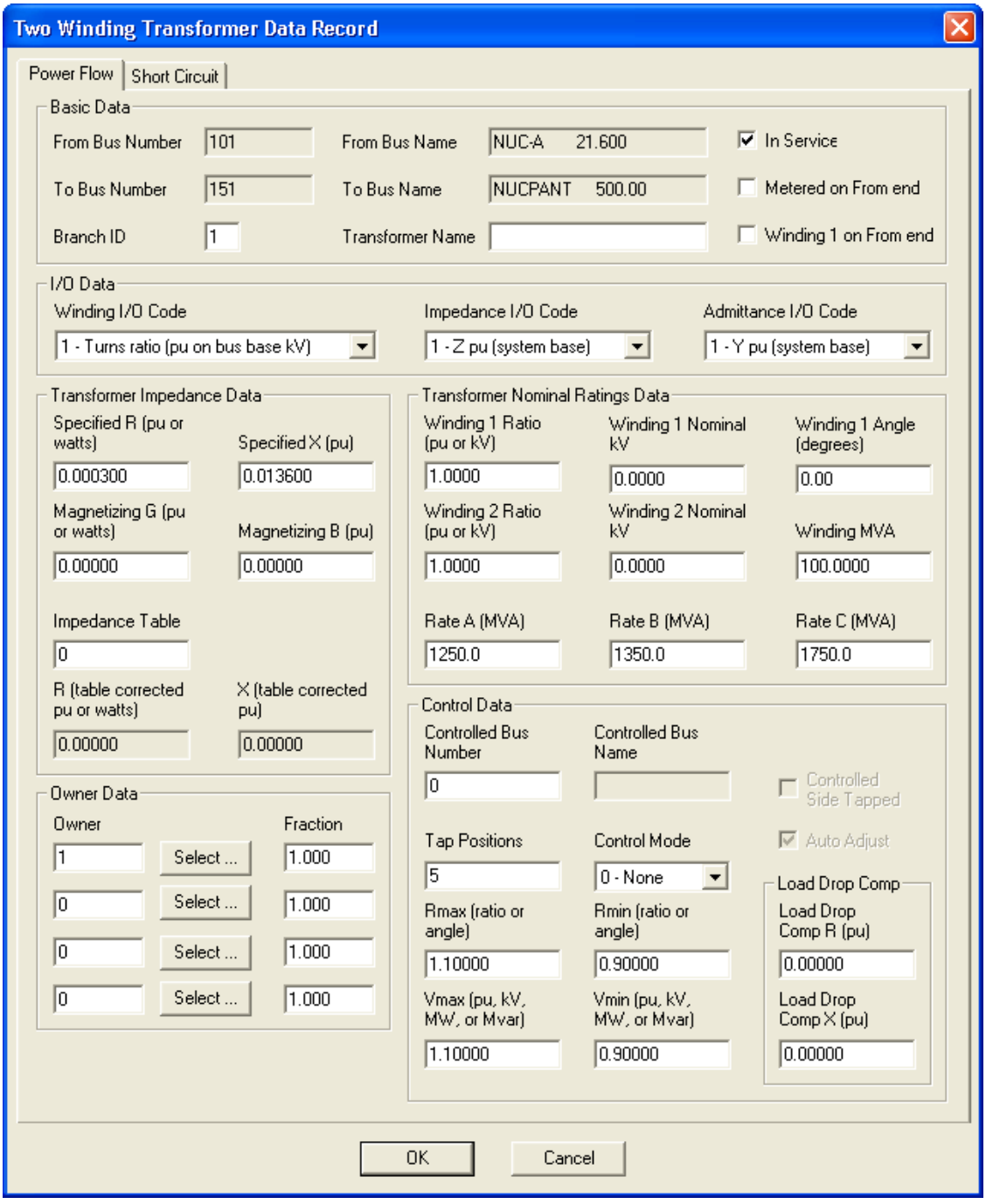


Figure B-9. Two Winding Transformer Data Record, Power Flow

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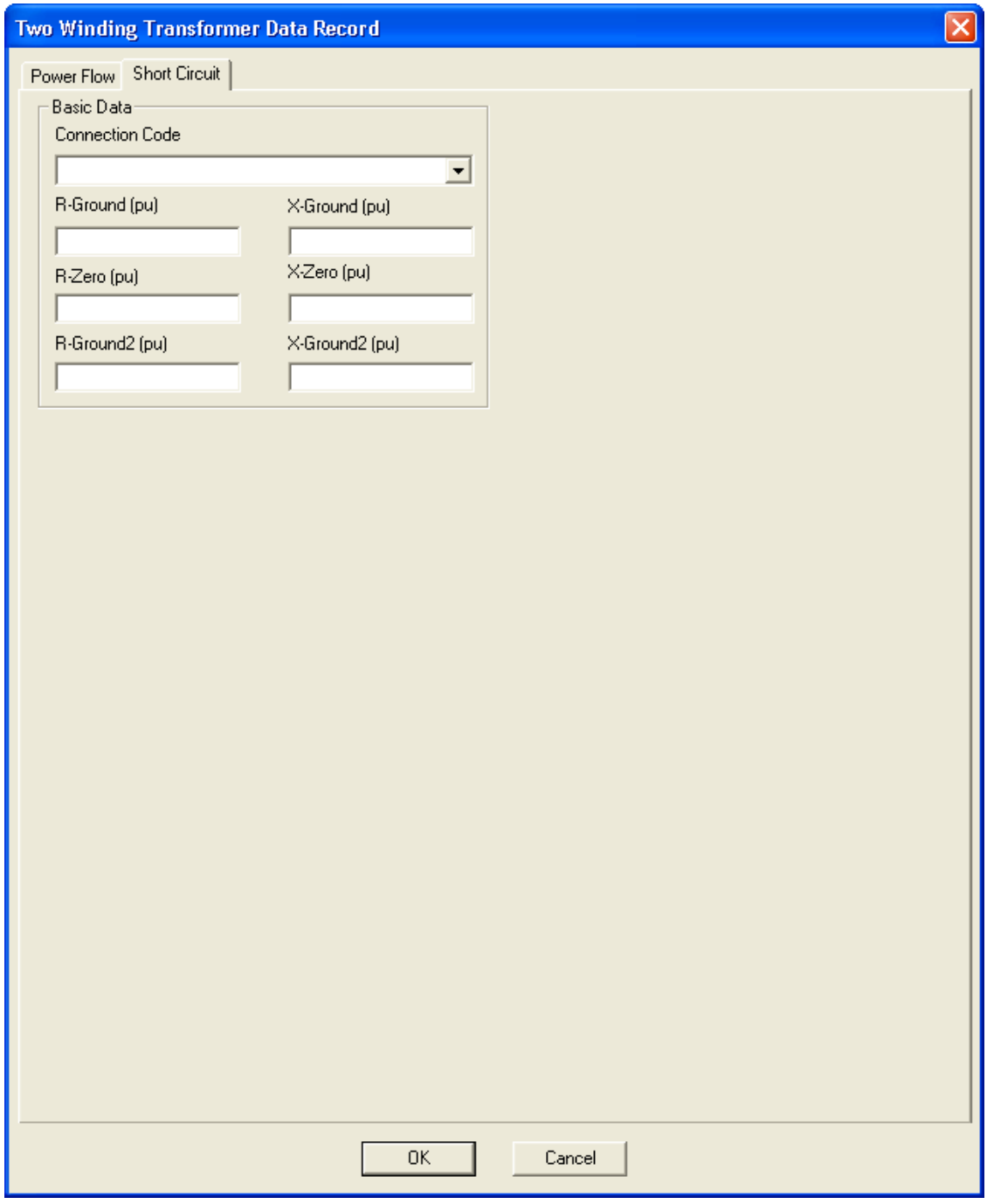


Figure B-10. Two Winding Transformer Data Record, Short Circuit

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